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Preface

The science of biology is truly one of the most dynamic and rapidly expanding human endeavors. Knowledge in the biological sciences is relevant not only to students' lives but also to the future of the humanity. An important aspect of learning biology is participating in the process of science and developing creative and critical reasoning skills. Our goal in choosing the experiments in this laboratory manual is to encourage participation in the scientific process and to make prediction before they initiate laboratory work. Students are required to experience their ability in doing experiments, the satisfaction of solving problems, synthesize results from their observations and experiments, and then draw conclusion from evidence and connecting concepts.

This laboratory manual studies the using of the microscope, different types of cells, macromolecules, Respiration, mitosis meiosis, Plant anatomy, and the anatomy of human body, ranged from cells up to the human system. So, this manual presents a wide range of laboratory experiences for students concentrating in nursing, physical and occupational therapy, pharmacology, agriculture, health, as well as biology and medical programs.

The purpose of writing a lab report is to determine how well you performed your experiment, how much you understood about what happened during the experimentation process, and how well you can convey that information in an organized fashion.

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Lab Safety, Scientific Writing, and Microscopy

A. Lab Safety

INTRODUCTION

Each laboratory is a restricted area. Enrolled students may work in a lab only when there are authorized personnel present. Friends of students in lab classes will not be allowed to "visit" inside the laboratory. Students are not permitted into the storage rooms or preparation areas unless given specific permission by their instructor or lab personnel.

Working in a laboratory environment requires that the student observe certain housekeeping procedures and safety precautions. These procedure and precautions are outlined below. The student needs to review and follow the procedure to ensure the safety of himself/herself, the instructor, fellow students, the equipment and the facilities.

Ensuring safety in the laboratory is the responsibility of everyone working in the lab. Please follow these guidelines carefully and be sure you understand each one. If you aren't sure, ASK!!!

Learning Objectives:

By the end of this lab activity, students should be able to:

1. Identify and apply the correct lab safety procedures to follow for a variety of scenarios.
2. List and explain the steps of the Scientific Method.
3. Identify and provide examples of questions that can be answered scientifically.
4. Explain what distinguishes a good scientific hypothesis.
5. Define, give examples of, and identify dependent, independent and standardized variables.
6. Explain the importance of control treatments and replication.
7. Explain the difference between quantitative and qualitative data.
8. Conduct a simple "Black Box" experiment using the Scientific Method

GENERAL GUIDELINES

1. USE COMMON SENSE WHEN WORKING IN THE LAB.
2. Be prepared for your work in the lab. Read all procedures thoroughly before entering the lab. Follow All written and verbal instructions carefully. If you do not understand a direction o r part of a procedure, ask the instructor before proceeding.
3. Do not eat, drink, or smoke in the lab. Do not use laboratory glassware as containers for food or beverages.
4. Always wear close-toed shoes in the lab.
5. Wear safety glasses or goggles at all times whenever working with chemicals or when there is an impact risk. DO NOT WEAR CONTACT LENS AT ANY TIME.
6. Wear the laboratory coat and it must be closed all lab. Period.
7. Long hair should be tied back when working with flames, chemicals or dissections.
8. Observe good housekeeping practices. Work areas should be kept clean and tidy at all times. Keep aisles clear. Push your chair under the desk when not in use.

9. No open flames are permitted in the laboratory unless specifically indicated by the instructor. When burners or hot plates are being used, caution should be exercised to avoid thermal burns. If you sustain a thermal burn, immediately flush the area with cold water and notify the instructor.
10. If there is a blood spill, immediately notify the instructor.
11. ANY ACCIDENTS OR INJURIES THAT OCCUR IN THE LAB MUST BE REPORTED TO THE INSTRUCTOR AT ONCE.
12. Broken glass is to be disposed of in the broken glass (sharps) container and reported to the instructor.
13. Keep hands away from face, eyes, mouth and body while using chemicals or preserved specimens. Wash your hands with soap and water after performing all experiments. Clean, rinse and wipe dry all work surfaces and apparatus at the end of the lab activity. Return all equipment to the proper area.
14. Handle all living organisms used in a lab activity in a human manner.
15. Never use mouth suction to fill a pipette. Use a rubber bulb or pipet pump or filler.
16. When removing an electrical plug from its socket, grasp the plug not the electrical cord. Hands must be completely dry before touching an electrical switch, plug or outlet.
17. Do not mix any chemicals except as instructed. Do not do unauthorized experiments.
18. Do not taste any chemical. Do not smell chemicals directly. Use your hand to waft the odor to your nose.
19. Do not pipet solutions by mouth. Use a rubber suction bulb or special pipet filler.
20. Do not work in the laboratory in the absence of your instructor or his authorized representative.
21. Handle glass tubing and thermometers with care.
22. NO chemicals are to be flushed down a drain unless specifically instructed to do so by the lab procedure.
23. Wastes are to be poured into the appropriately labeled waste container (e.g. solvent waste). halogenated solvent waste, etc.
24. DO NOT mix wastes from different categories.
25. Clean up solid and liquid spills immediately, but only after checking with your Laboratory instructor about possible safety hazards.
26. Read the label on chemical bottles carefully. Insure that you have the correct chemical.
27. Take no more of a chemical than an experiment requires.
28. Never return as unused chemical to a stock bottle. Dispose of it as waste.
29. Set up your glassware and apparatus away from the front edge of your laboratory bench.
30. For each solution or chemical use a separate pipette or dropper, Don't use A pipette or dropper for more than one solution to prevent cross contamination.
31. Treat any microorganism as pathogenic potential.
32. Follow any other housekeeping, safety, or disposal rules given by your instructor.

HANDLING CHEMICALS

1. Wear safety goggles whenever working with chemicals.
2. Chemicals and biological stains should be used with caution. Follow specific instructions regarding all chemicals used during lab. Check the label on chemical bottles twice before removing any of the contents. Take only as much chemical as you need. Do not carry a chemical stock bottle to your work station or remove it from the chemical station.
3. If your skin is exposed to any chemical, immediately flush the area with water for several minutes and notify the instructor.
4. Dispose of all chemical waste properly. Do not pour chemicals down the sink unless told to do so by your instructor. Check the label of all waste containers twice before adding your chemical waste to the container.

DISSECTIONS – Special Precautions

1. Students should consult with the instructor regarding the pros and cons of wearing contact lenses during dissections.
2. Safety glasses or other protective eyewear is recommended for all students performing dissections.
3. Protective gloves should be worn during dissections. If your skin comes in Contact with a chemical preservative, immediately run water over the area and notify the instructor.
4. Do not remove preserved specimens from the laboratory.
5. Preserved biological materials are to be treated with respect.
6. When using scalpels and other sharp instruments, always carry them with the tips and points pointing down and away. Notify your instructor of any cuts or other injuries.

Scientific Writing

Lab reports are an essential part of all laboratory courses and usually a significant part of your grade. This means that you have to complete biology lab reports.

The purpose of writing a lab report is to determine how well you performed your experiment, how much you understood about what happened during the experimentation process, and how well you can convey that information in an organized fashion.

The best way to prepare to write the lab report is to make sure that you fully understand everything you need to about the experiment. Obviously, if you don't quite know what went on during the lab, you're going to find it difficult to explain the lab satisfactorily to someone else. To make sure you know enough to write the report, complete the following steps:

1. Read your lab manual thoroughly, well before you start to carry out the experiment. Ask yourself the following questions:

   - What are we going to do in this lab? (That is, what's the procedure?)
   - Why are we going to do it that way?
   - What are we hoping to learn from this experiment?
   - Why would we benefit from this knowledge?

   Answering these questions will lead you to a more complete understanding of the experiment, and this "big picture" will in turn help you write a successful lab report.

2. Make use of your lab supervisor as you perform the lab. If you don't know how to answer one of the questions above.
3. Plan the steps of the experiment carefully with your lab partners.
4. Record the data carefully so you get them right. You won't be able to trust your conclusions if you have the wrong data.
5. Consult with your lab partners about everything you do. Collaborate with your partners, even when the experiment is "over." What trends did you observe? Was the hypothesis supported? Did you all get the same results? What kind of figure should you use to represent your findings? The whole group can work together to answer these questions.
6. Write the results you get from the experiments or draw what you see under the microscope. DO NOT DRAW AS MANUAL.
7. The student's report will be corrected if item " 6 " is achieved.

   Once you've completed these steps as you perform the experiment, you'll be in a good position to draft an effective lab report.

Lab Report Format:

A good lab report format includes six main sections:

- Title Page
- Title
- Introduction
- Materials and Methods
- Results
- Discussion and Conclusion
- Figures & Graphs
- References

Keep in mind that individual instructors may have a specific format that they require you to follow. Please be sure to consult your teacher about the specifics of what to include in your lab report.

Title Page:

Not all lab reports have title pages, if your instructor wants one, it would be a single page that states:

- o The title of the experiment.
- o Your name and the names of any lab partners.
- o Your instructor's name.
- o The date the lab was performed or the date the report was submitted

Title:

The title states the focus of your experiment. The title should be to the point, descriptive, accurate, and concise (ten words or less). If your instructor requires a separate title page, include the title followed by the name(s) of the project participant(s), class title, date, and instructors name. If a title page is required, consult your instructor about the specific format for the page.

Introduction:

The introduction of a lab report states the purpose of your experiment. Your hypothesis should be included in the introduction, as well as a brief statement about how you intend to test your hypothesis. To be sure that you have a good understanding of your experiment, some educators suggest writing the introduction after you have completed the methods and materials, results, and conclusion sections of your lab report.

Materials and Methods:

This section of your lab report involves producing a written description of the materials used and the methods involved in performing your experiment. You should not just record a list of materials, but indicate when and how they were used during the process of completing your experiment.

The information you include should not be overly detailed, but should include enough detail so that someone else could perform the experiment by following your instructions.

Results:

The results section should include all tabulated data from observations during your experiment. This includes charts, tables, graphs, and any other illustrations of data you have collected. You should also include a written summary of the information in your charts, tables, and/or other illustrations. Any patterns or trends observed in your experiment or indicated in your illustrations should be noted as well.

Discussion and Conclusion:

This section is where you summarize what happened in your experiment. You will want to fully discuss and interpret the information. What did you learn? What were your results? Was your hypothesis correct, why or why not? Were there any errors? If there is anything about your experiment that you think could be improved upon, provide suggestions for doing so.

Figures & Graphs:

Graphs and figures must both be labeled with a descriptive title. Label the axes on a graph, being sure to include units of measurement. Also, all parts must be labelled correctly.

Citation/References:

All references used should be included at the end of your lab report. That includes any books, articles, lab manuals, etc. that you used when writing your report.

The Microscope

Introduction

The microscope is an instrument designed to make small objects visible to human eyes. Early in the seventeenth century, Galileo Galilei arranged two lenses inside a cylinder to look at an insect and later described the eyes of insect. He became the first person to record a biological observation through a microscope as a non-biologist.

The microscope is an essential tool in modern biology. It allows us to view structural details of organs, tissue, and cells not visible to the naked eye. There are different types of microscopes, based on the physical principle used to make an image, these are:

A) Optical Microscopes: These microscopes use visible light (or UV light in the case of fluorescence Microscopy) to make an image. The light is refracted with optical lenses. This type of microscopes includes

- **Compound Microscope:**These microscopes are composed of two lens systems, an Objective and an ocular (eye piece). The maximum useful magnification of a compound microscope is about 1000x.

- **Stereo Microscope (Dissecting microscope):** These microscopes magnify up to about Maximum 100 X and supply a 3 - dimensional view of the specimen. They are useful for observing opaque objects.

- **Dark-field microscope**: used to observe live spirochetes, such as those that cause syphilis. This microscope contains a special condenser that scatters light and causes it to reflect off the specimen at an angle. A light object is seen on a dark background.

- **Phase-contrast microscope:** This microscope also contains special condensers that throw light “out of phase” and cause it to pass through the object at different speeds. Live, unstained organisms are seen clearly with this microscope, and internal cell parts such as mitochondria, lysosomes, and the Golgi body can be seen with this instrument.

- **Fluorescent microscope:** uses ultraviolet light as its light source. When ultraviolet light hits an object, it excites the electrons of the object, and they give off light in various shades of color. Since ultraviolet light is used, the resolution of the object increases. A laboratory technique called the fluorescent- antibody technique employs fluorescent dyes and antibodies to help identify unknown bacteria.

B) X-ray Microscope: As the name suggests, these microscopes use a beam of x-rays to create an image. Due to the small wavelength, the image resolution is higher than in optical microscopes. The maximum useful magnification is therefore also higher and is between the optical microscopes and electron microscopes. One advantage of x-ray microscopes over electron microscopes is, that it is possible to observe living cells.

C) Scanning acoustic microscope (SAM): These devices use focused sound waves to generate an image. They are used in materials science to detect small cracks or tensions in materials. SAMs can also be used in biology where they help to uncover tensions, stress and elasticity inside biological structure.

D) Scanning Helium Ion Microscope (SHIM or HeIM): As the name suggests, these devices use a beam of Helium ions to generate an image. There are several advantages to electron microscopes, One being that the sample is left mostly intact (due to the low energy requirements) and that it provides a high resolution.

E) Neutron Microscope: These microscopes are still in an experimental stage. They have a high resolution and may offer better contrast than other forms of microscopy.

F) Electron Microscopes: Modern electron microscopes can magnify up to 2 million times. This is possible, because the wavelength of high energy electrons is very small. At the same time, the high energy electrons are pretty tough on the sample observed. It may take a long time to completely dehydrate and specimen. Some biological specimens also need to be coated with a very thin layer of a metal before they can be observed. There are two types:

- **Transmission electron microscopy (TEM):** In this case, the electron beam is passed through the sample. The result is a two dimensional image.(Figure 1.1)

Figure 1.1: Transmission Electron Microscope. The transmission electron microscope produces an image using electrons that pass through the specimen. The image is then viewed on the monitor. This particular model magnifies from 8X up to 630,000X.

- **Scanning electron microscopy (SEM):** Here the electron beam is projected on the sample. The electrons do not go through the sample but bounce off. This way it is possible to visualize the surface structure of the specimen. The image appears 3-dimensional. (Fig.1.2)

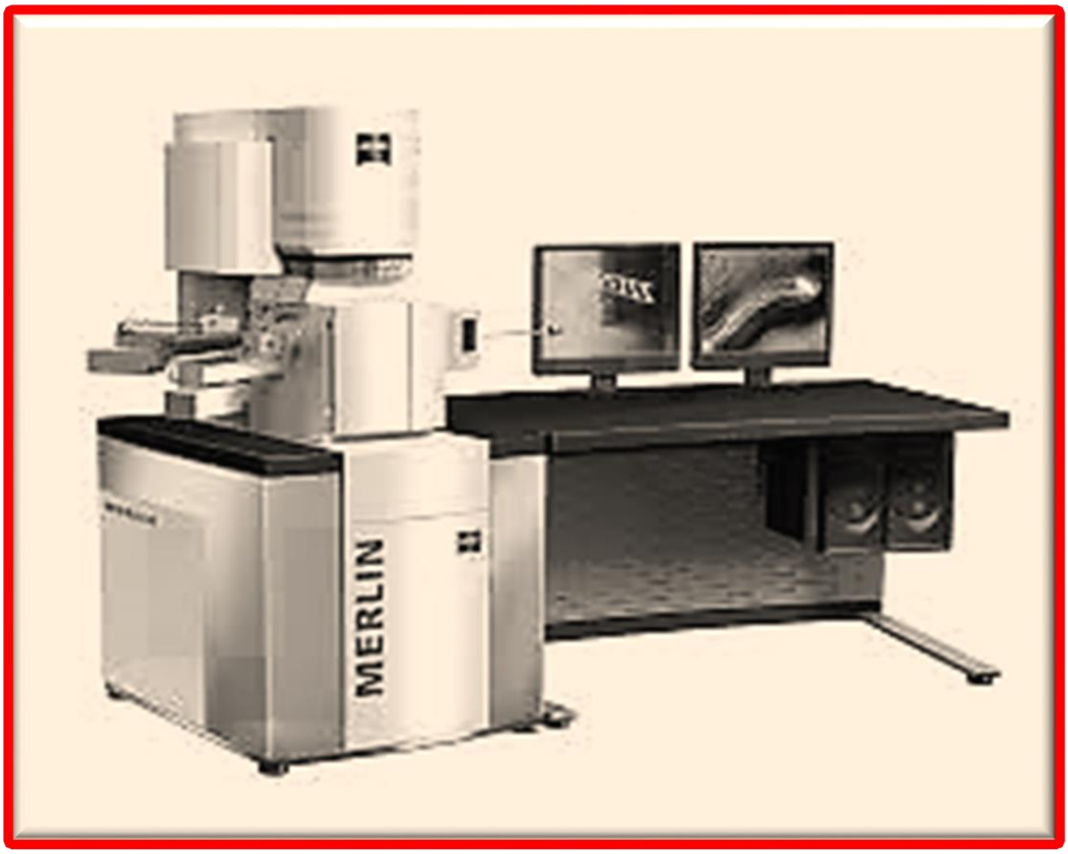


Figure1.2: Scanning Electron Microscope. This scanning electron microscope has the ability to magnify from 12X to 900,000X "resolving power of as little as 1.0 nm.

G) Scanning Probe Microscopes: It is possible to visualize individual atoms with these microscopes. The image of the atom is computer - generated, however. A small tip measures the surface structure of the sample by restoring over the surface. If an atom projects out of the surface, then a higher electrical current will flow through the tip. The amount of current is proportional to the height of the structure. A computer will then assemble the position data of the tip and the current to generate an image.

Conclusion: Microscopes can be classified based on the physical principle that is used to generate an image. Different microscopes visualize different physical characteristics of the sample (eg. elasticity can be visualized with acoustic microscopes). Image contrast, resolution (which determines magnification) and destructiveness of the sample are other relevant parameters.

Microscopy involve the following concepts:

- ❖ **Magnification**: refers to the size of an image. There is a set of three **objective lenses** on your microscope. The **magnification** (or power) of each objective lenses is engraved on the side of the objective. The ocular lens is also normally engraved with its magnification(typically 10x).

To determine the total magnification of a specimen, use the following formula:

Total Magnification = Ocular Magnification x Objective Magnification

- **Contrast:** the ability to see a particular detail against its background.
- **Resolving power Resolution**: Is the Clarity of an image or an actual measurement of how far apart two points must be in order for the microscope to view them as being separate.

- **Working distance:** is the distance between the stage and objective lens (Fig 1.3). Because objective lenses vary in lengths, the working distance will change as you switch from one objective lens to the next.

  In a microscope, as magnification increases, working distance decrease.

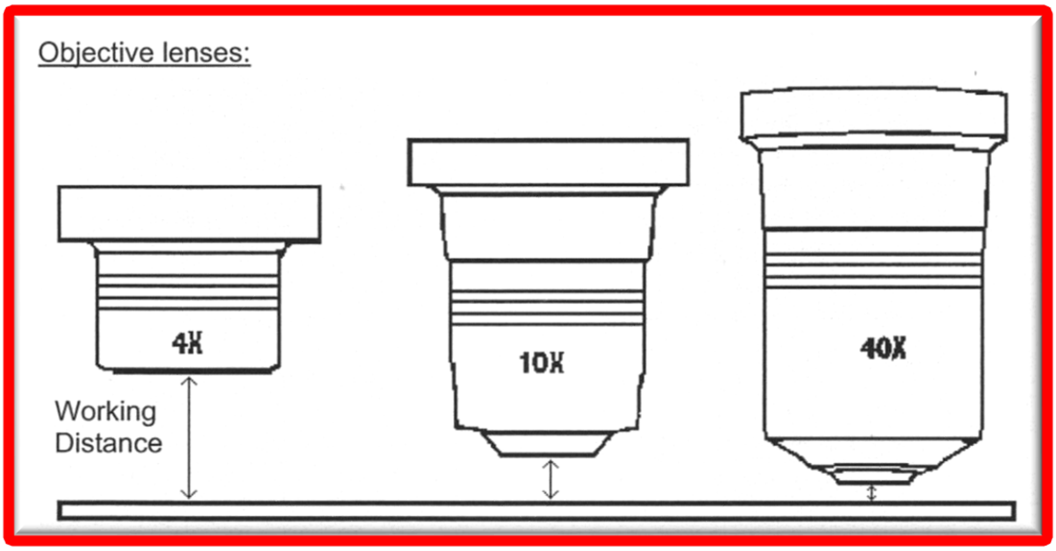


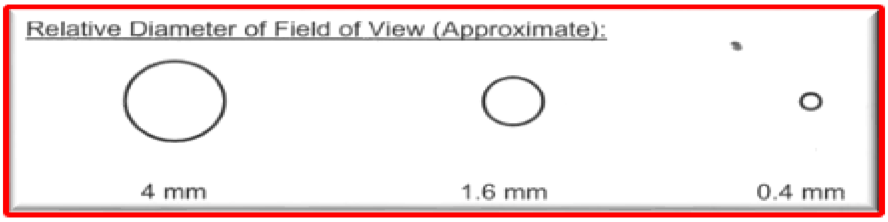
Fig. 1.3: Working Distances with Various Objective Lenses

- **Diameter of Field of View:** The approximate size of a specimen can be estimated if the **diameter of the field of view (DFV)** is known. In parfocal microscopes, if we know the magnification and DFV for one objective lens, we can calculate the DFV for a second objective on the same parfocal microscope using the following formula:

$$M_1 \times DFV_1 = M_2 \times DFV_2$$

where M_1 and DFV_1 = magnification and diameter of the field of view, respectively, of objective 1, M_2 and DFV_2 = magnification and diameter of field of view, respectively, of objective 2.

Magnification increases, the diameter of the field of view Decrease (Fig. 1.4)



4x 10x 40x

Fig. 1.4 Diameter of the Field of View (DFV) with Various Objective Lenses

PROCEDURE

Part A: Carrying of Microscope:

" Microscope is an expensive instrument so please handle and store it with care"

Care and Use of the Compound Microscope

- ALWAYS CARRY THE MICROSCOPE UPRIGHT WITH TWO HANDS, ONE ON THE BASE, THE OTHER ON THE ARM, IT MUST BE PARALLEL TO YOUR BODY
- MAKE SURE YOUR WORKBENCH IS FREE OF CLUTTER BEFORE YOU PLACE THE MICROSCOPE ON THE BENCH
- DO NOT DRAG OR SHOVE THE MICROSCOPE ACROSS THE LAB BENCH – ALWAYS LIFT TO MOVE OR TURN IT
- REMOVE THE DUST COVER AND CLEAN THE MICROSCOPE WITH LENS PAPER AS EXPLAINED BY YOUR INSTRUCTORE.
- PLACE THE MICROSCOPE ON THE TABLE WITH THE OCULAR TOWARD YOU.
- THE BACK OF THE BASE SHOULD BE AT LEAST 2 INCH FROM THE EDGE OF THE TABLE.
- POSITION YOURSELF AND THE MICROSCOPE SO THAT YOU CAN LOOK INTO THE MICROSCOPE COMFORTABLY AND WITHOUT TILTING THE MICROSCOPE.
- PLUG IN THE MICROSCOPE WITH ELECTRIC SOURCE.
- DON'T SWITCH ON THE MICROSCOPE TILL USE.

Part B: Parts of the compound microscope (Student Microscope):

The microscopes common used in our lab are compound binocular or monocular light microscopes. Binocular microscopes have two eyepieces. Monocular microscopes have one eyepiece. They are called compound microscopes because a specimen is viewed through a series of lenses.

Obtain a light microscope. The correct way of carrying it is to use one hand support the base and the other hand hold the arm. Then study the parts of the microscope depending on the following descriptions depicted in figure 1.5.

- **The eyepiece or ocular lens:** is located on the top of the microscope. It normally has a magnification of 10 (10 X, x means "times"). Be careful these lenses are not attached, and therefore may fall down unless the microscope is kept upright.
- **Body tube:** Carry the ocular lenses and joins them with nosepiece.
- **The resolving nosepiece:** is the revolving part to which the objective lenses are attached, uses to rotate the objectives in position. It must be firmly clicked to the position when the objective is changed.
- **Objectives are lenses:** located on the movable resolving nosepiece. They usually have different color strips.
  - The scanning objective is the shortest. Its magnification is 4 X.
  - The low-power objective is the second shortest lens. It has a magnification of 10 X.
  - The high- power objective (high-dry objective). It is normally marked 40x.
  - The oil immersion objective is the longest objective. It has the highest magnification(100 X). We only use oil immersion lens with immersion oil.

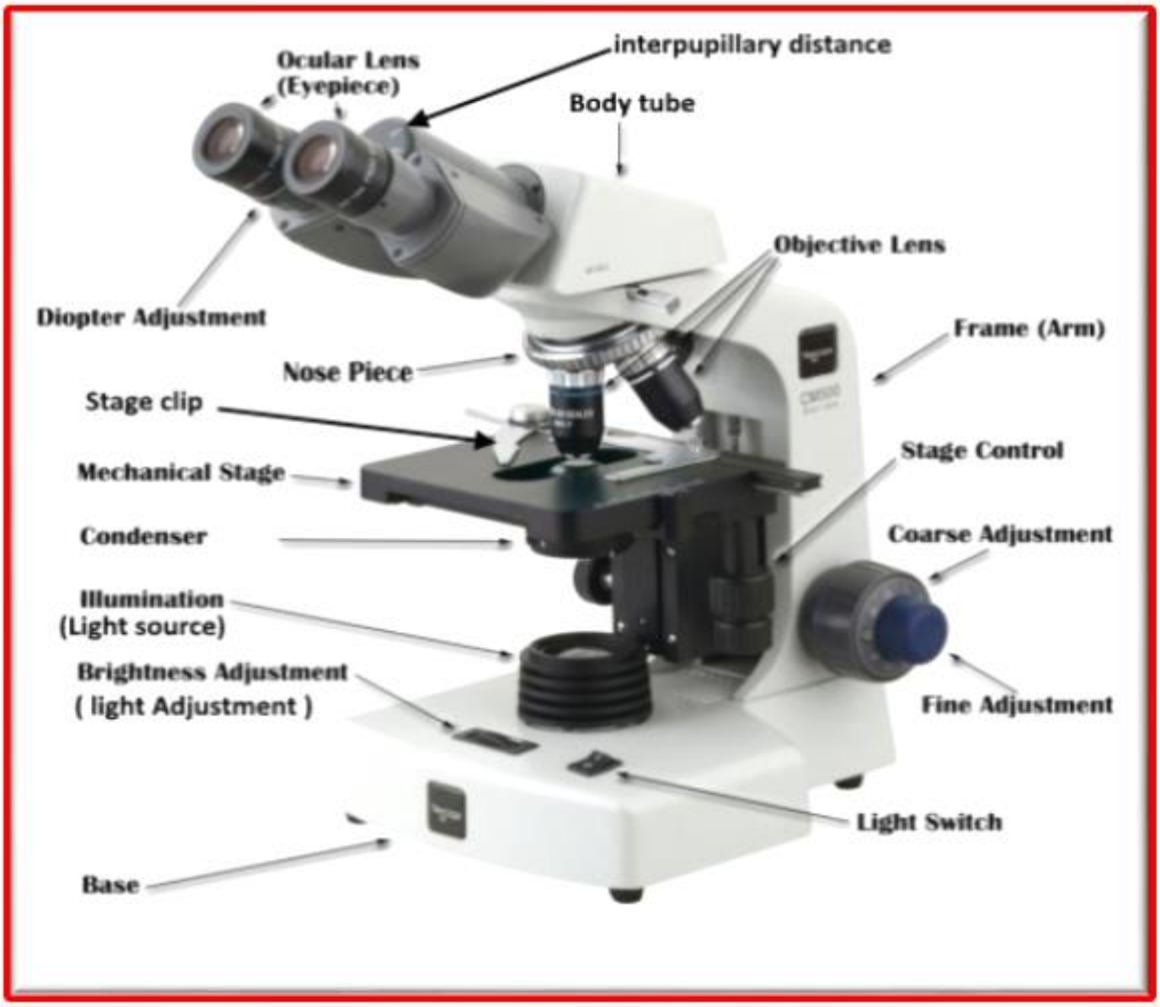


Fig1.5: The Compound Light Microscope

- **Stage:** Support the slide that is held to it by two stage clip, and has a hole so that light can pass through the specimen. Always center the specimen over this hole.
- **Fine focus adjustment knob:** makes final or fine focusing on a specimen.it uses with oil immersion lens (100X), and with high-dry objective (40 X) if there is image put not clear.
- **The coarse focus adjustment knob:** brings the specimen in focus under the scanning and low-power objectives (4X and 10X) and with high-dry objective (40 X) if there is no image at all.
- **Two stage adjustment knobs (mechanical stage control or adobe knobs):** control the movement of stage, therefore the movement of slides. When the stage move to the right the image move to the left, and vice versa.
- **The arm:** is the curved part of the frame. It supports upper parts of a microscope and provides a carrying handle.
- **The condenser lens:** is to concentrate and direct the light. There is an adjust knob to control the height of the condenser.
- **The iris diaphragm:** controls the amount of light passing through the specimen. It is control by a lever that move back and forth.
- **The base:** acts as a stand for the microscope and houses the lamp.
- **Light switch:** Turn on/off the illuminator of the microscope.
- **Light source:** usually a small electric light beneath the stage that controlled by light switch.

Part C : Viewing a Specimen with a Compound Light Microscope

Procedure

1. Clean the microscope with lens paper, plug in to the electrical source, and switch on the light source.
2. Use the **coarse focus adjustment knob** to maximize the **working distance** (the distance between the stage and the objective lens), this mean lower the stage to the lowest point.
3. Rotate the revolving **nosepiece** into position with the **scanning power (4x) objective** lens in the viewing position.
4. Center the slide holder of the **mechanical stage** on the microscope **stage**
5. Place the **slide** between the **stage clip** and push it all the way back to the bar
6. Using the **mechanical stage drive knobs**, center the coverslip and specimen over the stage aperture
7. While carefully watching the slide on the stage, use the **coarse focus adjustment knob** to move the specimen towards the **scanning objective lens** until it stops (raise the stage). The stage will come close to the lens but will not touch it
8. Adjust the **interpupillary distance** until you see a single circle while looking through the microscope with both eyes open. This circle of light is called the **field of view**
9. While looking through the **ocular lenses**, turn the **coarse and fine focus adjustment knobs** of the microscope until you see something you believe is the specimen. **Stop.** Move the **slide** back forth using the **mechanical stage drive knobs**. The item you thought was specimen should likewise be moving back and forth

11. Cover of close the eye that is not looking through the ocular containing the **diopter ring**. Viewing with only **that eye** focus using the **coarse focus adjustment knobs**. Adjust the light using the **iris diaphragm adjustment** lever and/or the **light adjustment**. Then close your other eye adjusting the **diopter ring** on that ocular lens to bring the object into focus

12. Adjust the **condenser** to the highest position

13. Using the **mechanical stage drive knobs**, center the specimen of choice in the viewing area

14. These microscopes are **parfocal** (if one lens is in focus, all other lenses are, at least, close to focus). In order to change to the next highest magnification, simply rotate the **nosepiece** to the **low power (10x) objective lens**

15. These microscopes are also **parcentral** (if an object is in the center of the field of view for one lens, it will be, at least, close to the center of the field of view at other lenses)

16. Using the **mechanical stage drive knobs**, re-center the specimen in the viewing area

17. With the **low power (10x) objective**, use the **coarse focus adjustment knobs** to focus the view of the specimen and the **iris diaphragm adjustment** lever to increase the light intensity on the specimen

18. Re-center the specimen in the field of view. Rotate the **nosepiece** to the **high power (40x) objective lens.** If **there Is :**

    a) image put not clear use the **FINE FOCUS ADJUSTMENT KNOB ONLY** to focus the specimen.
    b) No image use the **COARSE FOCUS ADJUSTMENT KNOB** to focus the specimen.

Then use the iris diaphragm adjustment lever to increase the light intensity on the specimen. If needed, use the light adjustment to provide additional light.

19. Re-center the specimen in the field of view. Rotate the nosepiece so no objective lens at the position. Add a drop of oil immersion on the slide at light spot region. Then rotate the nosepiece to oil immersion lens (100 X).

20. Use the **FINE FOCUS ADJUSTMENT KNOB ONLY** to focus the specimen.

21. When removing the slide, rotate the **nosepiece** so the **scanning power (4x) objective** is in the viewing position, then use the **coarse focus adjustment knob** to maximize the **working distance(lower the stage)**

22. After you have completed the laboratory activity, turn the light switch off. Clean all microscope lenses (objective and ocular) with **lens cleaner** and **lens paper**

23. Prepare the microscope for **storage** using the checklist below. Be sure:

a. The **scanning power (4x) objective** is in the viewing position
b. The **mechanical stage** has been positioned so the stage arm is flush with the right side of the stage
c. The cord is wrapped securely around the microscope arm
d. The **stage** has been adjusted all the way down
e. The **condenser** has been adjusted all the way up
f. The **light adjustment** is turned all the way down and the light is turned off

Part D: Viewing a prepared slide

Material:

1. Prepared slide of *Amoeba*

2. Compound microscope (Student Microscope)

Procedure:

1. Obtain a prepared slide of the single-celled organism, *Amoeba* (Figure 1.7).
2. Examine the prepared slide under the microscope using 4X, 10X, 40X, and oil immersion objective lenses as in part **Part C**
3. Use the formula to calculate the magnification of the microscope for each objective lens
4. Conclude the relationship between magnification and the microscopic field (DFV)

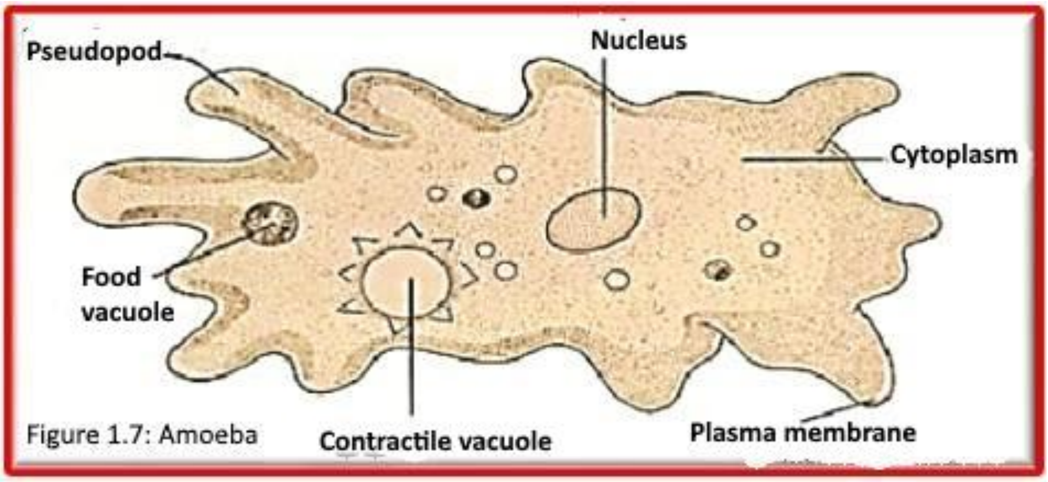


Figure 1.7: Amoeba

Part E: Preparing of a Wet Mount: (Figure 1.8)

Wet mount slide is a temporary slide in which cells are suspended in a drop of water on a slide and used to study fresh, living organisms. In this experiment, we need to prepare a wet mount slide for letter " **e** ".

Material:

1. Letter " e "
2. Cleaned Microscopic slide
3. Coverslip
4. Water
5. Dropper
6. Dissecting needle
7. Compound microscope

Procedure:

1. Add by dropper one or two drop of water on glass slide
2. Add the letter " **e** " to the water on the slide
3. Apply thin plastic coverslip over the letter " **e** " in water. Put the coverslip with an angle using dissecting Needle to prevent trapping of air bubbles.
4. Examine the prepared slide under the microscope using 4X, 10X, 40X, and oil immersion objective lenses as in part **C**
5. Use the formula to calculate the magnification of the microscope for each objective lens
6. Write your conclusion about the relationship between:
   a. Magnification and Microscopic field
   b. Magnification and resolution
7. Describe the image formed by the compound microscope.

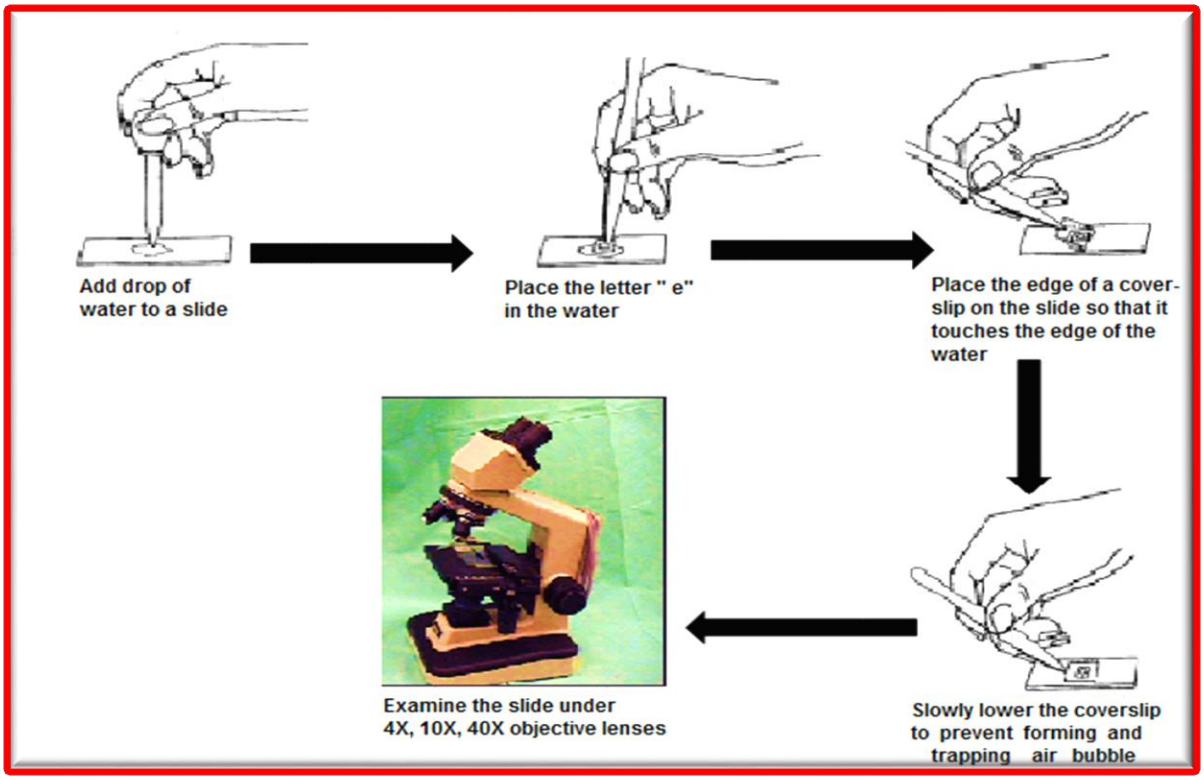


Figure 1.8 : Preparation of a Wet Mount

Part F: The Dissecting Microscope

It is possible to have too much magnification when viewing some specimens. For example, how would you use your compound light microscope to view an entire earthworm? Larger / or opaque specimens may require lower magnification. For this, biologists use dissecting microscopes (Fig. 1.9). These microscopes a large working distance between the specimen and the objective lens. They give 3-dimensional not inverted image.

Instructor will describe the use and features of this microscope.

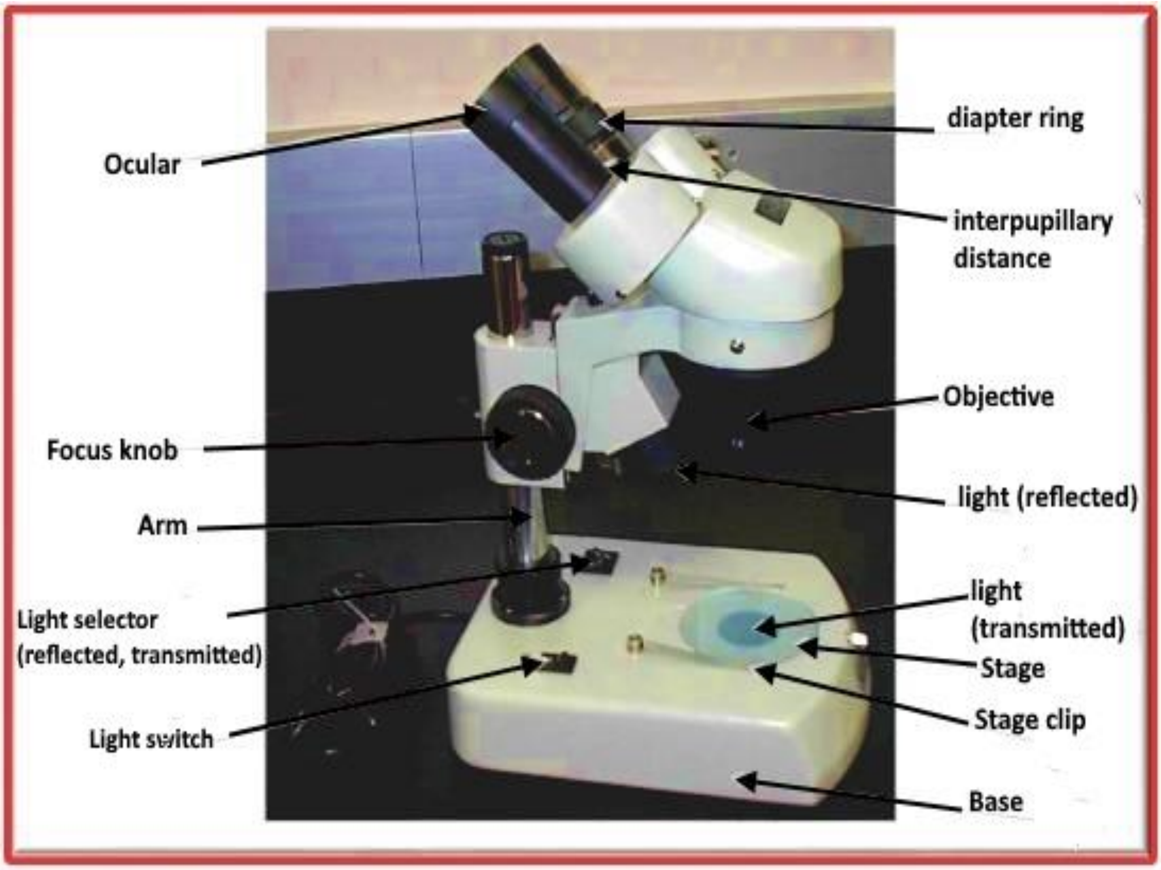


Fig. 1.9 The Dissecting Microscope

Procedure

1. Obtain a dissecting microscope using **two hands** to carry it
2. Identify the parts as per Fig. 1.6 and their functions
3. Observe the various objects (as insects) available in lab using the dissecting microscope

Experiment 2

Cell Structure and Function

Introduction

Cells are the basic structural and functional units of life, and they are fundamental to all organisms. Many forms of life exist as single-celled organisms which are either **Prokaryotic** or **Eukaryotic**. Larger and more complex organisms, including **plants** and **animals**, are multicellular; their bodies are collections of many kinds of specialized cells that could not survive for long on their own. All cells share certain basic features: they are all bounded by plasma membrane, Inside all cells is a semifluid substance called cytosol, in which subcellular components are suspended, and all cells contain chromosomes, which carry genes in the form of DNA.

Organisms of the domains **Bacteria** and **Archaea** consist of prokaryotic cells and they are usually **unicellular**. **Protists, fungi, animals**, and **plants** all consist of **eukaryotic cells** and they are either **unicellular** or **multicellular.**

Prokaryotic cells are characterized by having:

1. No nucleus
2. DNA in an unbound region called the **nucleoid**
3. No membrane-bound organelles
4. They are smaller than Eukaryotic cells.
5. **Cytoplasm** bound by the plasma membrane

Eukaryotic cells are characterized by having

1. DNA in a nucleus that is bounded by a membranous nuclear envelope
2. Membrane-bound organelles
3. Cytoplasm in the region between the plasma membrane and nucleus

PROCEDURE

I- Prokaryotic cell (bacterium)

Bacteria lack a nuclear membrane or other internal cell organelles (Figure 2.1)

- Observe a prepared slide under various magnifications to get a feel of the size of the bacteria and compare what you see to Figure 1.

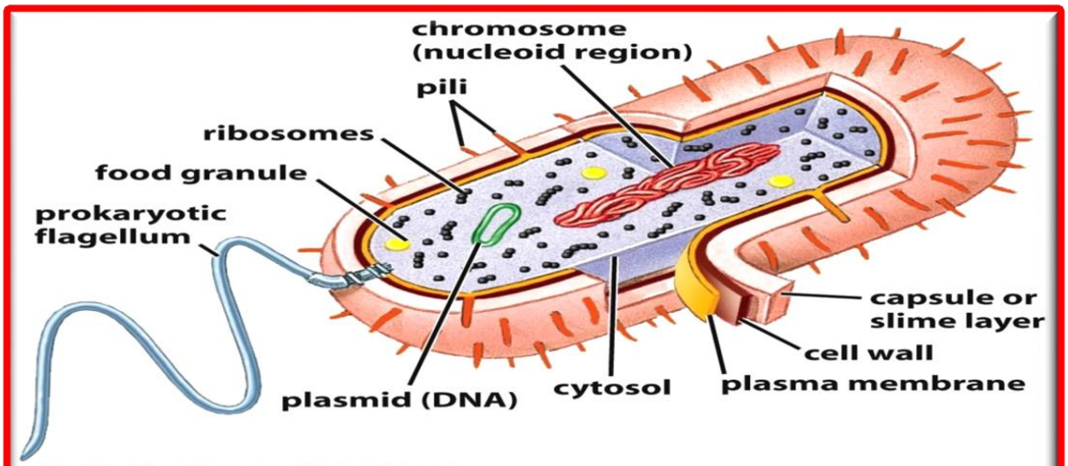


Fig. 2.1: The structure of bacterial cells

II- Eukaryotic cells

A. Plant cells

1. Epidermal cells of onion

- Cut an onion lengthwise into several parts according to the steps in Figure 2.2 A**,** and then observe it under the microscope.
- Add a small drop of a Lugol's iodine on the edge of the coverslip to stain cytoplasm and nucleus. Observe under the microscope and notice the cell wall, cytoplasm, vacuole and the nucleus. Compare to Figure 2.2 B

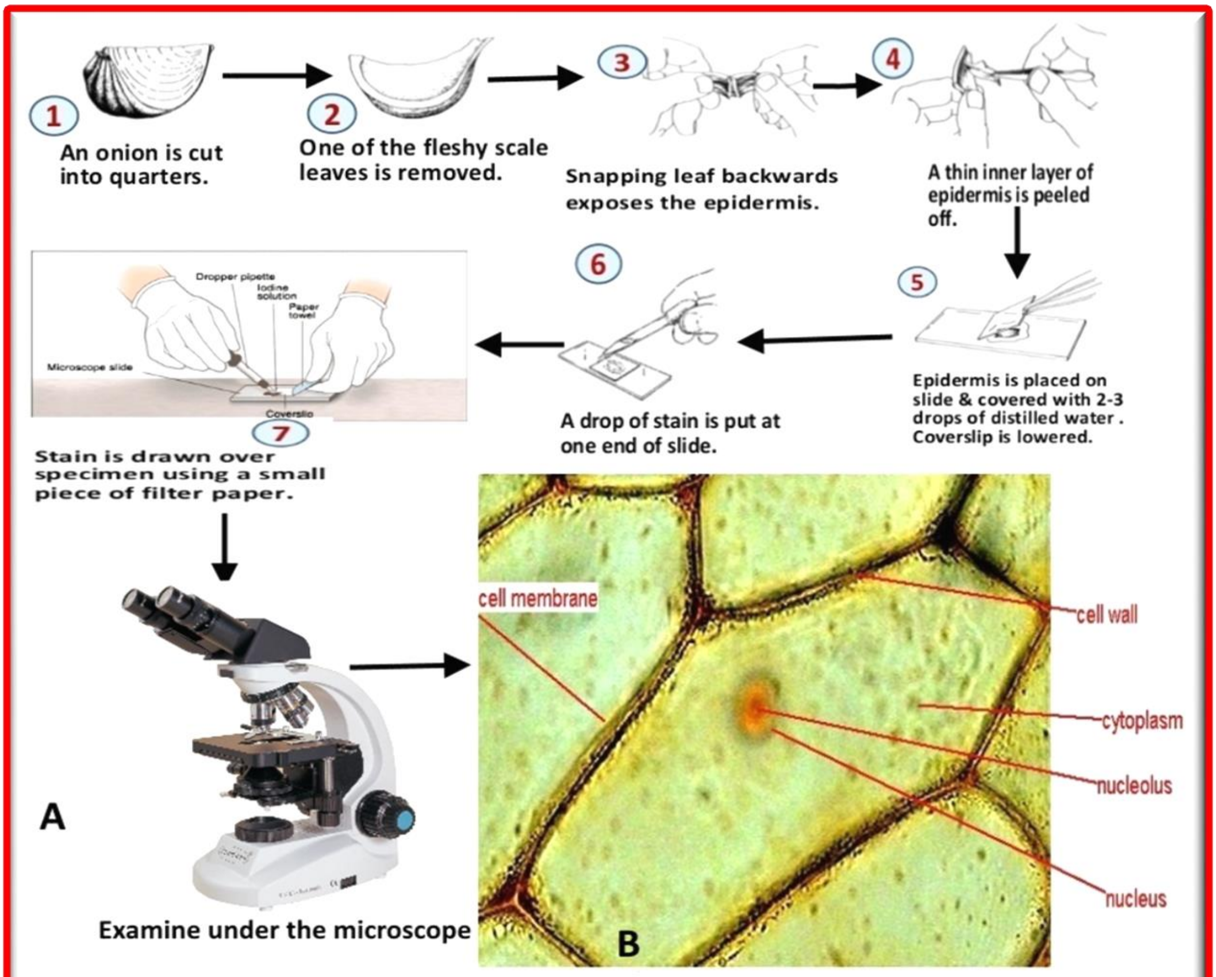


Fig. 2.2 A: Preparation of the wet mount slide of Onion epidermal cells B: Onion Epidermal Cells

Questions

1. What is the advantage of staining cells for microscopic examination?
2. Of what material is the cell wall composed?
3. Did staining influence the visibility of parts of the cell?

2- Leaf cells

- Ask for a ***Geranium*** leaf to prepare a wet mount slide. Observe under the microscope and notice Figure 2.3. Identify the chloroplasts (organelles that are important for photosynthesis), Guard cells, Stoma, and Epidermal cells.

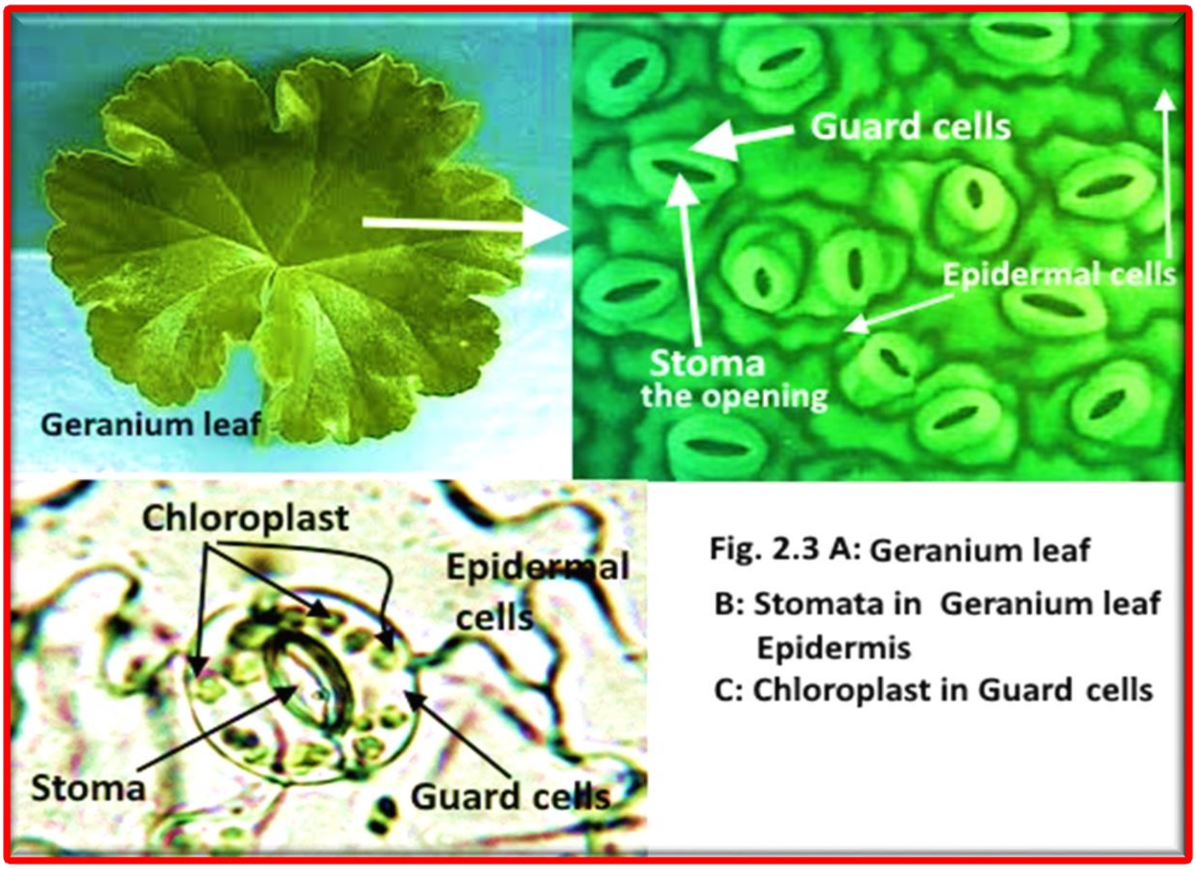


Fig. 2.3 A: Geranium leaf
B: Stomata in Geranium leaf Epidermis
C: Chloroplast in Guard cells

B. Animal cells

1. Epithelial cells of mouth

The cells lining the mucous membranes of your mouth are easily sloughed off. They are called squamous epithelial cells (cheek epithelium cells)(Figure 2.4 F):.
Make a wet mount slide of these cells as follows (Figure 2.4 G):

- Gently scrape the inside of the cheek surface with a clean toothpick.
- Tap the scrapings into a drop of water on a clean microscope slide.
- Add a drop of methylene blue to stain the cells.
- Cover the mixture with a coverslip and view under the microscope.

Notice the cell membrane, the cytoplasm and the nucleus.

Questions

1. Are all of the cells of the same general shape?
2. Why was the cell stained with iodine?
3. What is a tissue? Define a cell?
4. List three differences between animal and plant cells?

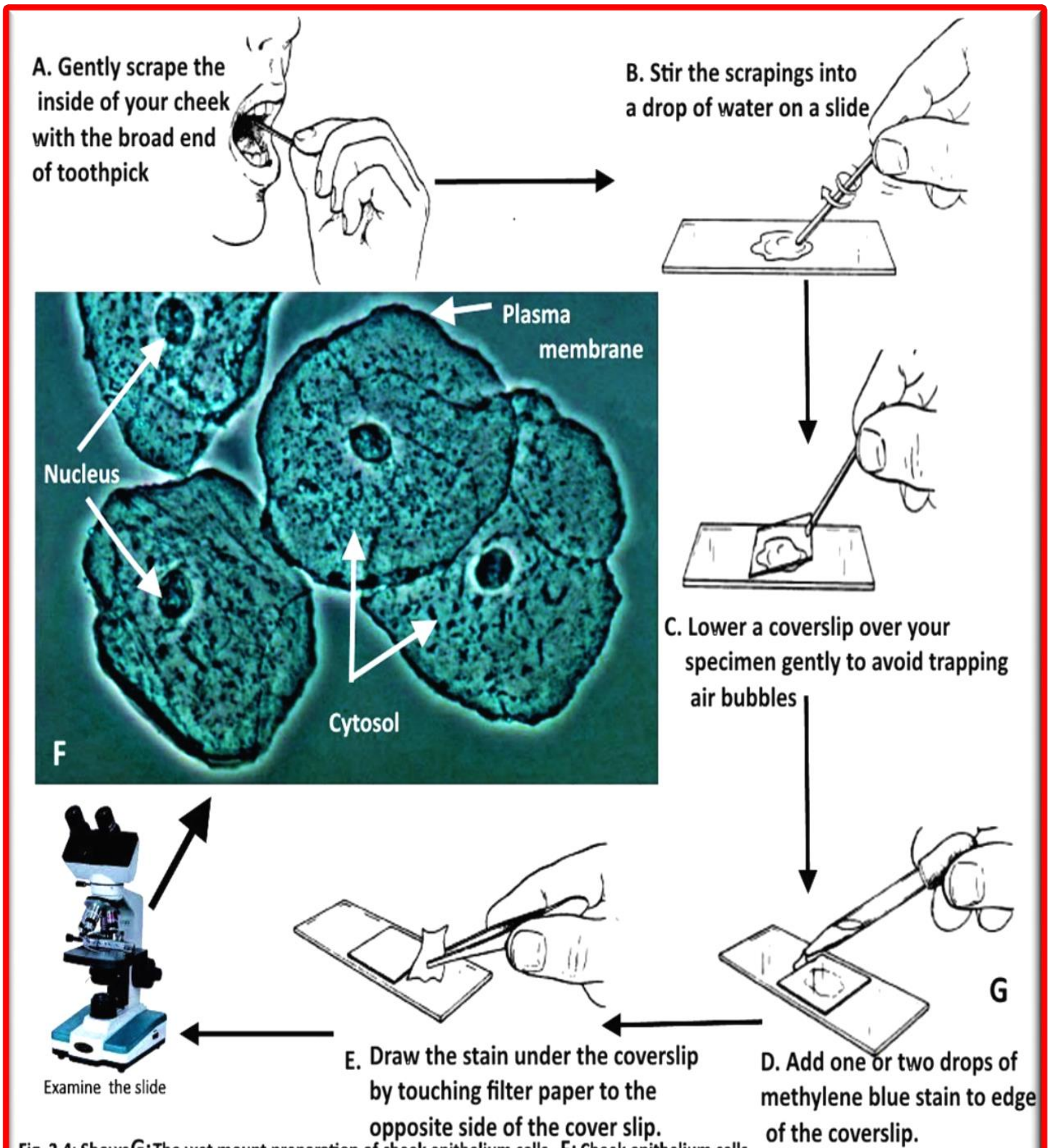


Fig. 2.4: Shows G: The wet mount preparation of cheek epithelium cells. F: Cheek epithelium cells

2. Animal and Plant cells Ultra structure

Study models of an animal (Figure 2.5) and Plant cells (Figure 2.6) that is on demonstration and identify most of the organelles.

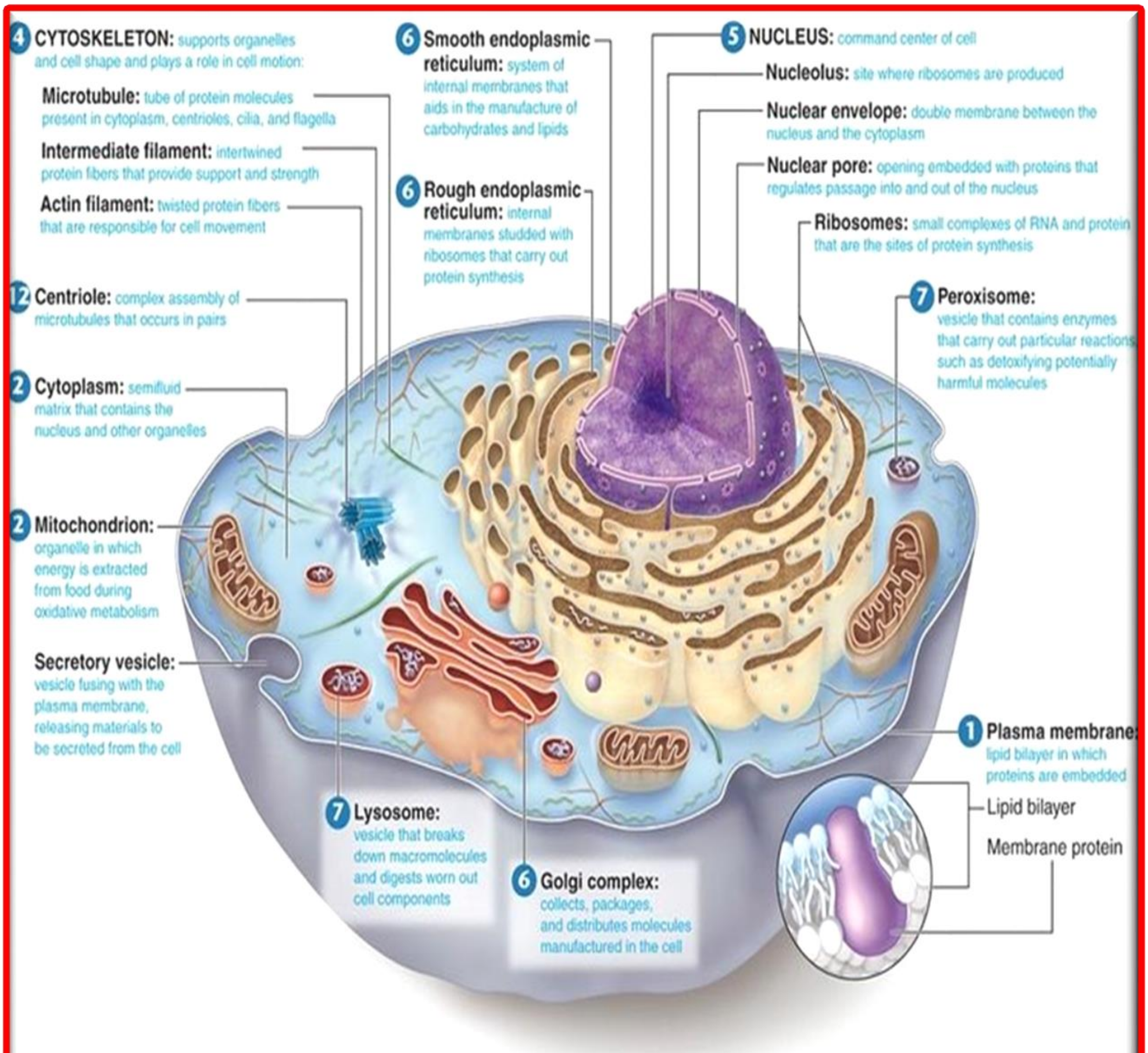


Fig. 2.5: Ultra structure of Animal cells with organelles' functions.

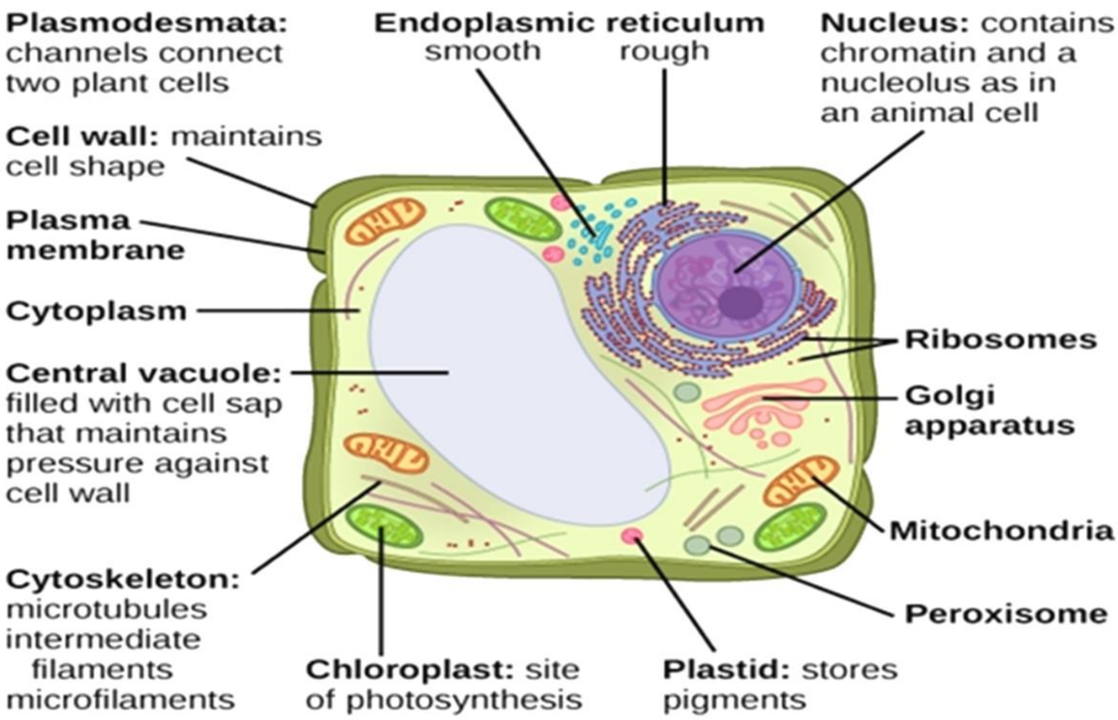


Fig. 2.6: Ultra structure of plant cells with qrganelles' functions

3. Paramecium

Paramecium is an example of a single-celled eukaryote organism, study the model available in the lab and identify the major organelles (Figure 2.7).

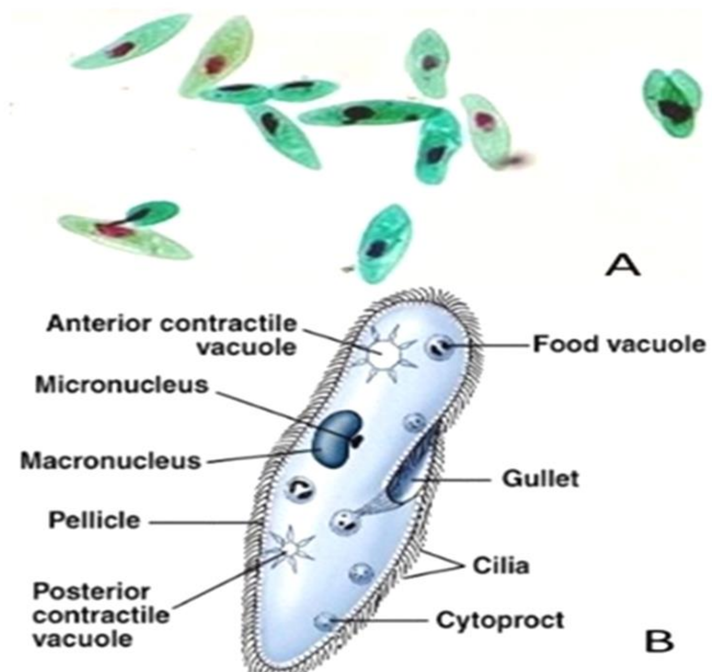


Fig.2.7: Paramecium, A: under low power
B: Ultrastructure

Experiment 3

Macromolecules and Living Things

Introduction

All living organisms consists of chemicals in many different forms. These molecules are made of atoms. Molecules could be small in size such as water molecules; other could be large such as proteins and polysaccharides. Essentially, four classis of molecules are related to living organisms, and makes up the majority of their structures. These classes are the carbohydrates, fats and lipids, proteins and nucleic acids.

Macromolecules are defined as polymers of similar molecules known as monomers. Polymers are formed by dehydration reactions, which will result in the formation of larger macromolecules and water. On the other hand, large macromolecules are broken down to their monomers by means of hydrolysis reactions. This will result in the formation of the building units (monomers) and in this case, water is required to complete this reaction. All these reactions require specific enzymes, either the dehydration or hydrolysis reactions.

In humans, over 1000 enzymes are known to mediate the different metabolic reactions as well as physiological, neurological reactions.

I) Carbohydrates

Carbohydrates are compounds that contain carbon, oxygen and hydrogen in certain proportions. They contain the same amount of carbon and oxygen and twice as much as hydrogen. In some instances, as in amino sugars, nitrogen is present as in the case of chitin.

Carbohydrates are divided into simples or complex carbohydrates. Simple carbohydrates include the monosaccharide, the monomers of carbohydrates, and the disaccharides. Complex carbohydrates are exemplified by polysaccharides.

- Monosaccharides could have either an aldehyde or a ketone group, known as aldoses or ketosis respectively (Figure 2.1A).
- Disaccharides consist of two monosaccharides linked together by glycosidic linkage (Figure 2.1 B).
- Polysaccharides are polymers of monosaccharides, linked together by either **α** or **β**- glycosidic linkage, to form starch or cellulose respectively (Figure 2.1C).

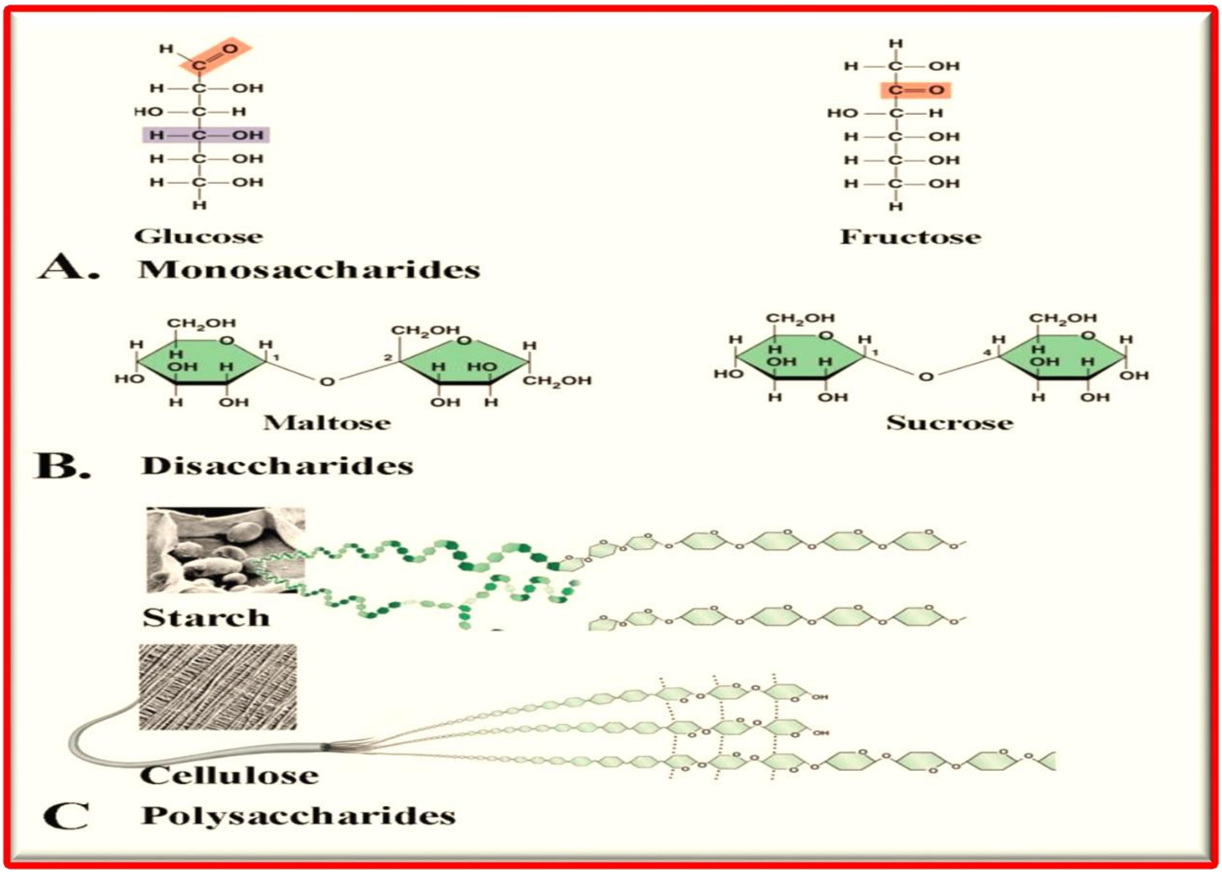


Figure 2.1: Classes of carbohydrates.

Identification of Carbohydrates

1. Benedict's Test for Reducing Sugars

Benedict's reagent is an aqueous solution of copper (II) sulphate, sodium carbonate and sodium citrate. This reagent is used to test for reducing sugars.

What is a reducing sugar?

A **reducing sugar** is any **sugar** that is capable of acting as a **reducing agent**. A sugar is classified as a reducing sugar only if it has an **open-chain** form with a free aldehyde group or a free ketone group. Along with some **disaccharides, oligosaccharides,** and **polysaccharides**, All **monosaccharides** are reducing sugars, because all of them have a aldehyde group (if they are aldoses) or can tautomerize in solution to form a aldehyde group (if they are ketoses). This includes common monosaccharides like **galactose, glucose, glyceraldehyde, fructose, ribose**, and **xylose**.

Disaccharides are formed from two monosaccharides and can be classified as either reducing or non-reducing. Non reducing disaccharides like **sucrose** and **trehalose** have **glycosidic bonds** between their **anomeric carbons** and thus cannot convert to an open-chain form with an aldehyde group; they are stuck in the cyclic form. Reducing disaccharides like **lactose** and **maltose** have only one of their two anomeric carbons involved in the glycosidic bond, meaning that they can convert to an open-chain form with an aldehyde group.

The aldehyde group of reducing sugar can be oxidized via a **redox reaction** and reduce **copper(II)** ions in **Benedict's reagent** (Cu^{2+} in aqueous sodium citrate) to copper(I), which then forms a brick red **copper(I) oxide** precipitate. Positive test will turn the color of the reducing sugar solution into orange to dark red, yellow, or green depending on the amount of reducing sugar (fig. 3.1).

Experiment:

a. Label 6 clean tubes from 1-6.
b. Add 1 ml (20 drops) of the following solutions using a dropper in the corresponding tubes.

1. Distilled water	4. Potato juice
2. 5% Glucose	5. 5% Starch
3. Onion juice	6. 5% Sucrose

c. Add I ml of Benedict's reagent to each tube.
d. Please the tubes in a rack in a boiling water bath for three minutes.
e. Record the results. Note changes in the colour for each tube.

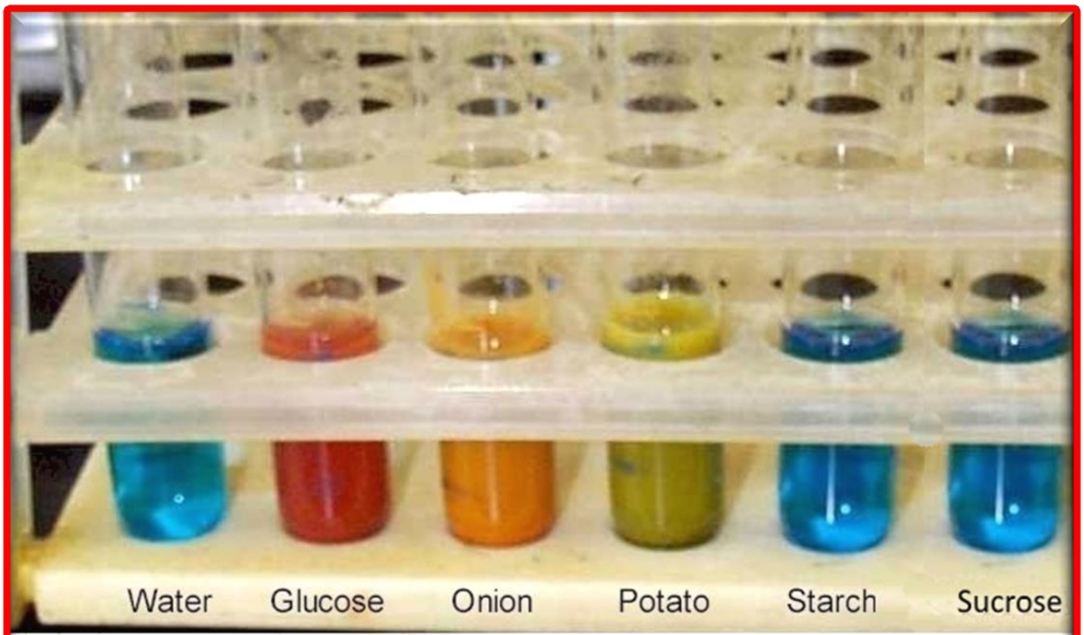


Fig. 3.1: Benedict's test

2. Lugol's Test for Starch

Lugol's test is used to identify starch. Lugol's solution consists of Iodine dissolved in an aqueous solution of potassium iodide. This complex will react with starch producing a deep blue-black color (fig. 3.2). This reaction is the result of the formation of polyiodide chains from the reaction of starch and iodine. The amylose, or straight chain portion of starch, causes the dark blue/black color. The amylopectin, or branched portion of starch, causes the formation of an orange/yellow hue color

Experiment:

a. Label 6 clean tubes from 1-6.
b. Add 1.0 ml of the following solutions using a pipette in the corresponding tubes.

1. Distilled water	**4. Potato juice**
2. 5% Glucose	**5.** Onion juice
3. 5% Starch	**6. 5% Sucrose**

c. Add 1.0 ml of lugol's solution to each test tube.
d. Record the result by Notice the changes in the colour of each test tube.

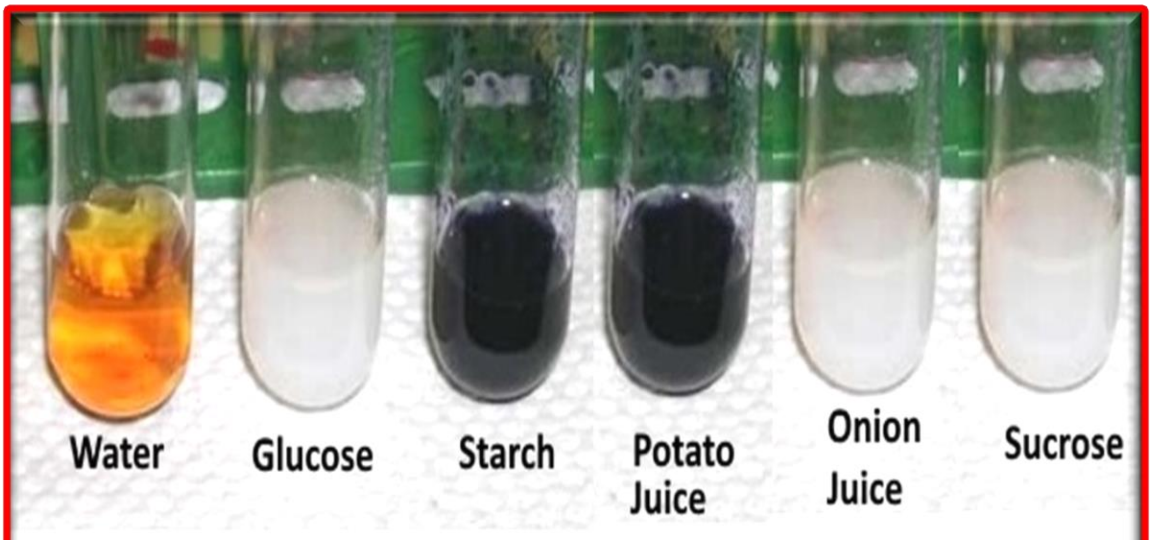


Fig. 3.2 : Lugol's test for starch

II) Fats and lipids

Lipids are used for structural purposes in cells (i.e., plasma membrane). They are also used to store excess energy in living organisms. Lipids are compounds that are insoluble in water, but dissolvable in non-polar solvents such as ether or chloroform. Lipids also differ from carbohydrates, proteins, and nucleic acids in that they are neither huge macromolecules nor polymers built from similar monomers. We will consider three types of lipids: fats, phospholipids, and steroids.

- **Fats:** A fat is a large lipid made from two kinds of smaller molecules: glycerol and fatty acids. Shown at the top in Figure 3.3 A, glycerol is an alcohol with three carbons, each bearing a hydroxyl group (-OH). A fatty acid consists of a carboxyl group (the functional group that gives these molecules the name fatty acid, -COOH) and a hydrocarbon chain, usually 16 or 18 carbon atoms in length. The non-polar hydrocarbon chains are the reason fats are hydrophobic. Figure 2.4 A also shows how one fatty acid molecule can link

  to a glycerol molecule by a dehydration reaction. Linking three fatty acids to glycerol produces a fat, as illustrated in Figure 3.4B. A synonym for fat is triglyceride, a term you may see on food labels or on medical tests for fat in the blood. Some fatty acids contain one or more double bonds, which cause kinks (or bends) in the carbon chain. See the third fatty acid in Figure 2.4 B. Such an unsaturated fatty acid has one fewer hydrogen atom on each carbon of the double bond. Fatty acids with no double bonds in their hydrocarbon chain have the maximum number of hydrogen atoms (are "saturated" with hydrogens) and are called saturated fatty acids. The kinks in un-saturated fatty acids prevent fats containing them from packing tightly together to solidify at room temperature.

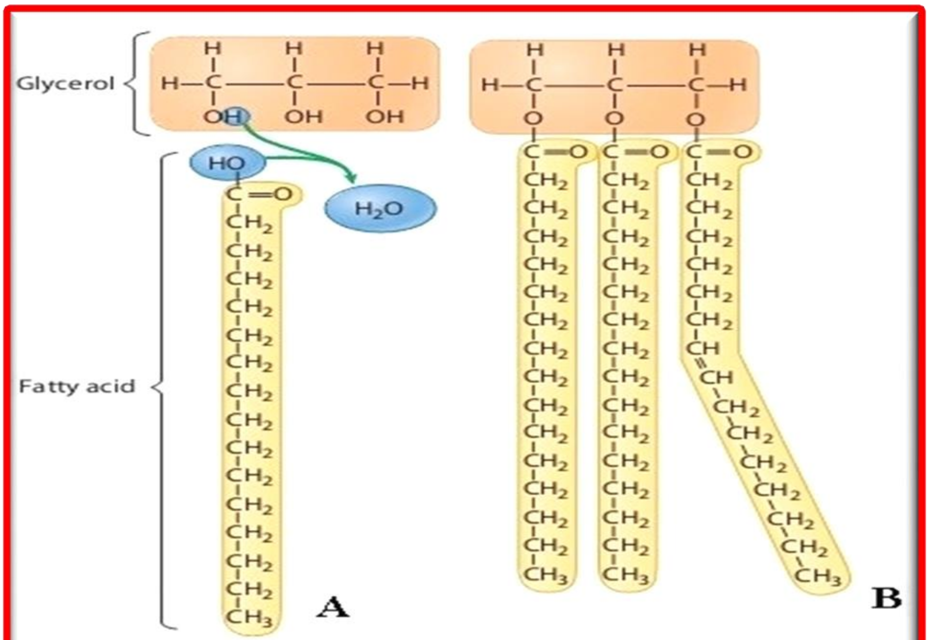


Figure 3.4: (A) a dehydration reaction linking a fatty acid molecule to a glycerol molecule. (B) A fat molecule (triglyceride) consisting of tree fatty acids linked to glycerol.

- **Phospholipids** are the major component of cell membranes. Phospholipids are structurally similar to fats, but they contain only two fatty acids attached to glycerol instead of three. As shown in Figure 3.5A, a negatively charged phosphate group is attached to glycerol's third carbon. The hydrophilic and hydrophobic ends of multiple molecules assemble in a bilayer of phospholipids to form a membrane (Figure 3.5B). The hydrophobic tails of the fatty acids cluster in the center, and the hydrophilic phosphate heads face the watery environment on either side of the membrane.

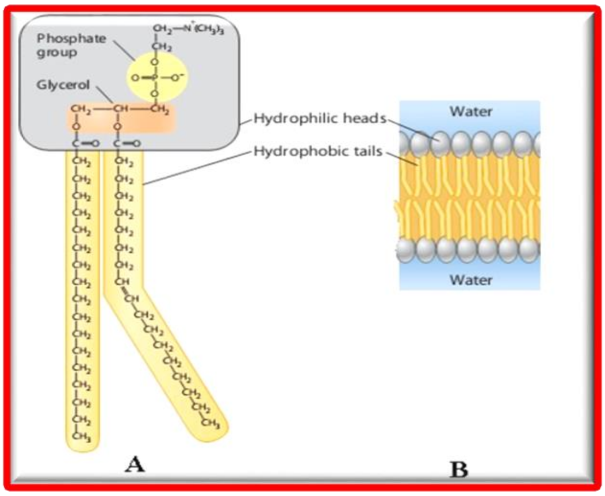


Figure 3.5: (A) Chemical Structure of a phospholipid molecule.
(B) Section of phospholipid membrane.

Steroids are lipids in which the carbon skeleton contains four fused rings, as shown in the structural formula of cholesterol in Figure 3.6.

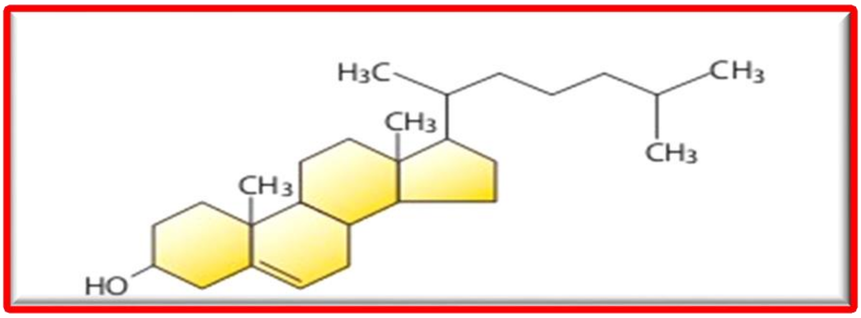


Figure 3.6: Cholesterols, a steroid

Identification of Fats and lipids Solubility test (Figure 3.6)

a. Label 5 clean tubes from 1-5.
b. Add 1 ml of the following solutions using a pipette in the corresponding tubes.

1. Ether	**4. Chloroform**
2. Acetone	**5. Distilled water**
3. Ethanol	

c. Add 1 ml of oil to each tube.
d. Mix and shake each tube vigorously.
e. Record the results in terms of solubility of the oil in the five tubes.

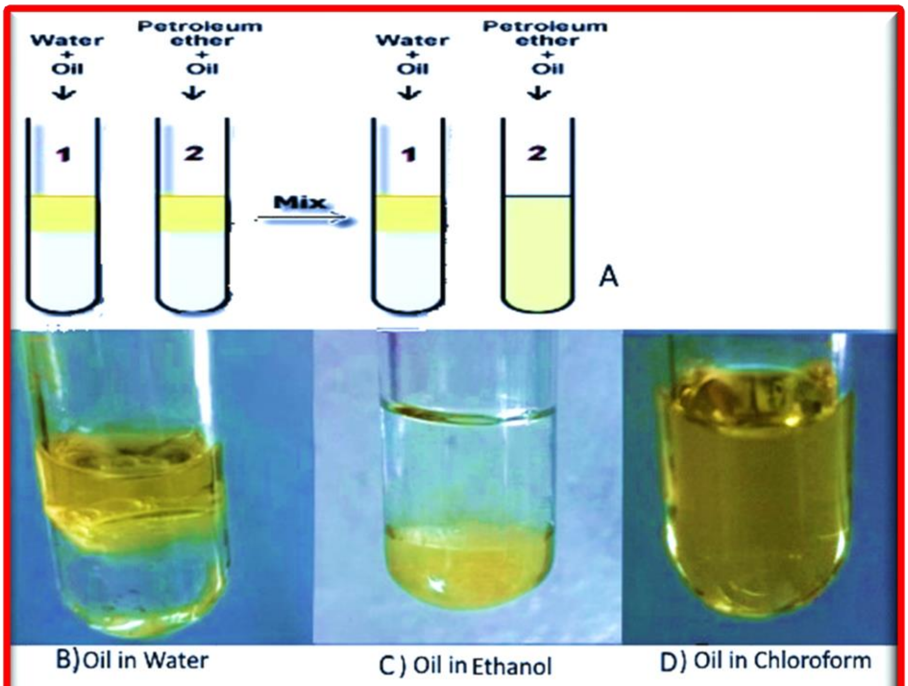


Fig. 3.7: Solubility of Oil in different solvent: A) The procedure; (B) & (C) Insoluble; (D) Soluble

III) Proteins

Proteins have a wide range of functions in living things. They act as structural components (i.e., in hair, muscle, skin, and other tissues) and as enzymes (catalysts that speed up biological reactions).

Protein is a polymer of amino acids. Protein diversity is based on differing arrangements of a common set of 20 amino acid monomers. Amino acids all have an amino group and a carboxyl group. As you can see in the general structure shown in Figure 3.8, both of these functional groups are covalently bonded to a central car-bon atom, called the alpha carbon. Also bonded to the alpha carbon is a hydrogen atom and a chemical group symbolized by the letter R.

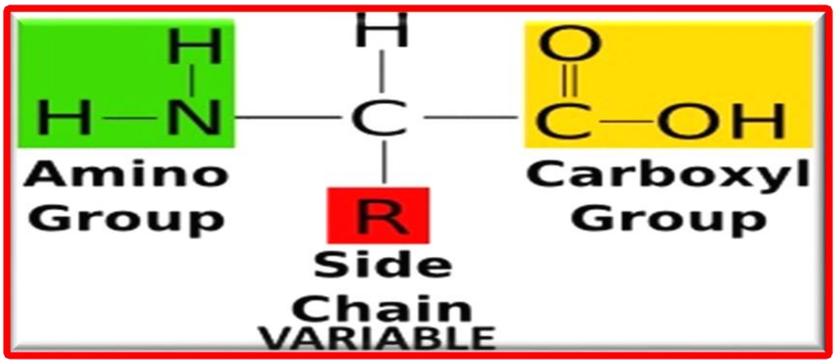


Figure 3. 8: General structure of an amino acid

Cells join amino acids together in a dehydration reaction that links the carboxyl group of one amino acid to the amino group of the next amino acid as a water molecule is removed (Figure 3.9). The resulting covalent linkage is called a peptide bond.

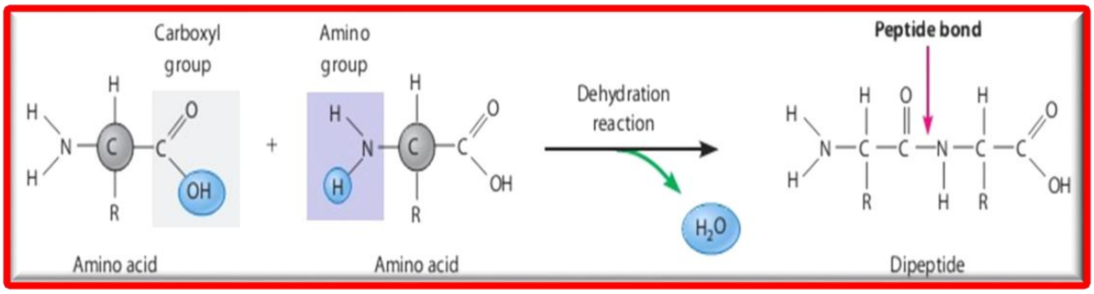


Figure 3.9: Peptide bond formation by dehydration reaction

Identification of peptide bonds in proteins Biuret Test:

Biuret reagent is made of sodium hydroxide and copper (II) sulphate. It is used for determining the presence of protein in a sample. The test relies on the reaction between copper ions and peptide bonds in an alkaline solution. A violet color indicates the presence of proteins. Proteins give a strong Biuret reaction because they contain a large number of peptide bonds. The blue reagent changes to pink when combined with short-chain polypeptides which contain smaller numbers of peptide bonds.

Experiment

a. Label 4 clean tubes from 1-4 (Figure 3.10).
b. Add 1 ml of the following solutions using a pipette in the corresponding tubes.

1. Egg Albumin	4. 5 % Glucose
2. 1 % Starch	5. Distilled Water

c. Add 1 ml of $CuSO_4$.
d. Add 1 ml of 10% sodium hydroxide solution (NaOH) to each tube.
e. Mix and shake each tube vigorously.
f. Record the results in terms of solubility of the oil in the five tubes.

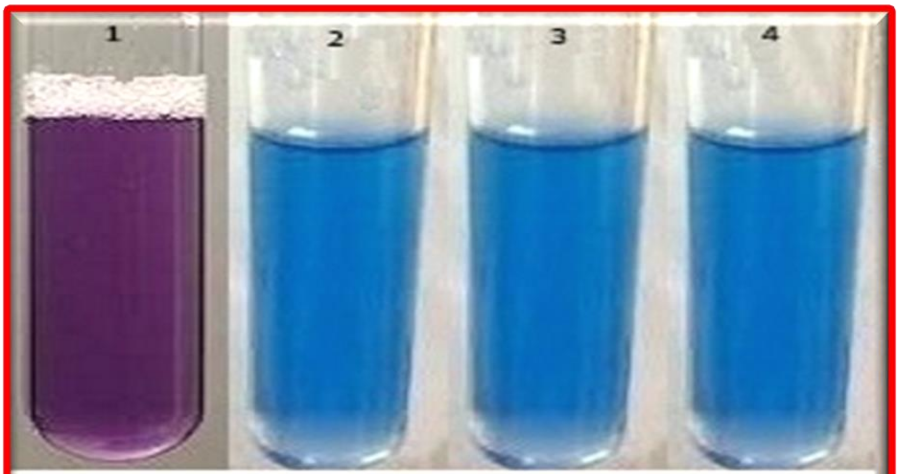


Fig. 3.10: Biuret test for 1: Egg albumin (+ve); 2: Starch (- ve) 3: Glucose (-ve); 4: D. H2O (-ve)

Vitamin C:

Ascorbic acid, commonly known as Vitamin C, is a water-soluble vitamin. It is important to the human diet. It helps the body in forming connective tissue, bone, teeth, blood vessel walls, and assists the body in assimilating iron and amino acids. A diet deficient in Vitamin C may cause a person to develop scurvy. Vitamin C is also an anti-oxidant. This means that it will help prevent oxidation. Consider foods that contain little or no Vitamin C, such as apples and bananas. After exposure to the air, these fruits turn brown. Now consider foods that contain a large amount of Vitamin C, such as oranges, lemons, and limes. When these fruits are exposed to the air, they do not turn brown. This is due to Vitamin C's ability as an anti-oxidant.

Plants and some animals make their own Vitamin C, but humans do not. For this reason, humans need to seek Vitamin C from other sources. The vitamin is present in foods such as citrus fruit, green vegetables, and potatoes, but it is easily deactivated by cooking or standing in the presence of air. The chemical instability of vitamin C acid is due to the fact that it is a strong reducing agent (Fig. 3.11) and can be deactivated by a wide range of oxidizing agents. For example, atmospheric oxygen oxidize the vitamin C in the juices that are stored in open containers and reduce its concentration in them.

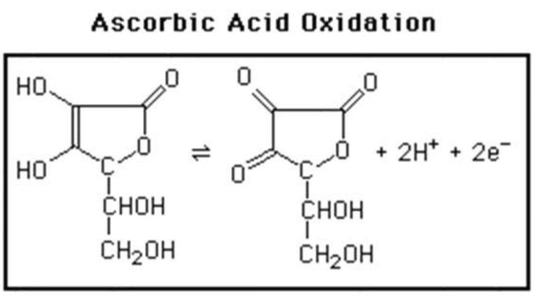


Fig. 3.11: Reduced and Oxidized forms of Indophenol

Experiment:

This experiment studies the concentration vitamin C in Lemon, Orange, and grapefruit juices chemically. Vitamin C is both a reducing agent and a weak acid, so, the vitamin C concentration in the juices will be determined using an oxidation and reduction titration. The titration reaction is:

$$H_2C_6H_6O_6 + HC_{12}H_6C_{12}O_2N \longrightarrow C_6H_6O_6 + HC_{12}H_8C_{12}O_2N.$$

(vitamin C) **(indophenol)**

In this reaction, the reactants and products are colorless or pale yellow with the exception of indophenol, which is either blue (pH > 4), or purple (pH < 4). Hence, the reaction between vitamin C and indophenol bleaches indophenol, and the color of the reaction mixture can be used to detect the end point of the titration then determine the vitamin C concentration in the juice.

Procedure (Fig. 3.12):

1. Add 1 ml of 1% indophenol (about 18 drops) in a cleaned test tube.

2- By a dropper, add juice drop by drop with shaking after each addition.

3- Continue adding the juice until the indophenol became colorless (the color of original juice you added)

4- Count How many drops you need from the juice to convert the indophenol to colorless.

5- Repeat step 1 – 5 using other juice

Note: the concentration of the vitamin C in the juice is inversely proportional with the number of juice drops needed to make the color of indophenol to disappear.

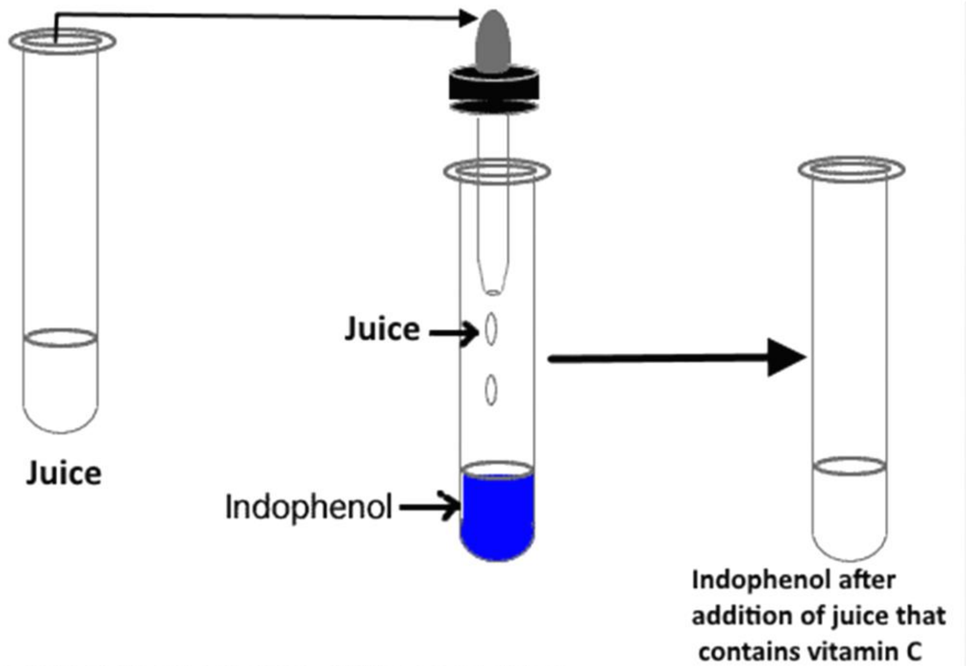


Fig. 3.12: Testing the presence of Vitamine C in juices of different fruits

Experiment 4

Enzyme Activity

Introduction

An enzyme is a protein molecule that serves as a biological catalyst or a substance that can increase the rate of a reaction without changing its own structure. Enzymes catalyze reactions that occur inside living things (i.e. metabolic reactions) by lowering the activation energy necessary to break the chemical bonds in reactants and form new chemical bonds in the products (Fig. 4.1).

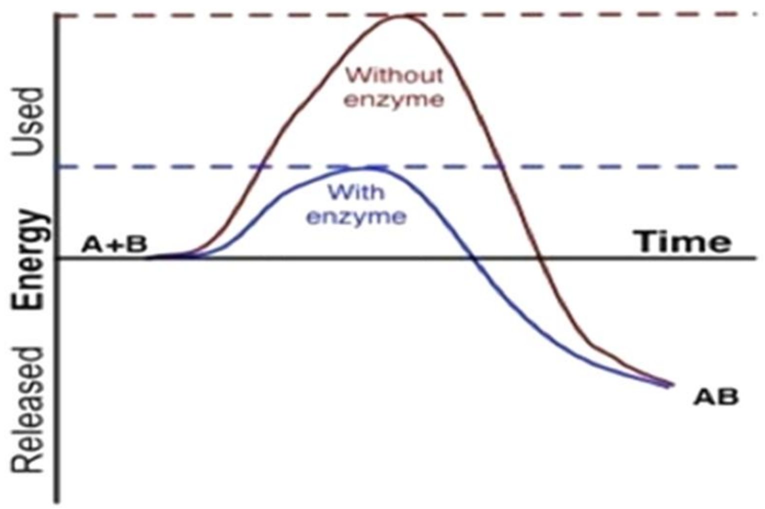


Figure 4.1: Enzymes catalyze chemical reactions by lowering the activation energy necessary for a chemical reaction to proceed.

Enzymes are proteins with 3-dimensional conformation work by binding to a substrate (reactant) at their active site to form an enzyme-substrate complex, and converting the substrate into product(s) of the chemical reaction **(Figure 4.2)**. Once the enzyme converts the substrate it releases the product(s) and leaves the reaction in the same exact form it was introduced. Because enzymes are not altered during the reactions, they can be used repeatedly.

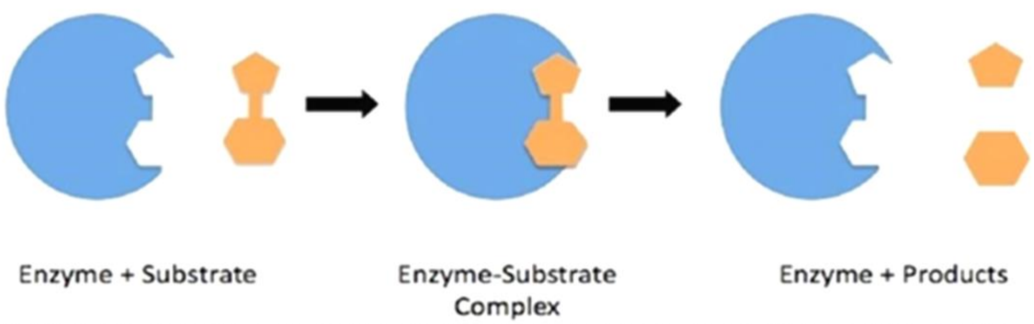


Figure 4.2: A diagram illustrating a substrate binding to the active site of an enzyme. After the product is formed and released from the active site, the enzyme can accept a new substrate molecule.

Enzymes are substrate specific, meaning that they catalyze only specific reactions. For example, proteases (enzymes that break peptide bonds in proteins) will not work on starch (which is broken down by the enzyme amylase). Notice that both of these enzymes end in the suffix -ase. This suffix indicates that a molecule is an enzyme.

The function of enzyme can be affected by several variables such as enzyme concentration, substrate concentration, cofactors, and environmental factors (Temperature, pH, Salt concentration (buffer) ……. etc.). Environmental factors may affect the ability of enzymes to function. Enzymes function best when they are operating within optimal environmental conditions such as within a specific range of

In this laboratory exercise, you will manipulate various factors (such as **substrate and enzyme concentration, temperature,** and **pH)** that affect **Catalase** activity , which is found in most living organisms. The enzyme **Catalase** decomposes **hydrogen peroxide (H_2O_2)**, a toxic compound, into water and oxygen as in the following reaction:

$$2H_2O_2 + \text{catalase} \longrightarrow 2\,H_2O + O_2$$

The amount of oxygen created is directly proportional to the rate of the enzymatic reaction. Therefore, measuring the amount of oxygen produced provides a measure of the speed at which the reaction is proceeding.

Isolation of the Catalase Enzyme: (This part most be daily done by technician)

1. Peel a potato and dice it into small pieces (approx. 2cm3).

2. Weigh out 500 g of the diced potato and place it in a blender along with 500 ml of tap water. Blend for
   about 10 minute until the mixture is smooth.

3. Place the extra, diced potato in a beaker and cover it with water to prevent it from browning.

4. Filter the slurry in the blender through several layers of cheesecloth held within a funnel into a flask.

5. This is the stock solution of the **catalase enzyme** that all the groups will use. If more enzyme solution is
   needed, the above steps should be repeated with the remaining diced potato.

6. Wash all the equipment used to prepare the enzyme solution immediately.

Part A: Enzyme Activity as a Function of Substrate Concentration (Fig. 4.3) :

An enzyme requires a substrate that it converts into product. The substrate is the substance that the enzyme acts upon. As substrate concentration **increases,** reaction rate will be increased up to certain limit, in present of sufficient concentration of enzyme.

Procedure

1. Obtain eight test tubes and a test tube rack per group
2. Add 20 drops of Potato extract to each of the eight tubes
3. Using a dropper, add 14 drops of H_2O and No H_2O_2 added to test tube #1
4. Using a dropper, repeat steps 3 to the remaining seven test tubes increasing by two drops the amount of H_2O_2 in each test tube. So, each test tube most receive the followings:

Test Tube	Potato extract (# drops)	H2O2 (#drops)
1	**20**	**0.0**
2	**20**	**2**
3	**20**	**4**
4	**20**	**6**
5	**20**	**8**
6	**20**	**10**
7	**20**	**12**
8	**20**	**14**

5. After timing for **60** seconds, mark the maximum height of the bubble column (Foam) by marker.
6. Upon completion, return to each test tube and measure in mm the height of the **foam** in each test tube **(Fig. 4.4)**.
7. Fill in Table 4.1 and plot the results as a bar chart

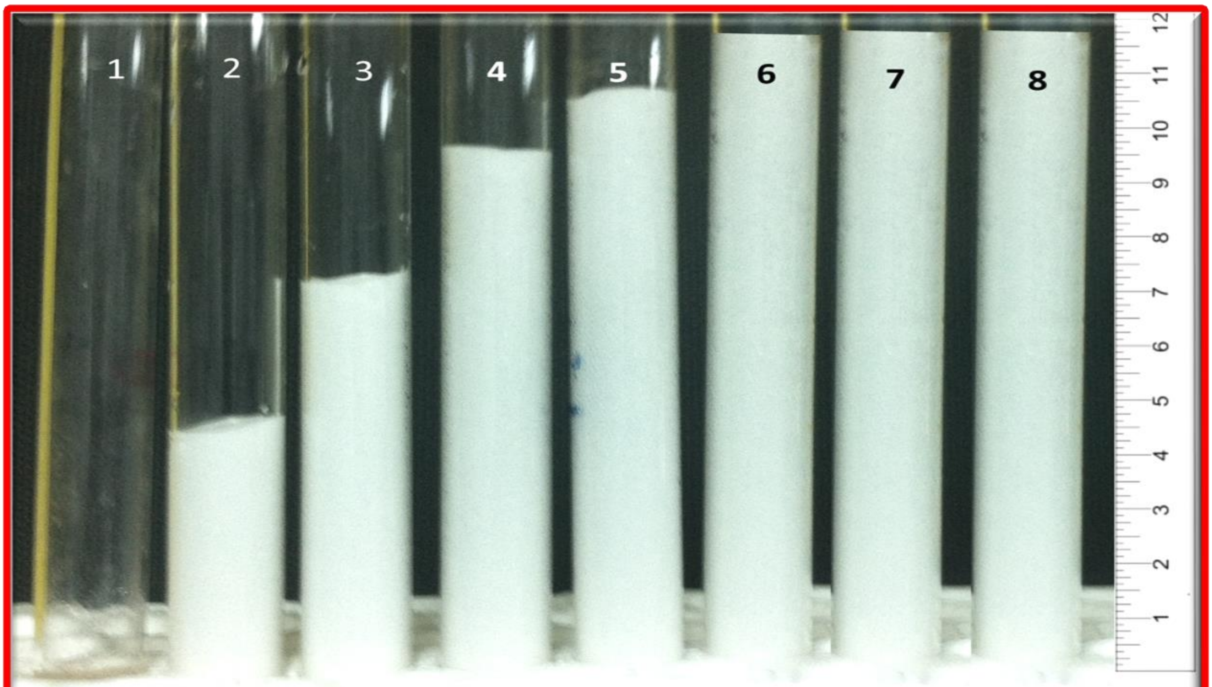


Fig. 4.4:Reaction rate for catalase at various substrate concentrations

Part B: Enzyme Activity as a Function of enzyme Concentration:

As enzyme concentration **increases,** reaction rate will be increased in present of sufficient concentration of substrate.

Procedure

1. Obtain six test tubes and a test tube rack per group
2. Add 20 drops of H2O2 to each of the six tubes
3. Using a dropper, add 25 drops of H2O and No Potato extract (No Enzymes) added to test tube #1
4. Using a dropper, repeat steps 3 to the remaining five test tubes increasing by three drops the amount of **Potato extract**. So, each test tube most receives the followings:

Test Tube	H_2O_2 (# drops)	Potato extract (#drops)
1	**20**	**0.0**
2	**20**	**3**
3	**20**	**6**
4	**20**	**9**
5	**20**	**12**
6	**20**	**15**

4. After timing for **60** seconds, mark the maximum height of the bubble column (Foam)with a marker
5. Upon completion, return to each test tube and measure in mm the height of the **foam** in each test tube **(Fig. 4.5)**.
6. Fill in Table 4.2 and plot the results as a bar chart

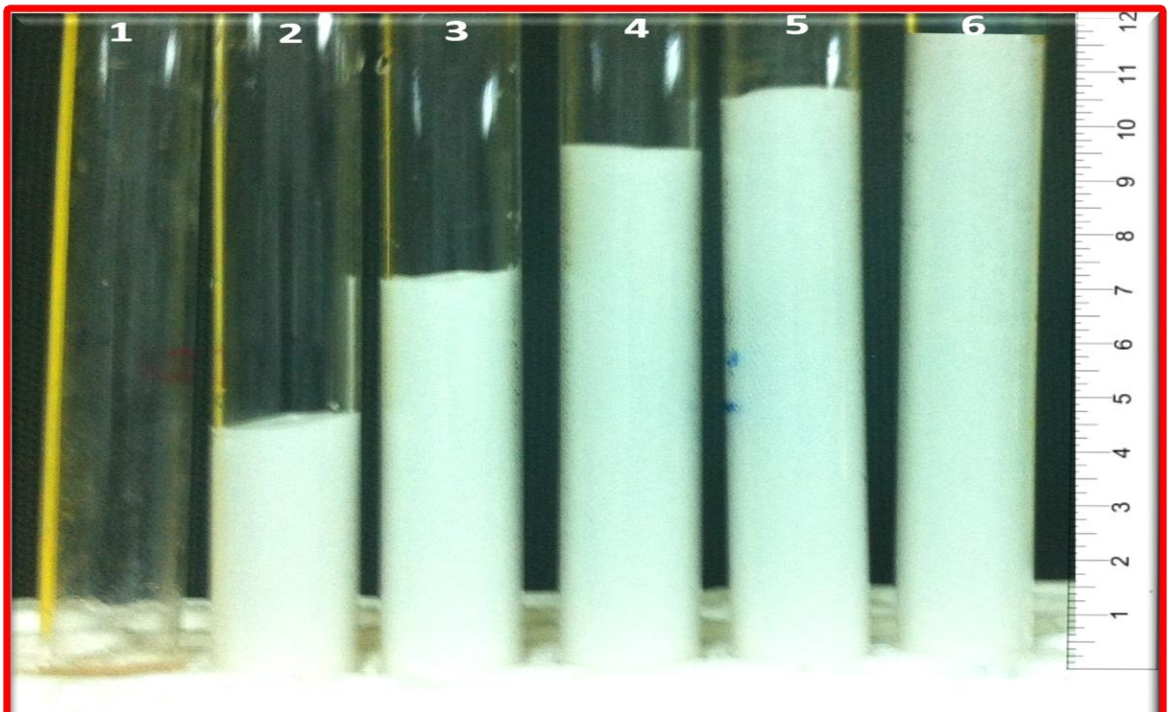


Figure 4.5: Reaction rate for catalase at various enzyme concentrations

Part C: Enzyme Activity as a Function of Temperature:

Temperature is a measure of the speed at which molecules are moving. As temperature increases, the molecular movement speed does so as well. Increasing temperature increases the probability and rate at which enzyme and substrate come together, thereby increasing the reaction rate. However, enzymes are subject to denaturation at excess temperatures. A denatured enzyme's active site conformation is changed not allowing the substrate to bind. The result is that at these temperatures, the reaction will decrease. An enzyme's **optimum** temperature is that point which has the greatest reaction rate but does not denature the enzyme.

Hypothesis: As temperature **increases,** reaction rate will _____________ and then _______________.

Procedure

1. Obtain **ten** test tubes and a test tube rack per group
2. Using a pipette, measure **1 ml (20) of H_2O_2** into **five of the ten** test tubes
3. Using a dropper, add **5 drops of Potato extract** to each of the **other five** test tubes
4. Place one test tube each of **H_2O_2** and catalase into each of the Ice pocket, Refrigerator, Room Temperature, Water bath (50 ^{O}C), and Boiling bath
5. Allow all tubes to acclimate for **10 minutes**
6. After 10 minutes, **proceeding one pair of tubes at a time**, pour the **H_2O_2** into the catalase which incubate under the same condition.
7. After timing for **60** seconds, mark the maximum height of the bubble column (Foam) with a marker
8. Upon completion, return to each test tube and measure in mm the height of the foam in each test tube by Ruler **(Fig. 4.6)**.
9. Fill in Table 4.3 and plot the results as a bar chart in Fig. 4.3

Condition	Test Tube	H_2O_2 (# drops)	Test Tube	Potato extract (# drops)
Ice pocket	1	**20**	6	5
Refrigerator	2	**20**	7	5
Room Temperature	3	**20**	8	5
Water bath (50 ^{O}C)	4	**20**	**9**	5
Boiling bath	5	**20**	**10**	5

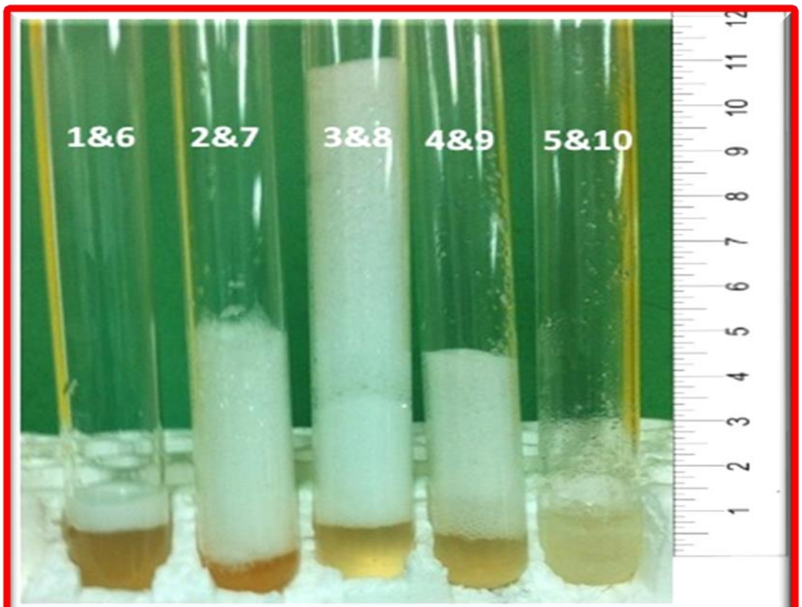


Figure 4.6: Reaction Rate of Catalase as a Function of Temperature

Part C: Enzyme Activity as a Function of pH :

Another variable that can affect enzyme conformation and therefore activity levels is pH. Like with temperature, enzymes also have an **optimum pH.**

Hypothesis: As pH moves **away from optimum,** reaction rate will _______________.

Procedure

1. Obtain **3** test tubes and a test tube rack per group
2. Using a pipette, add **15 drops Potato extract** to each of the three tubes.
3. Using a dropper, add 10 drops of lemon juice, H_2O, and NaOH to test tube 1, 2, 3 respectively
4. After 10 minutes , add **5** H_2O_2 to each test tube
5. After timing for **60** seconds, measure in mm the height of the foam in each test (Fig. 4.7)
6. Fill in Table 4.4

Test Tube	10 drops of	Potato extract (# drops)	After 10 minutes	H_2O_2 (#drops)
1	Lemon juice	15		5
2	H_2O	15		5
3	NaOH	15		5

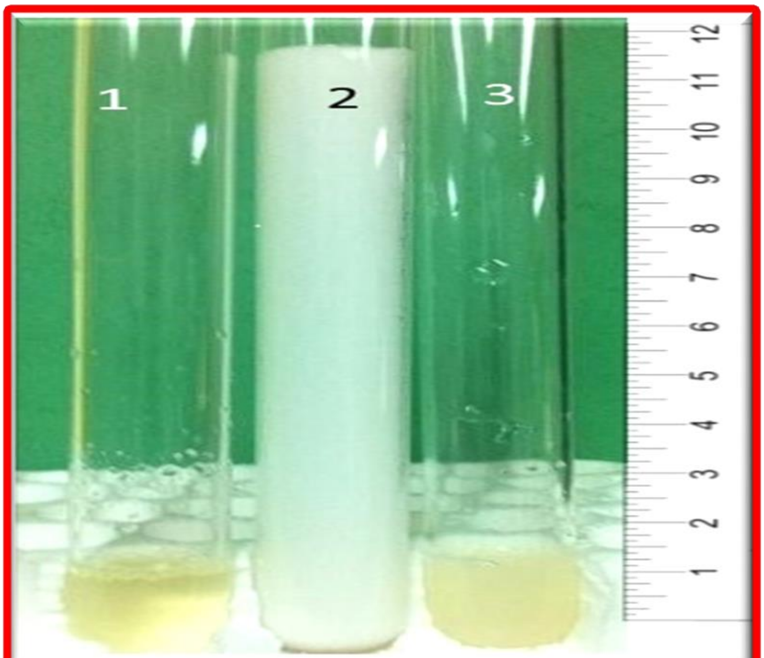


Figure 4.7: Reaction Rate of Catalase as a Function of PH.

Cellular Respiration and Fermentation

Introduction

Respiration is the process by which living organisms' gets energy (ATP) from their food molecules. Bacteria, protists, fungi, plants, and animals need ATP to stay alive. They need ATP for nearly every function that they perform. There are several types of cell respiration.

Aerobic Respiration occurs in several stages in cytoplasm and mitochondria of many organisms. Oxygen must be present for this process to run to completion. It involves glycolysis, pyruvate oxidation, Krebs cycle, and electron transport/chemiosmosis. The general equation for aerobic respiration is (**Fig.5.1)**:

$$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 34\text{-}36\ ATP$$

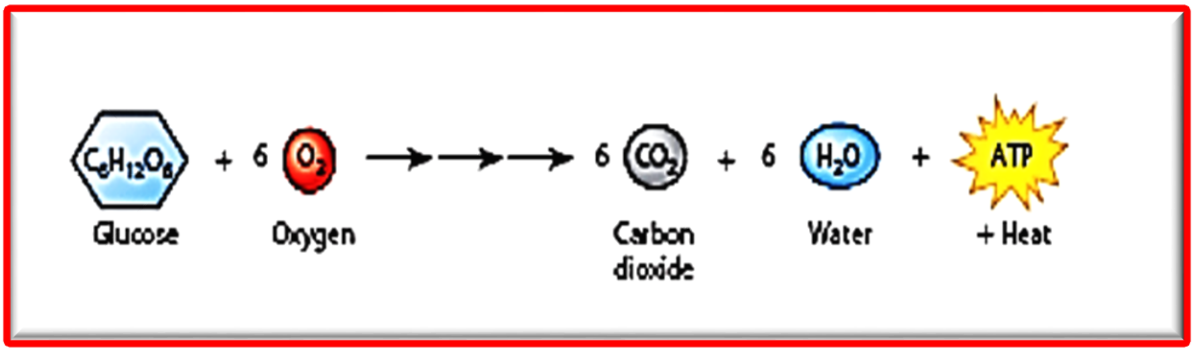


Figure 5.1: Summary equation for Cellular Respiration

Fermentation can occur when O_2 is lacking. Alcoholic fermentation produces 2 molecules of Ethanol, 2 molecules of CO_2, and 2 molecules of ATP for every molecules of glucose that enters glycolysis. During alcohol fermentation, the ethanol and carbon dioxide are considered "waste products." These waste products may be utilized in other pathways by other organisms. The general equation for alcohol fermentation (a type of anaerobic respiration) is (Fig.5.2) :

$$\underset{\text{Glucose}}{C_6H_{12}O_6} \longrightarrow \underset{\text{Ethanol}}{2C_2H_5OH} + \underset{\text{Carbon dioxide}}{2CO_2} + \underset{\text{Energy}}{2\ ATP}$$

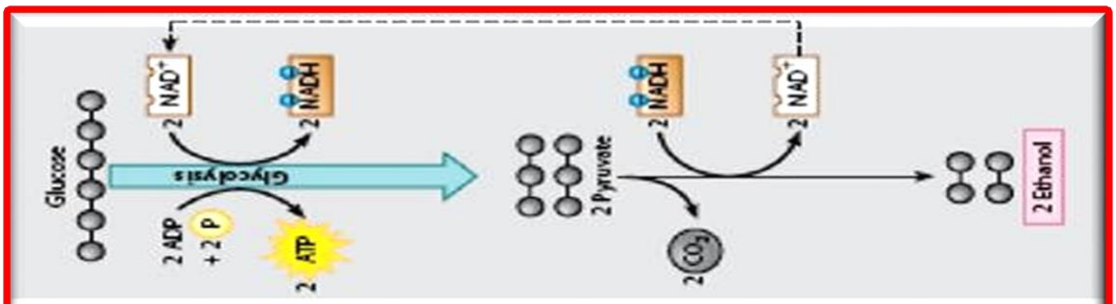


Figure 5.2: Alcohol fermentation

The sugar, glucose, is the initial reactant of both aerobic respiration and fermentation. Other carbohydrate molecules, such as more complex sugars and starches, must be converted to glucose and then broken down to release energy during respiration.

PART A: Carbon Dioxide (CO_2) Liberation:

In this exercise you will observe the reaction between CO_2 and barium hydroxide [Ba $(OH)_2$] during exhalation/inhalation. Barium carbonate will precipitate and water will be produced in this reaction **(Fig. 5.3).**

1. Take two bottles half filled with saturated barium hydroxide [Ba $(OH)_2$] solution. Label them as A (Exhale) and B (Inhale).
2. Take a deep breath then exhale in Bottle A (Exhale). Repeat this step 5-10 times.
3. Record the color of the solution after exhalation.
4. In Bottle B (Inhale), breath out, hold your nose and inhale air through your mouth from bottle B. repeat 10 times.
5. Record the color and explain your observation.

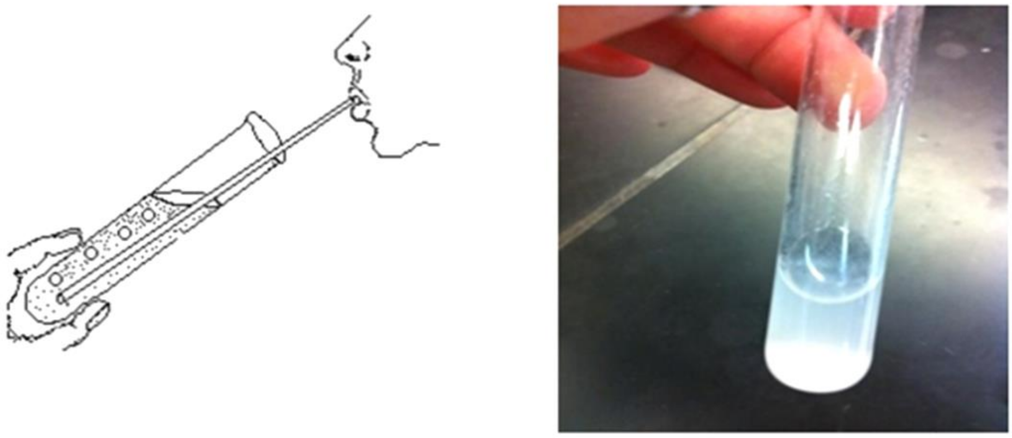


Figure 5.3: Formation of Barium Carbonate in Exhale Bottle

PART B: Water liberation

Take a clean empty glass test tube and exhale in it several times. Record your observation and explain it.

PART C: Heat of Respiration

In this exercise you will have two thermos respiratory chambers were already prepared (Chamber A and B) **(Fig.5.4).**

- Chamber A: contains actively *germinating* seeds were placed and stoppered with wet cotton to allow respiration.

- **Chamber B**: similar quantity of *killed* cooled seeds was placed and cotton stopper, a Thermometer was inserted inside each chamber Set up the apparatus as in the diagram below **(Fig.5.4)**.
- Record the temperature in each chamber at the end of your lab and explain your observation.

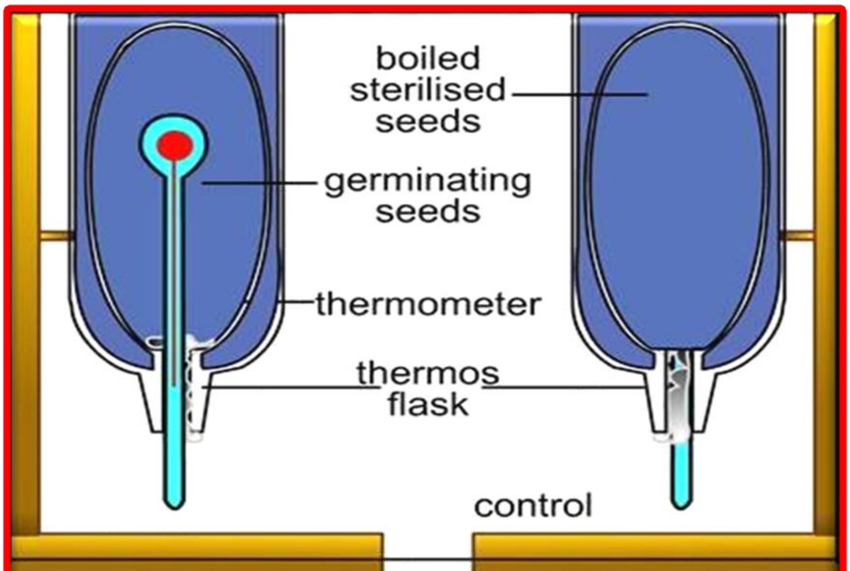


Figure 5.4: Heat Liberation during Cellular Respiration

PART D: Dehydrogenase activity

During respiration, hydrogen atoms are removed from glucose molecules by enzymes called dehydrogenases and passed to various chemicals called hydrogen acceptors **(Fig.5.5).** As the hydrogen atoms pass from one hydrogen acceptor to another, energy is made available for chemical reactions in the cell. In this experiment, a dye called methylene blue acts as an artificial hydrogen acceptor. When this dye is reduced by accepting hydrogen atoms it goes colorless (leukomethylene blue) **(Fig.5.6).**

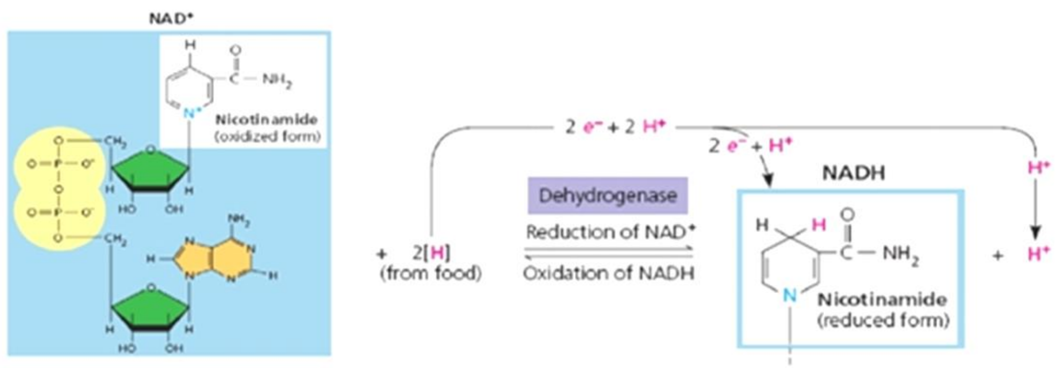


Figure.5.5: Role of Dehydrogenase enzyme during respiration process.

The practical

1. Label 3 test tubes, add the contents to each tube according to the following **(Fig.5.6)** :
   - Tube 1: 3 mL glucose + 1 mL methylene blue dye + 5 mL yeast suspension
   - Tube 2: 3 mL glucose + 1 mL methylene blue dye + 5 mL Boiled yeast suspension
   - Tube 3: 3 mL glucose + 1 mL methylene blue dye + 5 mL Distilled water
2. Incubate at 37 ºC for 15 minutes and record the color disappearance time for each tube.

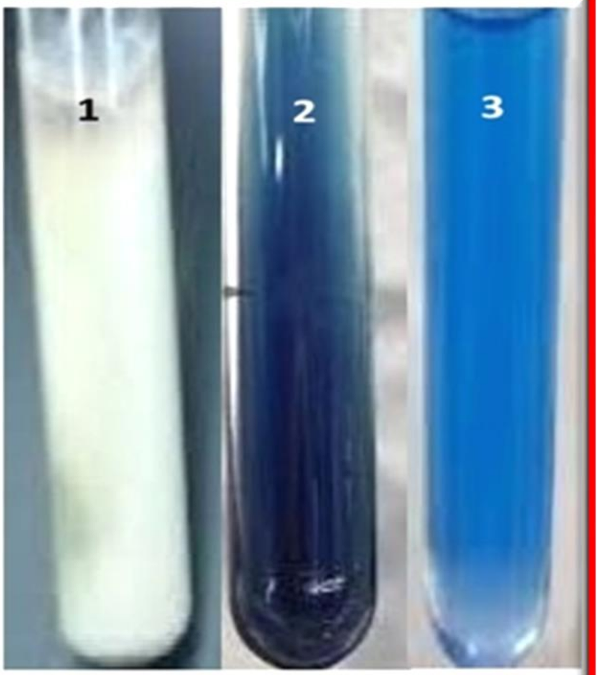


Fig. 5.6: Dehydrogenase test: (1) Positive (2) -ve, denatured enzyme; (3) -ve no enzyme

E. Anaerobic Respiration of yeast cells

Yeast cells are facultative anaerobes; they can use oxygen if it is present to metabolize sugars to CO_2 and water. If no oxygen is available, yeast cells metabolize pyruvate to Ethanol and CO_2. In this exercise you will place part of a yeast solution in anaerobic environments with various sugars and starch and observe the end product of yeast metabolism. You will use phenol red dye. This dye is a pH indicator that turns yellow in acidic solution and red in basic solution.

1. Label 4 test tubes, add the contents to each tube according to the following:
   - Tube 1: 5mL Distilled water + 5 mL yeast suspension
   - Tube 2: 5 mL of 20% glucose + 5 mL yeast suspension
   - Tube 3: 5 mL of 20% Sucrose + 5 mL yeast suspension
   - Tube 4: 5 mL of 20% Starch + 5 mL yeast suspension
2. Add a layer of oil at the top of each test tube to prevent the diffusion of oxygen from the air into the Yeast in Glucose solution, this ensuring **anaerobic** respiration will occur.

3. Wrapped each test tube with Aluminum foil (to prevent Photosynthesis)and Connect it with other test tube containing 5 ml lime water or phenol red as in the figure 5.7

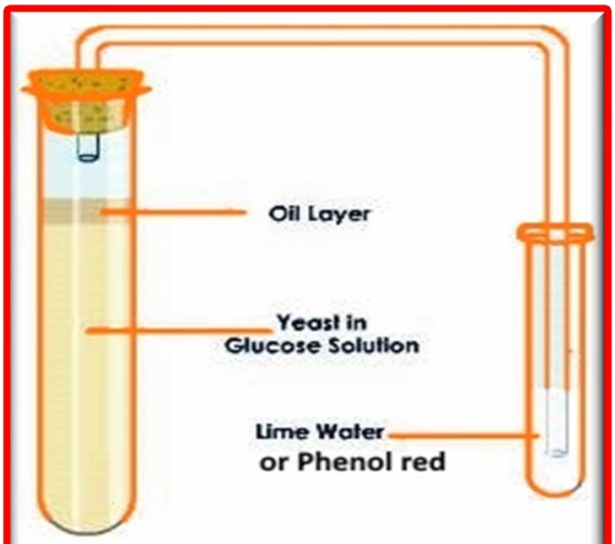


Fig. 5.7: The simplest set up that demonstrate Anaerobic respiration in yeast

4. Incubate the above system in water bath at 37^{O}C
5. Record the length of the air (mm) trapped in the vial in each test tube after 15 min.
6. Record the color of the solution in each tube after 15-20 min.

Diffusion and Osmosis

Introduction

The plasma membrane of the cell is selectively permeable membrane, which means that it allows the passage of some substances while it prevents the passage of others. Cells constantly move molecules across the plasma membrane, into and out of the cytoplasm. Some molecules are actively transported across the membrane, a process that requires transport proteins and energy input. Active transport occurs against a concentration gradient. In contrast to this active transport, diffusion and osmosis describe the movement of molecules along the concentration gradient in a passive process that does not require cellular energy.

Diffusion is the net movement of molecules or atoms from a region of high **concentration** to a region of low concentration. This is also referred to as the movement of a substance down a **concentration gradient.** A **gradient** is the change in the value of a quantity (e.g., **concentration, pressure, temperature**) with the change in another variable (e.g., distance). For example, a change in concentration over a distance is called a concentration gradient. Diffusion refers to the process by which molecules intermingle as a result of their **kinetic energy** of random motion and occurs even in the absence of a membrane **(Fig. 6.1).**

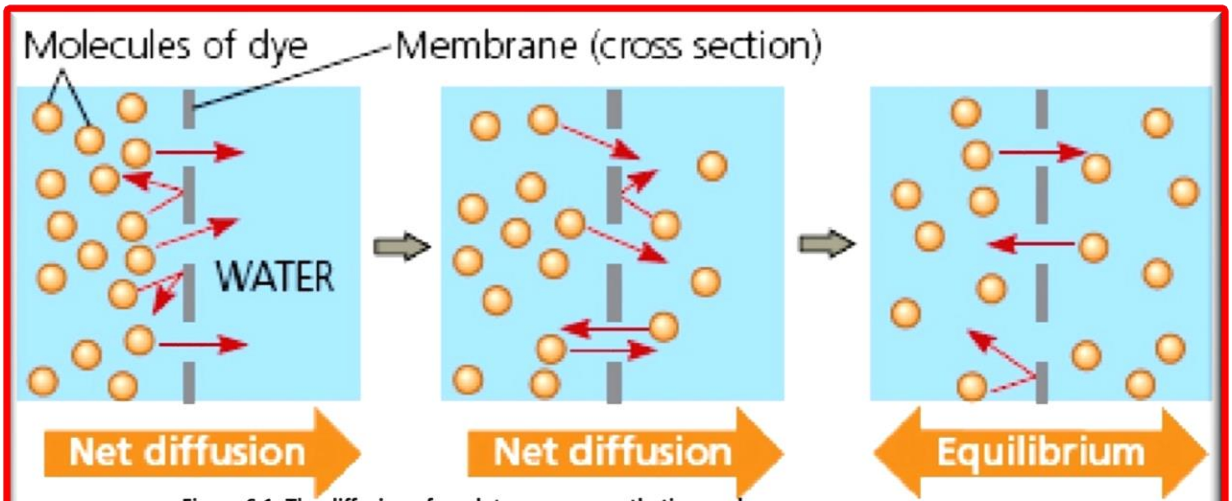


Figure 6.1: The diffusion of a solute across a synthetic membrane.

Osmosis is the net movement of solvent (e.g. water) molecules through a semi-permeable membrane from a region of low solute concentration (High free solvent concentration) into a region of higher solute concentration (Low free solvent concentration), in the direction that tends to equalize the solute concentrations on the two sides. Osmosis is a vital process in biological systems. The biological membranes are selective permeable membranes that allow some substances, including water, to pass freely through the membrane and block the passage of others **(Fig. 6.2).**

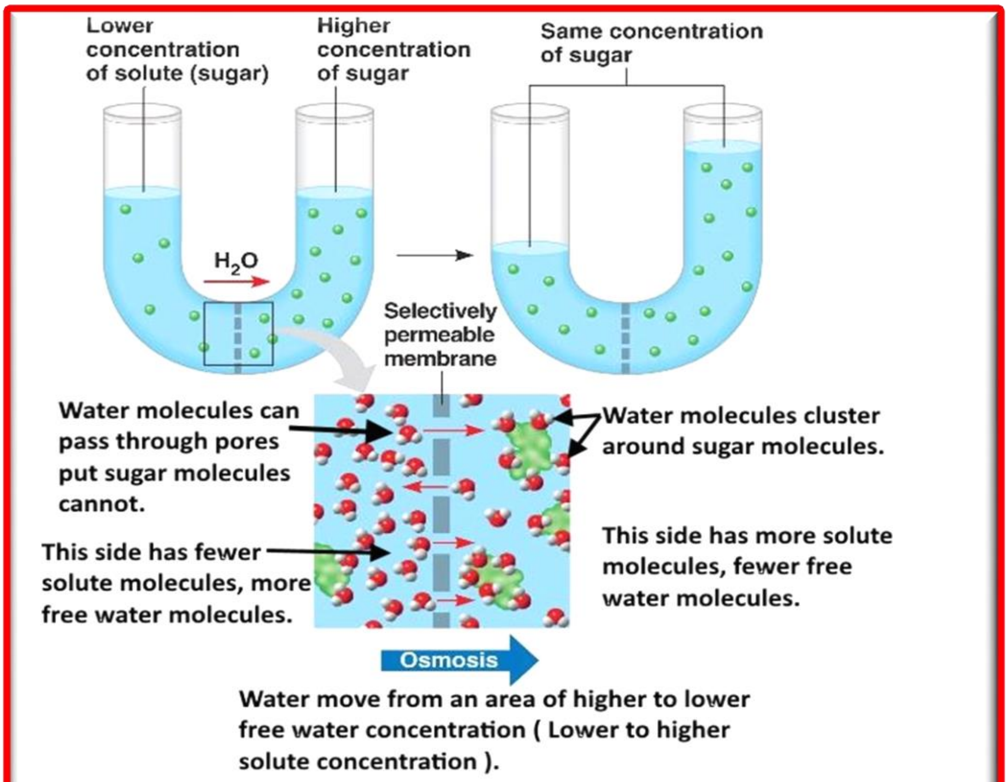


Figure 6.2: Osmosis. Two sugar solutions of different concentrations are separated by a membrane that the solvent (water) can pass through but the solute (sugar) cannot.

Tonicity is a description of one solution's solute concentration compared to that of another solution. The **hypertonic** solution has a greater concentration of Solute than the solution on the other side of the membrane. The **hypotonic** solution has a lower concentration of Solute than the solution on the other side of the membrane. When the two solutions are in **equilibrium**, the concentration of Solute being equal on both sides of the membrane, are said to be **isotonic**. The *net flow* of water is from the hypotonic to the hypertonic solution. When the solutions are isotonic, there is no net flow of water across the membrane.

Factors Influencing the Rate of Diffusion

The rate of diffusion (the distance moved in a given amount of time) is affected by factors such as temperature and the size of solute particles **(Fig. 6.3).**

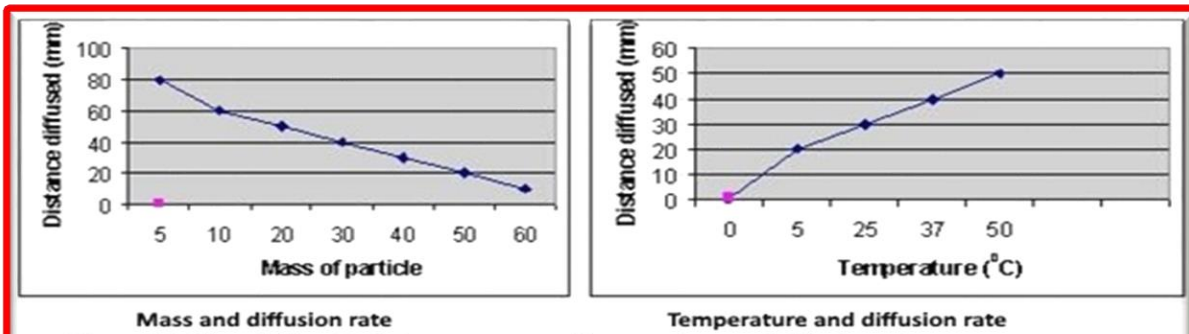


Figure 6.3: The relationship between mass and temperature and diffusion rate

PART A: Diffusion

In diffusion, there is a net movement of substances from a region of their greater concentration to a region of their lesser concentration. Diffusion can occur in solids, liquids, or gas, and across the plasma membrane of cells.

EXERCISE 1 : Diffusion of Gases

Hydrochloric acid (hydrogen chloride gas, HCl, dissolved in water) and ammonia solution, (ammonia gas, NH3, dissolved in water) consist of a volatile solute that can easily leave the solution and enter the gaseous state. When gaseous molecules of hydrogen chloride and ammonia collide they react according to the reaction below to form Ammonium chloride, NH4Cl, a white solid. A smoky white solid deposited on the glass tubing will provide evidence for the reaction.

$$HCl\ (g) + NH_3\ (g) \longrightarrow NH4Cl\ (s)$$

Procedure

1. Fix glass tubing with a clamp to a stand.

2. Saturate a piece of cotton with hydrochloric acid (HCl) and plug one end of the tube with it.

3. Saturate another piece of cotton with ammonium hydroxide (NH4OH) and plug the opposite end of the tube.

4. Observe what happen (Fig. 6.4).

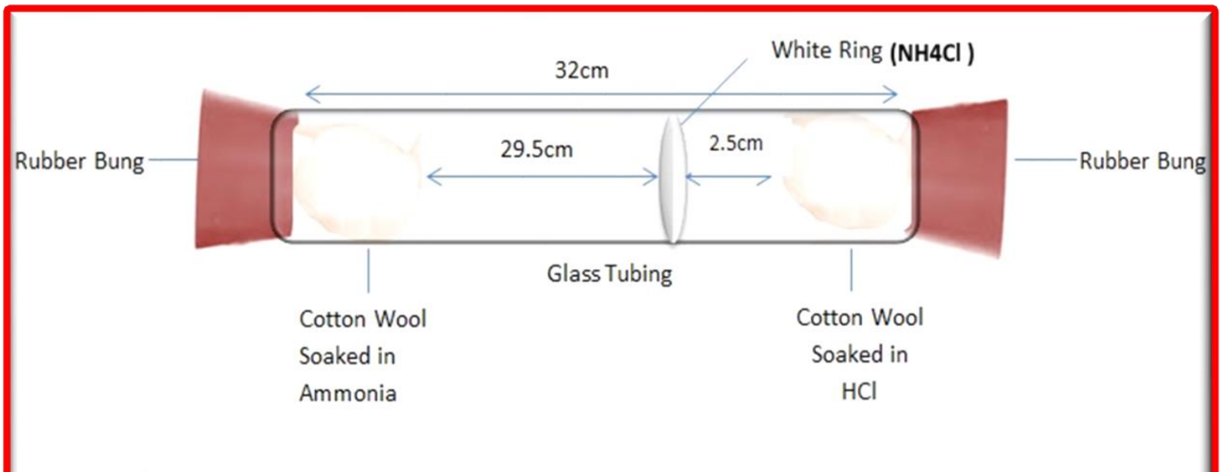


Fig. 6.4: Diagram showing the reaction of NH_3 and HCl (which formed a white ring)

EXERCISE 2: Diffusion of Molecules through a Selectively Permeable Membrane

If a membrane separates two solutions containing dissolved substances of different molecular size, some substances may readily pass through the pores of the membrane, but others may be excluded. In this exercise, you will test the permeability of the tubing to sulfate ion, starch, chloride ion and egg albumin.

Procedure (Fig.6.5 A):

1. Fold over 3 cm at the end of a 25 - to 30 – cm piece of dialysis tubing that has been soaking in water for a few minutes, pleat the folded end "accordion style," and close the end of the tube with the string or a rubber band, forming a bag. This procedure must secure the end of the bag so that no solution can seep through.
2. Slip Roll the opposite end of the bag between your fingers until it opens, and add 25 ml of a solution of 1% soluble starch in 1% sodium sulfate ($Na2SO4$).
3. Remove the bag from the funnel; and rinse its contents in a beaker filled with water.
4. Pour 250 ml of a solution of 1% egg albumin in 1% sodium chloride (NaCL) into a 600 ml beaker. Leave the bag in the beaker for 90 minutes then pour 20 ml of the beaker contents into a graduated cylinder. Label eight test tubes 1-8.
5. Pour out 5 ml from graduated cylinder into each of the first four test tubes. And do the following tests and record your results as (present=+, absent= -) **(Fig.6.5 B):**.
   a. Test for starch: Add several drops of iodine solution (I2KI) to test tube 1. The solution will turn blue-black if starch is present.
   b. Test for sulfate ions: Add several drops of 2% barium chloride (BaCL2) to test 2. A white precipitate will be formed, if sulfate ions are present.
   c. Test for chloride: Add several drops of 2% silver nitrate (AgNO3) to test tube 3. A milky white precipitate indicates the presence of chloride ions.
   d. Test for protein: Add several drops of NaOH then Biuret reagent to test 4. The solution will change from blue to pinkish-violet indicating the presence of protein.
6. Wash the graduated cylinder and rinse the bag in a beaker of distilled water then empty its contents into the graduated cylinder.
7. Decant 5 ml samples into each of the four remaining test tubes and perform the tests for Starch, sulfate ions, chloride ions and protein. Record the results of both series of tests.

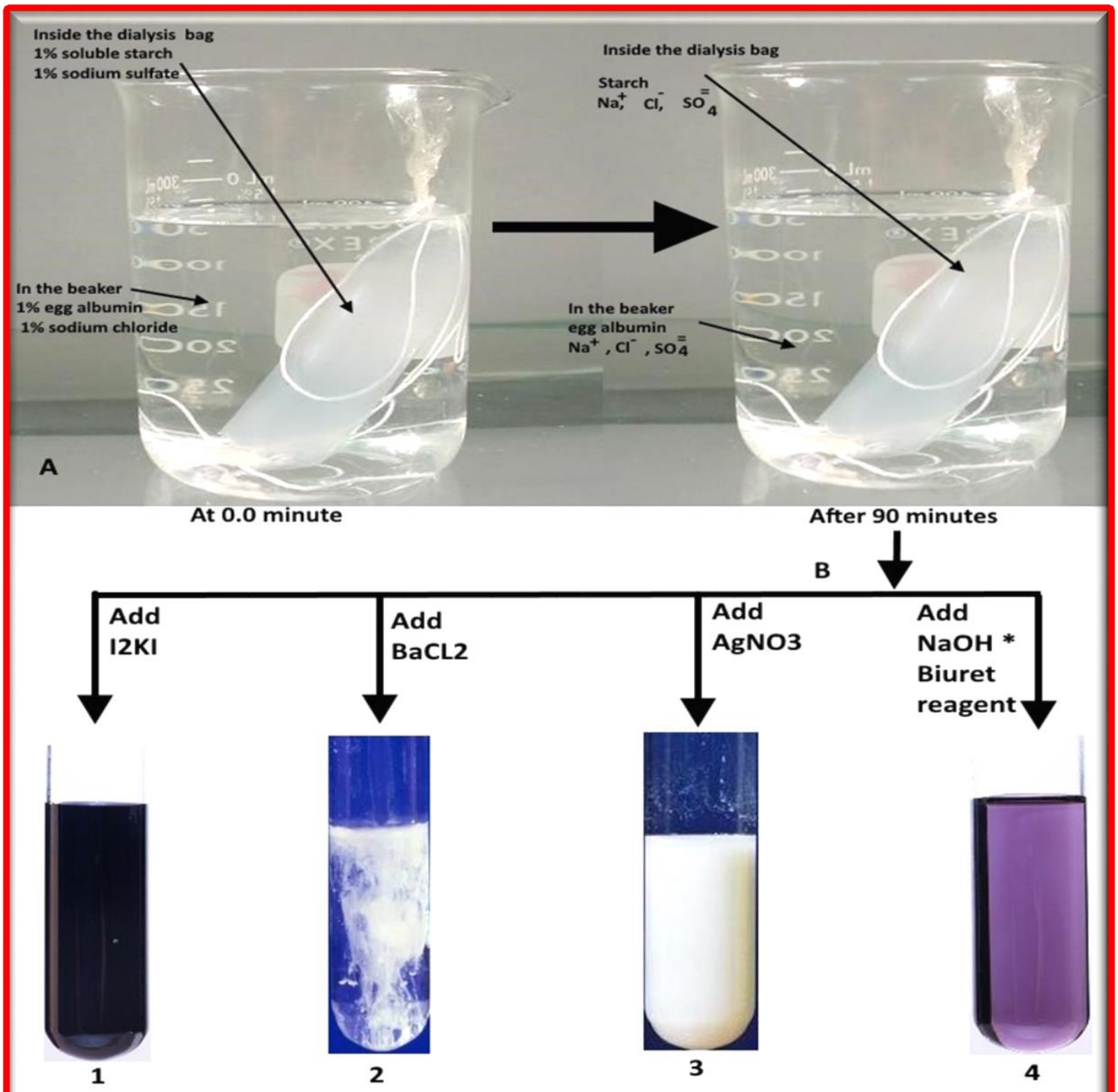


Fig. 6.5: A)Dialysis tubing bag. B) Positive result of : 1:starch 2: sulfate ions 3: chloride 4: protein

PART B: Osmotic Activity in Cells Exercise 1: Osmotic Behavior of Animal Cells

The shape of red blood cell is flattened and pinched inward into biconcave disk. When water moves into red blood cells placed in a hypotonic solution, the cells swell and the membranes burst, or undergo lysis. When water moves out of red blood cells placed in a hypertonic solution, the cells shrivel, or crenate (Fig. 6. 6A).

Exercise 3: Osmotic Behavior in plant Cells

The presence of a cell wall and a large central vacuole in a plant will affect the cell's response to solutions of differing **tonicity**. When a plant cell is placed in a hypertonic solution, water moves out of the cell; the protoplast shrinks and may pull away from the cell wall. This process is called **plasmolysis**, and the cell is described as **plasmolyzed** (Fig. 6.6B). In a hypotonic solution, the cell's

protoplast expands and at this point, the cell is **turgid**. If a plant's cells and their surroundings are isotonic, there is no net tendency for water to enter, and the cells become flaccid.

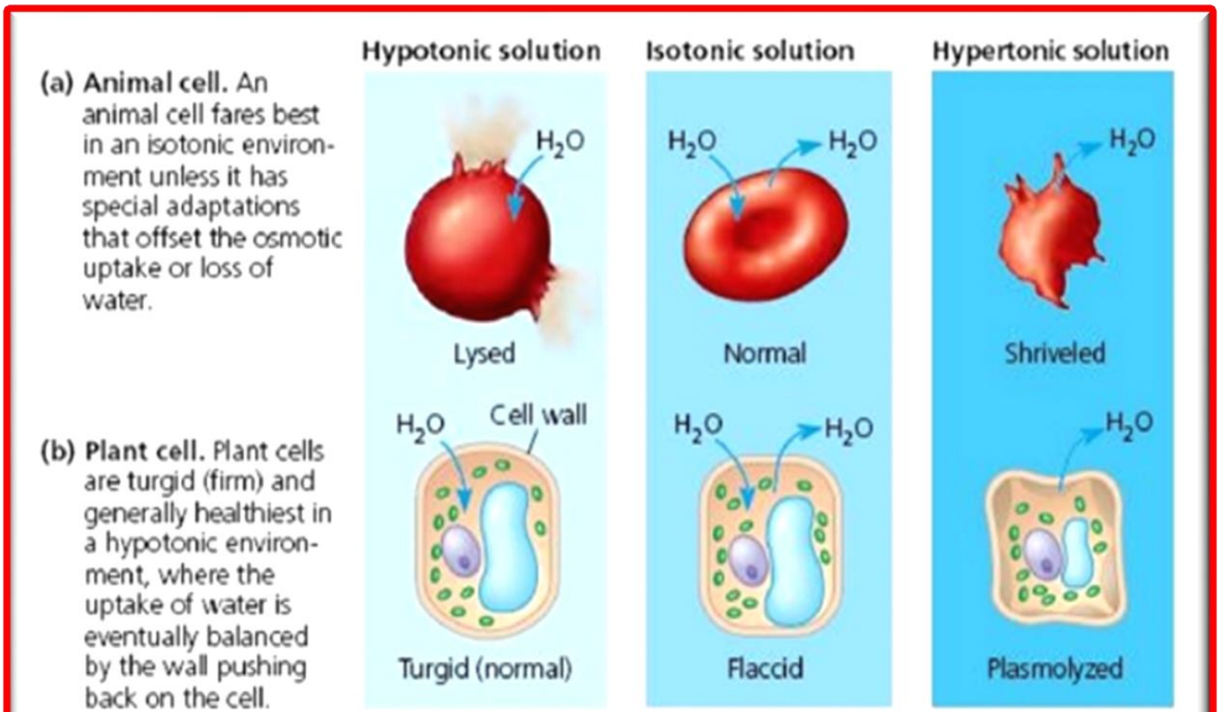


Figure 6.6: The water balance of living cells.

Procedure

To demonstrate the effects of Osmosis in plant cells, you will use onion cells and solutions with different tonicity, hypotonic, hypertonic and isotonic.

1. Prepare 3 wet mounts slides of onion epidermis, using a drop of (0.85 % NaCL solution to the first slide, 10% NaCL solution to the second slide and distilled water to the third.
2. Add one drop of lugols iodine stain for each slide wait 5 minutes.
3. Draw and describe the changes in the cell membrane under compound microscope.

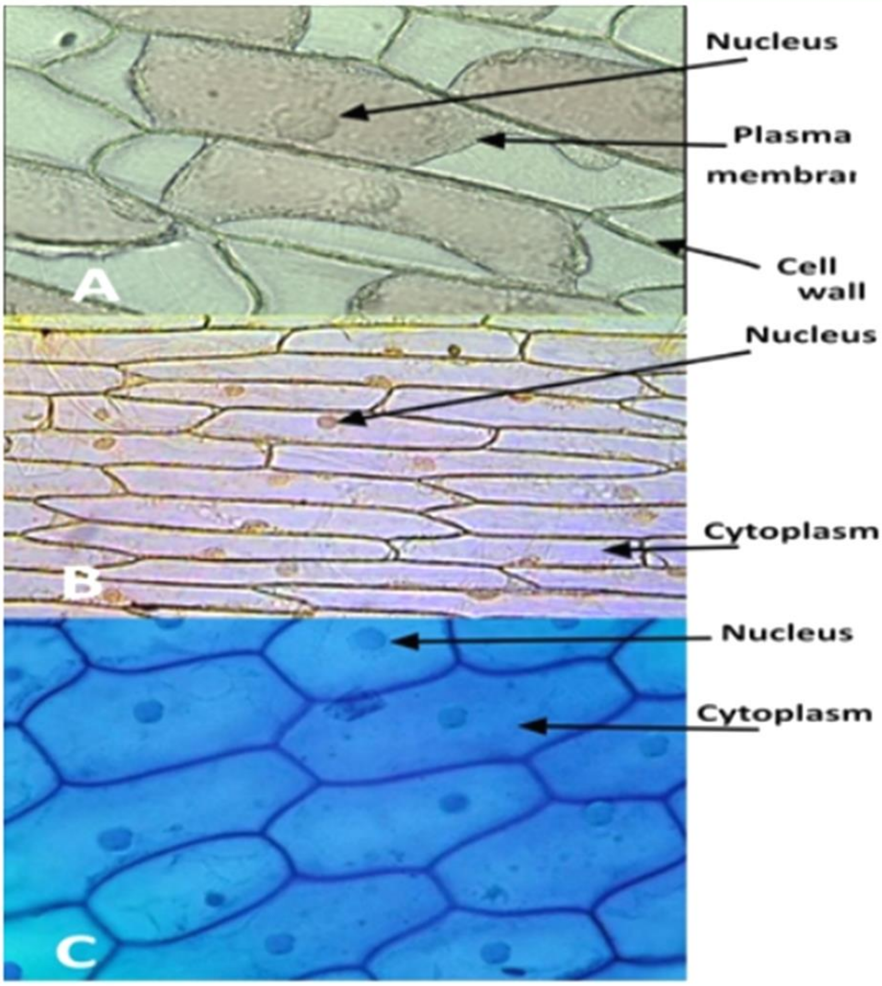


Fig. 6.7: plant cells in different solution, A: Plasmolyzed; B: Flaccid; C: Turgid (Deplasmolyzed)

Experiment 7
Mitosis and Meiosis

Introduction

Our human bodies are composed from trillions of cells that are specialized to perform specific functions. Most of these cells (**somatic cells**) that have the ability to divide to produce genetically identical daughter cells. However, our specialized **germ line cells** (sex cells) divide to produce the gametes. The continuity of life is based on the reproduction of cells, or **cell division**. Cell division plays several important roles in life. When a prokaryotic cell divides, it is actually reproducing, since the process gives rise to a new organism, while in multicellular eukaryotes, cell division is important for growth, development and regeneration.

Mitosis

Mitosis is the division of nucleus. The division of nucleus in somatic cells is preceded by DNA duplication, which guarantees that one copy is passed into each daughter cells. **Mitosis** is usually followed by **cytokinesis**, or cytoplasmic division, in which the other cell's constituents like organelles are divided.

The cell cycle

The cell cycle refers to the continuing series of divisions alternating with cell growth. Mitosis is just one part of the cell cycle (**Figure 7.1**). Mitotic (M) phase, which includes both mitosis and cytokinesis, is usually the shortest part of the cell cycle. The mitotic phase alternates with a much longer stage called interphase, which often accounts for about 90% of the cycle. Interphase can be divided into sub-phases: the G1 phase (first gap), the S phase (synthesis), and the G2 phase (second gap). During the interphase time, cells prepare itself for nuclear division as the following:

Gap 0 (G0): There are times when a cell will leave the cycle and quit dividing. This may be a temporary resting period or more permanent. An example of the latter is a cell that has reached an end stage of development and will no longer divide.

Gap 1 (G1): Cells increase in size in Gap 1, produce RNA and synthesize protein. An important cell cycle control mechanism activated during this period (G1 Checkpoint) ensures that everything is ready for DNA synthesis.

S Phase: To produce two similar daughter cells, the complete DNA instructions in the cell must be duplicated. DNA replication occurs during this S (synthesis) phase.

Gap 2 (G2): During the gap between DNA synthesis and mitosis, the cell will continue to grow and produce new proteins. At the end of this gap is another control checkpoint (G2 Checkpoint) to determine if the cell can now proceed to enter M (mitosis) and divide.

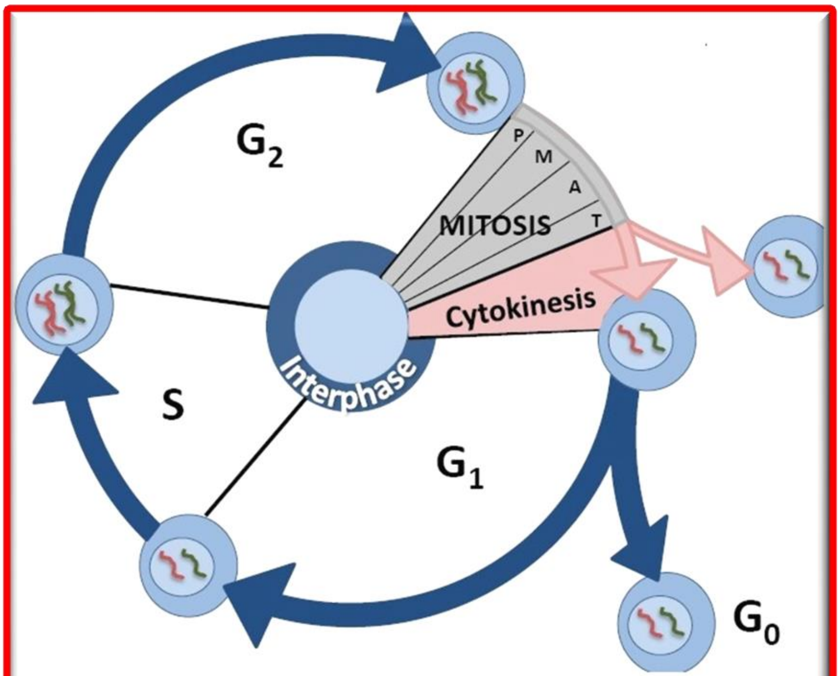


Figure 7.1. A typical cell cycle. The two gap periods (G1 and G2) are separated by the DNA synthesis (S) period, during which the DNA is replicated. Mitosis is followed by cell division (cytokinesis).

Stages of Mitosis

Mitosis is a type of cell division in which a parental cell produces two similar daughter cells that resemble the parental cell in terms of chromosomal number. This maintains constant number of chromosomes in each cell of successive generation. It occurs in somatic cells of the body. So, it is also called somatic cell **It occurs in four stages as follows** (Fig. 7.2):

I) Prophase

- It is the longest phase. During this phase the chromatin is organized into distinct chromosomes by coiling.
- The centrioles develop into asters and move towards the opposite poles of the cell to establish the plane of cell division. - Spindle apparatus begin to appear.
- Nucleolus and Nuclear membrane disintegrate and disappear -The chromosomes are set free in the cytoplasm.

II) Metaphase

- Spindle fibers are completely formed
- The chromosome become short and thick with two distinct chromatids each
- All the chromosomes move towards the center of the cell and arrange in the equatorial plane, right angles to the position of asters to form **metaphase plate**
- Chromosomes are attached to spindle fibers at their centromeres

III) Anaphase

- The centromere of all the chromosomes undergo longitudinal splitting and the chromatids of each chromosome separate to form daughter chromosomes
- The daughter chromosomes move toward the opposite poles from the equator by the activity of spindle fibers

V) Telophase

During this phase, the events of prophase will be reversed

- The daughter chromosomes reach the opposite poles
- The chromosomes undergo despiralization to form long, thin thread like structures called chromatin
- Nucleolus and nuclear membrane reappears
- The spindle fibers disappear

Cytokinesis

In animals, cytokinesis occurs by a process known as cleavage. It starts by the appearance of a cleavage furrow on the cell surface and goes inward. On the cytoplasmic side of the furrow is a contractile ring of actin microfilaments associated with the myosin protein. Cytokinesis in plant cells is different. There is no cleavage furrow. Instead, during telophase, vesicles derived from the Golgi apparatus fuse together to form the cell plate, the direction from inside to outward. Examine models of cytokinesis in plant and animal cells and compare them to Figure 7.3

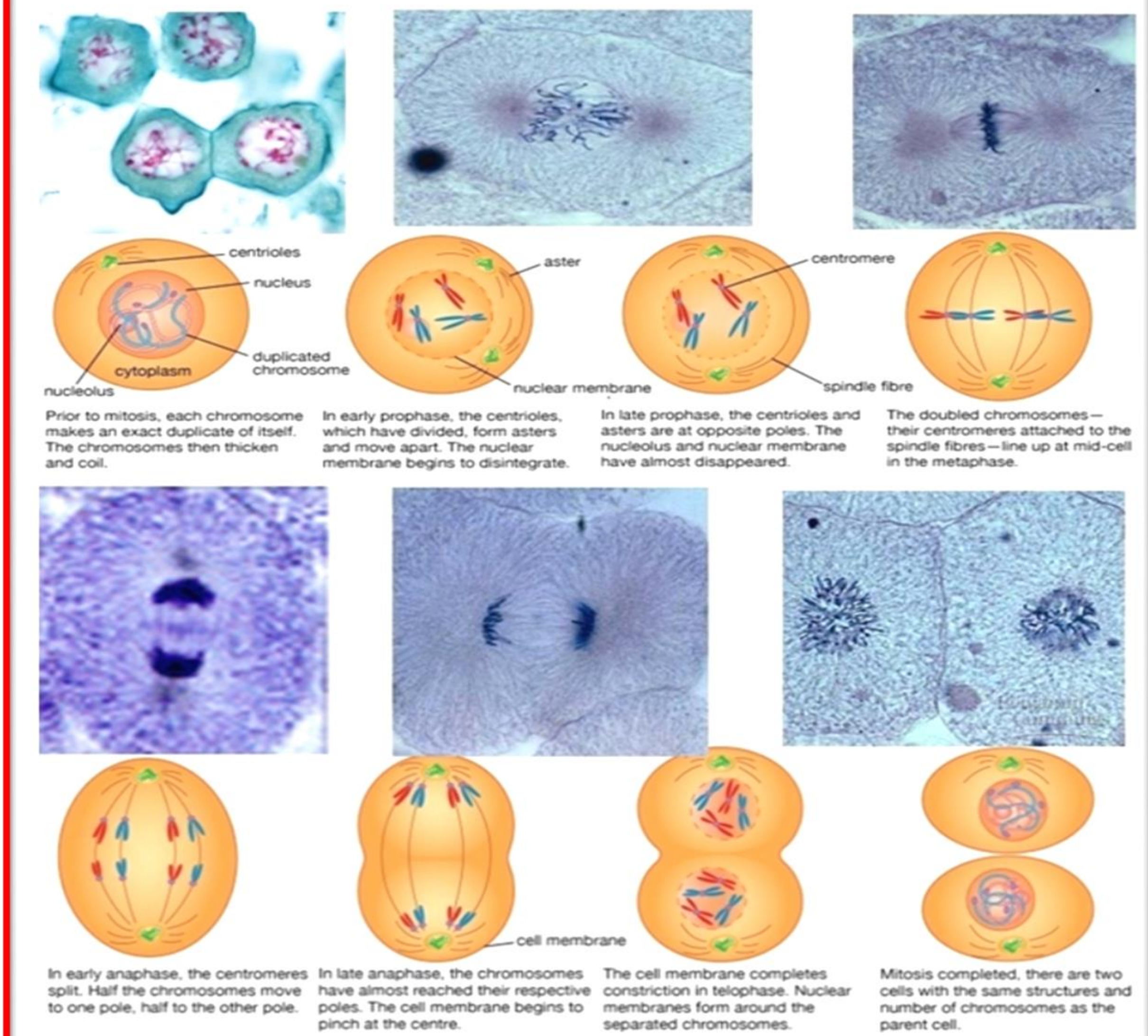


Fig.7.2 Different phases of Mitosis

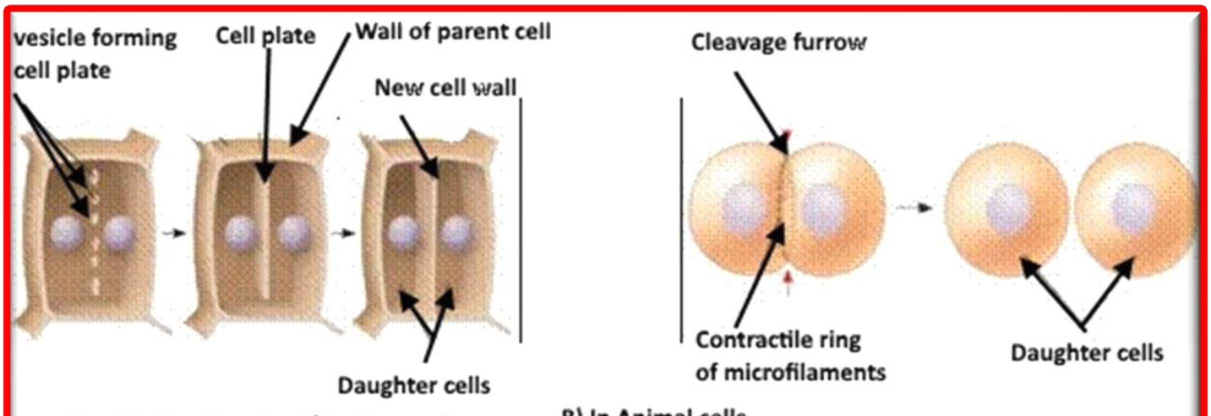


Fig. 7.3: Cytokinesis : A) In Plant cells B) In Animal cells

Procedure

I) The cell cycle in plant cells

Obtain a prepared slide of a longitudinal section of an *Allium* (onion) root tip. Examine the square cells just inside the root cap. This is the root **meristem** (embryonic tissue) where mitosis is occurring. Farther up the root is the elongation zone, where cells are long rectangles; these cells are not undergoing mitosis. Also, study models available for each stage of mitosis and compare to the different mitotic stages represented in **Figure 7.4.**

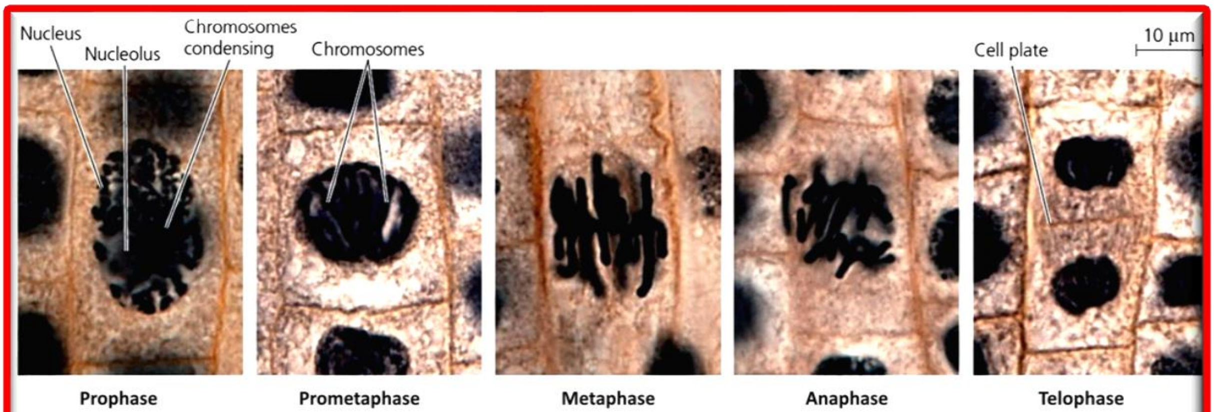


Fig. 7.4: Mitosis in onion root cells

II) The cell cycle in animal cells (Embryonic blastula stage of whitefish)

The developing embryo of any organism is a good tissue to examine for mitosis, since cells must divide at a high rate to transform a fertilized egg (single cell) into the trillions of cells of a viable organism. With several exceptions, mitosis in animals is like that in plants. One exception is that animal cells, unlike plant cells, contain centrioles, which are barrel-shaped structures that work as microtubules organizing centers.

Study the available slides and models to find representative cells in each of the phases of the cell cycle and compare with **figure 7.5**.

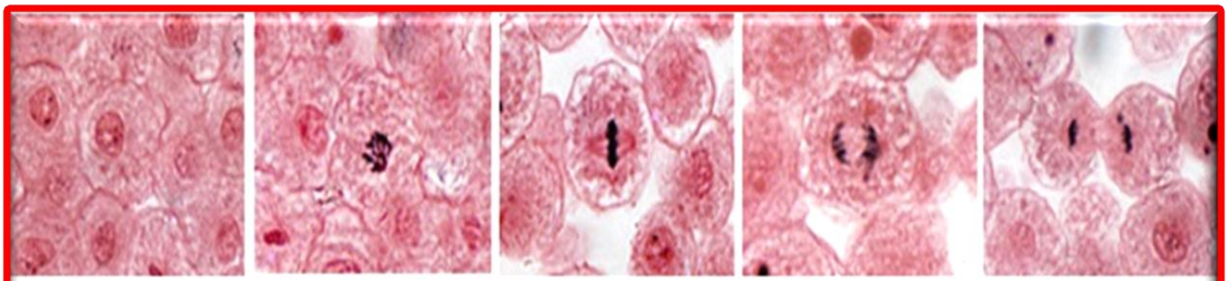


Figure 7.5: Mitosis in Whitefish blastula cells

Meiosis

The transmission of traits from one generation to the next is called inheritance, or **heredity**. These traits are transmitted from parents to offspring through a specialized nuclear division; meiosis. Meiosis is important for the production of gametes; the sperms and the oocytes, that's why meiosis is important for sexual reproduction. During meiosis the chromosome number is halved. In the body cells of most eukaryotes, chromosomes exist in pairs called homologous chromosomes. These are two chromosomes, which are physically similar and contain genetic information for the same traits. Cells that have two copies of each chromosome are called diploid (2N). Gametes that are produced at the end of meiosis have only one copy of homologs and are called haploid (1N).

Stages of Meiosis:

Meiosis consists of two divisions (Fig. 7.6):

A. During meiosis I homologous pairs of chromosomes are separated and cells become haploid.
B. During meiosis II the sister chromatids of each chromosome are separated (as in mitosis).

Meiosis I

At the start of meiosis, each chromosome consists of two sister chromatids, connected at the centromere. However, meiosis differs from mitosis, in that homologous pairs of chromosomes come together at the start of the meiosis, in a process called synapsis, forming a tetrad.

Prophase I:

- Just as in mitosis, the nucleolus and nuclear membrane disappear, and the centriole divides and forms the spindle. As in mitosis, the chromosomes coiled and become thick and visible, but with the chromosomes of each homologous pair tangled together. There are thus four chromatids in each pair of chromosomes.
- Portions of each chromatid may break off and reattach to an adjacent chromatid on the homologous chromosome. The places where this happens are called **chiasma** (pl. chiasmata). The whole process is called **crossing-over** and it results in unique combination of alleles on each chromatid. Crossing-over can only take place between genes. This is an essential part of the genetic **recombination** that takes place in meiosis **(Fig. 7.7).**

Metaphase I: Homologous pairs are still together and arranged on the equator of the cell.

Anaphase I: The homologous pairs of chromosomes separate from each other, as the spindle fibers pull one of each pair to opposite ends of the cell. The random separation of the homologous chromosomes that this produces is called **independent assortment**.

Telophase I: Cytokinesis takes place; and each new cell is haploid, containing one chromosome from each pair.

Meiosis II

- The chromosomes do not replicate before this, second, division, so the gametes formed will have only half the DNA of a normal cell.
- The second division of meiosis is essentially the same as mitosis.

Procedure

Study the charts and models available and show the major events that occur during each stage. Compare them to the events and stages of mitosis **(Fig7.8).** What are the major differences between meiosis and mitosis? Why meiosis is important for genetic variability?

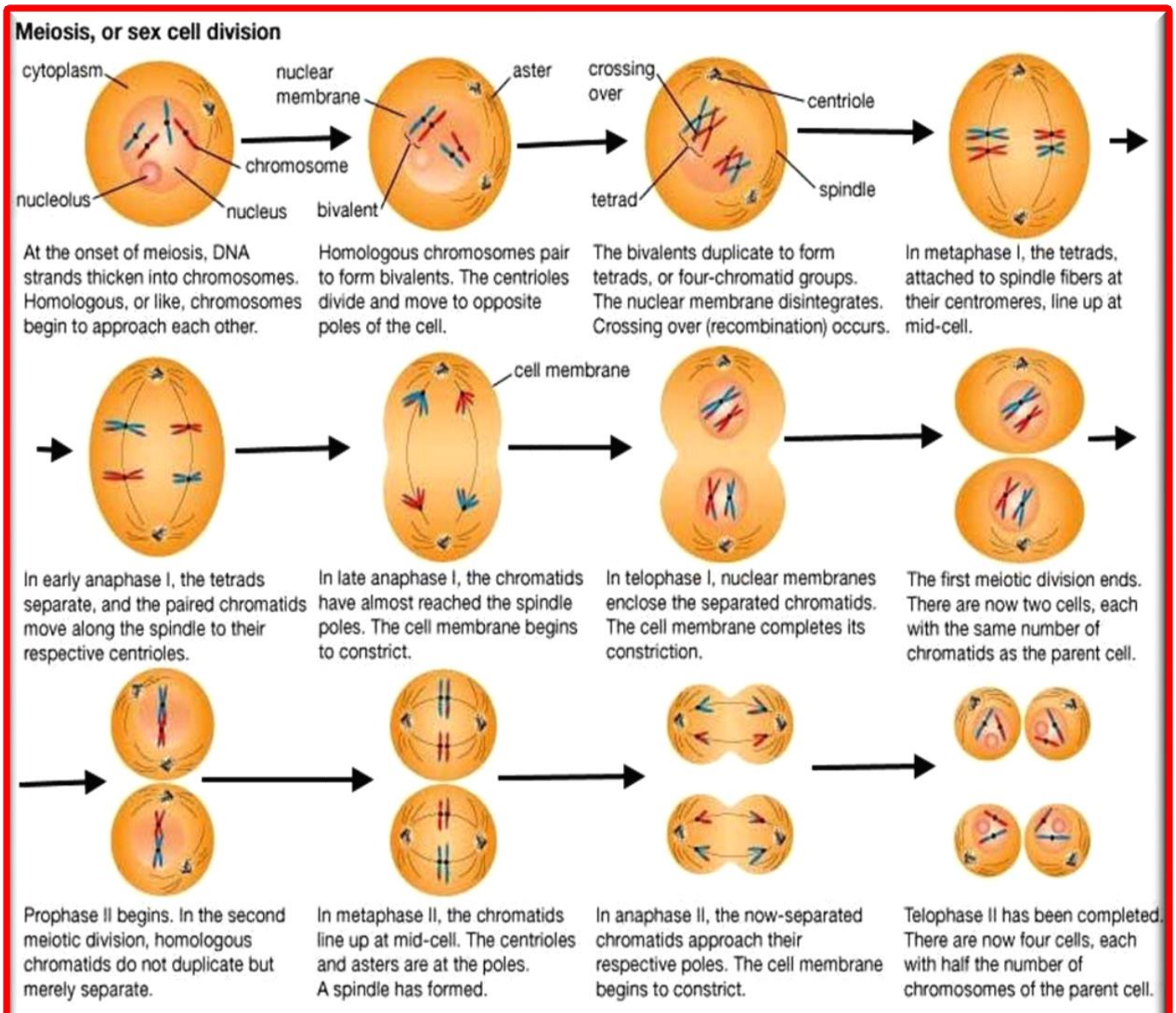


Figure 7. 6. Meiosis stages in eukaryotes. Meiosis is divided to two stages: I and II. During meiosis I homologous chromosomes segregate from each other while sister chromatids remain attached, while in meiosis II sister chromatids separate and segregate to the daughter cells.

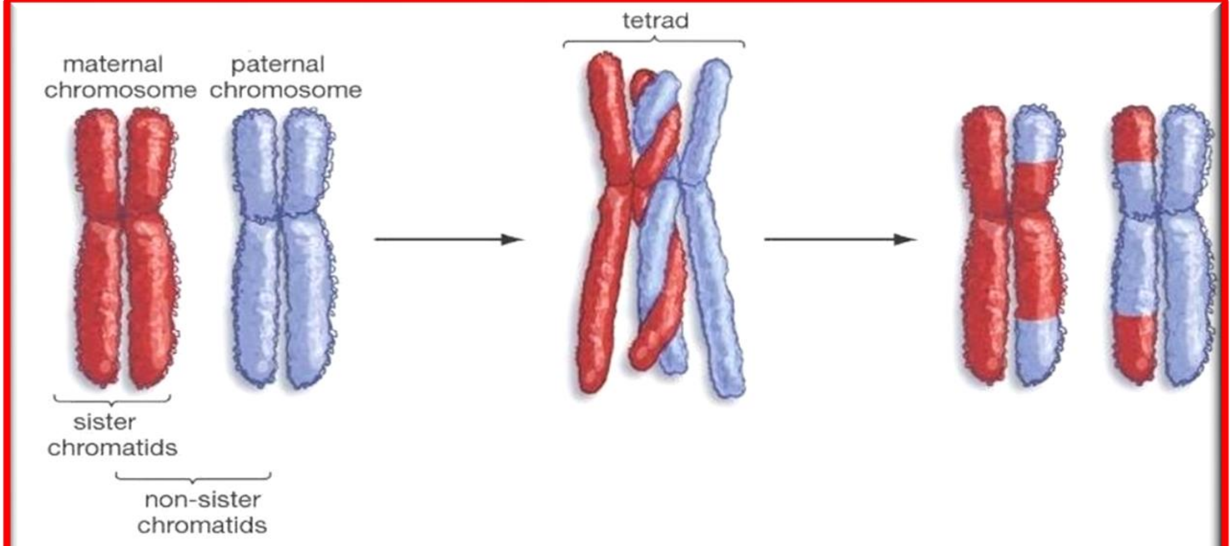


Figure 7.7: Crossing Over during prophase I

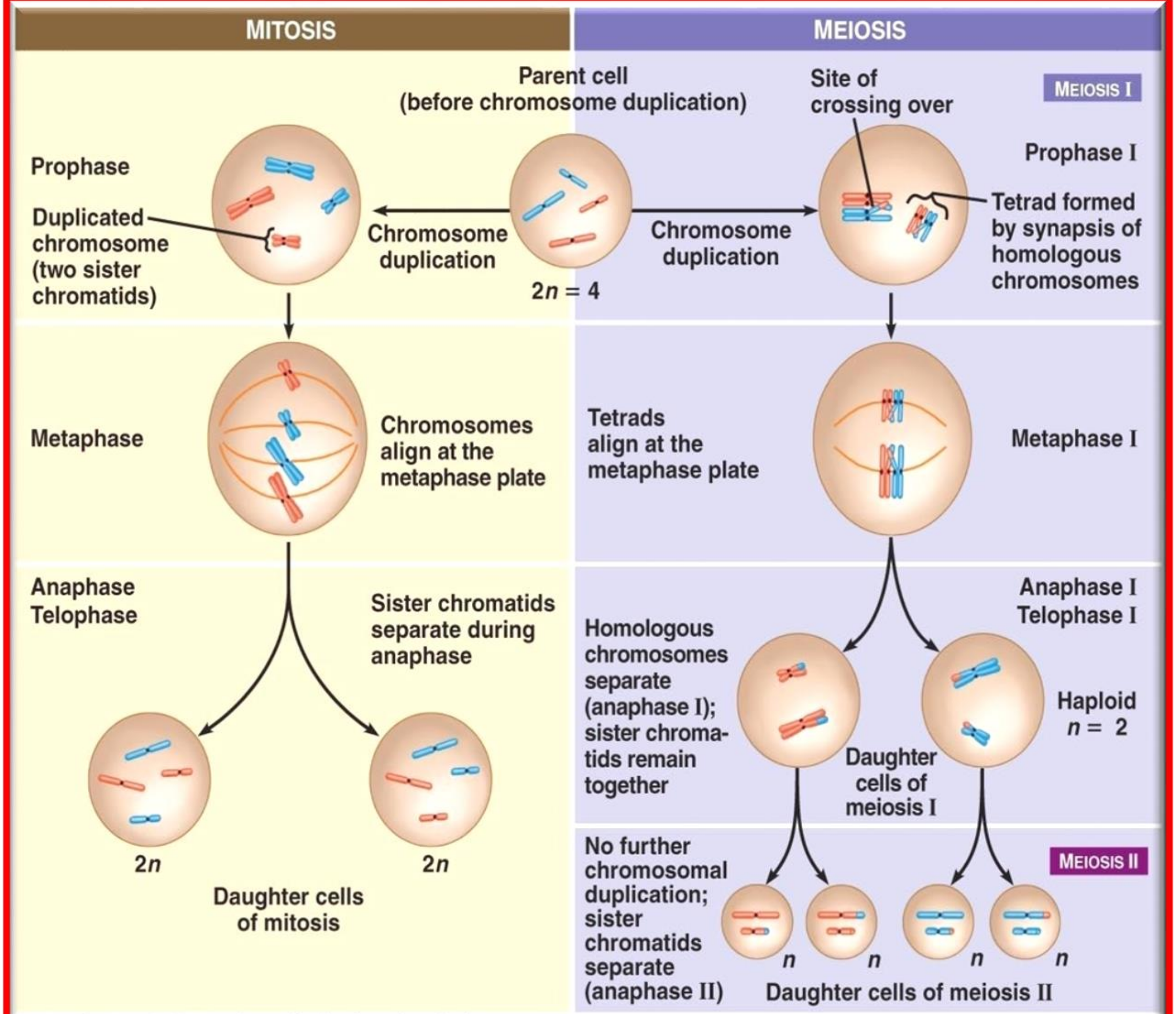


Figure 7.8: Comparison of Mitosis and Meiosis

Experiment 8
Human Genetics

Introduction

Genetics is the branch of science concerned with genes, heredity, and variation in living organisms. It seeks to understand the process of trait inheritance from parents to offspring. Our DNA is packaged into chromosomes which we inherit from each parent. Each body cell has a total of 46 chromosomes. One set of chromosomes (23) was received from the mother and another set of chromosomes (23) was received from the father. Diploid refers to cell that contain two copies of each chromosome. Each paired chromosomes with the same gene pattern are called homologous chromosomes (Fig. 8. 1A). These chromosomes have genes that provide the information to make the various types of proteins. A gene is a segment of DNA that contains instructions for making at least one protein. These proteins are important in carrying out the functions of the cells.

DNA Structure

DNA, deoxyribonucleic acid, is the blueprint of life. Watson and Crick proposed the “Double Helix” structure of DNA, where two DNA chains spiral around an axis with the sugar-phosphate backbones on the outside and pairs of bases (one from each chain) meeting in the middle. The two chains are antiparallel; each chain runs from 5’ to 3’ on opposite direction (Fig. 8.1B).

DNA is a long polymer composed of subunits known as nucleotides. Each nucleotide consists of a deoxyribose sugar, a phosphate, and one of four nitrogenous bases,(Fig. 8.1C &D and Fig. 8.2).

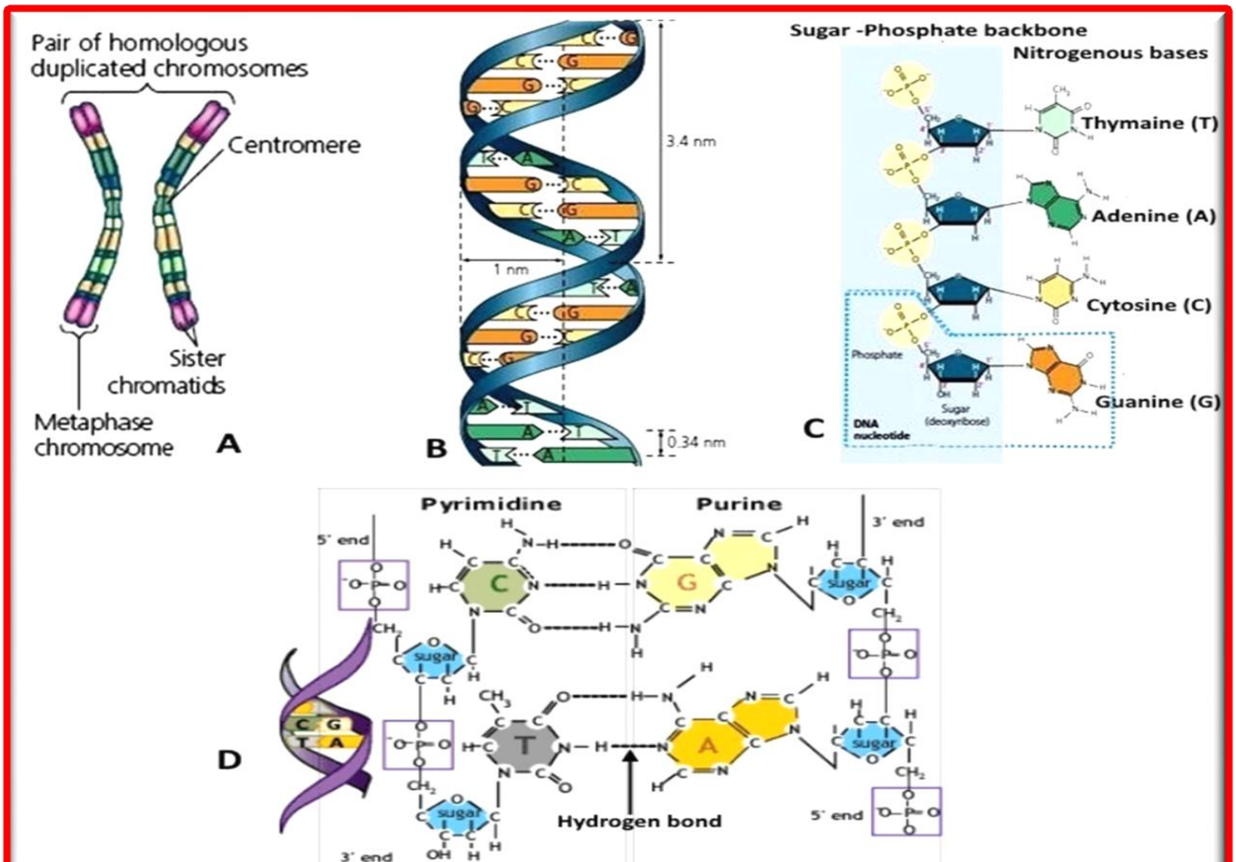


Figure 8.1. DNA structure. A) A pair of homologous chromosomes .B) DNA double helix model showing spiral double DNA strands. C&D) DNA chemical structure.

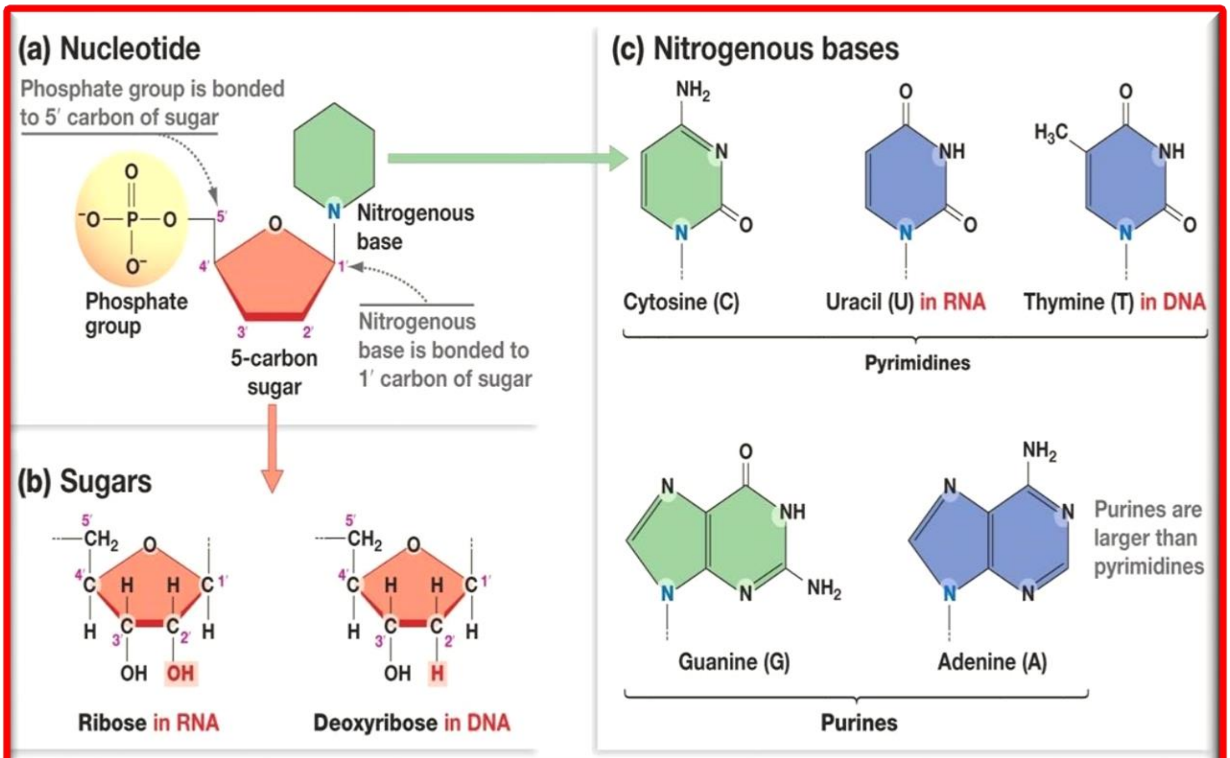


Figure 8.2: Nucluic Acids composition: A) Structure of nucleotide. B) Structure of both ribose and deoxyribose sugars. C) Structure of the five nitrogenous bases(C, U, T, G, A).

RNA Structure

Ribonucleic acid (**RNA**) is a composed of four types of bases: adenine (A), cytosine (C), guanine (G), and uracil (U). Unlike DNA, RNA is usually single-stranded and, contains ribose sugars rather than deoxyribose sugars, which makes RNA more unstable and more prone to degradation.

Dominant and Recessive Traits:

A gene that governs a particular trait may come in alternative forms called **alleles.** Not all of the alleles that we inherit are shown or expressed. Some alleles, the **dominant** ones, cover the expression of the **recessive** alleles. The recessive alleles don't disappear. They may show up in future generations when not paired with a dominant allele. When the same allele for a single gene is on both the paired **homologous chromosomes** the individual is **homozygous** for that trait. When the paired chromosomes carry different alleles for the same gene they are said to be **heterozygous**.

Single Gene Traits:

Hundreds of traits combine to determine appearance. A single gene determines some traits like tongue rolling. Others, like skin color, are determined by the combined effects of several genes.

Sex-Linked Traits

Color blindness and hemophilia are examples of genetic disorders that are due to a change in a gene on the X chromosome. Females rarely show these recessive disorders because a gene on their second X chromosome may cover the effects of the changed gene. Males have these disorders more often because the Y chromosome does not have a gene that can cover the effect of the changed X gene. Color blindness is an X-linked, recessive trait. Therefore, the possible genotypes are :

Females	Males
X^CX^C = normal vision	
X^CX^c = normal vision (carrier)	X^CY = normal vision
X^cX^c = colorblind	X^cY = colorblind

Procedure:

a- Study available model of DNA and Determine its structure

b- For each of the following physical traits, determine your **phenotype**, and your possible **genotypes** or the alleles in your genetic makeup.

1. Using **Figure 8.3**, have your lab partner help you determine your phenotype for each characteristic listed and *record your **phenotype***.
2. Next determine your possible **genotypes** for the traits listed and *record it*.

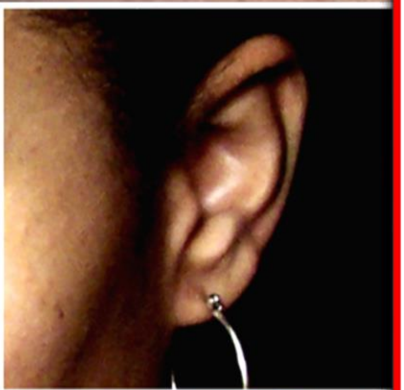
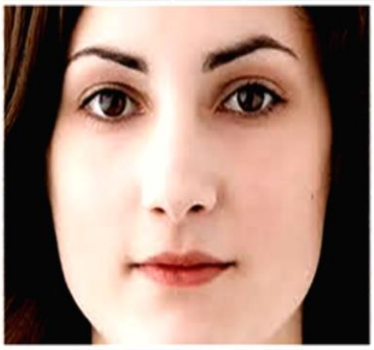
Trait	Description	Dominant	Recessive
Tongue roller:	Ability to curl up the sides of their tongue to form a tube shape.		
Widow's Peak	If your hairline forms a point at the center of the forehead, you have a widow's peak. If not, you have a straight hairline.		
Attached earlobes	If earlobes hang free, they are detached. If they connect directly to the sides of the head, they are attached.		
Freckles	Freckles are small, concentrated spots of a skin pigment called melanin. Freckles show a dominant inheritance pattern		
Has dimples	Dimples are small, natural indentations on the cheeks. They can appear on one or both sides		

Figure 8.3: Single gene human traits

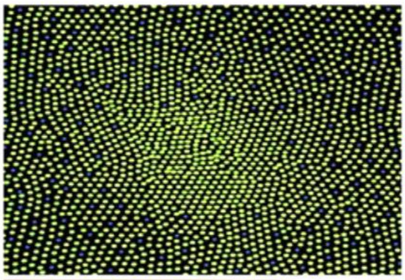
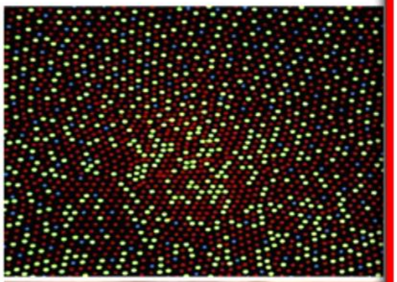
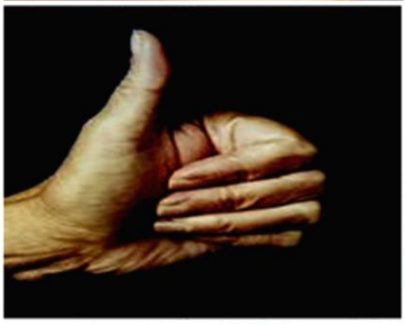
Trait	Description	Dominant	Recessive
Color blind	Red-green able to see red and green, it is caused by a single gene located on the X-chromosome. Control by recessive Gene.		
Cleft chin	small, natural indentations on the chin, control by dominant gene		
Hitchhiker's thumb	Ability to bent the thumb backward called Hitchhiker's thumb, which control by recessive single gene. Un able to do that called straight little finger (dominant)		
baldness Hair loss	the gene for baldness dominant in male, but recessive in female if it's heterozygotes (need one gene to appear in male and two gene to appear in female).		
Bent Pinky	Dominant: Bent Little Finger; Straight Little Finger is recessive		

Figure 8.3: Single gene human traits

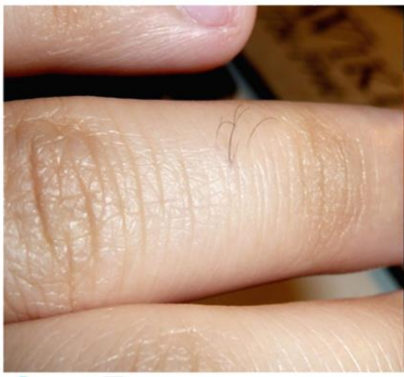
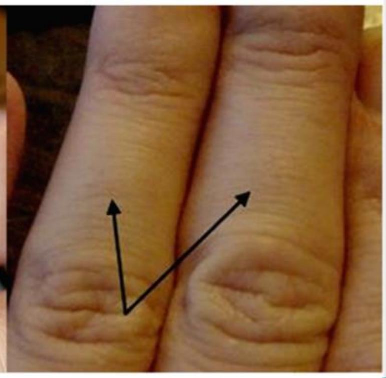
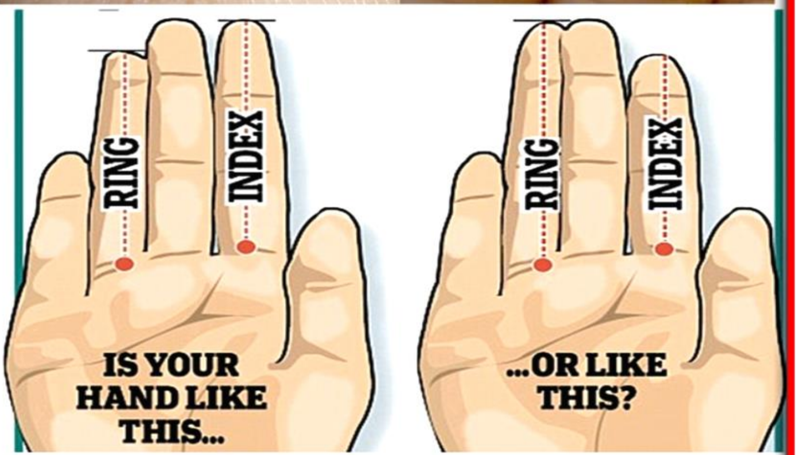
Trait	Description	Dominant	Recessive
Mid-digital hair	Hair growing from the mid –digit of each finger Mid- digit hair dominant over no hair in mid digit		
Index finger shorter than ring finger	The index (2nd finger, counting the thumb as number one) finger is shorter than the ring(4th) finger.The expression of this gene is sex influenced. The phenotype is *usually* dominant in males and recessive in females.		

Figure 8.3: Single gene human traits

Multiple Alleles: Human Blood Groups

Red blood cells have molecules called **antigens** on their surface. An antigen is a molecule that causes the immune system to produce **antibodies** against it. When a foreign antigen enters the body the immune system will build antibodies against it. **ABO** antigens are attached to human red blood cells (**Fig.8.4)**. The surface antigens on red blood cells are coded for by one gene that has three different alleles. This is an example of a **multiple allelic system**. The I^A allele determines the **A** antigen, the I^B allele determines the **B** antigen, and the "**i**" allele determines no antigens (type O).

An individual carries a matched pair of chromosomes and thus has two alleles for the ABO blood groups. Two alleles may be expressed at the same time. If an individual has I^A and I^B, they will have type **AB** blood. Since both alleles are expressed, this is an example of **codominance.** The possible genotypes and phenotypes are listed in **Table 8.1.** If red blood cells with foreign antigens on them enter the body the antibodies produced against them will cause the blood to clump (not clot—clotting is something quite different) (**Fig.8.6)**. A normal person never makes antibodies against his own antigens. If this occurred it would be disastrous! Imagine a person with **A** antigen making antibodies against **A** and clumping his own blood cells. Instead, a person with **A** would make antibodies against foreign antigens such as **B** (anti-B antibodies). This is why only a blood type that is compatible can be used during blood transfusions **(Fig.8.4)**. For example, type **A** blood can only be given to persons with type **A** or type **AB** blood (**Table 8.1**).

The **Rh** system is completely separate, but works in much the same way. If you have the **Rh** antigen present on your red blood cells, you are **Rh**$^+$. If it is absent, you are **Rh**$^-$.

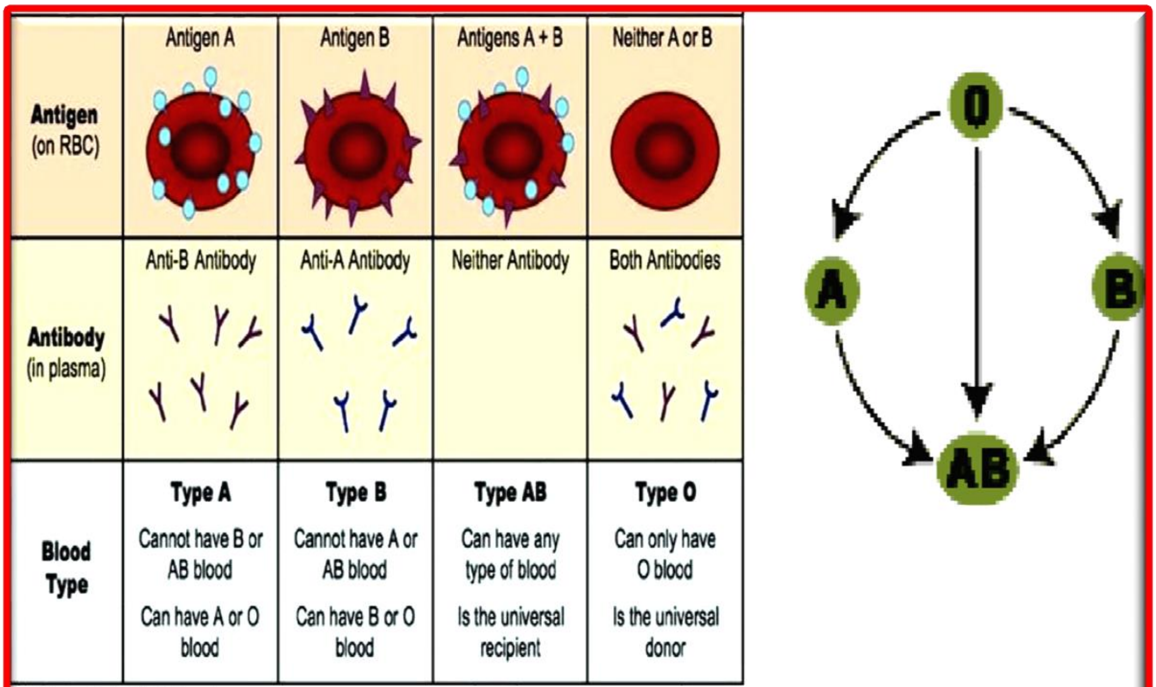


Figure 8.4: ABO blood groups

Table 8.1. Human Blood Groups			
Genotype	**Phenotype (Blood Type)**	**Antigen present**	**Antibody produced**
I^AI^A	A	A	Anti-B
I^Ai	A	A	Anti-B
I^BI^B	B	B	Anti-A
I^Bi	B	B	Anti-A
I^AI^B	AB	A & B	None
ii	O	None	Anti-A & Anti-B
Rh^+, Rh^+	Rh^+	Rh	None
Rh^+, Rh^-	Rh^+	Rh	None
Rh^-, Rh^-	Rh^-	None	Anti-Rh (after exposure)

Determining Your ABO Blood Type:

In this procedure, you will use a serum with antibodies in it against a specific antigen. Conduct a blood typing exercise as the following (**Fig.8.5)**:

1. Wash your hands before carrying out the test and again after carrying out the test.
2. Set the table with all the materials required. Remember to place the Monoclonal Antibody (Mab) kit in an Ice tray.

3. Open an Alcohol swab, and rub it at the area from where the blood will be sampled (fingertip). (Discard the swab)

4. Open the Lancet cover, put pressure at the tip of the finger from where blood will be sampled (Maintain it). Prick the fingertip with the opened Lancet.(Discard the Lancet)

5. Massage the finger from the bottom to the top to encourage blood flow. Press the blood towards fingertip. Repeat pressing until a drop with a 3 to 4 mm (1/8 inch) diameter is seen.

6- As blood starts oozing out, make 1 drop of blood fall in each of the three depressions (circle) of the blood typing slide.

7. Once the blood added to the three depressions, place a cotton ball at the site where it was pricked. Using the thumb, put pressure on the area to stop blood flow.

8. Add the Anti- antigen:

   a- Take the Anti-A (blue) bottle, suspend the content and use the dropper to place a drop of the Mab in the 1st spot. Place the bottle back in ice.
   b. Take the Anti-B (yellow) bottle, suspend the content and use the dropper to place a drop of the Mab in the 2nd spot. Place the bottle back in ice.
   c. Take the Anti-D (colorless) bottle, suspend the content and use the dropper to place a drop of the Mab in the 3rd spot. Place the bottle back in ice.

9. Take a tooth pick and mix the content in each well. Discard the tooth pick after using in one well (take a new one for the next well).

10. After mixing, wait for a while tilting the blood typing slide then observe where the agglutination takes place.

11. Record the results using **Table 8.2 and Figure 8.6**.

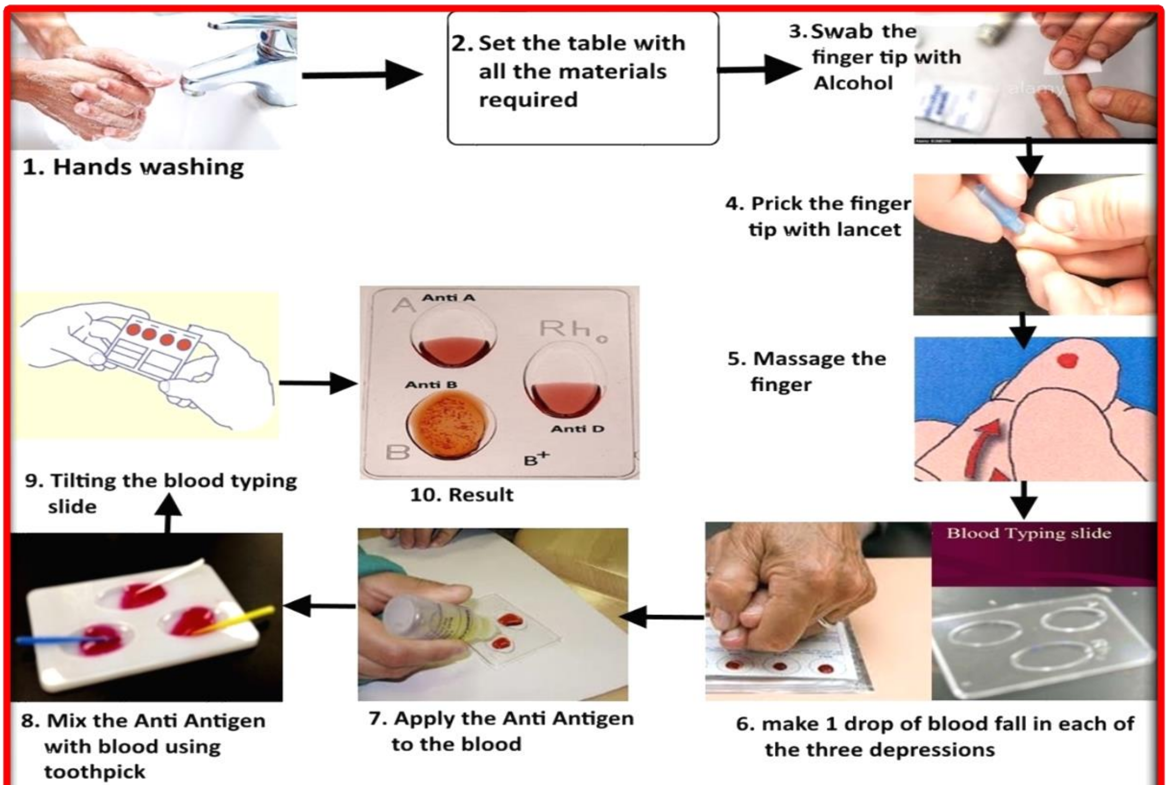


Fig.8.5: Procedure of Blood typing(grouping)

Table 2. Blood Antigen Test

Red Blood Cell Antigen	Antiserum Added	Results
A	Anti-A	clump
B	Anti-B	clump
AB	Anti-A & Anti-B	both clump
O	Anti-A & Anti-B	no clump with both
Rh	Anti-Rh	clump

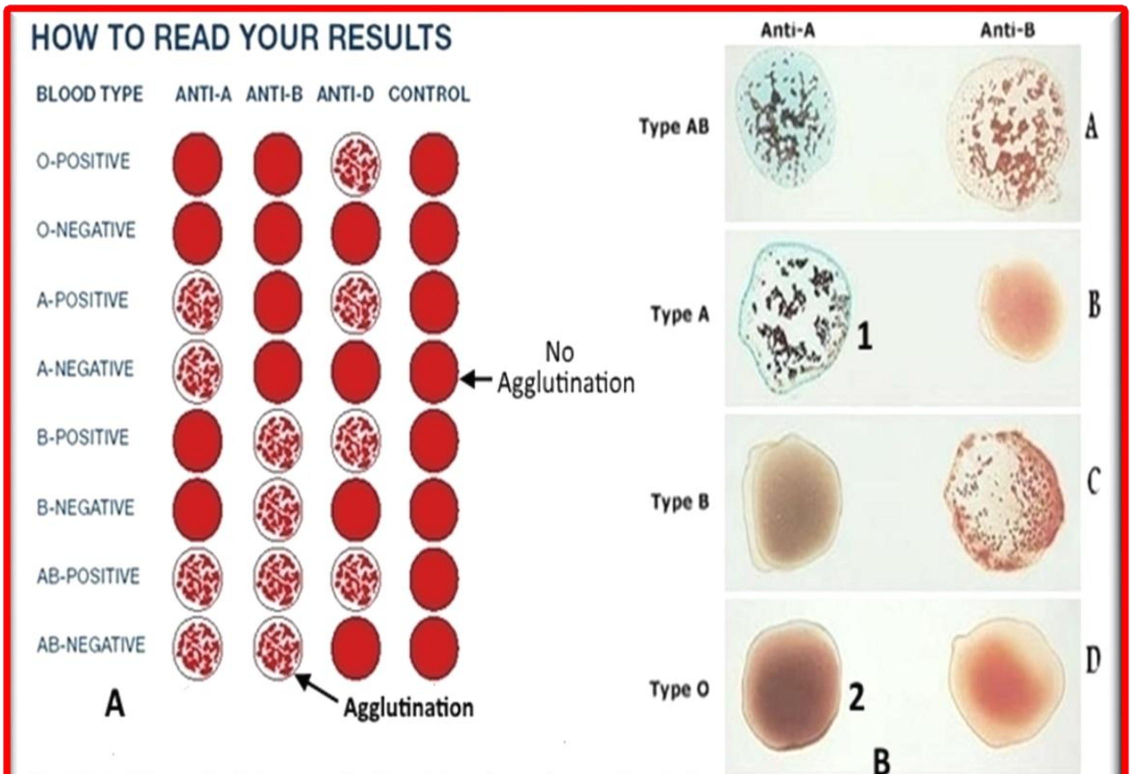
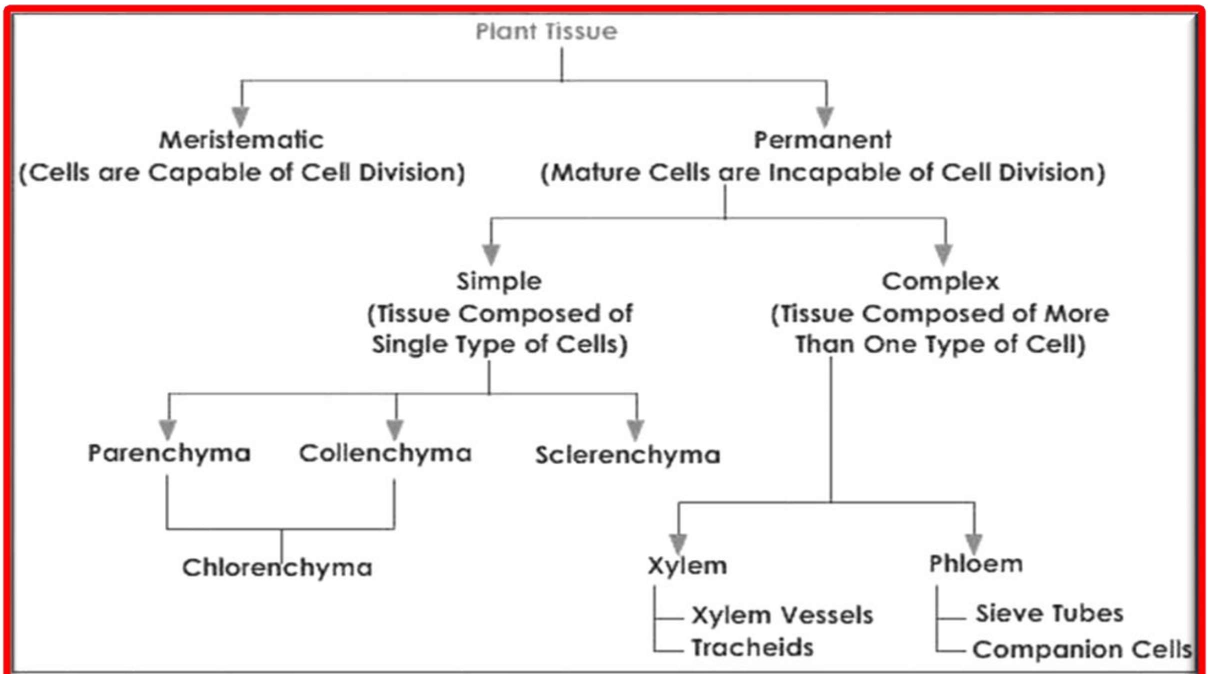


Fig. 8.6 A: Scheme to determine the Blood type (group) according to its, reaction with Anti Antigen; B: Real experiment showing the clumping (1) and No clumping (2)

Experiment 9
Plant Tissues and Organ

I) Plant tissue:

All plants are classified as *producing seeds* or *not producing seeds*. Those that produce seeds are divided into flowering (angiosperms) and non-flowering (gymnosperms). Flowering plants are further divided into *monocotyledonous* and *dicotyledonous* (monocot and dicot) plants. The plant body is made up of different kinds of tissues. Classification of tissues is presented in this schema (Scheme 9.2)



Scheme 9.1: Classification of plant Tissues

I) Meristematic tissues

Meristematic tissue consists of actively dividing cells, and leads to increase in length and thickness of the plant. Cells in these tissues are roughly spherical or polyhedral, to rectangular in shape, and have thin cell walls. They are compactly arranged without inter-cellular spaces between them. Each cell contains a dense cytoplasm and a prominent nucleus with small or no vacuoles. they are classified as:

a) Apical Meristem - It is present at the growing tips of stems and roots and increases the length of the them. This meristem is responsible for the linear growth of an organ.

b) Lateral Meristem - This meristem consist of cells which mainly divide in one plane and cause the organ to increase in diameter and growth. Lateral Meristem usually occurs beneath the bark of the tree in the form of Cork Cambium and in vascular bundles of dicots in the form of vascular cambium. The activity ofthis cambium results in the formation of secondary growth.

c) **Intercalary Meristem** - This meristem is located in between permanent tissues. It is usually present at the base of node, inter node and on leaf base. They are responsible for growth in length of the plant and increasing the size of the internode, They result in branch formation and growth.

Permanent tissues:

The meristematic tissues that take up a specific role lose the ability to divide and differentiate to form different types of permanent tissue. There are 3 types of permanent tissues these are:

1. **Simple permanent tissues**: A group of cells that are similar in origin, structure, and function. They are of four types:

a. Parenchyma (fig. 9.1A)
Parenchyma is the bulk of a substance. In plants, it consists of relatively unspecialized living cells with thin cell walls that are usually loosely packed so that large spaces between cells (intercellular spaces) are found in this tissue. This tissue provides support to plants and also stores food. In some situations, a parenchyma contains chlorophyll and performs photosynthesis; in this case it is called a **chlorenchyma**. Chlorenchyma found in palaside layer of leaf.

b. Collenchyma (fig. 9.1D)
Collenchyma is a living tissue of primary body like Parenchyma. Cells are thin-walled but possess thickening of cellulose, water and pectin substances (pectocellulose) at the corners where number of cells join. This tissue provides mechanical support, elasticity, and tensile strength to the plant body and the cells are compactly arranged and have very little inter-cellular spaces. It occurs chiefly in hypodermis of stems and leaves. It is absent in monocots and in roots.

c. Sclerenchyma (fig. 9.1B)
Sclerenchyma consists of thick-walled, dead cells. These cells have hard and extremely thick secondary walls due to uniform distribution of lignin. Sclerenchyma cells are closely packed without inter-cellular spaces between them. Sclerenchyma cells mainly occur in hypodermis, pericycle, secondary xylem and phloem. They also occur in endocarp of almond and coconut. It is made of pectin, lignin, and protein. The main function of Sclerenchyma tissues is to give support to the plant. Provides hardness and protective covering to seed and nuts.

d. Epidermis
The entire surface of the plant consists of a single layer of cells called epidermis or surface tissue. Most of the epidermal cells are relatively flat. The outer and lateral walls of the cell are often thicker than the inner walls. The cells forms a continuous sheet without inter cellular spaces. It protects all parts of the plant.

2. Complex permanent tissues:

The complex tissue consists of more than one type of cells, which work together as a unit. Complex tissues help in the transportation of organic material, water and minerals up and down the plants. That is why it is also known as conducting and vascular tissue. The common types of complex permanent tissue are: Xylem or wood and Phloem or bast. Xylem and phloem together form vascular bundles.

a. Xylem (fig. 9.1E) is a chief, conducting tissue of vascular plants. It is responsible for conduction of water and mineral ions/salt and consists of a combination of parenchyma cells, fibers, vessels, tracheid and ray cells. Tracheid have thick secondary cell walls and are tapered at the ends. They do not have end openings such as the vessels. The tracheid ends overlap with each other, with pairs of pits present. The pit pairs allow water to pass from cell to cell.

b. Phloem (fig. 9.1F) carries dissolved food substances throughout the plant. This conduction system is composed of sieve-tube member and companion cells that are without secondary walls. The parent cells of the vascular cambium produce both xylem and phloem. Sieve tubes are formed from sieve tube members laid end to end. The end walls are full of small pores where cytoplasm extends from cell to cell. These porous connections are called sieve plates. In spite of the fact that their cytoplasm is actively involved in the conduction of food materials, sieve-tube members do not have nuclei at maturity. The companion cells are nestled between sieve-tube members that function in some manner bringing about the conduction of food.

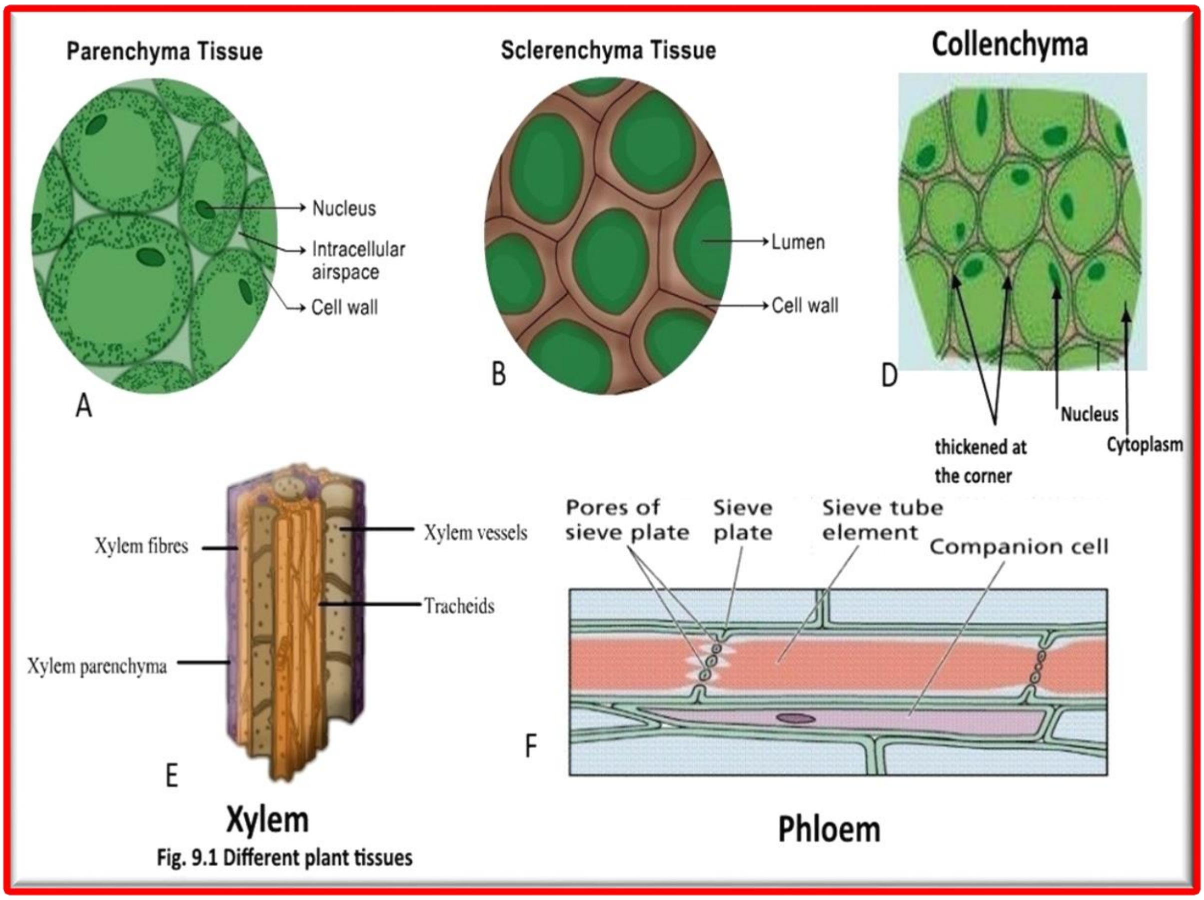


Fig. 9.1 Different plant tissues

II) Plant Organ

Plant tissues are arranged or combined into **organs** that carry out life functions of the organism. There are two types of plant organs: vegetative organs (**leaf**, **stem**, and **root**) and **reproductive organs (flower, fruit, and seeds)** (fig. 9.2).

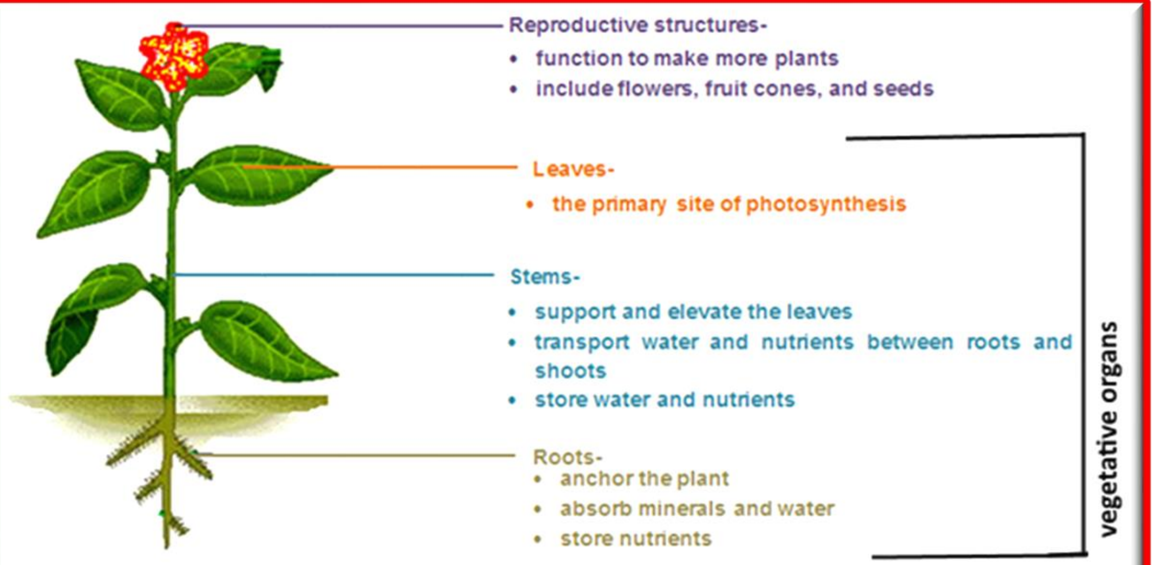


Fig. 9.2: Plant organs

A- Vegetative organs:

1) The Root

In **vascular plants,** the root is best defined as the non-leaf, non-nodes bearing parts of the plant's body that typically lies below the surface of the soil. However, roots can also be aerial or aerating (growing up above the ground or especially above water). Root arise from radicle and has four major functions, these are:

1) Absorption of water and inorganic nutrients
2) Anchoring of the plant body to the ground, and supporting it.
3) Storage of food and nutrients.
4) Vegetative reproduction.

In response to the concentration of nutrients, roots also synthesize cytokinin, which acts as a signal as to how fast the shoots can grow. Roots often function in storage of food and nutrients.

Structure of the Root:

I) A longitudinal section through the Root (fig.9.3A) shows that it divide into several regions:

1. **The root cap:** The cells that are formed by the lower portion of the apical meristem differentiate or change into cells, which are arranged and form the Root cap. Root cap direct the growing of the root downward by sense the gravity, secrete a slimy mucous-like substance that lubricates the root for movement between the soil particles, and protect the meristematic tissue.

2. **Apical meristematic:** Compose of dividing cells that form the tissue of other root region.

3. **Region of elongation:** Behind the apical meristem of the root is an area known as the region of elongation. It is in this zone that the root cells begin to elongate primarily by filling their **cell vacuoles** with water. The growth in the root is due to a combination of the increase in the number of cells that takes place in the apical meristem in roots, and in the size of the individual cells, that takes place in the region of elongation.

4. R**egion of maturation**: Behind the region of elongation is a **region of differentiation**. In this region of the root, the cells mature and begin to change their structure and their function. This process is known as differentiation. The epidermal cells of this zone produce long extension cells - **root hairs** (fig.9.3B), which increase the surface area for water and nutrients absorption by root.

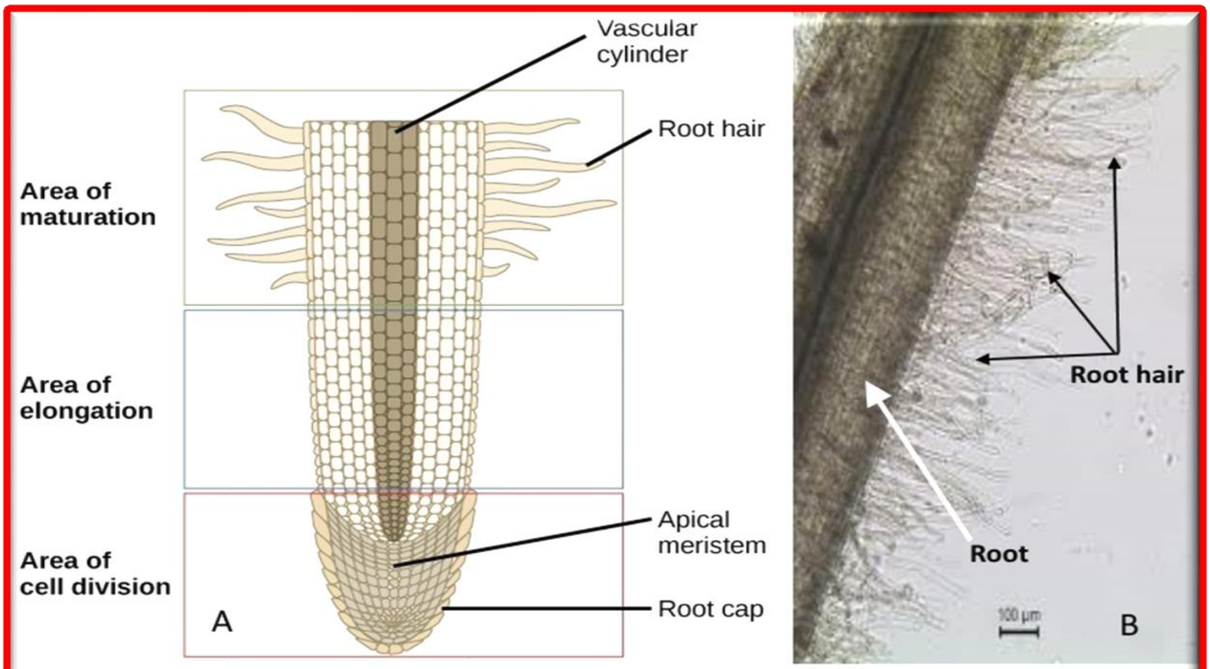


Fig. 9.3 A: Longitudinal section of the root B: Root with Root hair

II) Cross section of Dicotyledonous Root:

The different tissues in the root have a distribution which is common to all dicotyledonous plants and is shown in fig.9.34A.

The epidermis is a single layer of cells on the outside that protects the inner tissues. The epidermal layer of the root has no waterproof cuticle. The cells of the root hair is an out ward growth of epidermal cells.

The cortex consists of *parenchyma* cells. These cells are large which enables them to store water and food. They also facilitate the movement of water from the root hair cells on the outside of the plant to the xylem on the inside of the plant.

The endodermis forms the innermost layer of the cortex. It is a layer of tightly-packed, modified parenchyma cells. The radial and transverse cell walls are thickened with a water-impermeable, waxy suberin layer, known as the Casparian strip. This layer helps to regulate the flow of water from the cortex into the stele, rather than allowing the water to spread to all the root cells.

The stele, **or vascular cylinder** (fig.9.4B) (responsible for transporting water and minerals), consists of the pericycle, phloem, cambium and xylem. **The pericycle** is the outermost layer of the stele, and consists of one or more rows of thin-walled parenchyma cells. It is in close contact with the xylem and phloem tissues of the root. Its functions are the formation of lateral roots and the formation of a lateral meristem to allow secondary growth (thickening) to occur in the root.

The Xylem in dicot root arrange in an (X arch) shape and The phloem tissues lay between the arms of the xylem. **The cambium** separates the xylem and phloem tissues from each other. This is the area where secondary growth of xylem and phloem tissues occur. Sometimes there is Pith in the center of the stele, which made up from Parenchyma cells.

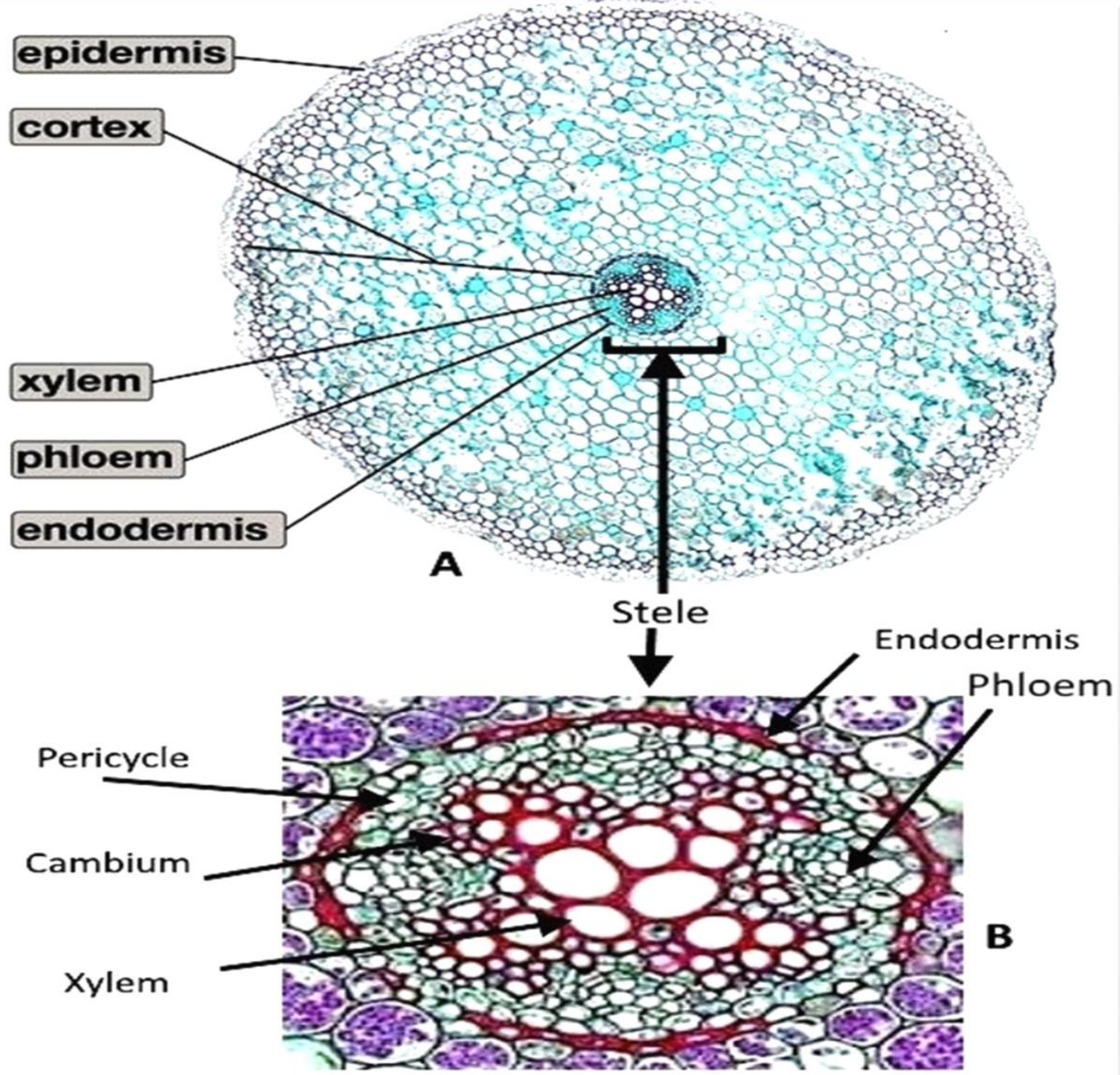


Fig.9.4 A: Cross Section of Dicot Root; B: The structure of the stele

II) Cross section of Monocot Root(fig.9.5A):

Ground tissue differentiate into: Cortex, Endodermis, pericycle and pith. The vascular bundle made up from 8 or more alternate bundles of xylem and phloem arrange in an alternate axis (fig.9.5B).

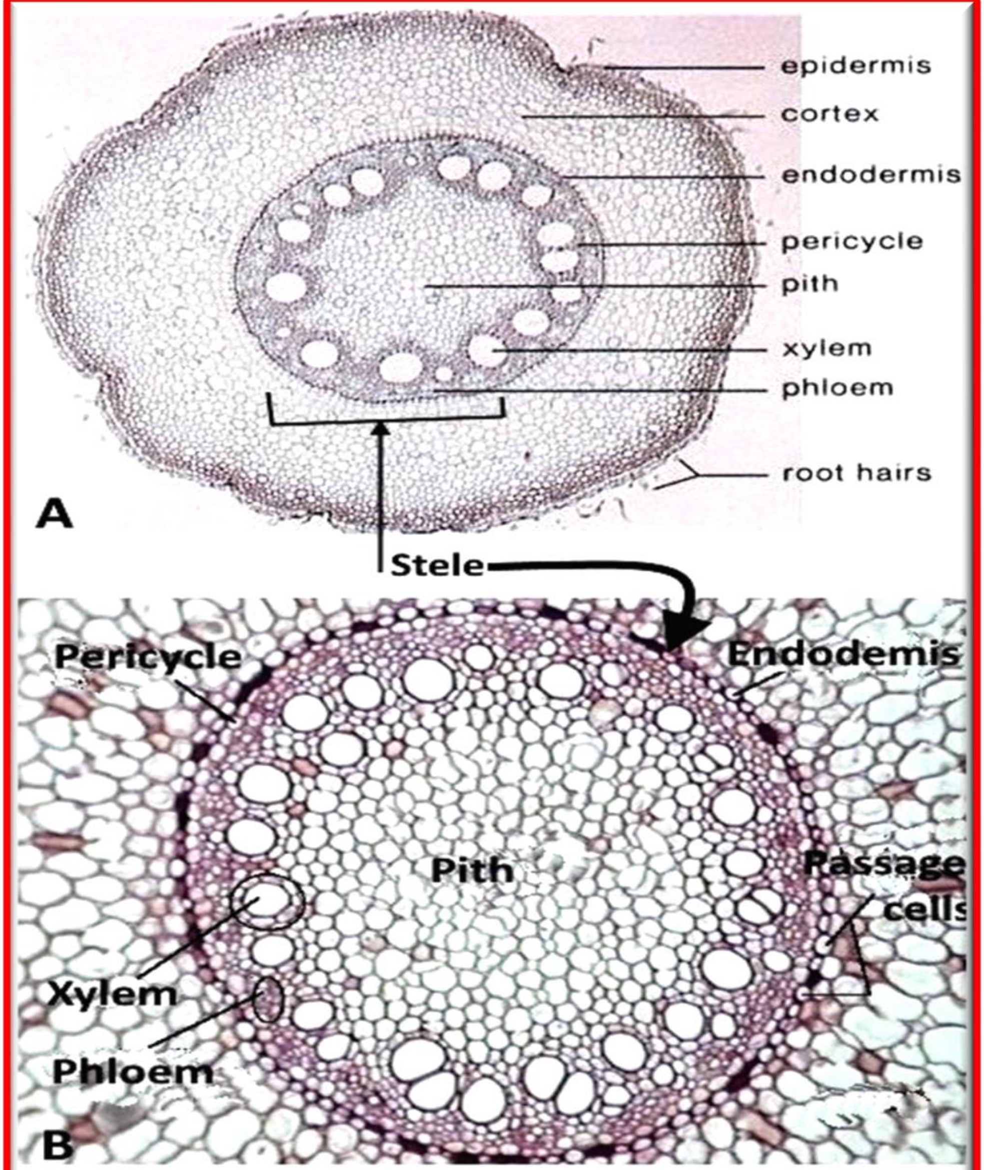


Fig. 9.5 A: Cross section of monocot Root; B: The Stele

2) Stem anatomy

The main stem develops from the plumule of the embryo and the lateral branches develop from the buds. Nodes and internodes are regions found on the stem. Nodes are the regions from which leaves and lateral branches develop, and the regions between nodes are known as internodes (shown in Figure 9.6). Stems usually grow above the soil surface and towards the light from the sun. Depending on the hardness of the stem, we can distinguish between herbaceous stems, which are leafy non-woody structures, and woody stems. Woody stems are harder than herbaceous stems. Stems have four main functions:

- Support for the plant as it holds leaves, flowers and fruits upright above the ground. Stems keep the leaves in the light and provide a place for the plant to keep its flowers and fruits.
- Transport of fluids between roots and shoots in the xylem and phloem.
- Storage of nutrients.
- Production of new living tissue: stems contain meristematic tissue that generates new tissue.

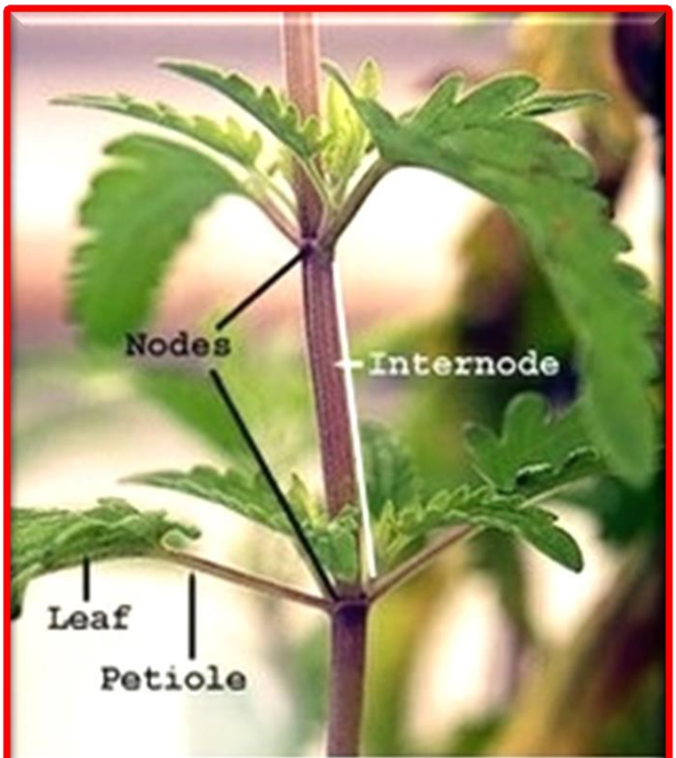


Fig.9.6 Node and Internode

I) Cross section of Dicotyledonous stem :

From the study of the transverse section of the dicotyledonous stem you will identify the following four regions of tissues (Figure 9.7A):

1- The epidermis consists of a **single layer of living cells** which are closely packed. The walls are **thickened** and covered with a **thin waterproof layer** called the **cuticle** . **Stomata** with **guard cells** are found in the epidermis. In some stems either **unicellular or multicellular hair-like outgrowths** appear from the epidermis.

Functions

- The epidermis *protects the underlying tissues*.
- The cuticle *prevents the desiccation of inner tissues* and thus *prevents water loss* .
- The stomata allows *gaseous exchange* for the processes of respiration and photosynthesis.

2- Hypodermis is a few layers located beneath the epidermis of the dicot stem. It is made up from collenchyma cells

3- The Ground tissue in the dicot differentiate into cortex, endodermis, pericycle, medullary rays and Pith.

4- Vascular Cylinder or Stele (Figure 9.7B)**:** This region comprises the **pericycle, vascular bundles and pith (medulla)**.

- **The pericycle** is made up of **sclerenchyma cells** which are **lignified, dead fiber cells** . These cells have **thick, woody walls** and **tapering ends**. The function of Pericycle are: **Strengthens for the stem** and **Provides protection for the vascular bundles**.

- **Vascular Bundles**

  The vascular bundles are situated in a **ring on the inside** of the pericycle of the plant. This distinct ring of vascular bundles is a **distinguishing characteristic** of dicotyledonous stems. A mature vascular bundle consists of three main tissues - **xylem, phloem and cambium**. The **phloem** is located towards the **outside** of the bundle and the **xylem** towards the **center**. The **cambium separates** the xylem and phloem, which bring about **secondary thickening**.

- **Pith (Medulla)**

  The pith occupies the **large central part** of the stem. It consists of **thin-walled parenchyma cells** with **intercellular air spaces**. Between each vascular bundle is a **band of parenchyma, the medullary rays**, continuous with the cortex and the pith.

  Functions of the Pith or Medulla

  - The cells of the pith store water and starch.
  - They allow for the exchange of gases through the intercellular air spaces.
  - The medullary rays transport substances from the xylem and phloem to the inner and outer parts of the stem.

II) Cross section of Monocotyledonous stem :

The tissues of dicots and monocots are basically the same (Figure 9.8A&B). However, there are essential differences in the arrangement of the **epidermis, ground tissue and vascular tissue** .

1- Epidermis

The structure and functions of this tissue are the same as those of the epidermis of the stem of a dicotyledonous plant. The walls are **thickened** and covered with a **thin waterproof layer** called the **cuticle**. **Stomata** with **guard cells** are found in the epidermis. In some stems either **unicellular or multicellular hair-like outgrowths** appear from the epidermis. Beneath the epidermis, a multilayer of sclerenchyma cells called hypodermis.

2- Ground Tissue

This region is composed of **small, thick-walled sclerenchyma** on the inside of the epidermis. These layers of cells are followed by larger thin-walled parenchyma cells. Intercellular air spaces are found in the parenchyma. **A cortex or pith** is absent.

Functions

- Sclerenchyma tissue strengthens the stem.
- Parenchyma tissue stores synthesised organic food such as starch.
- Intercellular air spaces allow the exchange of gases.

3- Vascular Bundles

The vascular bundles are found **scattered throughout the ground tissue**. The vascular bundles **occurring nearer the rind of the stem** are smaller and are **closer to one another. The vascular bundles contain no cambium and consequently secondary thickening does not occur. The vascular bundle is composed of the following parts**(Figure 9.8C)**:**

- **Sclerenchyma sheath:** Thick-walled sclerenchyma fibres surround the vascular bundle, which *protect the vascular* bundles and *strengthen* the stem.
- **Xylem:** Large xylem vessels are found within an irregular intercellular air space. This space is surrounded by thin-walled parenchyma cells.
- **Phloem:** Phloem is composed of thin-walled cells, sieve tubes and companion cells.

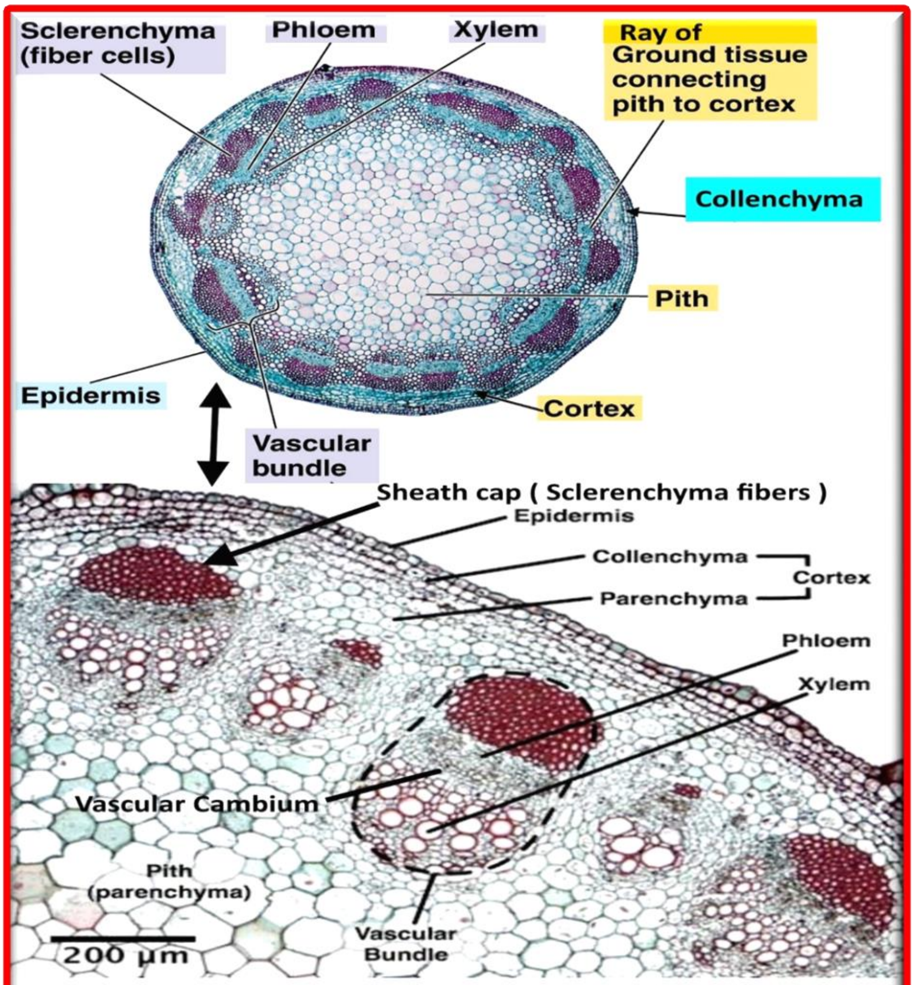


Fig. 9.7 A: Internal structure of Dicot stem; B: Structure of Vascular bundle

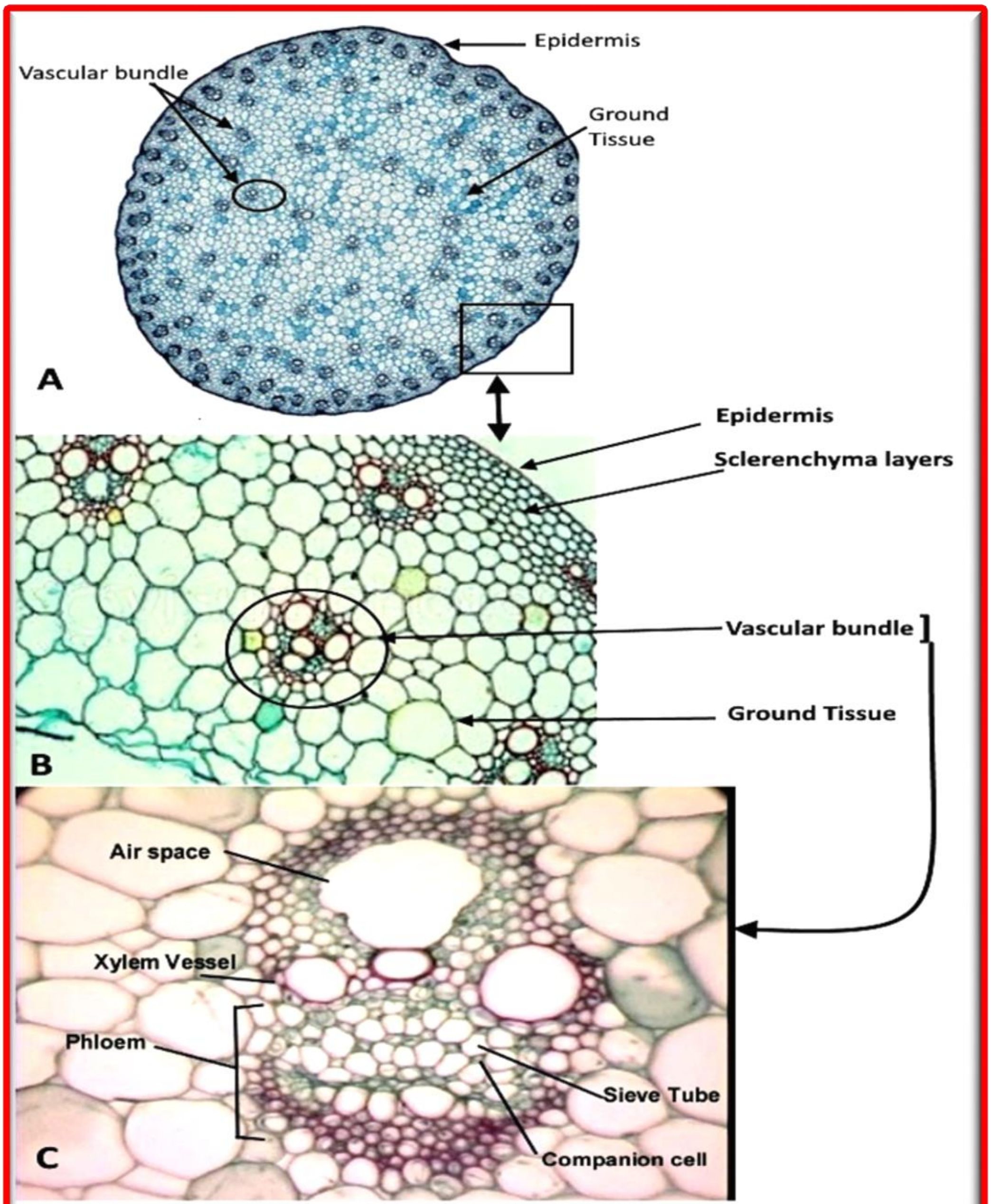


Fig. 9.8 A: Internal structure of Monocot Stem; B: Magnified part of cross section of Monocot Stem; C: Vascular bundle of Monocot Stem

III) The Woody Dicot Stem

A transverse section of a woody stem of 3-years old basswood tree shows that it consists the following (fig.9.9):

1- **The bark** includes Periderm (cork, cork cambium, and Phelloderm), cortex, and phloem

2- **Vascular cambium** is one of the two lateral meristems. The New vascular tissue is added at the **vascular cambium**, which is one of the two lateral meristems. The vascular cambium continually adds new **secondary xylem** to the outside of the existing xylem (just inside the vascular cambium), so the xylem area grows thicker and thicker. The new **secondary phloem** cells are added to the inner part of the ring of phloem. The second type of lateral meristems is **cork cambium, which divide to form the protective cork**

3- **Xylem**: Each year, the cambium produces a layer of secondary xylem and a layer of secondary phloem. Because more of the derivatives mature toward the inside, the layer of secondary xylem each year is much thicker than the layer of secondary phloem. Xylem makes up the **wood** region and arrange in rings called **annual ring**. The annual ring represent one year and each one contains two sub ring, **the spring and summer rings**. The xylem vessels made in the spring, when water is plentiful, have larger diameters than those made later in the season. No xylem is made during the dormant season.

4- **Pith:** in young stems, the pith, which is made of parenchyma, stores food. The pith disappears in old woody stem

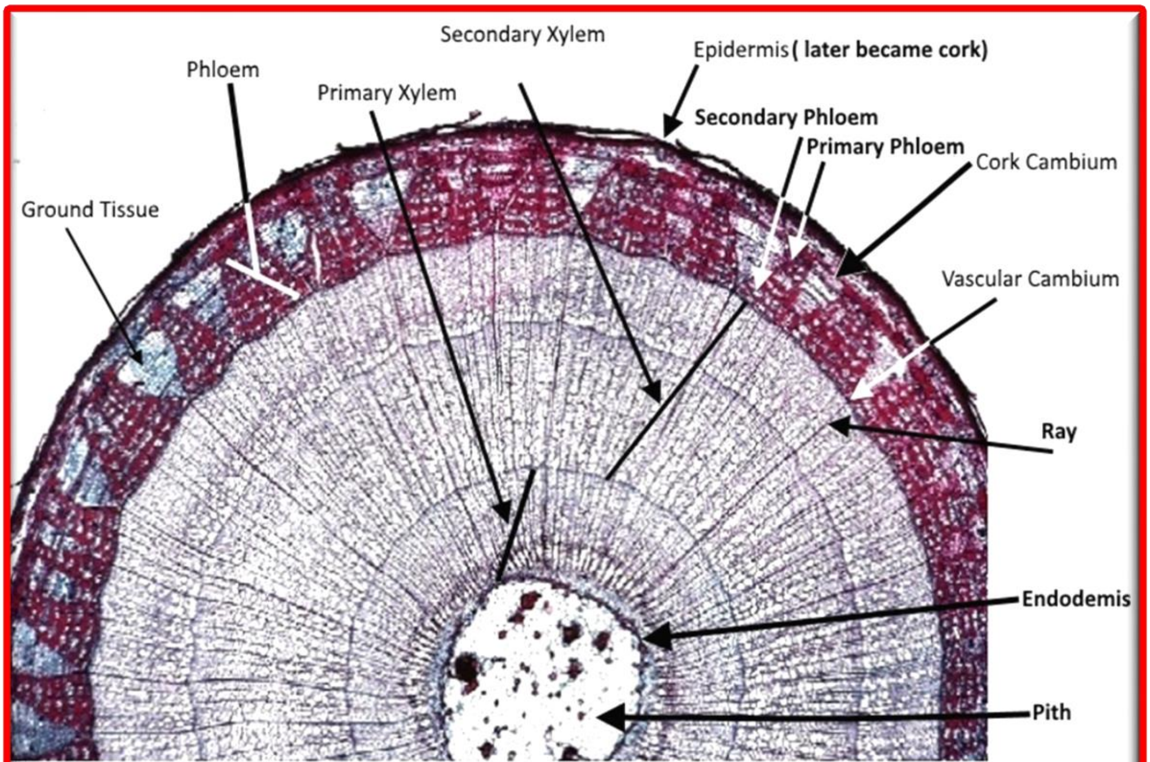


Fig. 9.9 : The anatomy of 3-years woody stem

3) Leaf Anatomy

The leaf is the primary photosynthetic organ of the plant. It consists of a flattened portion, called the blade (a lamina), that is attached to the plant by a structure called the petiole (leaf stalk), and **stipules** (small structures located to either side of the base of the petiole) (fig.9.10a). Sometimes leaves are divided into two or more separated parts called leaflets. Leaves with a single undivided blade are called simple, those with two or more leaflets are called compound. The leaflet of the compound leaf attached to the stem by **rachis** (fig.9.10b).

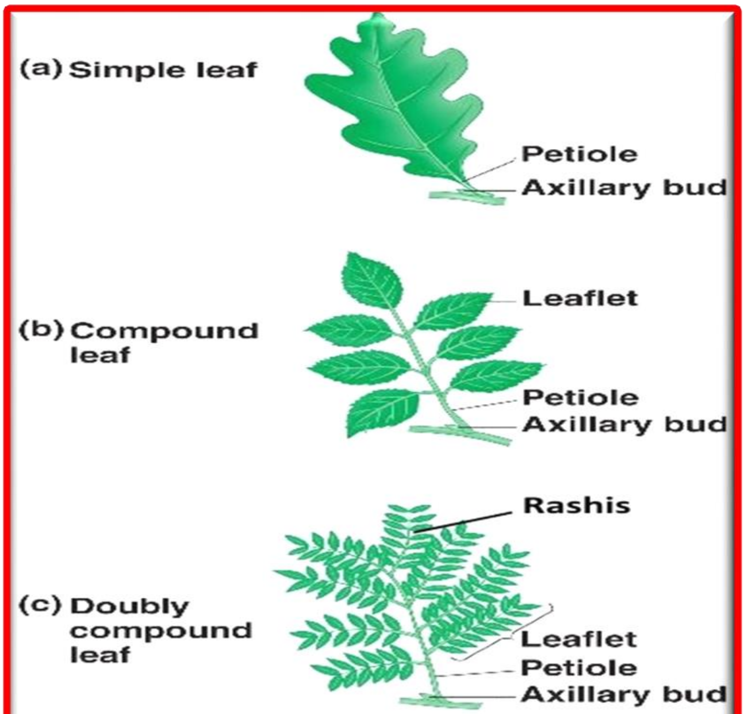


Fig. 9.10: Types of leaves with its structure

Internal structure of the leaf:

A transverse section passing through the midrib region of the leaf reveals the following structure:

1- Epidermis

The epidermis is the outer layer of cells covering the leaf. It is covered with a waxy cuticle which is impermeable to liquid water and water vapor and forms the boundary separating the plant's inner cells from the external world. The cuticle is in some cases thinner on the lower epidermis than on the upper epidermis. The epidermis serves several functions: protection against water loss by way of transpiration, regulation of gas exchange, secretion of metabolic compounds, and (in some species) absorption of water. The epidermis tissue includes several differentiated cell types; epidermal cells, epidermal hair cells (trichomes), cells in the stomatal complex; guard cells and subsidiary cells. The epidermal cells are the most numerous, largest, and least specialized and form the majority of the epidermis. These are typically more elongated in the leaves of monocots than in those of dicots.

The stomatal pores perforate the epidermis and are surrounded on each side by chloroplast-containing guard cells, and two to four subsidiary cells that lack chloroplasts, forming a specialized cell group known as the stomatal complex. The opening and closing of the stomatal aperture regulates the exchange of gases and water vapor between the outside air and the interior of the leaf. In a typical leaf, the stomata are more numerous over the lower epidermis than the upper epidermis.

2- Mesophyll

Most of the interior of the leaf between the upper and lower layers of epidermis is a *parenchyma* (ground tissue) or chlorenchyma tissue called the **mesophyll**. In ferns and most flowering plants, the mesophyll is divided into two layers:

- An upper **palisade layer** of vertically elongated cells, one to two cells thick, directly beneath the upper epidermis. Its cells contain many more chloroplasts than the spongy layer. These long cylindrical cells are regularly arranged in one to five rows. Cylindrical cells, with the *chloroplasts* close to the walls of the cell, can take optimal advantage of light.

- Beneath the palisade layer is the **spongy layer**. The cells of the spongy layer are not so tightly packed, so that there are large intercellular air spaces between them for oxygen and carbon dioxide to diffuse in and out of during respiration and photosynthesis. These cells contain fewer chloroplasts than those of the palisade layer. The pores or *stomata* of the epidermis open into sub stomatal chambers, which are connected to the air spaces between the spongy layer cells.

3- Veins

The **veins** are the vascular tissue of the leaf and are located in the spongy layer of the mesophyll. The pattern of the veins is called venation. The venation is typically parallel in monocotyledons and forms an interconnecting network in dictocotyledons.

A vein is made up of a vascular bundle. At the core of each bundle are clusters of two distinct types of conducting cells: Xylem and phloem. A sheath of ground tissue made of lignin surrounding the vascular tissue. This sheath has a mechanical role in strengthening the rigidity of the leaf.

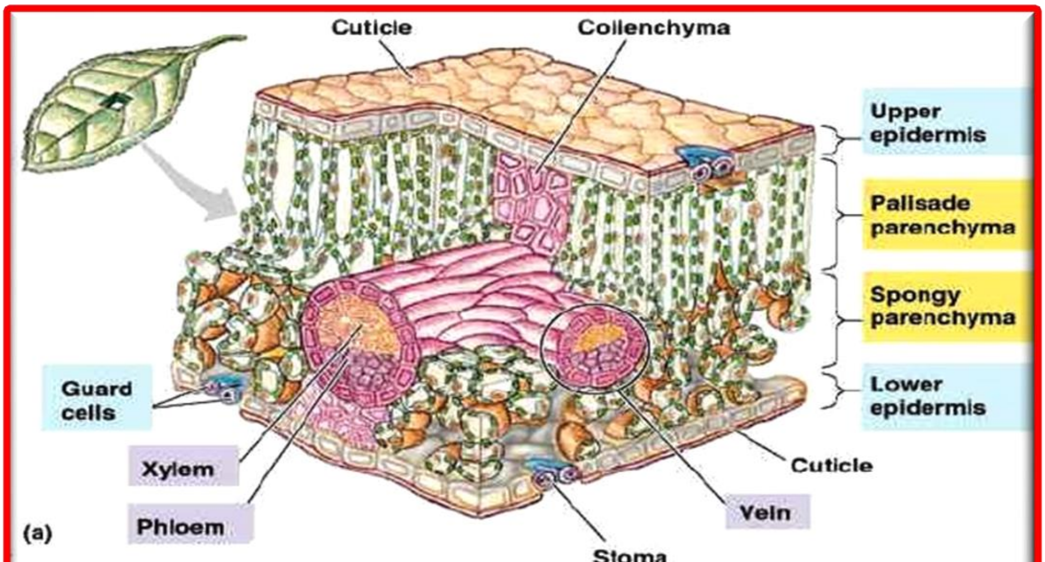


Fig. 9.11: The anatomy of the dicot leaf

Monocot leaf:

Figure 9.12 shows the structure of monocot leaf.

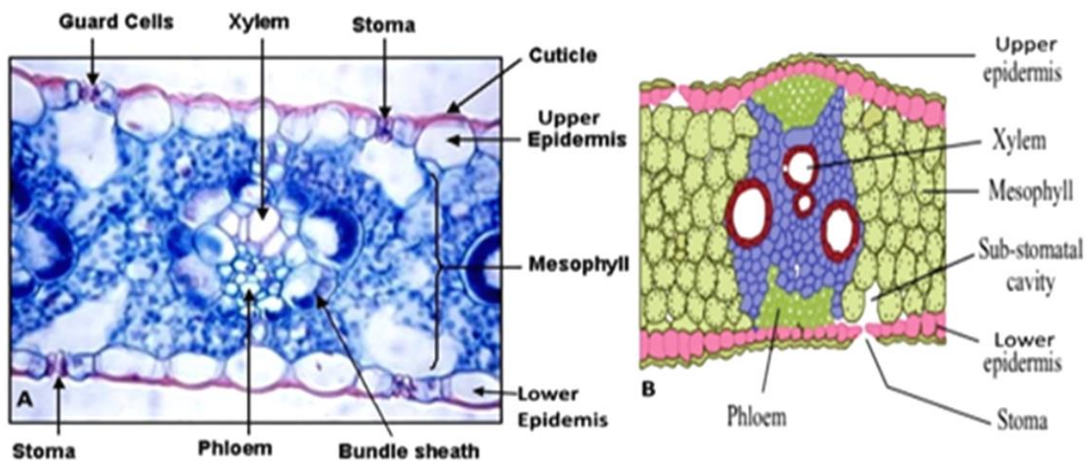


Fig.9.12: Anatomy of Monocot stem; A: Cross section; B : Sketch

The following table (Table 9.2) shows the main differences between Dicot and Monocot leaves:

Characters	Dicot leaf	Monocot leaf
1. Nature of orientation	Typically dorsi-ventral	Typically iso-bilateral
2. Stomata	Hypostomatic	Amphistomatic
3. Motor cells	Absent	Present in the upper epidermis
4. Mesophyll	Differentiated into palisade and spongy parenchyma	Undifferentiated
5. Veins	Irregularly scattered	Parallely arranged
6. Xylem vessels	Many protoxylem and metaxylem vessels in each bundle	Two protoxylem and two metaxylem vessels in each bundle
7. Bundle sheath extensions	Made up of collenchyma	Made up of sclerenchyma

Table 9.2 : shows the differences between Dicot and Monocot leaves

Reproductive organs of the plants:

The reproductive organs include **Flowers, Fruits, and seeds**, which are responsible about the continuation of a given species.

The flowers:

The flower is the reproductive unit found in flowering plants (Angiosperms). Flowers have many different shapes and sizes, and there are many variations in color, number of flower parts and the arrangements of these parts.

a- **Types of Flowers: There are many different types of the flowers, these are:**

1- Perfect flowers: have both male and female reproductive organs, and the plant is said to be <u>monoecious</u> plant.

2- **Imperfect flowers**: have only male reproductive organs or only female reproductive organs, and the plant said to be **dioecious** plant. A flower that lacks stamens **is pistillate, or female,** while one that lacks pistils is said to be **staminate, or male.**

3- Complete flowers: have stamens, a pistil, petals, and sepals.

4- **Incomplete flowers**: lack any parts of complete flower.

b- The function of the flower:

The biological function of a flower is to effect reproduction, usually by providing a mechanism for the union of sperm with eggs. In addition to that, flowers have long been admired and used by humans to beautify their environment, and also as objects of romance, ritual, religion, medicine and as a source of food.

Floral parts

Basically, each flower consists of a floral axis upon which are borne the essential organs of **reproduction (stamens and pistils)** and usually **accessory organs (vegetative**), which consists **perianth (sepals and petals)**; the latter may serve to both attract pollinating insects and protect the essential organs. The floral axis is a greatly modified stem; unlike vegetative stems, which bear leaves, it is usually contracted, so that the parts of the flower are crowded together on the stem tip, the receptacle. A stereotypical flower consists of four kinds of structures attached to the tip of a short stalk. Each of these kinds of parts is arranged in a **whorl** on the **receptacle**. The four main whorls (starting from the base of the flower or lowest node and working upwards) are as follows:

a- Vegetative part (Perianth)

Perianth consists Collectively the calyx and corolla.

1- **Calyx**: the outermost whorl consisting of units called **sepals**; these are typically green and enclose the rest of the flower in the bud stage, however, they may be absent or prominent and petal-like in some species.

2- **Corolla**: the next whorl toward the apex, composed of units called **petals**, which are typically thin, soft and colored to attract animals (Insect) that help the process of **pollination**.

b- Reproductive parts:

There are two parts: Male and Female parts. The male parts of the flower, comprise the **stamens**, each of which consists of a supporting **filament** and an **anther**, in which **pollen** is produced. The Female part of the flower, comprise the **pistils**, each of which consists of an **ovary**, with an upright extension, the **style**, on the top of which rests the **stigma**, the pollen-receptive surface. The **ovary** encloses the ovules, or potential seeds. A **pistil** may be simple, made up of a single **carpel**, or ovule-bearing modified leaf; or compound, formed from several carpels joined together.

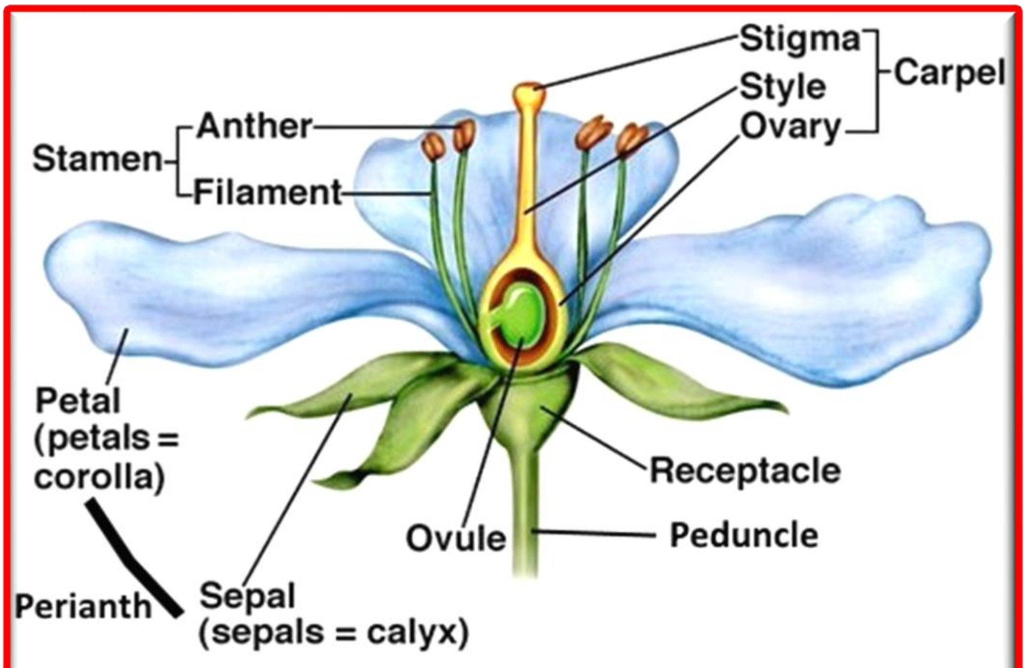


Fig. 9.13: Anatomy of the flower.

Practical:

1- Study available model of Dicot Root, Dicot Stem, Monocot Stem, Vascular bundle (Xylem), Dicot leaf, and Flower

2- Study available prepared slide of Dicot and Monocot Root, Dicot and Monocot Stem, Root hair, Old woody Stem and Dicot Leaf.

3- Identify all parts you see in the models and prepared slides, and then identify the Plant tissue made up these parts.

4- Study a fresh stem to determine all of its external parts.

5- Study compound and simple fresh leaves.

6- Study a fresh flower and determine all parts.

Experiment 10
Animal Tissue

Introduction

The study of tissue is known as **histology**. Tissue is a cellular organizational level intermediate between cells and a complete organ. A tissue is an ensemble of similar cells from the same origin that together carry out a specific function. **Organs** are then formed by the functional grouping together of multiple tissues.

Animal tissue (Fig. 10.1) grouped in to four basic types: **connective, muscle, nervous**, and **epithelial**. Multiple tissue types compose organs and body structures. While all animals can generally be considered to contain the four tissue types, the manifestation of these tissues can differ depending on the type of organism. For example, the origin of the cells com prizing a particular tissue type may differ developmentally for different classifications of animals.

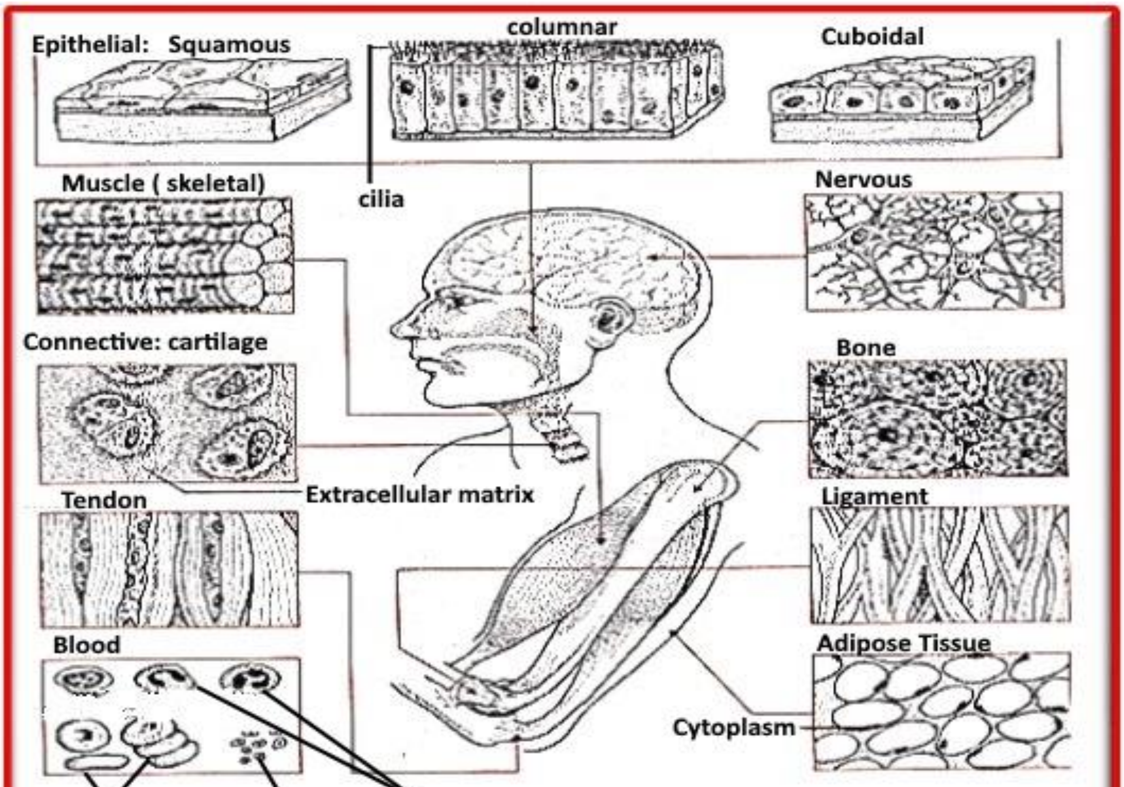


Fig. 10.1: Human Tissue through body

I) Epithelium

The **epithelium** form the **endothelium**, a specialized type of epithelium that composes the **vasculature**. By contrast, a true **epithelial tissue** is present only in a single layer of cells held together via **tight junctions**, to create a selectively permeable barrier. This tissue covers all organismal surfaces that come in contact with the external environment such as the **skin**, the airways, and the digestive tract. It serves functions of protection, secretion, and absorption, and is separated from other tissues below by a **basal lamina (basement membrane)**. Epithelial layers contain no blood vessels (non -vascularized), so they must receive nourishment via diffusion of substances from the underlying connective tissue, through the **basement membrane.**

In general, these tissues are classified by the morphology of their cells, and the number of their layers (table.10. 1). Epithelial tissue that is only one cell thick and every cell is in direct contact with the underlying **basement membrane** is known as **simple epithelium**, while Stratified epithelium made up from two or more layers.

Simple epithelial tissues (fig.10.2): are classified by the shape of their cells. The four major classes are:

(1) **Simple squamous** (fig.10.2D): which is found lining areas where passive diffusion of gases occur. e.g. walls of capillaries, linings of the pericardial, pleural, and peritoneal cavities, as well as the linings of the alveoli of the lungs.

(2) **Simple cuboidal** (fig.10.2A): these cells may have secretory, absorptive, or excretory functions. examples include small collecting ducts of kidney, pancreas and salivary gland.

(3) **Simple columnar** (fig.10.2B): found in areas with extremely high secretive (as in wall of the stomach), or absorptive (as in small intestine) areas. They possess cellular extensions (e.g. microvilli in the small intestine, or cilia found almost exclusively in the female reproductive tract).

(4) **Pseudostratified epithelium** (fig.10.2C): this is also called **respiratory epithelium** as it is almost exclusively confined to the larger respiratory airways of the nasal cavity, trachea and bronchi

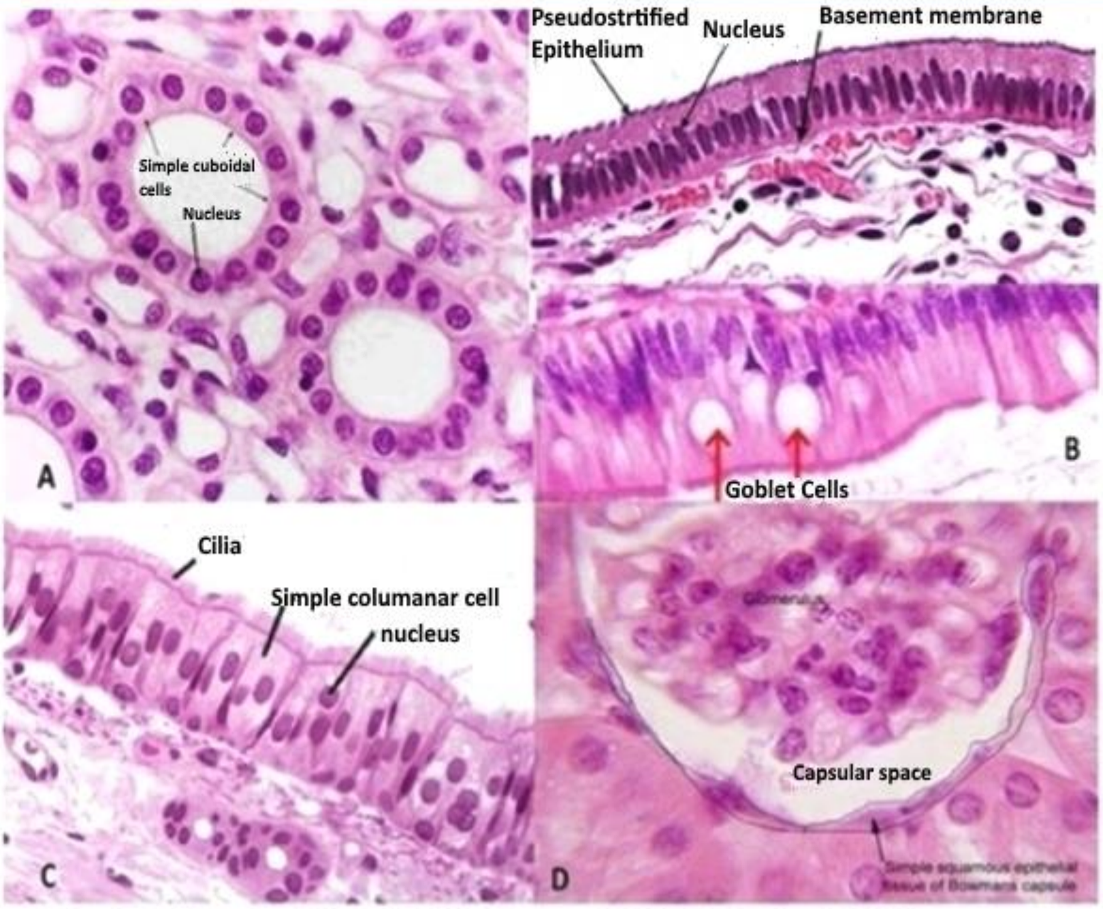


Fig. 10.2 Types Simple epithelium tissue : A) Simple cuboidal; B) Simple columnar with Goblet cells; C) Pseudostratified; D) Simple squamous

Stratified epithelium:

Stratified epithelium **(fig. 10.3B)** differs from simple epithelium in that it is multilayered. It is therefore found where body linings have to withstand mechanical or chemical insult such that layers can be abraded and lost without exposing sub epithelial layers. Cells flatten as the layers become more apical, though in their most basal layers the cells can be squamous, cuboidal or columnar. Stratified epithelia (of columnar, cuboidal or squamous type) can have the following specializations:

Specialization	Description
Keratinized	The most apical layers (exterior) of cells are dead and lose their nucleus and cytoplasm, instead contain a tough, resistant protein called keratin. It is found in the mammalian skin.
Transitional	Transitional epithelia are found in tissues that stretch and it can appear to be stratified cuboidal when the tissue is not stretched or stratified squamous when the organ is distended and the tissue stretches. It is almost exclusively found in the bladder, ureters and urethra
Squamous	Squamous cells have the appearance of thin, flat plates. They fit closely together in tissues; providing a smooth, low-friction surface over which fluids can move easily. Squamous cells tend to have horizontally flattened, oval or shaped like an egg nuclei. Squamous epithelia are found lining surfaces utilizing simple passive diffusion such as the alveolar epithelium in the lungs.
Cuboidal	Cells are roughly cuboidal in shape, appearing square in cross section. Each cell has a spherical nucleus in the centre. Cuboidal epithelium is commonly found in secretive or absorptive tissue: for example the (secretive) exocrine gland, the pancreas and the (absorptive) lining of the kidney tubules as well as in the ducts of the glands.
Columnar	Cells are elongated and column-shaped. Their nuclei are elongated and are usually located near the base of the cells. Columnar epithelium forms the lining of the stomach and intestines. Goblet cells (unicellular glands) are found between the columnar epithelial cells of the duodenum. They secrete mucus, which acts as a lubricant.
Pseudostratified	These are simple columnar epithelial cells whose nuclei appear at different heights, giving the misleading impression that the epithelium is stratified when the cells are viewed in cross section. Pseudostratified epithelium can also possess fine hair-like extensions of their apical (luminal) membrane called cilia. In this case, the epithelium is described as "ciliated" pseudostratified epithelium. In the respiratory tract the wafting effect produced causes mucus secreted locally by the goblet cells (to lubricate and to trap pathogens and particles) to flow in that. Ciliated epithelium is found in the airways (nose, bronchi), but is also found in the uterus and Fallopian tubes of females, where the cilia propel the ovum to the uterus.

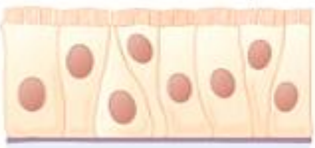
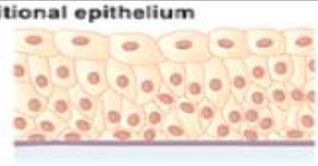
Cells	Location	Function
Simple squamous epithelium 	Air sacs of lungs and the lining of the heart, blood vessels, and lymphatic vessels	Allows materials to pass through by diffusion and filtration, and secretes lubricating substance
Simple cuboidal epithelium 	In ducts and secretory portions of small glands and in kidney tubules	Secretes and absorbs
Simple columnar epithelium 	Ciliated tissues are in bronchi, uterine tubes, and uterus; smooth (nonciliated tissues) are in the digestive tract, bladder	Absorbs; it also secretes mucous and enzymes
Pseudostratified columnar epithelium 	Ciliated tissue lines the trachea and much of the upper respiratory tract	Secretes mucus; ciliated tissue moves mucus
Stratified squamous epithelium 	Lines the esophagus, mouth, and vagina	Protects against abrasion
Stratified cuboidal epithelium 	Sweat glands, salivary glands, and the mammary glands	Protective tissue
Stratified columnar epithelium 	The male urethra and the ducts of some glands	Secretes and protects
Transitional epithelium 	Lines the bladder, uretha, and the ureters	Allows the urinary organs to expand and stretch

Table . 10.1 : Different types of epithelial tissue, their location, and functions

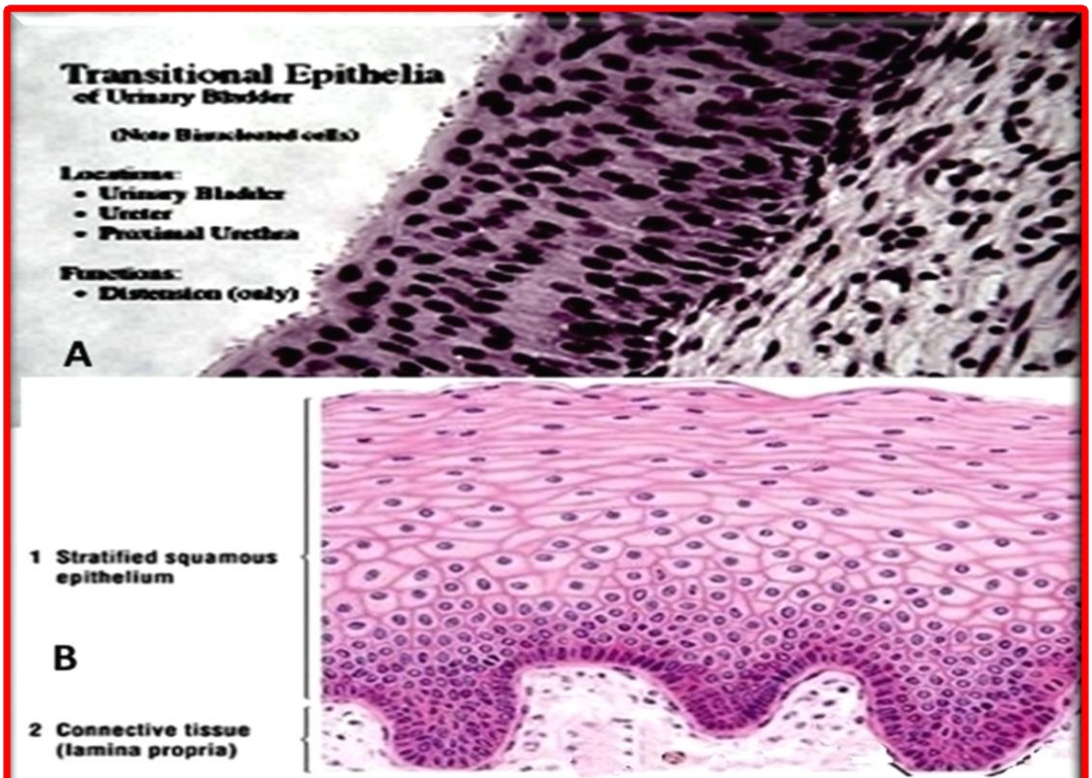


Fig. 10.3: A) Transitional Epithelia; B) Stratified Squamous Epithelium

Practical : Study available slide of simple columnar (villi of intestine), Simple cuboidal (kidney tubule), Pseudostratified (Trachea), and Stratified squamous epithelium (wherever the skin is folded).

II) Connective tissue (CT):

Connective tissues are fibrous tissues. They are made up of cells separated by non-living material, which is c allied an **extracellular matrix**. It supports, connects, or separates different types of tissues and organs in the body. Connective tissue is found in between other tissues everywhere in the body, including the **central nervous system**. The outer membranes, the **meninges**, that cover the brain and spinal cord are composed of connective tissue.

All connective tissue apart from **blood** and **lymph** consists of three main components: fibers (**elastic** and **collagenous fibers), ground substance** and **cells**. Blood and lymph lack the fiber component. All are immersed in the **body water**. The cells of connective tissue include **fibroblasts, adipocytes, mast cells, macrophages** and **leukocytes.** Connective tissue can be broadly subdivided into:

1- **Connective tissue proper:** consists of:

a- Loose (Areolar)connective tissue(fig. 10.4): is the most common type of connective tissue in vertebrates. It is strong enough to bind different tissue types together, yet soft enough to provide flexibility and cushioning. It holds **organs** in place and attaches **epithelial tissue** to other underlying tissues. It also surrounds the **blood vessels** and **nerves**. Cells called **fibroblasts** are widely dispersed in this tissue; There are three main types of connective tissue fiber: **Collagenous, Elastic, and Reticular fibers.**

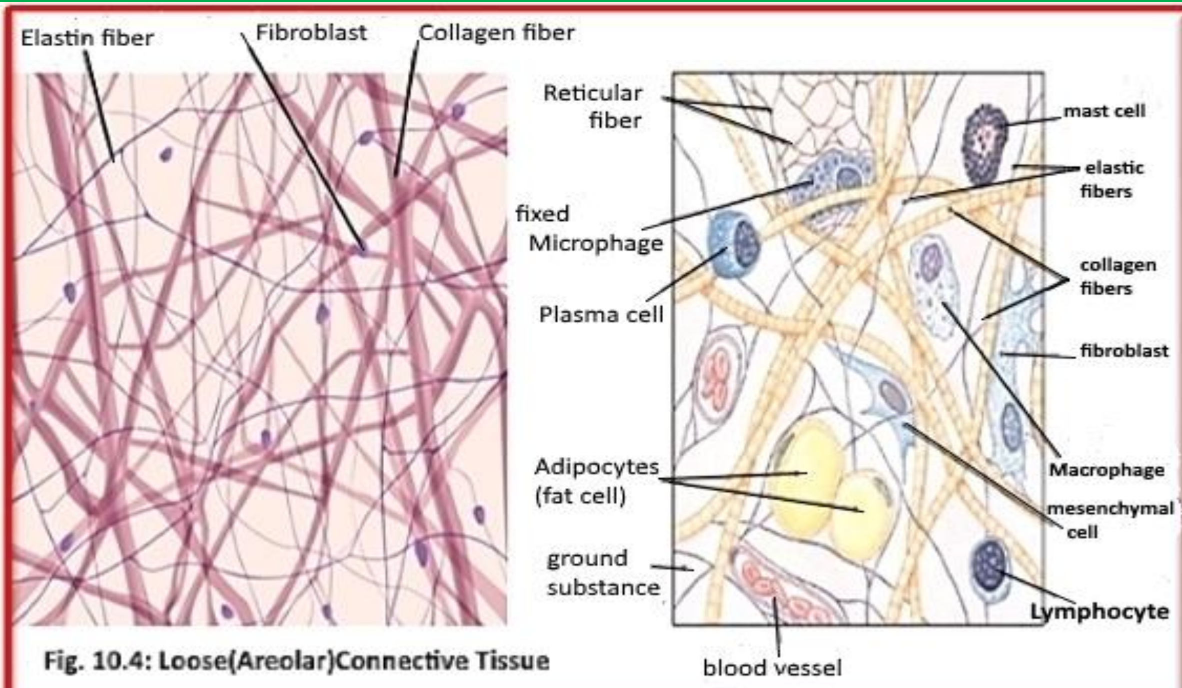


Fig. 10.4: Loose(Areolar)Connective Tissue

b- Dense connective tissue: is mainly composed of collagen fibers. Crowded between the collagen fibers are rows of **fibroblasts**, fiber-forming cells, that manufacture the fibers. Dense connective tissue forms strong, rope-like structures such as **tendons** and **ligaments**. Dense connective tissue also make up the lower layers of the **skin** (dermis), where it is arranged in sheets. There are two types:

** **Dense irregular connective tissue** **(fig. 10.5):** collagen bundles and matrix are distributed in irregular patterns, sometimes it is found in the form of a network.

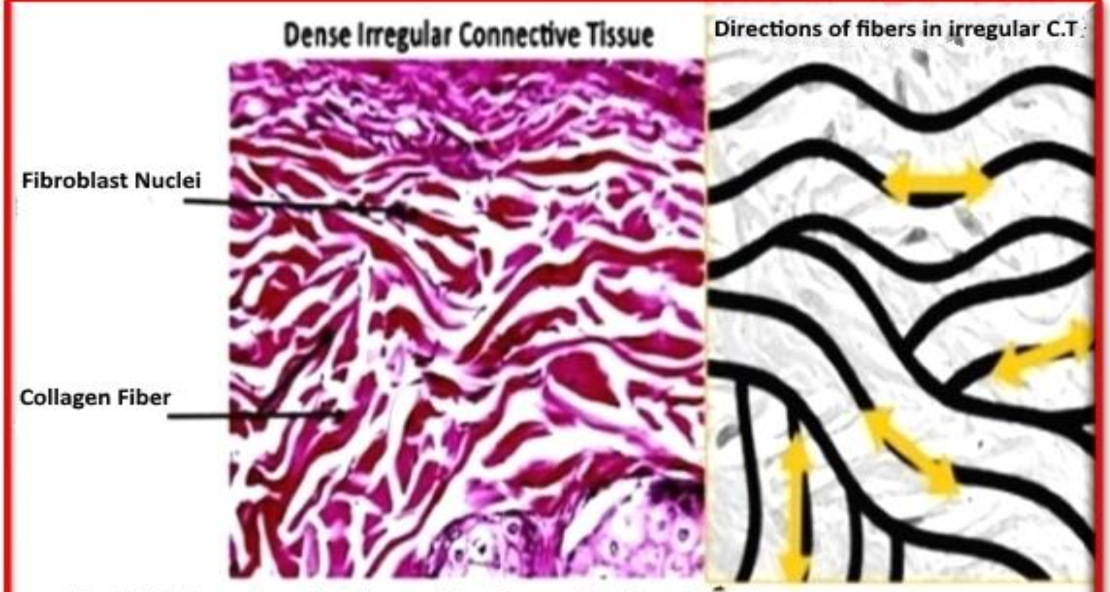


Fig. 10.5: Dense irregular Connective Tissue with fibers direction

** **Dense regular connective tissue** (fig. 10.6): The **<u>collagen</u>** fibers in dense regular connective tissue are bundled in a parallel fashion. In this kind of tissue, elastic and reticular fibers are completely absent.

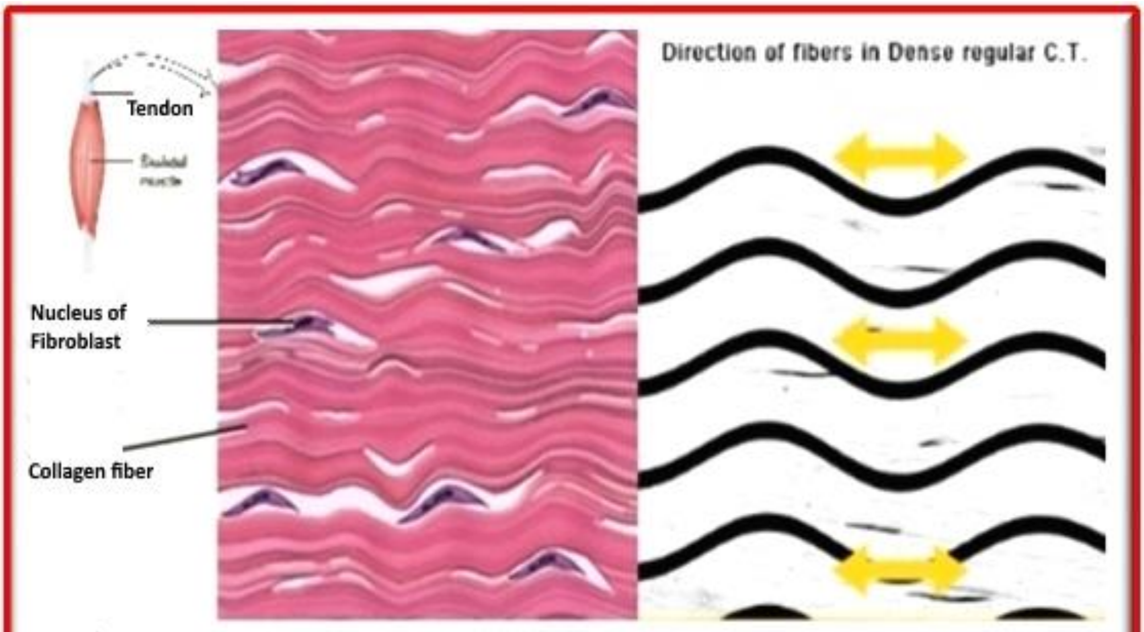


Fig. 10.6:Dense Connective Tissue with fibers directions

2- **Special connective tissue:** consists of :

a- Reticular connective tissue (fig. 10.7): is a type of **connective tissue**. It has a network of **reticular fibers**, made of **collagen** fibers. Reticular fibers are synthesized by special **fibroblasts** called **reticular cells**, The fibers are thin branching structures. The fibers form soft skeleton (**stroma**) to support the **lymphoid organs (lymph node stromal cells, red bone marrow**, and (**spleen**), also held **Adipose tissue.**

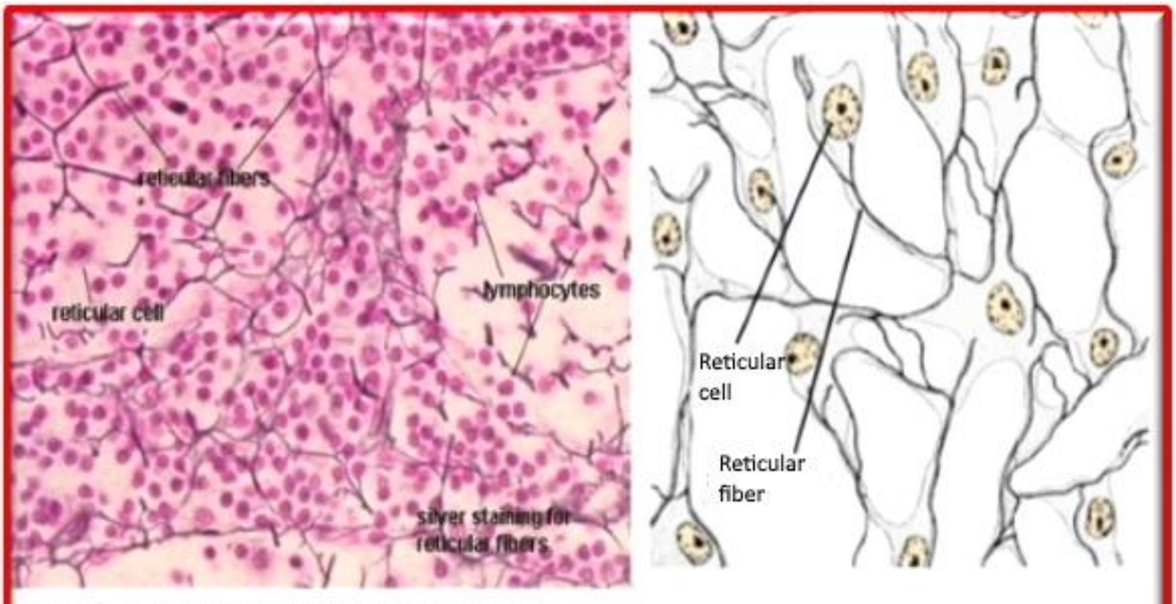


Fig. 10.7: Reticular Connective Tissue

b- Adipose tissue (fig. 10.8): **body fat connective tissue** composed mostly of **adipocytes, fibroblasts, vascular endothelial cells and adipose tissue macrophages**. Its main role is to store energy in the form of **lipids**, although it also cushions, **insulates** the body, and it produces **hormones** such as **leptin**. The two types of adipose tissue are **white adipose tissue** (WAT), which stores energy, and brown adipose **tissue** which generates body heat.

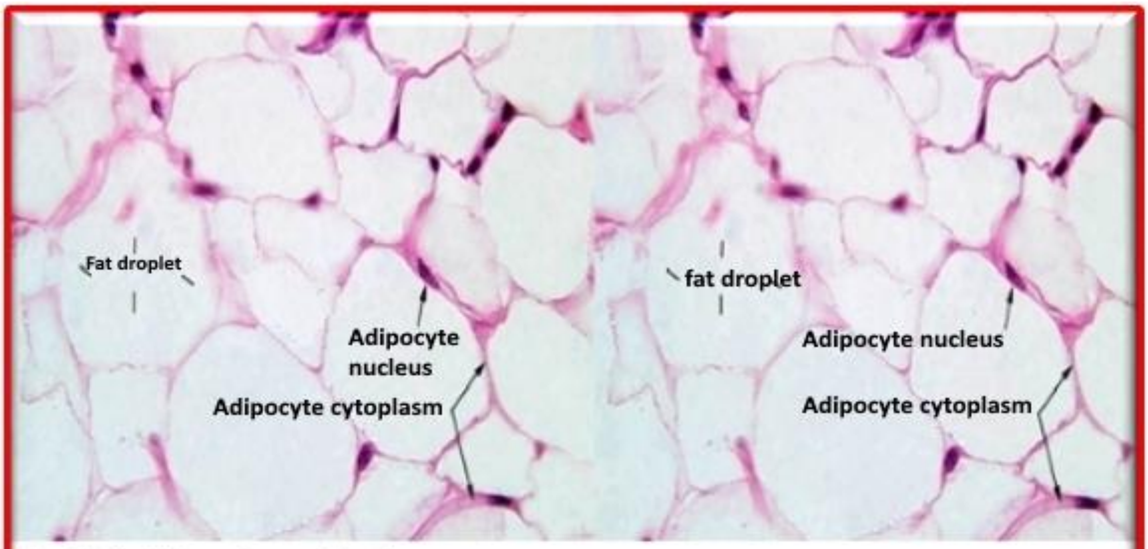


Fig. 10.8: Adipose Connective Tissue

c- Cartilage (fig. 10.9) : is a flexible **connective tissue** in animals which composed of specialized cells called **chondrocytes** that produce a large amount of collagenous **extracellular matrix,** abundant ground **substance** that is rich in **proteoglycan** and **elastin** fibers. Cartilage is classified in three types, **elastic cartilage, hyaline cartilage** and **fibrocartilage**, which differ in relative amounts of cartilage. All of these types does not contain **blood** vessels (avascular) or **nerves** (aneural). The chondrocytes are supplied by **diffusion**. Because of its rigidity, cartilage often serves the purpose of holding tubes open in an animal's body, so it is found in **joints** between **bones**, the **rib cage**, the **ear**, the **nose**, the **bronchial tubes** and the **intervertebral discs.**

Cartilages in the Adult Body

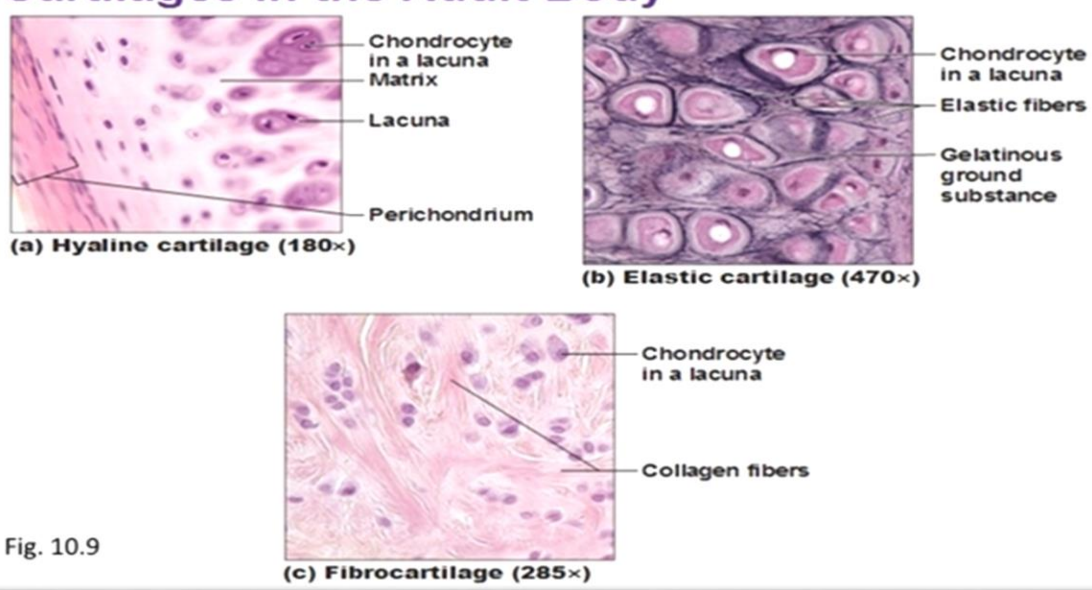


Fig. 10.9

d- Bones (fig.10.10) : consist of living cells embedded in a mineralized organic matrix. This matrix consists of organic components, The inorganic composition of bone **(bone mineral)** is primarily formed from salts of **calcium** and **phosphate**. There are two types of bone tissue: **cortical bone** and cancellous **bone**. Cortical bone consists of a repeating structure called an **osteon** or Haversian system, which is the primary anatomical and functional unit. Each osteon has concentric **lamellae** (layers) of mineralized matrix, which are deposited around a central canal, known as the haversian canal, each containing a blood and nerve supply. Lacunae contains the bone cells (Osteocyte).

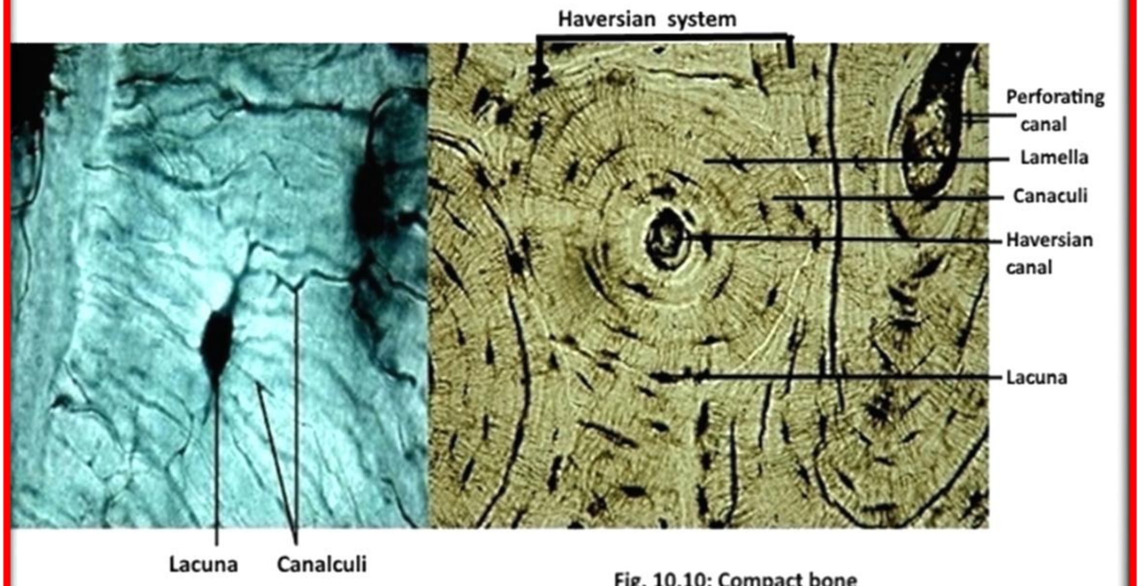


Fig. 10.10: Compact bone

e- Blood (fig.10.11) : is a **bodily fluid** in animals that delivers necessary substances such as **nutrients**. and **oxygen** to the cells and transports **metabolic waste** products away from those same cells. In human it is composed of **blood cells** suspended in **blood plasma**. Plasma, which constitutes 55% of blood fluid is mostly water (92% by volume), the blood cells are mainly **red blood cells** (also called RBCs or erythrocytes), **white blood cells** (also called WBCs or leukocytes) and **platelets**. The most abundant cells in vertebrate blood are red blood cells.

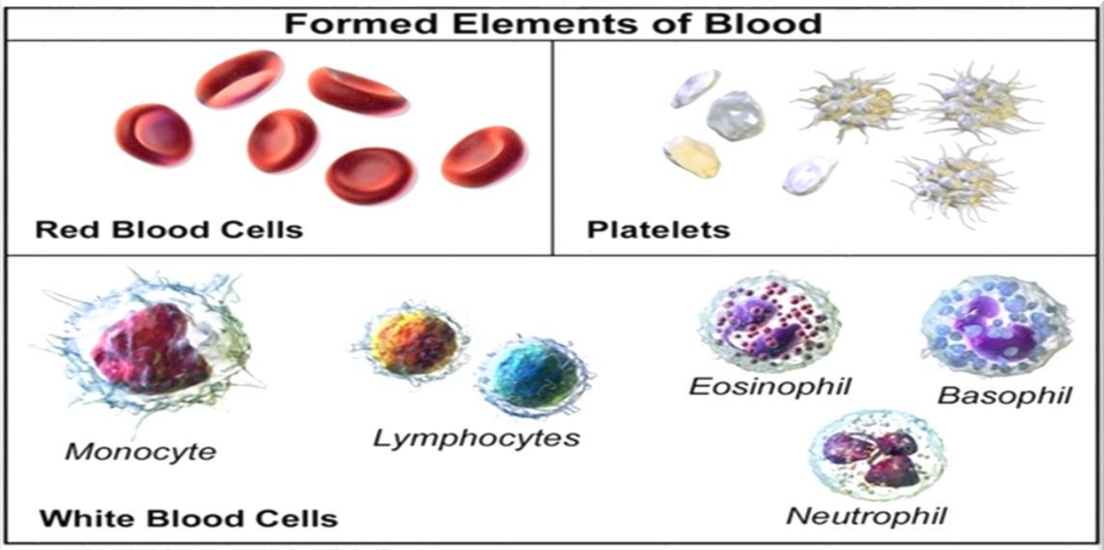


Fig.10.11: Human Blood Cells

Practical : Study available slide of Reticular C.T., Adipose C.T., Adipose C.T., Bone, Blood smear, Blood smear. Hyaline cartilage.

III) Muscle tissue (fig.10.12):

Muscles is a **soft tissue** that composes **muscles** in animal bodies, and gives rise to muscles' ability to contract. Muscle tissue varies with function and location in the body. In mammals, the three types are: **skeletal the or striated muscle**; **smooth** or non-striated muscle; and **cardiac muscle**, which is sometimes known as semi- striated.

- **Skeletal muscle (fig. 10.12A)** : striated in structure and under voluntary control (without conscious intervention), is anchored by **tendons** to **bone** and is used to effect **skeletal** movement such as **locomotion** and to maintain posture. An average adult male is made up of 42% of skeletal muscle and an average adult female is made up of 36% (as a percentage of body mass).[3] It also has striations unlike smooth muscle.
- **Smooth muscle (fig. 10.12B)**: neither striated in structure nor under voluntary control, is found within the walls of organs and structures such as the **esophagus, stomach, intestines, bronchi, uterus, urethra, bladder, blood vessels**, and the **arrector pili** in the skin (in which it controls erection of body hair). These muscle types may be activated both through interaction of the **central nervous system** (CNS) as well as by receiving innervation from peripheral **plexus** or **endocrine** (hormonal) activation.
- **Cardiac muscle** (myocardium) **(fig. 10.12C)**, found only in the heart, is a striated muscle similar in structure to skeletal muscle but not subject to voluntary control.

Cardiac and skeletal muscles are "striated" in that they contain **sarcomeres** and are packed into highly regular arrangements of bundles; smooth muscle has neither. While skeletal muscles are arranged in regular, parallel bundles, cardiac muscle connects at branching, irregular angles (called intercalated discs). Striated muscle contracts and relaxes in short, intense bursts, whereas smooth muscle sustains longer or even near-permanent contractions. The following Table shows a comparison between different types of muscles (Table: 10.2):

Anatomy	smooth muscle	cardiac muscle	skeletal muscle
Neuromuscular junction	none	none	present
Fibers	fusiform, short (<0.4 mm)	branching	cylindrical, long (<15 cm)
Mitochondria	few	numerous	many to few (by type)
Nuclei	1	1	few
Sarcomeres	none	present, max. length 2.6 µm	present, max. length 3.7 µm
Syncytium	none (independent cells)	none (but functional as such)	present
Self-regulation	spontaneous action (slow)	yes (rapid)	none (requires nerve stimulus)
Response to stimulus	unresponsive	"all-or-nothing"	"all-or-nothing"

Table: 10.2 : A comparison between different types of muscles

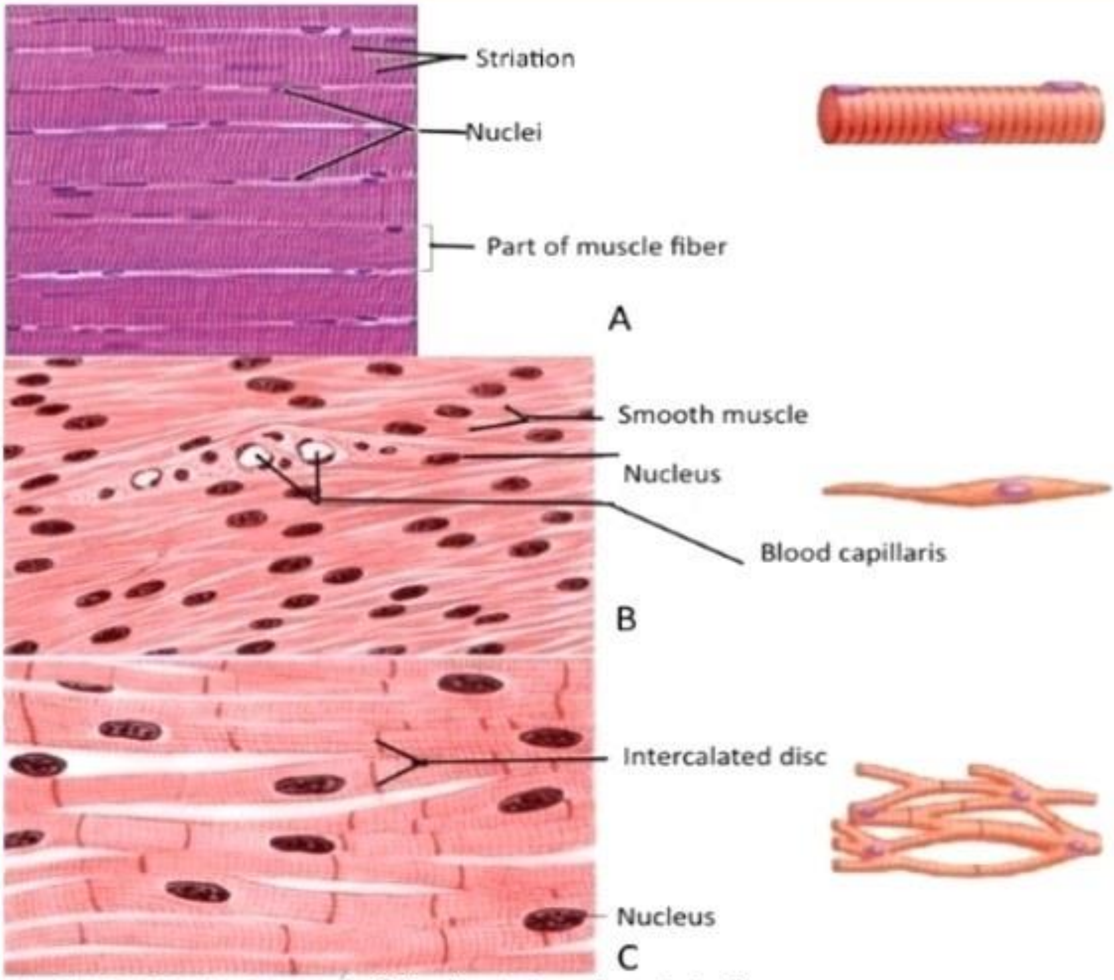


Fig. 10.12 : Muscles Tissue **A: Skeletal** **B: Smooth** **C: Cardiac**

Practical: Study available slides of Smooth, Skeletal, and cardiac muscle.

IV) Nervous tissue (fig.10. 13):

Nervous tissue is the main component of the two parts of the **nervous system**; the **brain** and **spinal cord** of the **central nervous system** (CNS), and the branching peripheral nerves of the **peripheral nervous the system** (PNS), which regulates and controls bodily functions activity. It is composed of **neurons**, or nerve cells, which receive and transmit impulses, and **neuroglia**, also known as glial cells or more commonly as just *glia*, which assist the propagation of the **nerve impulse** as well as providing nutrients to the neuron.

Generally, nervous tissue is categorized into four types of tissue. In the **central nervous system** (CNS), the tissue types found are **grey matter** and **white matter**.

1- **Gray matter** (**fig.10.13C**) is composed of cell bodies, dendrites, unmyelinated axons, protoplasmic astrocytes, microglia, and very few myelinated axons.

2- **White matter** (**fig.10.13C**) is co mposed of myelinated axons, fibrous astrocytes, myelinating oligodendrocytes, and microglia. In the **peripheral nervous system** (PNS), the tissue types are **nerves** and **ganglia.**

Functions of the nervous system are sensory input, integration, control of **muscles** and **glands, homeostasis** , and **mental activity**.

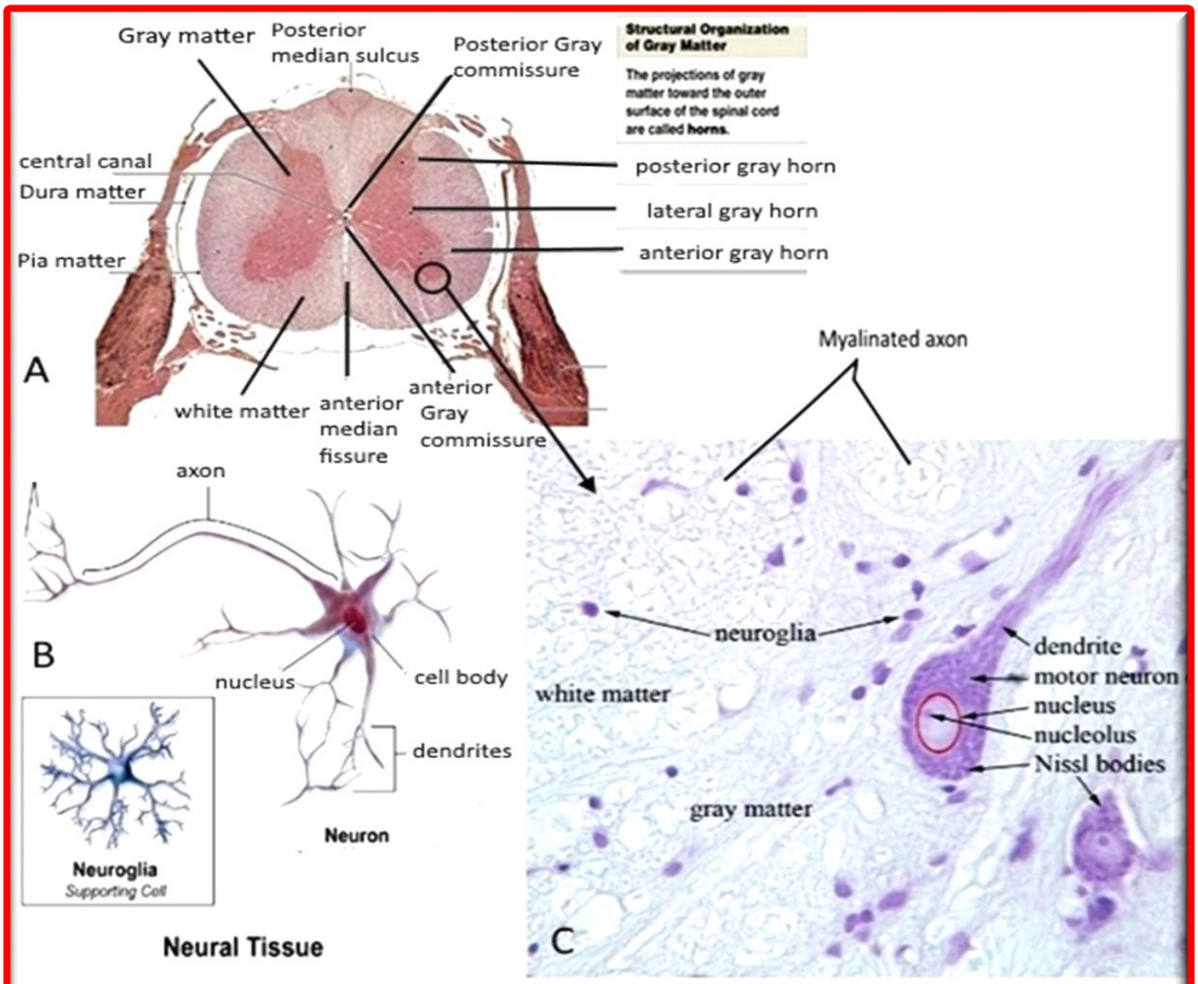


Fig. 10.13: Nervous Tissue, A: Cross section of Spinal cord; B: Neuron C: Components of Gray matter and white matter

Practical: **Study Available slide and model of spinal cord, determine the Gray matter, White matter, Neuron, Nucleus of the neuron in Gray matter, and Axon in White matter.**

Experiment 11
Histology of Human organs

Similar tissue that cooperate with each other to carry certain function form an organ. The Science which study the organs is called Histology. In This Lab we will study the Histology of Human skin, Intestine, Kidney, liver, and spinal cord

HUMAN SKIN

The **human skin** is the outer covering of the body. In **humans**, it is the largest organ of the **integumentary system.** The skin has multiple layers of **ectodermal tissue** and guards the underlying **muscles, bones, ligaments** and **internal organs**. Human skin is similar to that of most other **mammals**, except that it is not protected by a fur. Though nearly all human skin is covered with **hair follicles**, it can appear hairless. There are two general types of skin, hairy and **glabrous skin.**

Because it interfaces with the environment, skin plays a key roles in Protection, Sensation, Heat regulation, Control of evaporation, Aesthetics and communication, Storage and synthesis, Excretion, Absorption, and Water resistance.

Skin is composed of three primary layers (Fig. 11.1):

I) Epidermis (Fig. 11.1B): is the outermost layer of the skin. It forms the waterproof, protective wrap over the body's and is made up of **stratified squamous epithelium** with an underlying **basal lamina.** The epidermis contains no **blood vessels**, and cells in the deepest layers are nourished almost exclusively by diffused oxygen from the surrounding air and to a far lesser degree by blood capillaries extending to the upper layers of the dermis. Epidermis is divided into the following 5 sublayers or strata (beginning with the outermost layer): ***Stratum corneum*** **and** ***Stratum lucidum*** (only in palms of hands and bottoms of feet), ***Stratum granulosum, Stratum spinosum, Stratum germinativum*** (also called "stratum basale"). Cells are formed through mitosis at the basale layer. The daughter cells move up the strata changing shape and composition as they die due to isolation from their blood source. The cytoplasm is released and the protein **keratin** is inserted. They eventually reach the corneum and slough off (**desquamation**). This process is called **"keratinization"** . This keratinized layer of skin is responsible for keeping water in the body and keeping other harmful chemicals and pathogens out, making skin a natural barrier to infection.

II) Dermis (Fig. 11.1A) : is the layer of skin beneath the **epidermis** that consists of **connective tissue** and cushions the body from stress and strain. The dermis is tightly connected to the epidermis by a **basement membrane**. It also harbors many **nerve endings** that provide the sense of touch and heat. It contains the **hair follicles, sweat glands, sebaceous glands, apocrine glands, lymphatic vessels** and **blood vessels**. The blood vessels in the dermis provide nourishment and waste removal from its own cells as well as from the Stratum basale of the epidermis.

The dermis is structurally divided into two areas:

a- Papillary region (Fig. 11.1A): a fingerlike projections which composed of loose **areolar connective** tissue. Its provide the dermis with a "bumpy" surface that interdigitates with the epidermis, strengthening the connection between the two layers of skin.

b- Reticular region **(Fig. 11.1A & c):** The reticular region lies deep in the papillary region and is usually much thicker. It is composed of dense irregular connective tissue, and receives its name from the dense concentration of **collagenous, elastic**, and **reticular fibers** that weave throughout it. These **protein fibers** give the dermis its properties of strength, extensibility, and elasticity. Also located within the reticular region are the **roots of the hair, sebaceous glands, sweat glands, receptors, nails,** and **blood vessels. Sebaceous glands** produce a substance called **sebum**, a naturally healthy skin lubricant. When the skin produces excessive **sebum**, it becomes heavy and thick in texture.

III) Hypodermis (**subcutaneous adipose layer**) **(Fig. 11.1A) :** The hypodermis is not part of the skin, and lies below the dermis. Its purpose is to attach the skin to underlying bone and **muscle** as well as supplying it with blood vessels and nerves. It consists of loose connective tissue and elastin. The main cell types are **fibroblasts, macrophages** and **adipocytes** (the hypodermis contains 50% of body fat). Fat serves as padding and insulation for the body.

Practical : Study available model and prepared Slide to identify all parts of Human Skin.

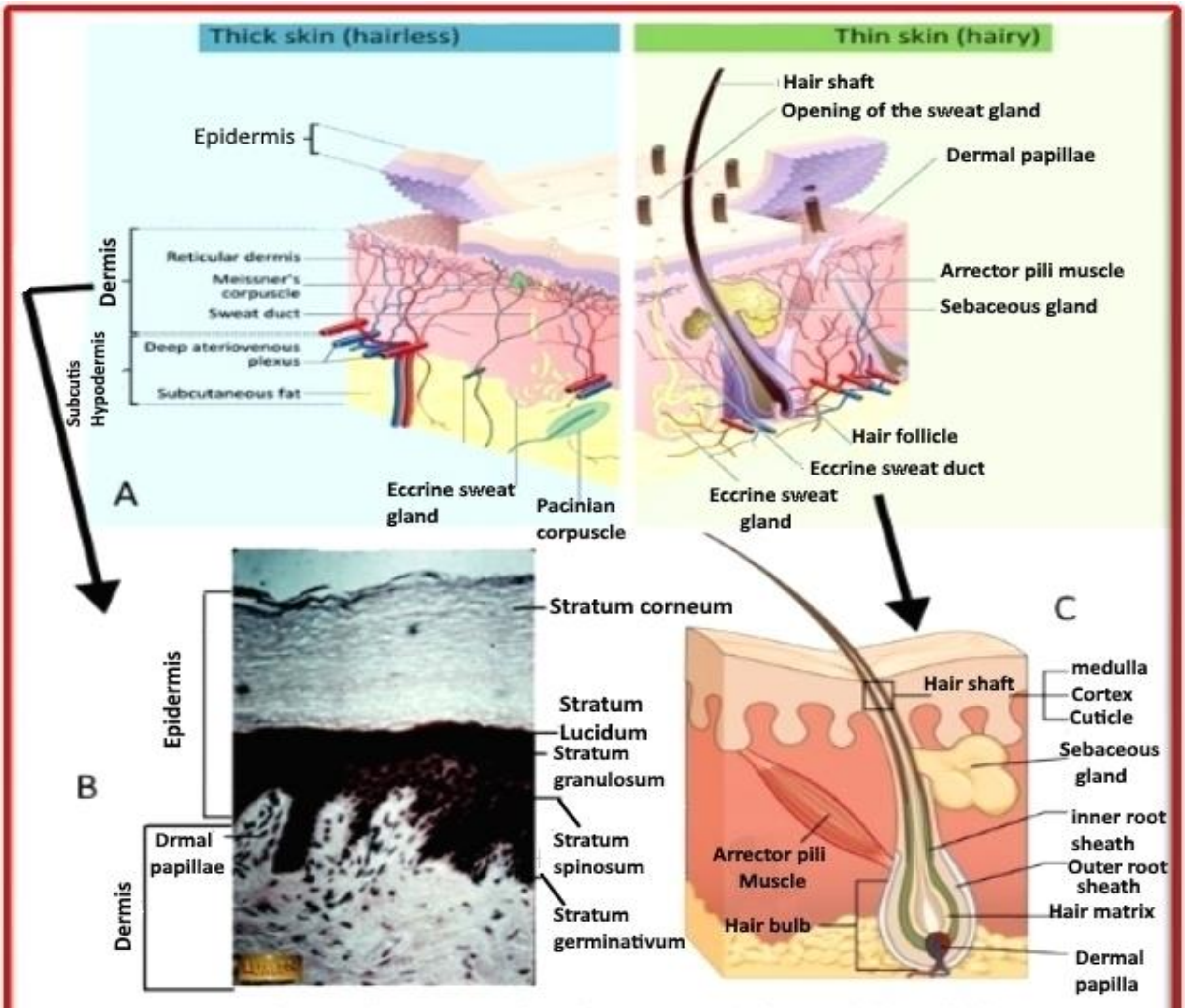


Fig. 11.1 A: Structure of Human Skin; B: Layers of Epidermis; C: Hair Shaft, Hair Follicle and bulb.

Human Intestines:

The intestines are vital organs in the gastrointestinal tract of our digestive system. Their functions are to digest food and to enable the nutrients released from that food to enter into the bloodstream. Our intestines consist of two major subdivisions: the small intestine and the large one. The small intestine is much smaller in diameter, but is much longer and more massive than the large intestine. Together the intestines take up most of the space within the abdominal body cavity and are folded many times over to pack their enormous length into such a small area.

The basic pattern and arrangement of layers in the intestinal wall are seen in both the duodenum and jejunum. In each, there is a mucosa, submucosa, muscularis, and an adventitia or a serosa. (Figure 2)

The innermost layer of the intestines is the mucosa (Figure 2B). The mucosa has finger-like projections (Fig.11.2C), the villi, lined by simple columnar epithelium tissue containing mucus-secreting **goblet cells**. Mucus produced by the mucosa lubricates the interior of the intestines to prevent friction from food passing through the lumen, or hollow portion of the intestine. The epithelial cells of the mucosa absorb the nutrients and water from digested food and transfer these substances to blood in nearby capillaries. Surface of the epithelium has a striated border (microvilli) to increase its absorptive surface and The **nucleus of epithelium cells** is Ovoid, located in the lower half of the columnar cell **Basement membrane :**A delicate membrane that supports the epithelium. Composed primarily of reticular fibers embedded in an amorphous protein polysaccharide ground substance. **Central lacteal** (Fig.11.2C)**:** A lymph vessel situated near the center of the villus. Note its endothelial lining. The lacteals become distended during absorption of fat . **Capillary :**Capillaries of the villus form a network that lies underneath the basement membrane of the lining epithelium .Villi of the duodenum tend to be flattened, whereas those of the jejunum are more rounded. The core of the villus is composed of loose connective tissue, blood vessels, a lymphatic vessel, smooth muscle fibers, and other cells of the connective tissue .This portion of the mucosa is named the lamina propria. The lamina propria terminates at the muscularis mucosae, which is composed of a band of smooth muscle fibers a few layers thick. Located within the mucosa are simple tubular intestinal glands, the crypts of Lieberkühn, and lymphatic nodules. Lymphatic nodules are found frequently in the jejunum than in the duodenum.

The submucosa (Fig.11.2C) : is composed of loose connective tissue and contains, in the duodenum but not the jejunum, the compound tubular mucous glands of Brunner. They are a continuation of the pyloric glands found in the stomach .

The muscularis (Fig. 11.2A&B) : contains an inner circular and an outer longitudinal layer of smooth muscle fibersto move the intestines. Intestinal movements such as peristalsis and segmentation help to move food through the intestine and churn the food so that it has contact with the intestinal walls. The two layers are separated by reticular and collagenous connective tissue containing nerve fibers and parasympathetic ganglion cells.

The outermost layer of the intestines is the serosa, which consists primarily of loose connective tissue containing nerves, blood and lymphatic vessels, and a mesothelium. Wherever the intestine is not bound to the posterior abdominal wall, i.e., retroperitoneal, the intestine has a suspending mesentery covered with mesothelium. When the intestine is retroperitoneal, it does not have a mesentery or a mesothelial covering and the outermost layer is called adventitia. Most of the duodenum is retroperitoneal, whereas the entire jejunum is intraperitoneal.

Practical : Study available model and prepared Slide of ileum to identify all parts of Human ileum.

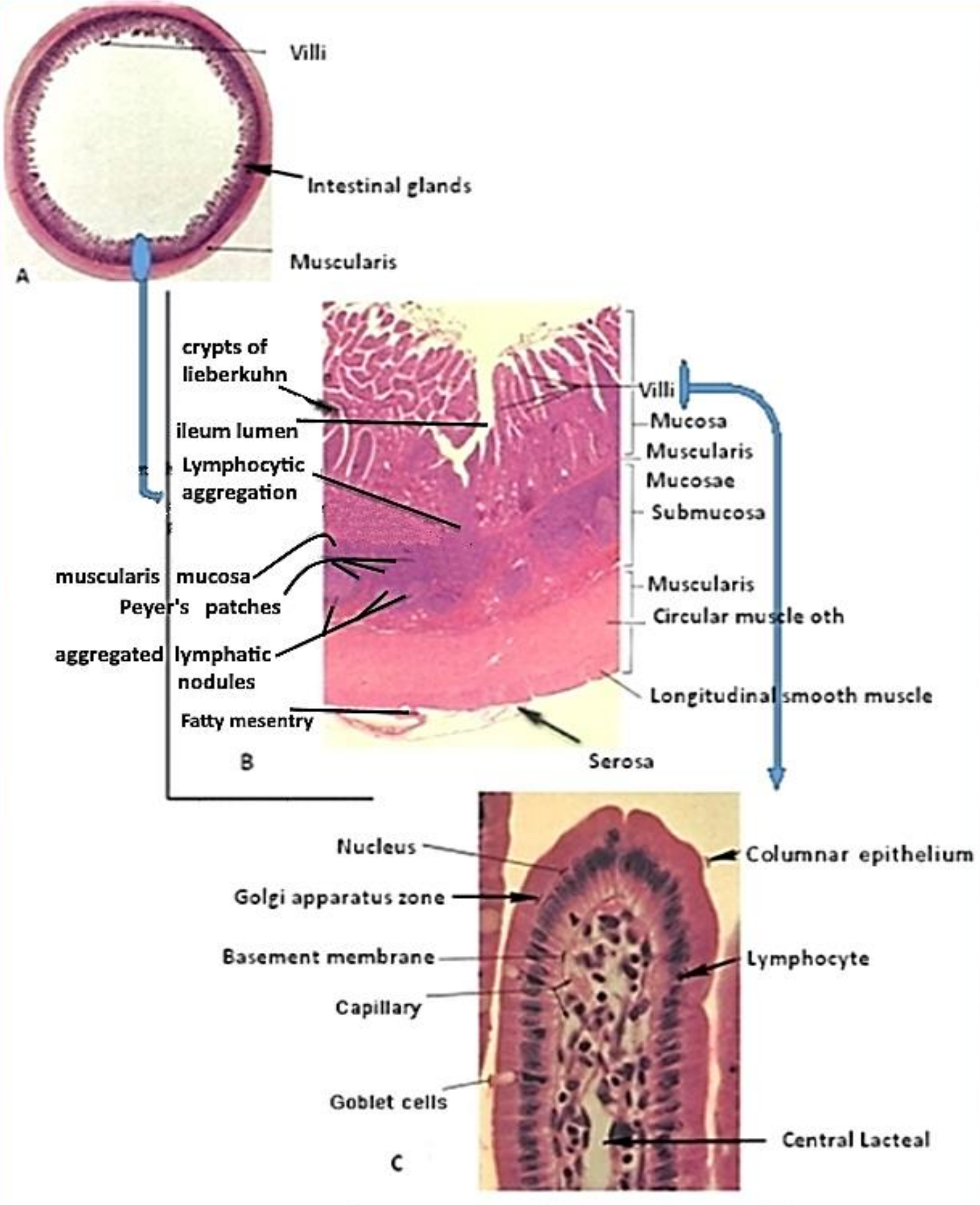


Fig. 11.2: Shows A: cross section of ileum B: Layers of ileum, C: structure of villus

Human Kidney:

The human body comprises two bean-shaped organs that are located on either side of the body underneath the diaphragm. A Cross-section of a Human Kidney (Fig. 11.3) shows that the vital structural components of a kidney are enclosed in a smooth but tough fibrous capsule called renal capsule. Inside this capsule, two distinct regions can be observed: a pale outer region called renal cortex, and a dark inner portion called renal medulla.. Inside the medulla are conical bodies known as renal pyramids. The renal pelvis joins the concave part of the kidney with the ureter. Each kidney contains about one million nephrons, which are tube-like structures that carry out the task of the filtration of blood. Resting on top of each kidney is an adrenal gland. Each kidney together with its adrenal gland is surrounded by two layers of fat (the **perirenal** and **pararenal fat**) and the **renal fascia.**

The Medulla (Fig. 11.3 & Fig. 11.3B)

The medulla is defined as the region lacking renal corpuscles. The medulla contains most of the length of the loops of Henle and most of the length of the collecting ducts (collecting tubules). The medulla of the kidney is formed of pyramidal structures known as renal pyramids. The medulla lies near the concave side of the kidney. The apex of a pyramid is known as the papilla. The papilla meets the calyx, a branch of the renal pelvis. The basal portion of the pyramidal structures elongate and expand, as they grow towards the cortex. The spaces between the renal pyramids are known as renal columns.

The Cortex (Fig. 11.3 & Fig. 11.3A)

The cortex is defined as the area containing renal corpuscles, also contains proximal and distal convoluted tubules, descending and ascending thick limbs of the loops of Henle, and the initial parts of the collecting ducts. The cortex of the kidney defined is composed of two different types of tissues, the medullary rays and the 'labyrinth of Ludwig' or 'cortical substance proper'. The medullary rays, also known as 'Henle', are cylindrical in shape and are aligned parallel to each other. The medullary rays are the extensions of pyramidal structures, while the cortical substance proper is interspersed between them. The labyrinth of Ludwig contains small structures known as glomeruli or Malpighian tubules. The space between the cortex and medulla contains blood vessels that form an arcade shape. These blood vessels run parallel to the surface of the cortex.

Capsule (Fig. 11.3)

The outer thin capsule of dense FECT contains relatively large collage fibers arranged in a two dimensional array. The outer capsule is covered by mesothelium on surfaces exposed to the peritoneal cavity. A layer of myofibroblasts lies between the dense FECT outer capsule and the parenchyma of the cortex in the human kidney but may be absent in smaller mammals. The myofibroblast layer is sometimes called the inner layer of the capsule and the outer dense FECT is called the outer layer of the capsule.

Renal Pelvis or Artery

The renal artery enters the concave side of the kidney through the hilum. The renal pelvis branches out as it moves towards the cortical portion of the kidney. The artery branches out in right angles or in an oblique manner. The branches at the base of the renal pelvis are known as major calyx, whereas the branches that are smaller in diameter and away from it are called minor calyx.

Connective Tissues

The material present between the main parts of the kidney is formed of blood vessels, stroma, and collecting tubules. These parts of the kidney appear more like a colloidal substance altogether.

Renal Tubules (Fig. 11.5)

The renal tubules are small tubes which have a diameter of 0.2 mm. The tubes either follow a straight path or twist around themselves. Every renal tubule originates from a sac-like structure present around the glomerulus. The structure is known as the Bowman's capsule.

Nephrons (Fig. 11.5)

Each kidney contains more than 800,000 nephrons, each of which serves as the basic functional unit by performing three essential processes: filtration, reabsorption and secretion. The nephron essentially comprises renal corpuscle and renal tubule. Renal corpuscle (Fig. 11.5C) is made up of a network of capillaries called glomerulus which arise from the afferent arteriole, and exit the renal corpuscle as efferent arteriole; and a cup-like sac called Bowman's capsule that surrounds the glomerulus. The initial step of filtration occurs in the renal corpuscle. Then ions and water molecules are forced into the Bowman's capsule which has outer or parietal layer compose of a simple squamous epithelium. The filtrate passes from the Bowman's capsule into the renal tubule, Renal tubule is a specialized tubular structure made up of proximal convoluted tubule, a U-shaped tube called Loop of Henle, and a distal convoluted tubule. The Loop of Henle is three parts: Descending thick limb, Thin limb, and Ascending thick limb. All tubular structure, except Thin limb, are lined by simple cuboidal epithelium, the thin limb is lined by simple squamous epithelium. Capillaries called peritubular capillaries that arise from the efferent arterioles surround the renal tubule.

The nephrons regulate blood pressure and blood volume. They are controlled by the endocrine system. The hormones such as aldosterone, parathyroid hormone, and antidiuretic hormone help in the regulation of the functioning of nephrons.

Practical : Study available model and prepared Slide to identify all parts of Human Kidney.

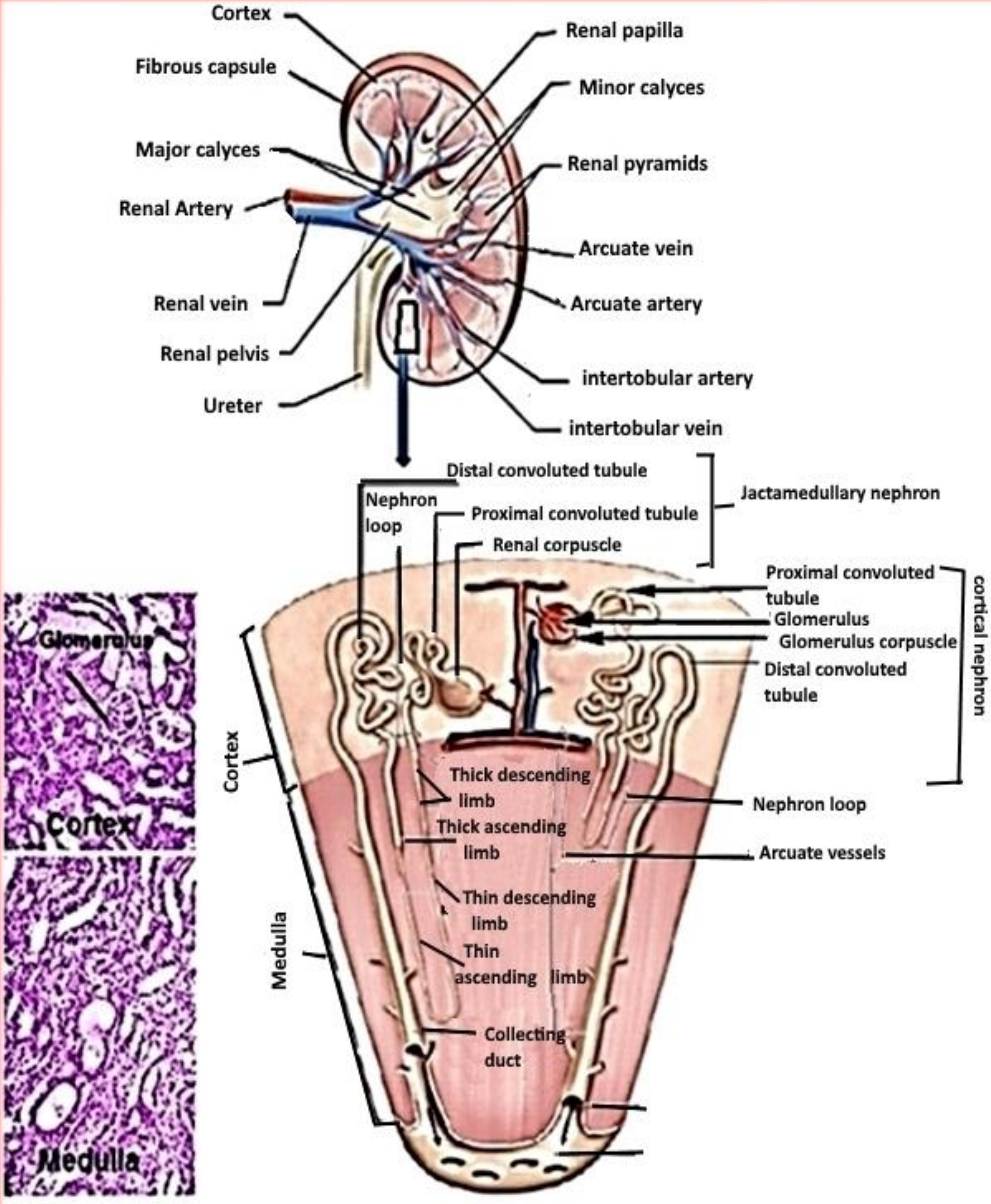


Fig. 11.3 : Cross section of Human kidney; Different layers of the kidney and how we can differentiate between Cortex and Medulla layers.

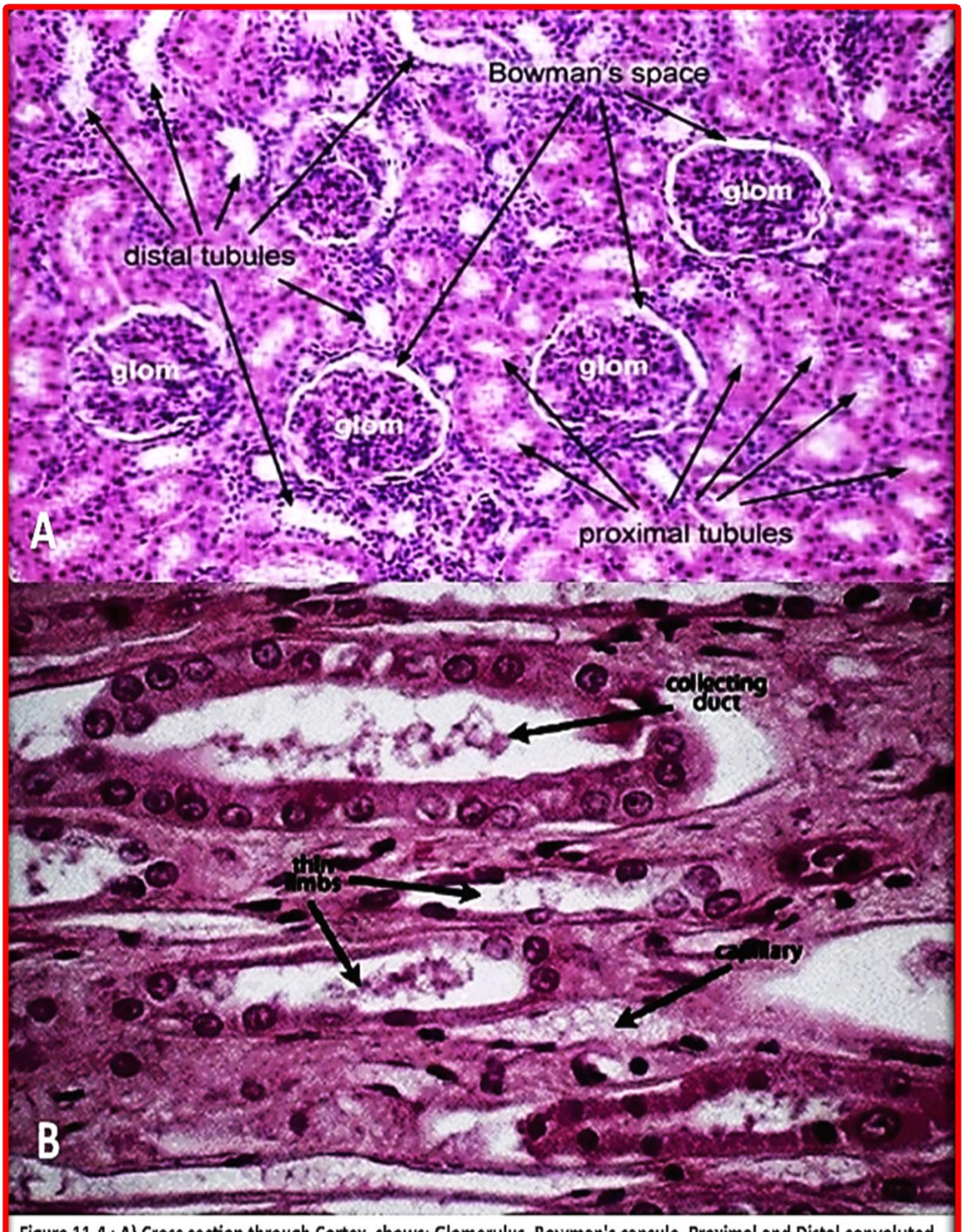


Figure 11.4 : A) Cross section through Cortex, shows: Glomerulus, Bowman's capsule, Proximal and Distal convoluted tubule

B) Cross section through Medulla, shows: Thin limbs (Loop of Henle), Collecting duct and Capillaries

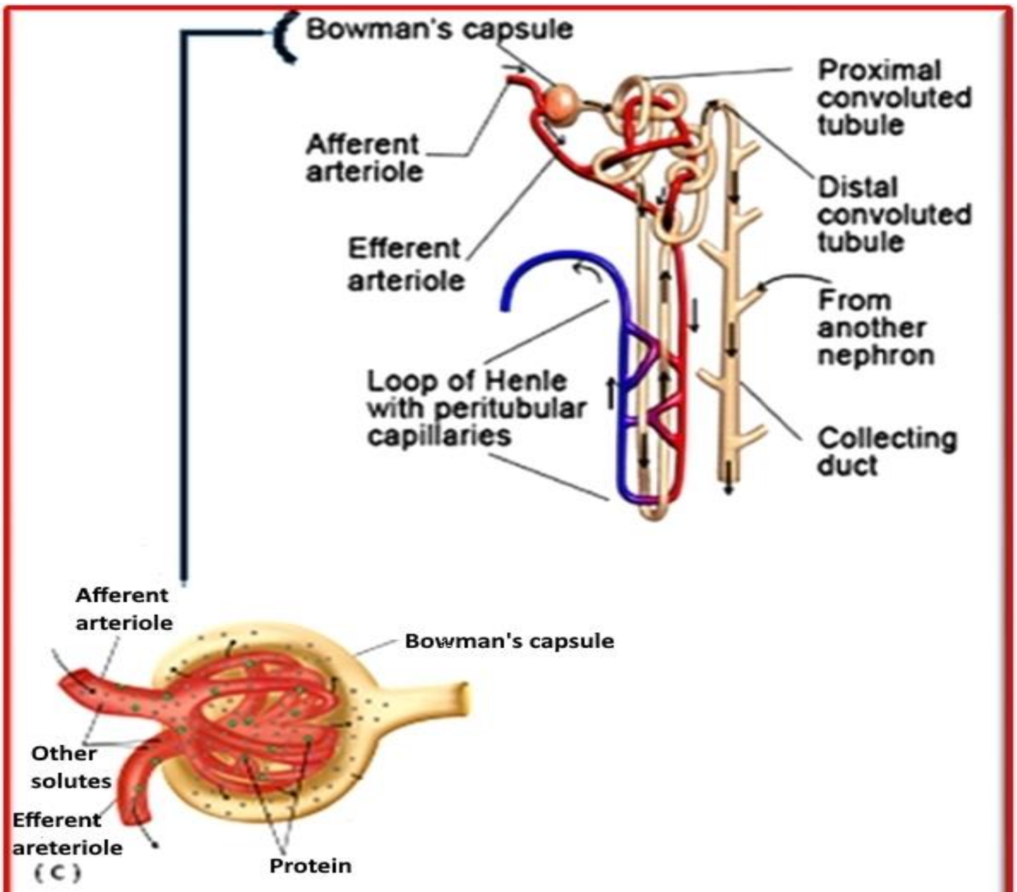


Figure 11.5: Shows the structure of functional unit of Human kidney; C: The structure of Glomerulus

Human Liver:

The liver is essential to life, and, although it is the largest gland in the body, only a fraction of its total mass is required. The liver can be considered both an exocrine gland, secreting bile via a system of bile ducts into the duodenum, and an endocrine gland, synthesizing and releasing a variety of organic compounds into the blood stream. A cross section of the liver shows it is compose from the followings (Fig. 11.6):

Hepatic lobule: the functional unit of the liver.

Central vein : Forms the axis of the hepatic lobule. Receives blood from the hepatic sinusoids and drains into intercalated (sub lobular) vein.

Liver cells (acini): Polyhedral cells with a round central nucleus. Arranged in cords and plates radiating in a spoke-like manner from the central vein.

Sinusoids : Form an extensive fenestrated system of vascular channels that radiate from the enteral vein. Lined with endothelial cells and Kupffer phagocytic cells. Receive blood from the interlobular branches of the portal vein and hepatic artery at the periphery of the lobule. Blood flows toward the center of the lobule and is drained by the central vein.

Hepatic cords : Sheets or plates of hepatic cells, one cell thick. Note the large polyhedrall liver cells with round nuclei and prominent nucleoli .

Bile canaliculi : Seen in cross section, these are channels between rows of cells within hepatic cords. Bile flows toward the periphery of the lobule to enter the system of bile ducts and gallbladder. Bile is secreted by the hepatic cells and is composed of water, bile salts, bile pigments, cholesterol, lecithin, fat, and inorganic salts. Bile secreted into the duodenum produces an acceleration in the action of pancreatic and intestinal lipases and facilitates the absorption of fats from the intestine. Bile salts are absorbed during digestion and returned to the liver for reutilization. It is believed that the bile salts are secreted twice during the digestion of a single meal (enterohepatic circulation).

A hepatocyte: is the main tissue cell of the liver, making up 70-80% of the liver's cytoplasmic mass. Hepatocytes contain large amounts of rough endoplasmic reticulum and free ribosomes. Hepatocytes are involved in protein synthesis, protein storage, transformation of carbohydrates, synthesis of cholesterol, bile salts and phospholipids, and detoxification, modification, and excretion of exogenous and endogenous substances.

Portal space: The space between hepatic lobule. It contains a branch of the bile duct, a branch of hepatic Artery, and a branch of portal Vein.

Practical : Study a prepared Slide to identify all parts of Human Liver.

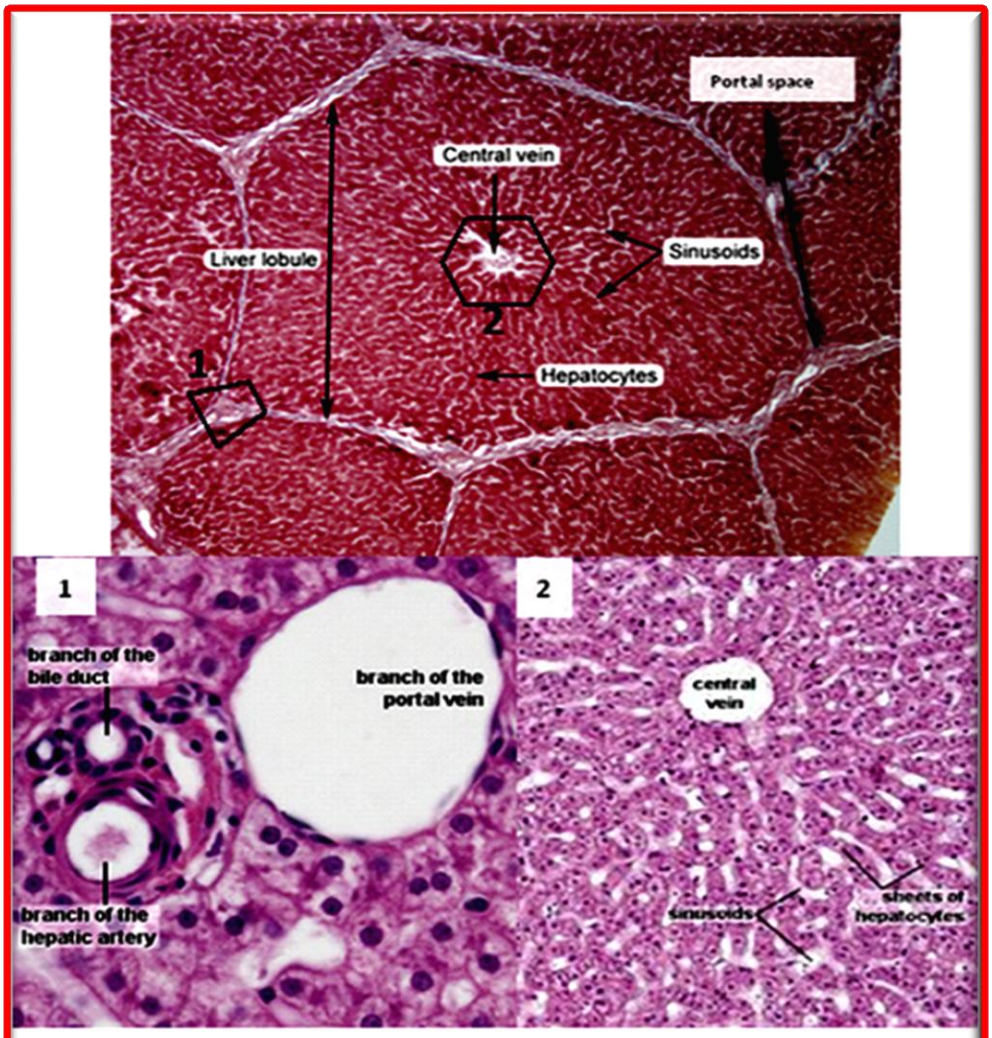


Figure 11.6 :Cross section of Human liver shows the Liver lobule, 1): Portal space; 2): Central vein

Human Spinal Cord:

Along with the brain, the spinal cord makes up the central nervous system. It is divided into 31 pairs of nerves, making 62 nerves composed of sensory and motor neurons. The nerves are named off of where they leave the spine. They are divided into 5 groups, cranial, thoracic, lumbar, sacral, and coccygeal.

- 8 pairs in the cervical region
- 12 pairs in the thoracic region
- 5 pairs in the lumbar
- 5 pairs in the sacral
- 1 pairs in the coccygeal.

The spinal cord (Fig. 11.7) contains both white and grey matter. The white matter travels in tracts to and from the brain and it contains myelinated axon. The grey matter forms a sort of H, with 3 horns on either side of the spinal cord. The ventral horn contains somatic motor nuclei, the lateral horn contains autonomic motor nuclei. The dorsal horn is divided, with the farthest back being the somatic sensory portion and the other portion being the visceral sensory portion. Grey matter contains: neurons, glial cells, and non-myelinated axon.

Practical : Study available model and prepared Slide to identify all parts of Human Spinal cord.

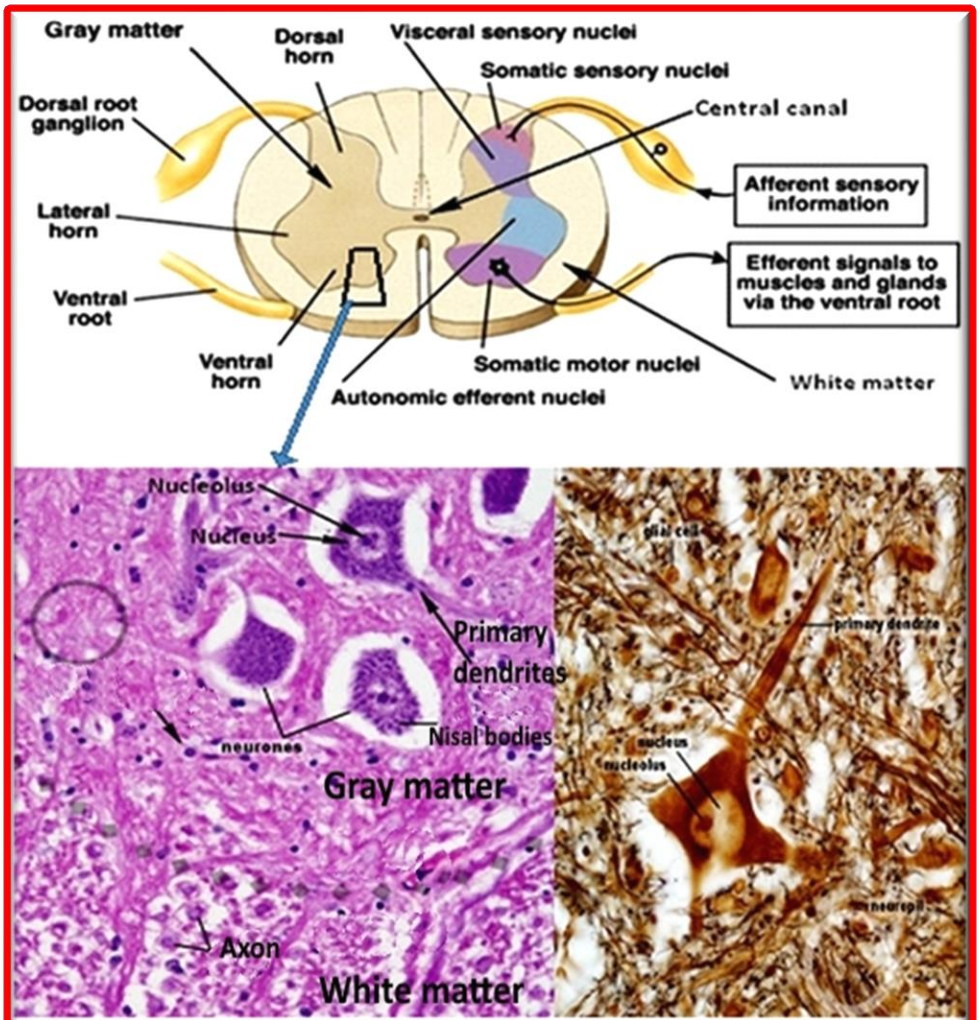


Fig. 11.7 :Cross section of the Human Spinal cord, also, shows the structure of Grey and white matter)

Human Trachea:

The trachea, commonly known as the windpipe, is a tube about 4 inches long and less than an inch in diameter in most people. The trachea begins just under the larynx (voice box) and runs down behind the breastbone (sternum). The trachea then divides into two smaller tubes called bronchi: one bronchus for each lung.

The trachea is composed of about 20 rings of tough cartilage. The back part of each ring is made of muscle and connective tissue. Moist, smooth tissue called mucosa lines the inside of the trachea. The trachea widens and lengthens slightly with each breath in, returning to its resting size with each breath out.

Viewed in cross section, the trachea is about one inch (2.6 cm) in diameter. It has a thin, membranous wall with C-shaped rings of cartilage embedded into it. Between sixteen and twenty cartilage rings are stacked along the length of the trachea, with narrow membranous regions spaced between the cartilage rings. The open ends of the cartilage rings face the posterior of the trachea near the esophagus.

Four layers of tissues make up the walls of the trachea (Fig. 11.8 A & B)**:**

- **The mucosa** is the innermost layer (Fig. 11.8B & fig.11.9) and consists of ciliated pseudostratified columnar epithelium with many goblet cells. Goblet cells produce sticky mucus to coat the inner lining of the trachea and catch any debris present in inhaled air before it reaches the lungs. On the surface of the columnar cells, long, hair-like cilia beat together to push mucous away from the lungs like a microscopic conveyor belt. Mucus from the trachea, along with any trapped contaminants, makes its way to the larynx, where it is either expelled during coughing or swallowed and digested in the stomach.

- Deep to the mucosa is **the submucosa layer** (Fig. 11.8B & fig.11.9), which is made of areolar connective tissue containing blood vessels and nervous tissue. Many collagen, elastin and reticular protein fibers give soft support and elasticity to the wall of the trachea, while blood vessels and nerves support the other layers of the tracheal wall. Longitudinal smooth muscle fibers are present in the posterior trachea between the ends of the cartilage rings. This smooth muscle tissue allows the trachea to adjust its diameter as needed.

- Surrounding the submucosa is a layer of C-shaped cartilages (**hyaline cartilage**) (Fig. 11.8B) that forms the supportive rings of the trachea. Hyaline provides a strong, yet flexible structure that maintains an open airway and is resistant to external stresses. It consist lacunae where the chondrocyte (cartilage cell) or chondrocyte set located.

- The outermost layer of the trachea is **the adventitia** (Fig.11.8B), a layer of areolar connective tissue that loosely anchors the trachea to the surrounding soft tissues.

Practical : Study a prepared Slide to identify all parts of Human Trachea.

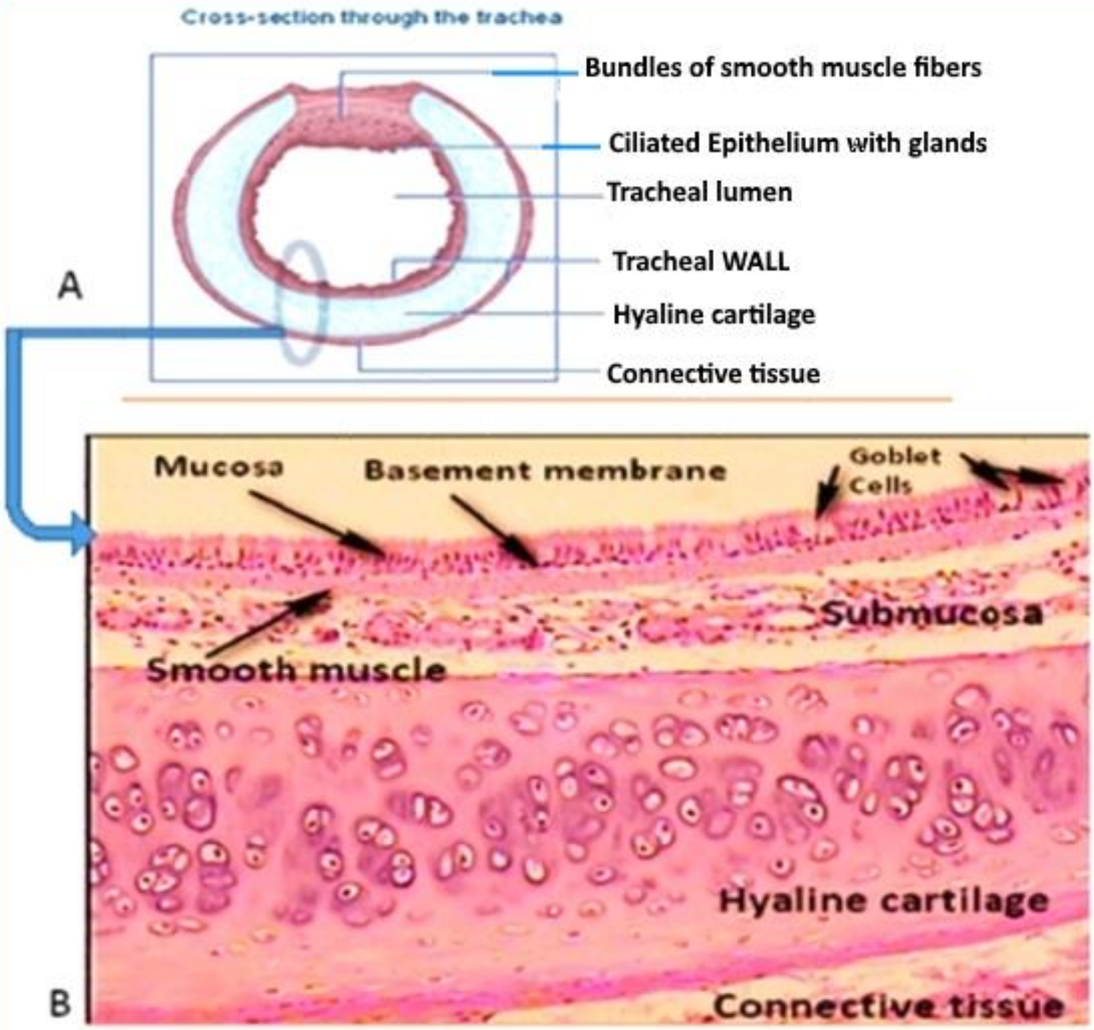


Fig. 11.8 : Cross section of Human Trachea, Shows different layer of Human Trachea.

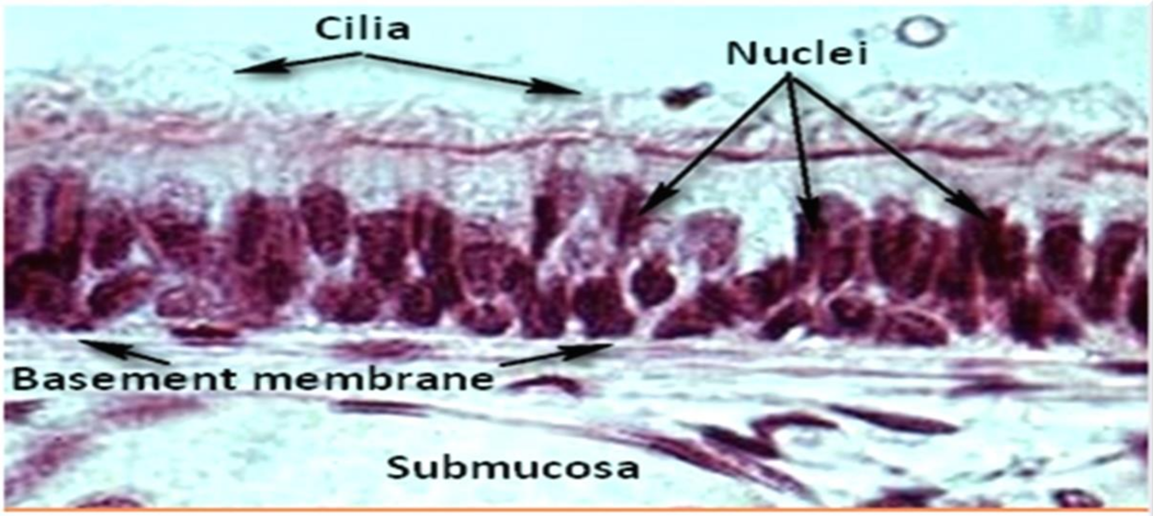


fig. 11.9: Shows the Mucosa layerof the trachea (Pseudostratified epithelium with multilayer nuclei, cilia, basement membrane), and Submucosa layer

Experiment 12

Human System

The human body consists of many interacting systems that carry out specific functions necessary for everyday living. Each system contributes to the maintenance of homeostasis, of itself, other systems, and the entire body. A system consists of two or more **organs,** which are functional collections of tissue. Systems do not work in isolation, and the well-being of the person depends upon the well-being of all the interacting body systems. Our bodies consist the following biological systems:

I) Cardiovascular System (Circulatory system):

The cardiovascular system consists of the heart, blood vessels, and the approximately 5 liters of blood that the blood vessels transport. Responsible for transporting oxygen, nutrients, hormones, and cellular waste products throughout the body, the cardiovascular system is powered by the body's hardest - working organ — the heart, which is only about the size of a closed fist. Even at rest, the average heart easily pumps over 5 liters of blood throughout the body every minute. Cardiovascular system includes two parts:

a) The Heart

The **heart** (fig. 12.1) is a muscular pumping organ located medial to the lungs along the body's midline in the thoracic region. The bottom tip of the heart, known as its apex, is turned to the left, so that about 2/3 of the heart is located on the body's left side with the other 1/3 on right. The top of the heart, known as the heart's base, connects to the great blood vessels of the body: the **aorta**, vena cava, pulmonary trunk, and pulmonary veins.

b) Blood Vessels

Blood vessels (fig.12.2& fig.12.2A) are the body's highways that allow blood to flow quickly and efficiently from the heart to every region of the body and back again. The size of blood vessels corresponds with the amount of blood that passes through the vessel. All blood vessels contain a hollow area called the lumen through which blood is able to flow. Around the lumen is the wall of the vessel, which may be thin in the case of capillaries or very thick in the case of arteries. Blood vessels are two primary circulatory loops in the human body: the pulmonary circulation loop and the systemic circulation loop:

1. **Pulmonary circulation loop:** transports deoxygenated blood from the right side of the heart to the **lungs**, where the blood picks up oxygen and returns to the left side of the heart. The pumping chambers of the heart that support the pulmonary circulation loop are the right atrium and right ventricle.

2. **Systemic circulation loop**: carries highly oxygenated blood from the left side of the heart to all of the tissues of the body (with the exception of the heart and lungs). Systemic circulation removes wastes from body tissues and returns deoxygenated blood to the right side of the heart. The left atrium and left ventricle of the heart are the pumping chambers for the systemic circulation loop.

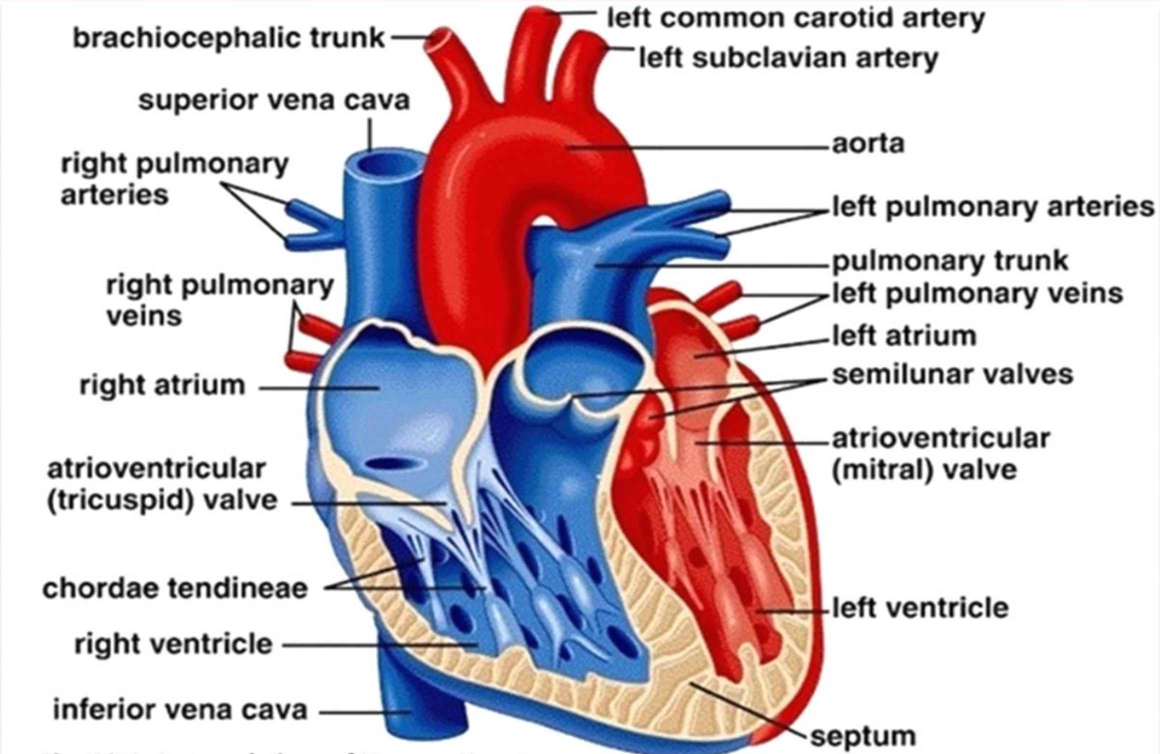


Fig. 12.1: Internal view of Human Heart

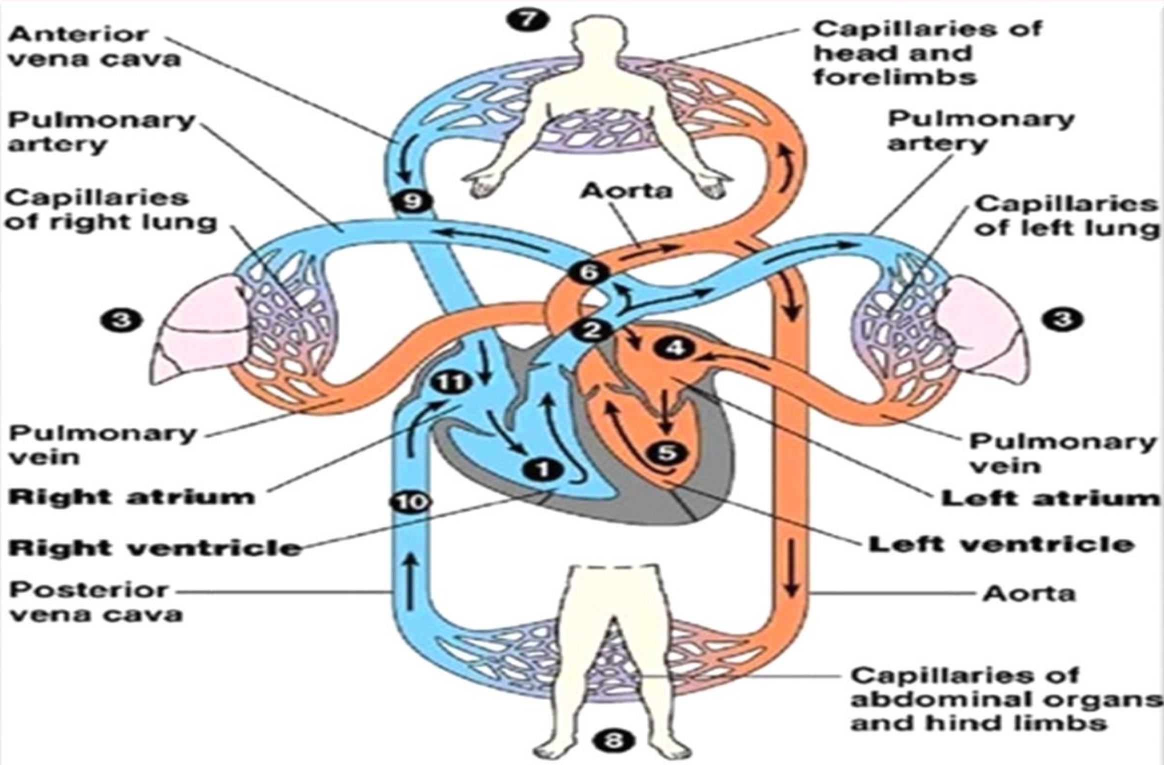


Fig. 12.2: Human Circulatory Loops

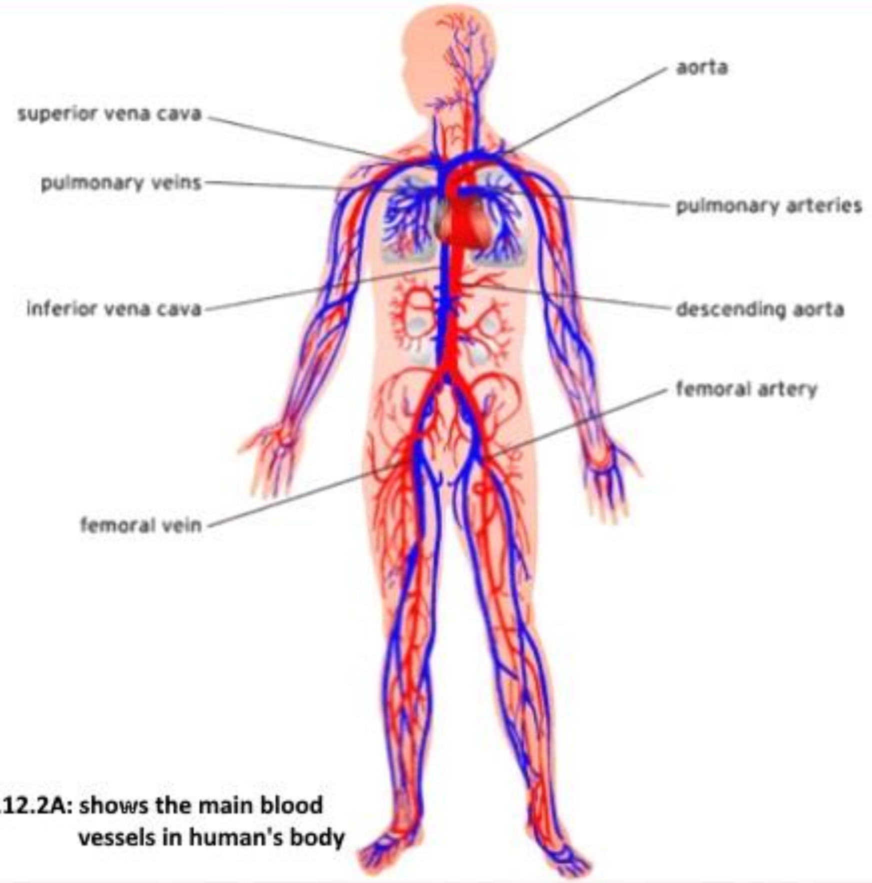


Fig.12.2A: shows the main blood vessels in human's body

II) Digestive System

The digestive system (Fig. 12.3 & Fig.12.4) is a group of organs working together to convert food into energy and basic nutrients to feed the entire body. Food passes through a long tube inside the body known as the alimentary canal or the gastrointestinal tract (GI tract). The alimentary canal is made up of the oral cavity, pharynx, esophagus, stomach, small intestines, and large intestines. In addition to the alimentary canal, several important accessory organs help your body to digest food but do not have food pass through them. Accessory organs of the digestive system include the teeth, tongue, salivary glands, liver, gallbladder, and pancreas. To achieve the goal of providing energy and nutrients to the body, six major functions take place in the digestive system: Ingestion, secretion, Mixing and movement, Digestion, Absorption, Excretion.

Mouth

Food begins its journey through the digestive system in the mouth, also known as the **oral cavity**. Inside the mouth are many accessory organs that aid in the digestion of food—the tongue, teeth, and salivary glands.

Pharynx

The pharynx, or throat, is a funnel-shaped tube connected to the posterior end of the mouth. The pharynx is responsible for the passing of masses of chewed food from the mouth to the esophagus.

Esophagus

The **esophagus** is a muscular tube connecting the pharynx to the stomach that is part of the **upper gastrointestinal tract**. It carries swallowed masses of chewed food along its length up into the stomach.

Stomach

The **stomach** is a muscular sac that is located on the left side of the abdominal cavity, just inferior to the **diaphragm**. This major organ acts as a storage tank for food. The stomach also contains hydrochloric acid and digestive enzymes that continue the digestion of food that began in the mouth.

Duodenum

The first or proximal portion of the small intestine, about 25 cm (10 inches) long, extending from the pylorus to the jejunum. It plays an important role in digestion of food because both the common bile duct and the pancreatic duct empty into it.

Liver and Gall bladder

The **liver** is a roughly triangular accessory organ of the digestive system located to the right of the stomach, just inferior to the diaphragm and superior to the small intestine. The main function of the liver in digestion is the production of bile and its secretion into the small intestine. The **gallbladder** is a small, pear-shaped organ located just posterior to the liver. The gallbladder is used to store and recycle excess bile from the small intestine so that it can be reused for the digestion.

Pancreas

The **pancreas** is a large gland located just inferior and posterior to the stomach. It is connected to the duodenum. The pancreas secretes digestive enzymes into the small intestine to complete the chemical digestion of foods.

Small Intestine

The **small intestine** is a long, thin tube about 1 inch in diameter and about 10 feet long that is part of the **lower gastrointestinal tract**. The entire small intestine is coiled like a hose and the inside surface is full of many ridges and folds. These folds are used to maximize the digestion of food and absorption of nutrients.

Large Intestine

The **large intestine** is a long, thick tube about 2 ½ inches in diameter and about 5 feet long. The large intestine absorbs water and contains many symbiotic bacteria that aid in the breaking down of wastes to extract some small amounts of nutrients. Feces in the large intestine exit the body through the anal canal. Large intestine divide into three parts: Ascending, Transverse, and Descending colon.

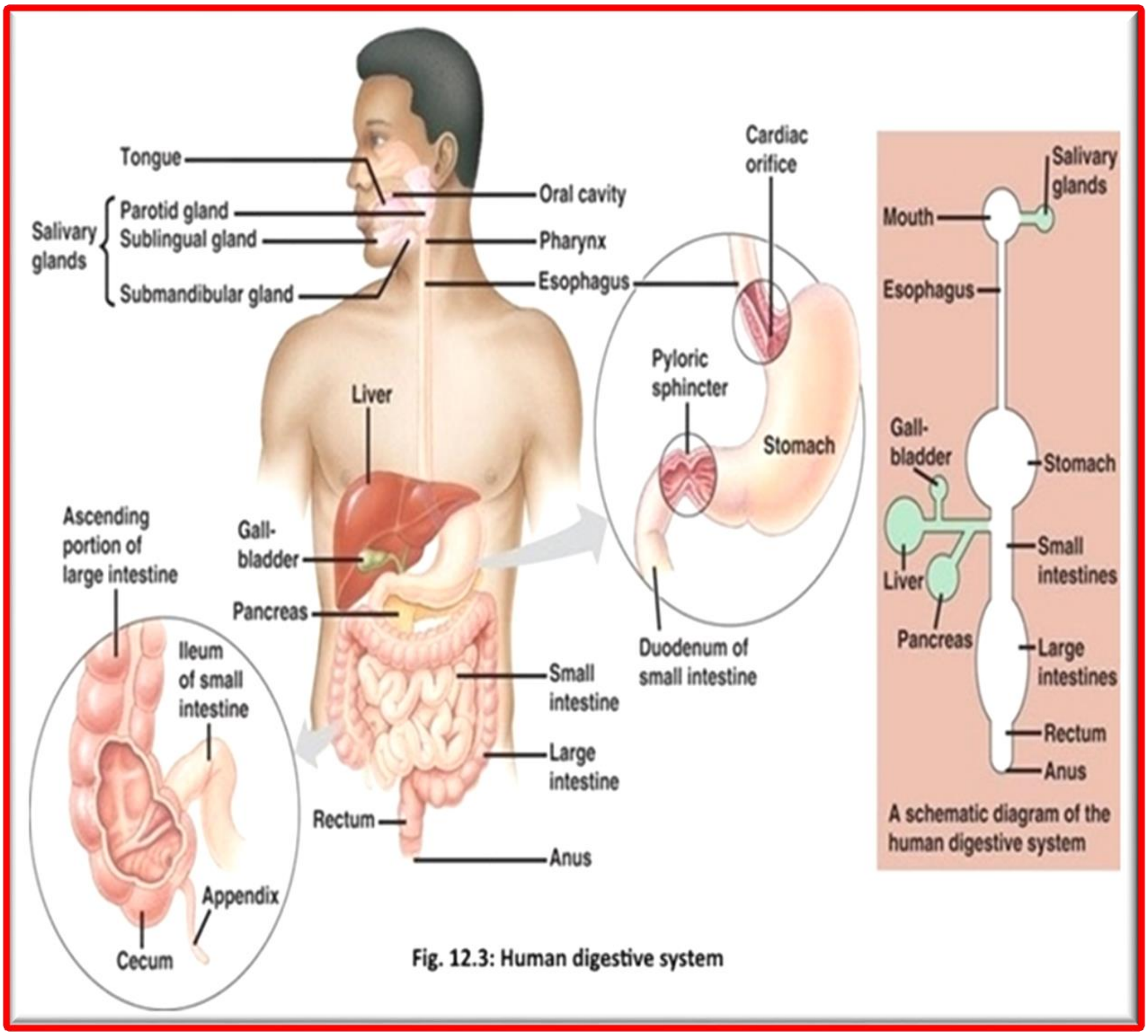


Fig. 12.3: Human digestive system

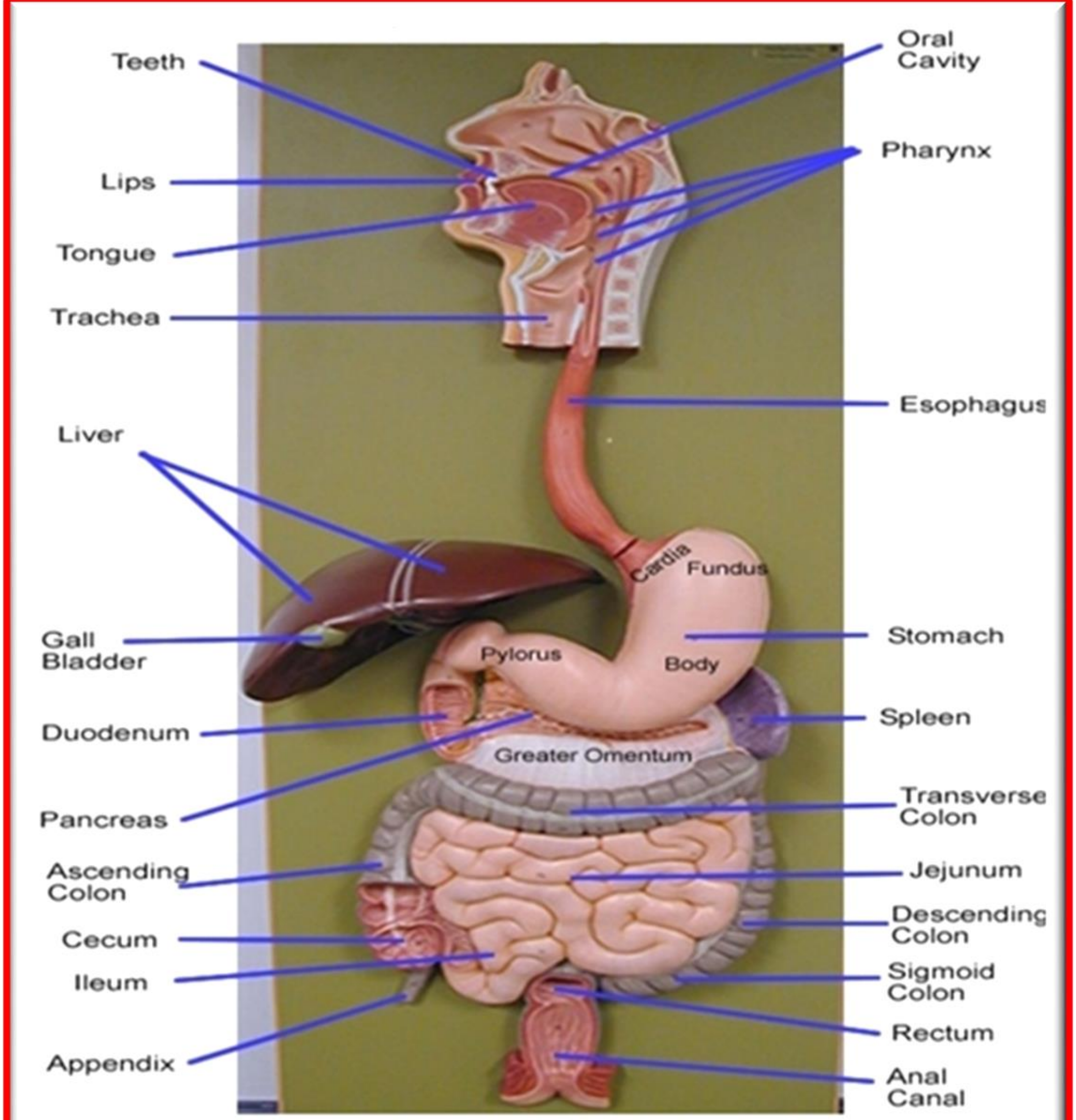


Fig. 12.4: Anatomy of Human digestive system

III) Reproductive System

The **reproductive system** or **genital system** is a collection of internal and external organs — in both males and females — that work together for the purpose of **sexual reproduction**. Many non-living substances such as fluids, **hormones**, and **pheromones** are also important accessories to the reproductive system. Unlike most **organ systems**, its anatomy differ between male and female. These differences allow for a combination of genetic material between two individuals, which allows for the possibility of greater **genetic fitness** of the **offspring.**

Female Reproductive System

The female reproductive system(fig.12.5) includes the ovaries, fallopian tubes, uterus, vagina, vulva, mammary glands and breasts. These organs are involved in the production and transportation of gametes and the production of sex hormones. The female reproductive system also facilitates the fertilization of ova by sperm and supports the development of offspring during pregnancy and infancy.

Ovaries

The **ovaries** are a pair of small glands about the size and shape of almonds, located on the left and right sides of the pelvic body cavity lateral to the superior portion of the uterus. Ovaries produce female sex hormones such as estrogen and progesterone as well as ova (commonly called "eggs.

Fallopian Tubes:

The **fallopian tubes** are a pair of muscular tubes that extend from the left and right superior corners of the uterus to the edge of the ovaries. The fallopian tubes end in a funnel-shaped structure called the infundibulum, which is covered with small finger-like projections called fimbriae. The **fimbriae** swipe over the outside of the ovaries to pick up released ova and carry them into the infundibulum for transport to the uterus.

Uterus

The **uterus** is a hollow, muscular, pear-shaped organ located posterior and superior to the urinary bladder. Connected to the two fallopian tubes on its superior end and to the vagina (via the **cervix**) on its inferior end, it surrounds and supports the developing fetus during pregnancy.

Vagina

The **vagina** is an elastic, muscular tube that connects the **cervix of the uterus** to the exterior of the body. It is located inferior to the uterus and posterior to the **urinary bladder**. The vagina functions as the receptacle for the **penis** during sexual intercourse and carries sperm to the uterus and fallopian tubes.

Vulva

The **vulva** is the collective name for the external female genitalia located in the pubic region of the body. The vulva surrounds the external ends of the urethral opening and the vagina and includes the mons pubis, labia majora, labia minora, and clitoris.

Breasts and Mammary Glands

The **breasts** are specialized organs of the female body that contain mammary glands, milk ducts, and adipose tissue. The two breasts are located on the left and right sides of the thoracic region of the body. In the center of each breast is a highly pigmented **nipple** that releases milk when stimulated. The areola, a thickened, highly pigmented band of skin that surrounds the nipple, protects the underlying tissues during breastfeeding. The **mammary glands** are a special type of sudoriferous glands that have been modified to produce milk to feed infants. Within each breast, 15 to 20 clusters of mammary glands become active during pregnancy and remain active until milk is no longer needed. The milk passes through milk ducts on its way to the nipple, where it exits the body.

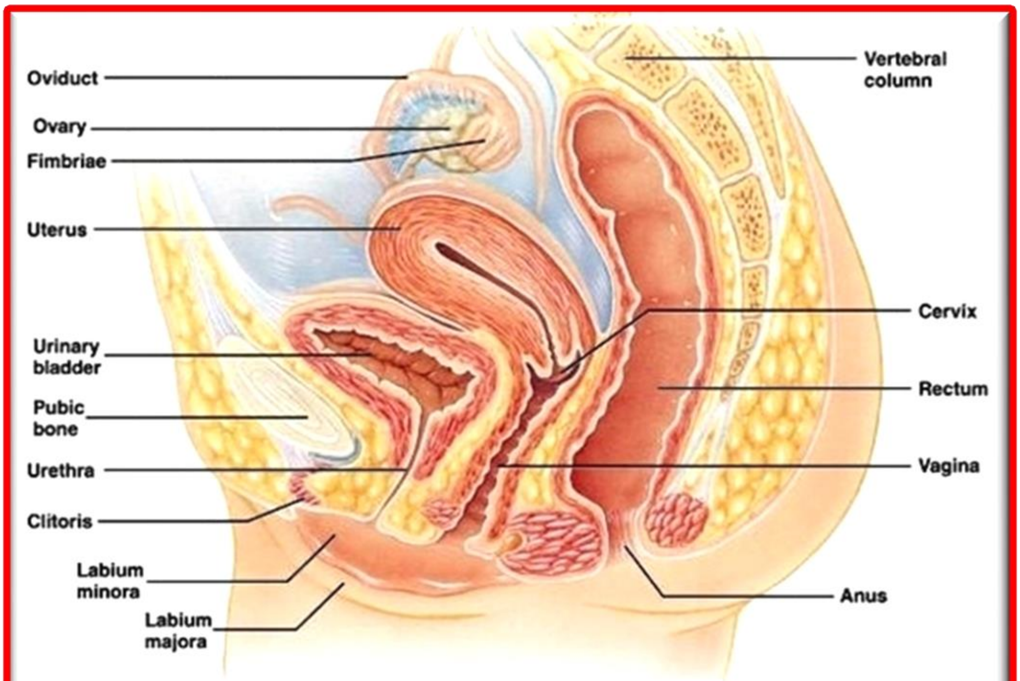


Fig. 12.5: Female Reproductive System of Human

Male Reproductive System

The male reproductive system (fig.12.6) includes the scrotum, testes, spermatic ducts, sex glands, and penis. These organs work together to produce sperm, the male gamete, and the other components of semen. These organs also work together to deliver semen out of the body and into the vagina where it can fertilize egg cells to produce offspring.

Scrotum

The scrotum is a sac-like organ made of skin and muscles that houses the testes. The smooth muscles that make up the scrotum allow it to regulate the distance between the testes and the rest of the body in order to regulate the testes temperature.

Testes

The 2 **testes**, also known as testicles, are the male gonads responsible for the production of sperm and testosterone.

Epididymis

The **epididymis** is a sperm storage area that wraps around the superior and posterior edge of the testes. Sperm produced in the testes moves into the epididymis to mature before being passed on through the **male reproductive organs**.

Spermatic Cords and Ductus Deferens

Within the scrotum, a pair of spermatic cords connects the testes to the abdominal cavity. The spermatic cords contain the ductus deferens along with nerves, veins, arteries, and lymphatic vessels that support the function of the testes.

The **ductus deferens**, also known as the vas deferens, is a muscular tube that carries sperm superiorly from the epididymis into the abdominal cavity to the ejaculatory duct. The smooth muscles of the walls of the ductus deferens are used to move sperm towards the ejaculatory duct through peristalsis.

Seminal Vesicles

The **seminal vesicles** are a pair of lumpy exocrine glands that store and produce some of the liquid portion of semen. The liquid produced by the seminal vesicles contains proteins and mucus and has an alkaline pH to help sperm survive in the acidic environment of the vagina. The liquid also contains fructose to feed sperm cells so that they survive long enough to fertilize the oocyte.

Ejaculatory Duct

The ductus deferens passes through the prostate and joins with the urethra at a structure known as the ejaculatory duct. The **ejaculatory duct** contains the ducts from the seminal vesicles as well. During ejaculation, the ejaculatory duct opens and expels sperm and the secretions from the seminal vesicles into the urethra.

Urethra

Semen passes from the ejaculatory duct to the exterior of the body via the urethra; the urethra passes through the prostate and ends at the **external urethral orifice** located at the tip of the penis. Urine exiting the body from the urinary bladder also passes through the urethra.

Prostate

The **prostate** is a walnut-sized exocrine gland that produces a large portion of the fluid that makes up semen. This fluid is milky white in color and contains enzymes, proteins, and other chemicals to support and protect sperm during ejaculation.

Cowper's Glands

The **Cowper's glands**, are a pair of pea-sized exocrine glands located inferior to the prostate and anterior to the anus. The Cowper's glands secrete a thin alkaline fluid into the urethra that lubricates the urethra and neutralizes acid from urine remaining in the urethra after urination.

Penis

The **penis** is the male external sexual organ located superior to the scrotum and inferior to the umbilicus. The penis is roughly cylindrical in shape and contains the urethra and the external opening of the urethra.

Semen

Semen is the fluid produced by males for sexual reproduction and it contains sperm, the male reproductive gametes, along with a number of chemicals suspended in a liquid medium. The chemical composition of semen gives it a thick, sticky consistency and a slightly alkaline pH.

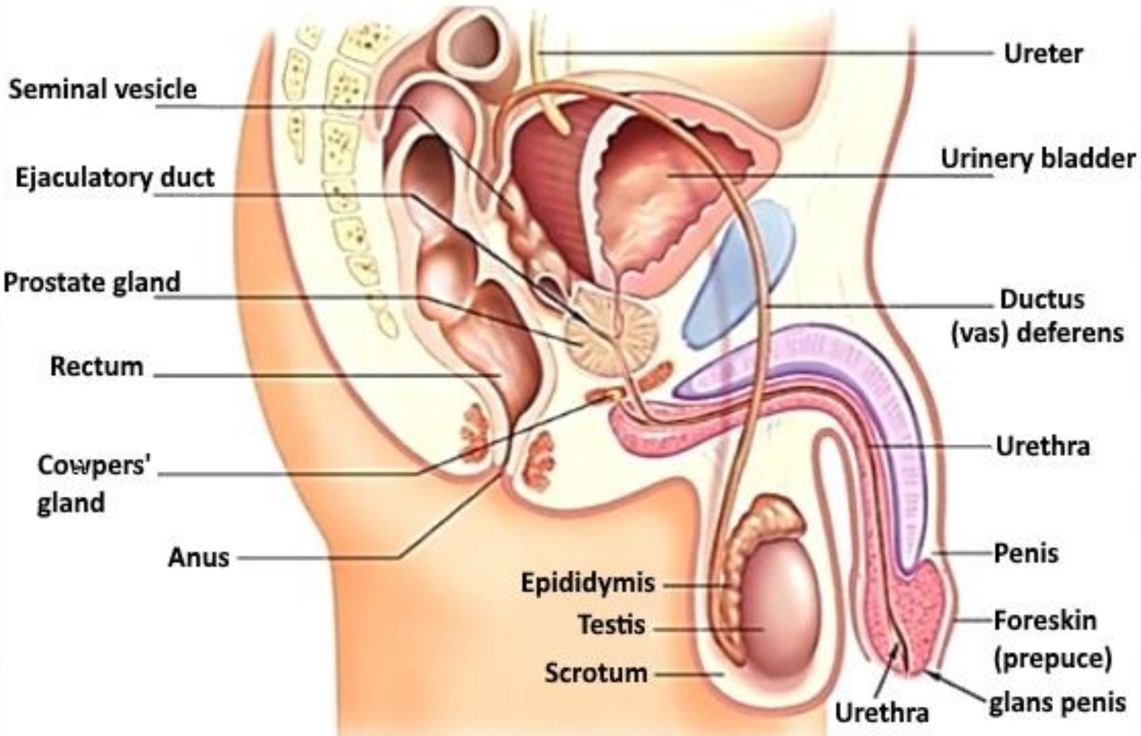


Fig. 12.6: Male Reproductive System of Human

IV) Muscular System

The muscular system (fig.12.7) is responsible for the movement of the human body. Attached to the bones of the skeletal system are about 700 named muscles that make up roughly half of a person's body weight. Each of these muscles is a discrete organ constructed of skeletal muscle tissue, blood vessels, tendons, and nerves. Muscle tissue is also found inside of the heart, digestive organs, and blood vessels. In these organs, muscles serve to move substances throughout the body....

Muscle Types

There are three types of muscle tissue: Visceral, cardiac, and skeletal.

1. **Visceral Muscle:** Visceral muscle is found inside of organs like the **stomach**, intestines, and blood vessels. Visceral muscle makes organs contract to move substances through the organ.
2. **Cardiac Muscle**: Found only in the **heart**, cardiac muscle is responsible for pumping blood throughout the body.
3. **Skeletal Muscle**: Skeletal muscle is the only voluntary muscle tissue in the human body—it is controlled consciously. The function of skeletal muscle is to contract to move parts of the body closer to the bone that the muscle is attached to.

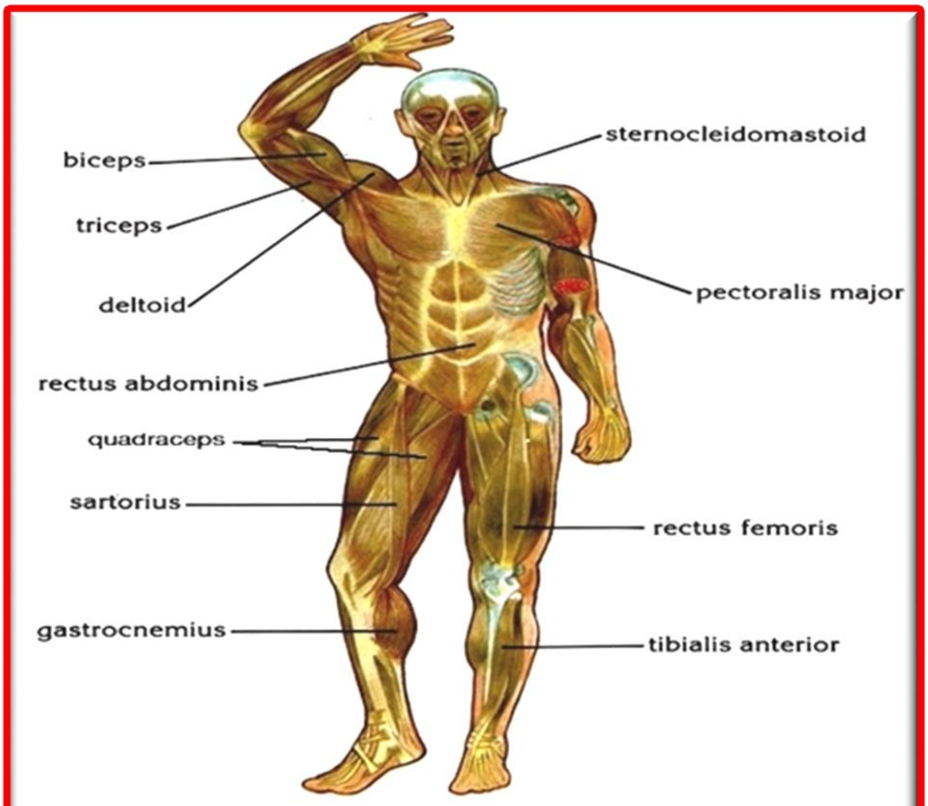


Fig. 12.7: Human Muscular System

V) Respiratory System

The cells of the human body require a constant stream of oxygen to stay alive. The respiratory system (fig. 12.8) (called also respiratory apparatus, ventilator system) is a biological system consisting of specific organs and structures used for the process of respiration in an organism, the intake and exchange of oxygen and carbon dioxide between an organism and the environment. provides oxygen to the body's cells while removing carbon dioxide, a waste product that can be lethal if allowed to accumulate. There are 3 major parts of the respiratory system: the airway, the lungs, and the muscles of respiration. The airway, which includes the nose, mouth, pharynx, larynx, trachea, bronchi, and bronchioles, carries air between the lungs and the body's exterior. The lungs act as the functional units of the respiratory system by passing oxygen into the body and carbon dioxide out of the body. Finally, the muscles of respiration, including the diaphragm and intercostal muscles, work together to act as a pump, pushing air into and out of the lungs during breathing.

Nose and Nasal Cavity

The **nose and nasal cavity** form the main external opening for the respiratory system. The nasal cavity is a hollow space within the nose and **skull** that is lined with hairs and mucus membrane. The function of the nasal cavity is to warm, moisturize, and filter air entering the body before it reaches the lungs by trapping dust, mold, pollen and other environmental contaminants.

Mouth

The mouth, also known as the **oral cavity**, is the secondary external opening for the respiratory tract. Most normal breathing takes place through the nasal cavity, but the oral cavity can be used to supplement or replace the nasal cavity's functions when needed. Because the pathway of air entering the body from the mouth is shorter than the pathway for air entering from the nose.

Pharynx

The pharynx, also known as the throat, is a muscular funnel that extends from the posterior end of the nasal cavity to the superior end of the **esophagus** and larynx. The **epiglottis** is a flap of elastic cartilage that acts as a switch between the trachea and the esophagus.

Larynx

The larynx is located in the anterior portion of the neck. The epiglottis is one of the cartilage pieces of the larynx and serves as the cover of the larynx during swallowing. Inferior to the epiglottis is the **thyroid cartilage**, **thyroid, and t**he vocal folds.

Trachea

The trachea, or windpipe, is a 5-inch long. The trachea connects the larynx to the bronchi and allows air to pass through the neck and into the thorax. The rings of cartilage making up the trachea allow it to remain open to air at all times.

The main function of the trachea is to provide a clear airway for air to enter and exit the lungs. In addition, the epithelium lining the trachea produces mucus that traps dust and other contaminants and prevents it from reaching the lungs.

Bronchi and Bronchioles

At the inferior end of the trachea, the airway splits into left and right branches known as the primary bronchi. The left and right bronchi run into each lung before branching off into smaller secondary bronchi. The secondary bronchi in turn split into many smaller tertiary bronchi within each lobe. The **tertiary bronchi** split into many smaller bronchioles that spread throughout the lungs. Each bronchiole further splits into many smaller branches less than a millimeter in diameter called terminal bronchioles. Finally, the millions of tiny terminal bronchioles conduct air to the alveoli of the lungs. The main function of the bronchi and bronchioles is to carry air from the trachea into the lungs.

Lungs

The **lungs** (fig. R1) are a pair of large, spongy organs found in the thorax lateral to the **heart** and superior to the diaphragm. The left lung is therefore slightly smaller than the right lung and is made up of 2 lobes while the right lung has 3 lobes.
The interior of the lungs is made up of spongy tissues containing many capillaries and around 30 million tiny sacs known as **alveoli**. The alveoli are lined with thin simple squamous epithelium that allows air entering the alveoli to exchange its gases with the blood passing through the capillaries.

Muscles of Respiration (fig. 12.8)

Surrounding the lungs are sets of muscles that are able to cause air to be inhaled or exhaled from the lungs. The principal muscle of respiration in the human body is the diaphragm, a thin sheet of skeletal muscle that forms the floor of the thorax. Between the ribs are many small **intercostal muscles** that assist the diaphragm with expanding and compressing the lungs.

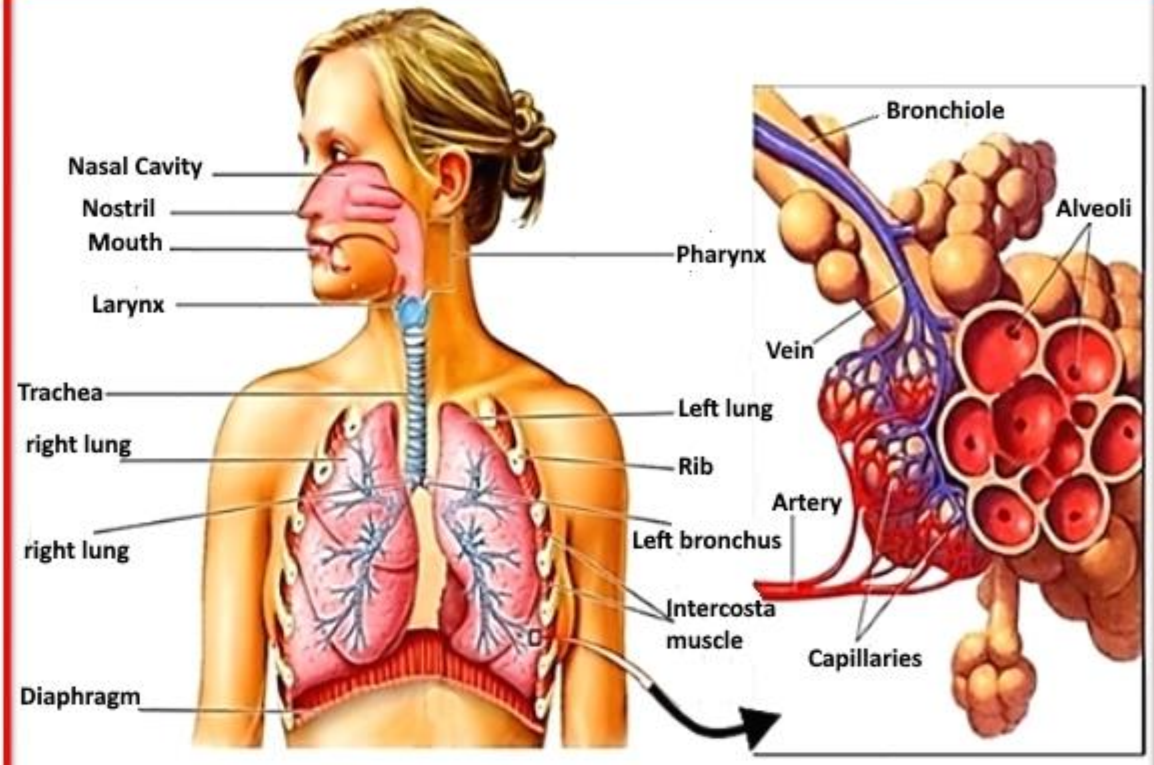


Fig. 12.8: Human Respiratory System

VI) Nervous System

The nervous system consists of the brain, spinal cord, sensory organs, and all of the nerves that connect these organs with the rest of the body. Together, these organs are responsible for the control of the body and communication among its parts. The brain and spinal cord form the control center known as the central nervous system (CNS), where information is evaluated and decisions made. The sensory nerves and sense organs of the peripheral nervous system (PNS) monitor conditions inside and outside of the body and send this information to the CNS. Efferent nerves in the PNS carry signals from the control center to the muscles, glands, and organs to regulate their functions. The Two main parts of Nervous system are: Central Nervous system (CNS) and peripheral nervous system (PNS).

Central Nervous system (CNS)

The brain (fig.12.9) and spinal cord (fig.12.10) together form the central nervous system (CNS), where information is processed and responses originate. The brain, a soft, wrinkled organ that weighs about 3 pounds, is located inside the cranial cavity, and protect by the skull, Cerebrospinal fluid, and meninges membrane. The brain, the seat of higher mental functions such as consciousness, memory, planning, and voluntary actions, also controls lower body functions such as the maintenance of respiration, heart rate, blood pressure, and digestion.

The spinal cord is a long, thin mass of bundled neurons that carries information through the vertebral cavity of the spine beginning at the medulla oblongata of the brain on its superior end and continuing inferiorly to the lumbar region of the spine. The white matter of the spinal cord functions as the main conduit of nerve signals to the body from the brain. The grey matter of the spinal cord integrates reflexes to stimuli.

The peripheral nervous system (fig.12.10) are bundles of axons called nerves that act as information highways to carry signals between the brain and spinal cord and the rest of the body. There are different nerves types:

- **Afferent Nerves**: carry information from sensory receptors to the central nervous system
- **Efferent Nerves**: carry signals only from the central nervous system to effectors such as muscles and glands.
- **Mixed Nerves**: afferent axons act as lanes heading toward the central nervous system and efferent axons act as lanes heading away from the central nervous system.
- **Cranial Nerves**: Extending from the inferior side of the brain are 12 pairs of cranial nerves.
- **Spinal Nerves**: Extending from the left and right sides of the spinal cord, they are 31 pairs of spinal nerves.

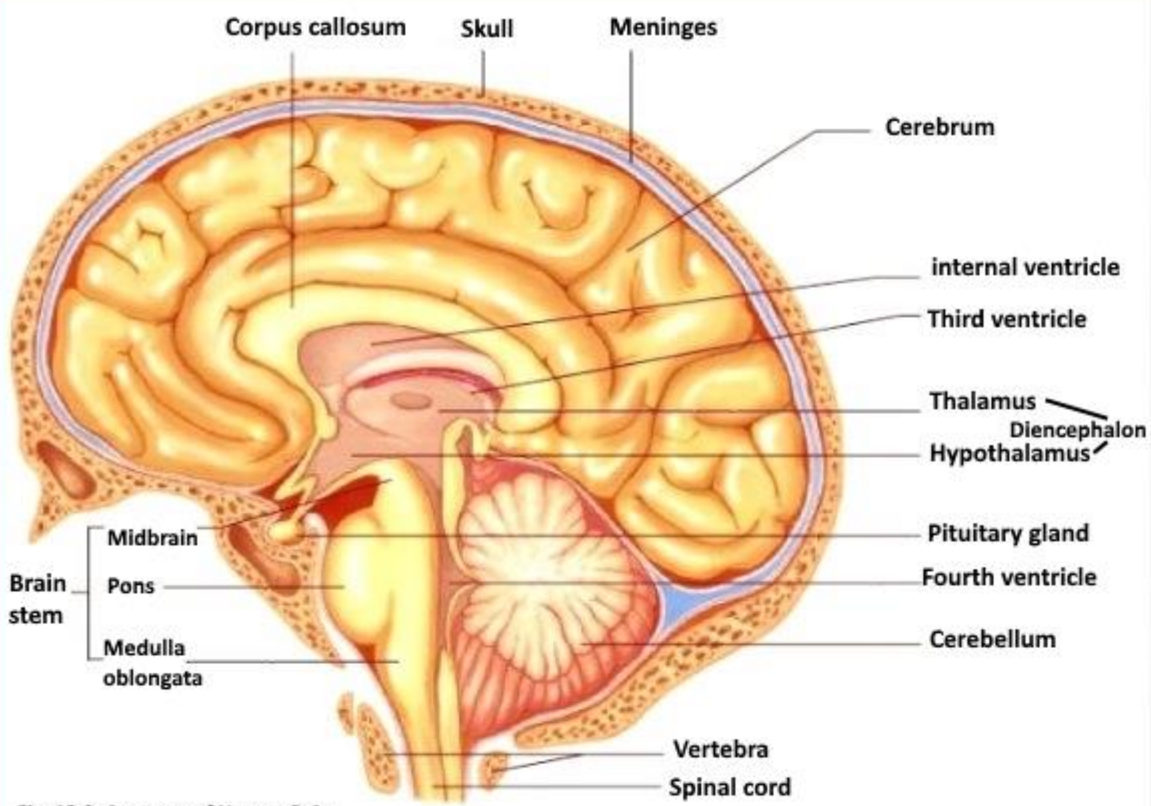


Fig. 12.9: Anatomy of Human Bain

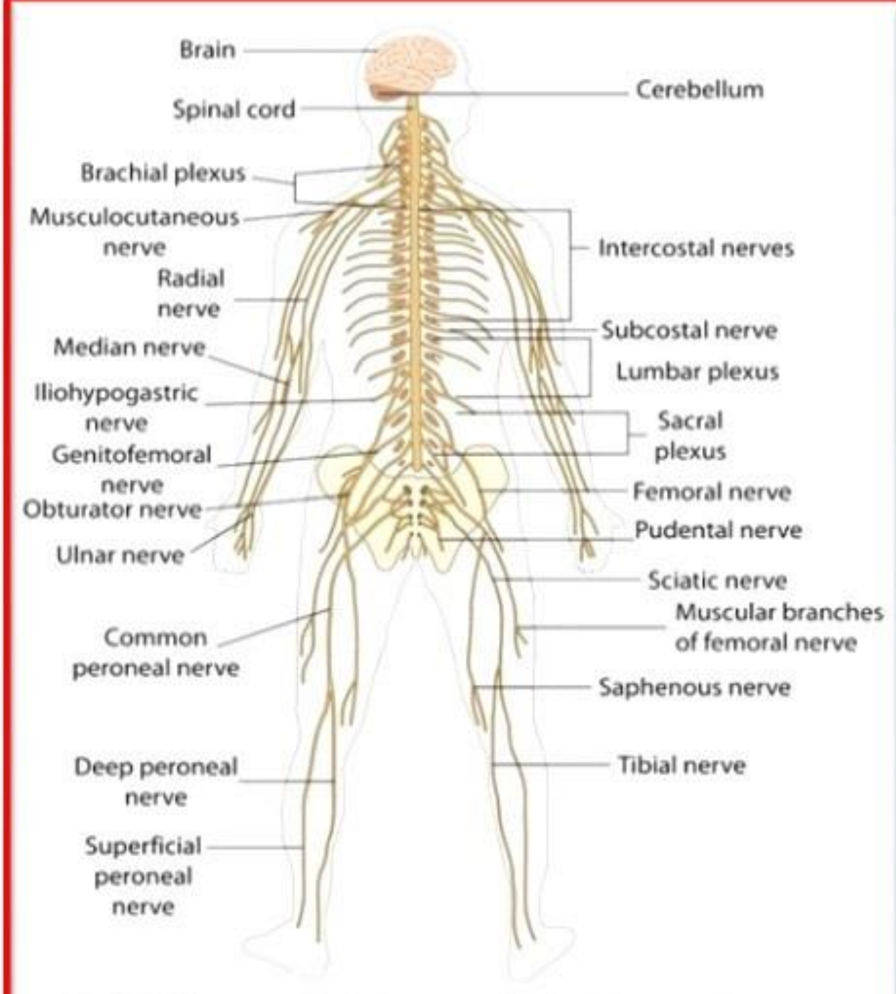


Fig. 12.10: Shows: A: Brachial plexus, Femoral and Tibial nerves, and sciatic nerve

VII) Skeletal System

The skeletal system includes all of the bones and joints in the body. Each bone is a complex living organ that is made up of many cells, protein fibers, and minerals. The skeleton acts as a scaffold by providing support and protection for the soft tissues that make up the rest of the body. The skeletal system also provides attachment points for muscles to allow movements at the joints. New blood cells are produced by the red bone marrow inside of our bones. Bones act as the body's warehouse for calcium, iron, and energy in the form of fat. Finally, the skeleton grows throughout childhood and provides a framework for the rest of the body to grow along with it.

Skeletal System Anatomy:

The skeletal system in an adult body(fig.12.11A&B) is made up of 206 individual bones. These bones are arranged into two major divisions: **the axial skeleton** and **the appendicular skeleton**. The axial skeleton runs along the body's midline axis and is made up of 80 bones in the following regions:

• **Skull** (fig.12.11B)
• **Hyoid**
• **Auditory ossicles**
• **Ribs**
• **Sternum**
• **Vertebral column**

The appendicular skeleton (fig.12.11A)**:** made up of 126 bones in the following regions:
• **Upper limbs**
• **Lower limbs**
• **Pelvic girdle**
• **Pectoral (shoulder) girdle**

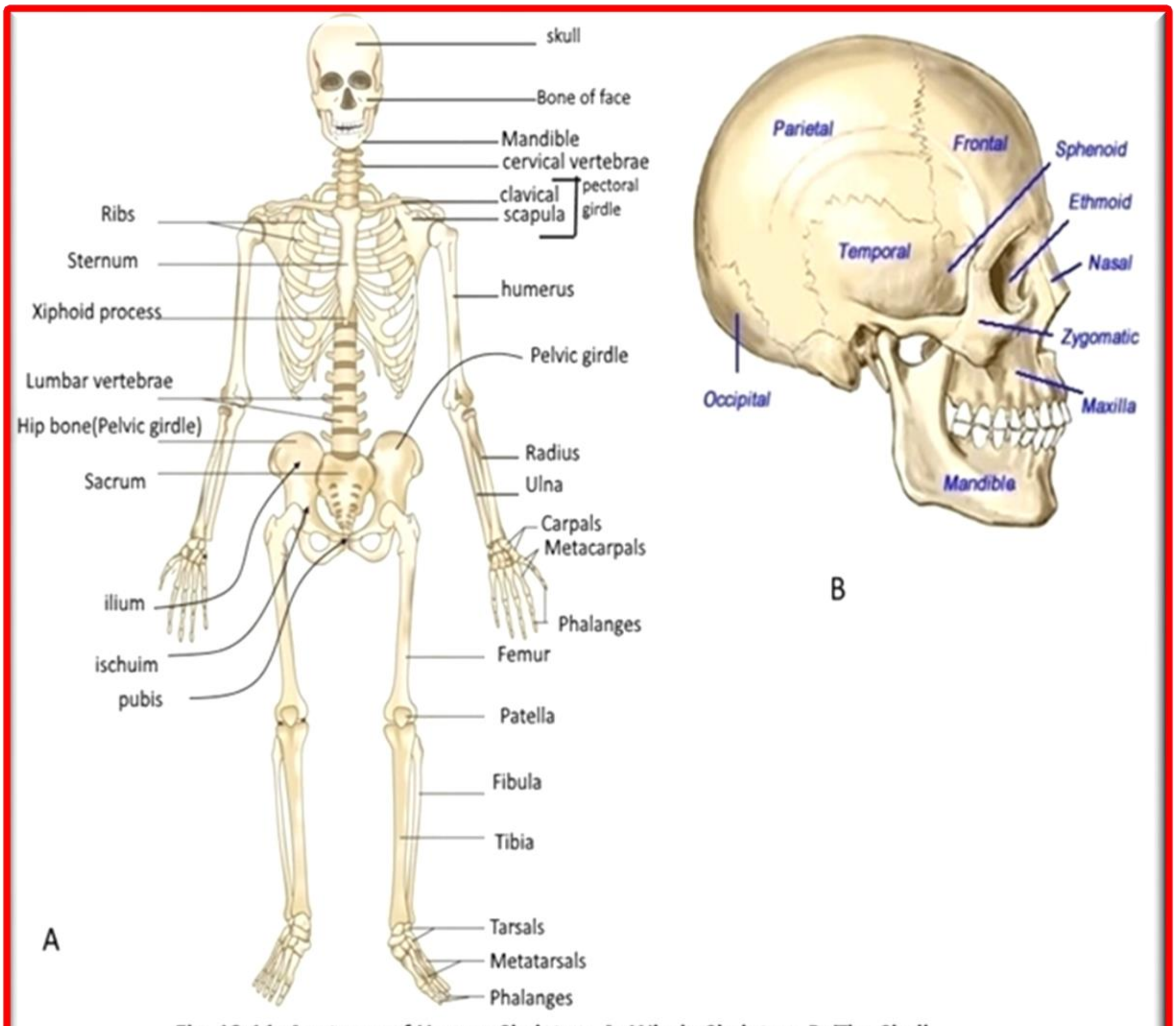


Fig. 12.11: Anatomy of Human Skeleton, A: Whole Skeleton, B: The Skull

VIII) Urinary System

The urinary system (fig.12.12A&B) consists of the kidneys, ureters, urinary bladder, and urethra. The kidneys filter the blood to remove wastes and produce urine. The ureters, urinary bladder, and urethra together form the urinary tract, which acts as a plumbing system to drain urine from the kidneys, store it, and then release it during urination. Besides filtering and eliminating wastes from the body, the urinary system also maintains the homeostasis of water, ions, pH, blood pressure, calcium and red blood cells.

Urinary System Anatomy (fig.12.12A&B):

Kidneys

The kidneys are a pair of bean-shaped organs found along the posterior wall of the abdominal cavity. The left kidney is located slightly higher than the right kidney because the right side of the liver is much larger than the left side. The kidneys filter metabolic wastes, excess ions, and chemicals from the blood to form urine.

Ureters

The ureters are a pair of tubes that carry urine from the kidneys to the urinary bladder. These valves prevent urine from flowing back towards the kidneys.

Urinary Bladder

The urinary bladder is a sac-like hollow organ used for the storage of urine. The walls of the bladder allow it to stretch to hold anywhere from 600 to 800 milliliters of urine.

Urethra

The urethra is the tube through which urine passes from the bladder to the exterior of the body. The female urethra ends inferior to the clitoris and superior to the vaginal opening. In males, the urethra ends at the tip of the penis. The urethra is also an organ of the male reproductive system as it carries sperm out of the body through the penis.

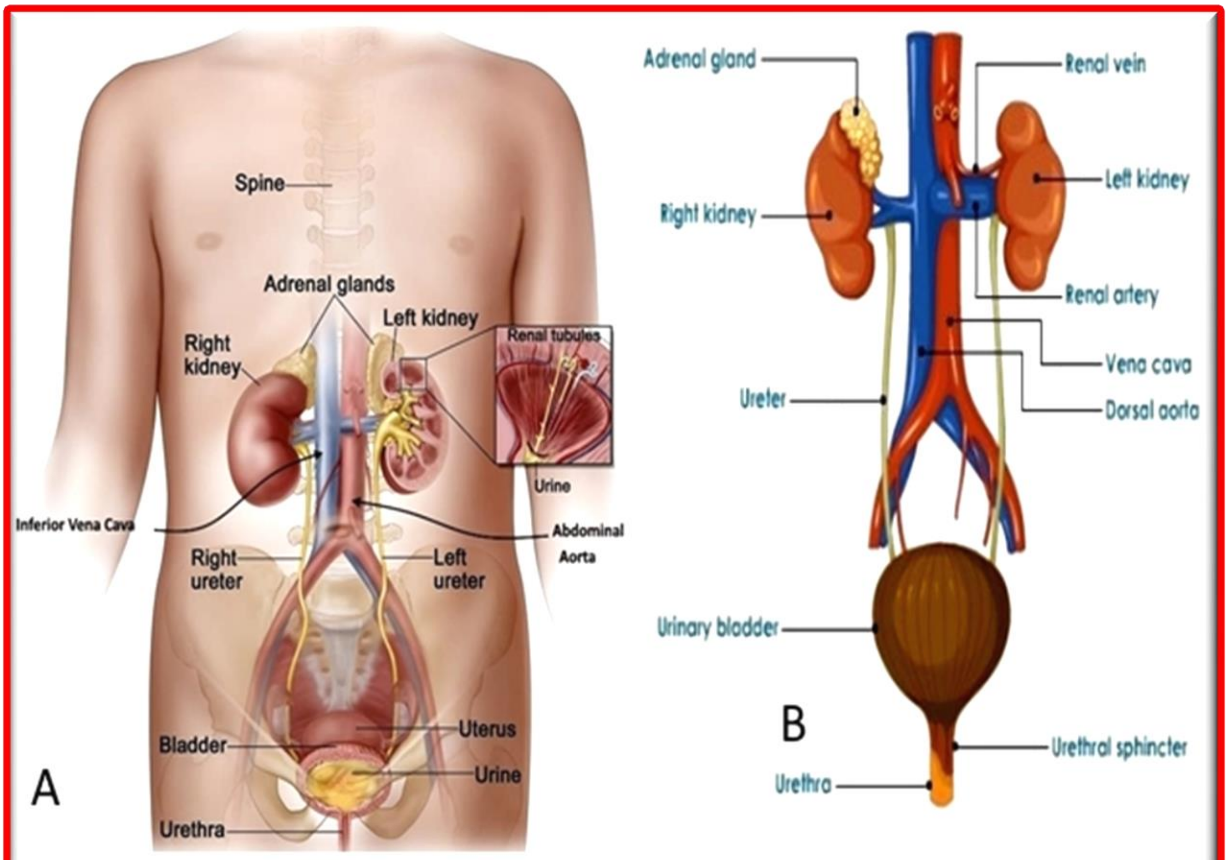


Fig. 12.12: Human Urinary System A: For Female B: For Man

IX) Sense Organs

A **sense** is "A system that consists of a group of sensory cell types that responds to a specific physical phenomenon, and that corresponds to a particular group of regions within the brain where the signals are received and interpreted."

In addition to sight, smell, taste, touch, and hearing, humans also have awareness of pressure, temperature (thermoception), pain (nociception), vibration (mechanoreception), and various internal stimuli (e.g. the different **chemoreceptors** for detecting salt and carbon dioxide concentrations in the blood) balance (equilibrioception). The sense of balance is maintained by a complex interaction of visual inputs, the proprioceptive sensors (which are affected by gravity and stretch sensors found in muscles, skin, and joints), the inner ear vestibular system, and the central nervous system. Disturbances occurring in any part of the balance system, or even within the brain's integration of inputs, can cause the feeling of dizziness or unsteadiness. In this lab we will study the Organ of sight (the eyes) and the organ of hearing (the ear).

The Eye:

The eye (fig. 12.13) is the organ of vision. It has three coats, enclosing three transparent structures. The outermost layer is composed of the **cornea** and **sclera.** The cornea is the transparent circular part of the front of the human eyeball. It has the important optical function of **refracting** light entering the eye through the **pupil** and onto a transparent lens that focuses light on the retina.

The middle layer consists of the **choroid, ciliary body,** and **iris**. **The choroid** is the layer of the eyeball located between the **retina** and the **sclera**. It is a thin, highly vascular (i.e. it contains **blood vessels**) membrane that is dark brown in colour and contains a pigment that absorbs excess light and so prevents blurred vision. The **ciliary body** alters the curvature of the **lens** of the eye (accommodation). **The iris** is the *coloured part of the human eye. It is a thin circular contractile 'curtain' located in the **aqueous humour,** which is in front of ('anterior to') the **lens** but behind ('posterior to') the **cornea**. It contains a circular aperture called the **pupil** through which light passes into the eye. The size of the **pupil**, and therefore the amount of light that is admitted into the eye, is regulated by the **pupillary reflex** - which is also known as the '**light reflex**'.

The innermost is the **retina.** The **retina** is a light-sensitive structure lining the interior of the eye. It contains photosensitive cells called **rods** and **cones.** The cone cells are sensitive to color and are located in the part of the retina called the fovea, where the light is focused by the lens. The rod cells are not sensitive to color, but have greater sensitivity to light than the cone cells. These cells are located around the fovea and are responsible for peripheral vision and night vision. The eye is connected to the brain through the **optic nerve**. The point of this connection is called the "blind spot" because it is insensitive to light. **The optic nerve** is the route by which information is sent from the eye for processing by the **brain**. The optic nerves progress from the posterior of the eyeball, into the skull, through the **optic chiasma,** then on to the cortex of the occipital lobe on each side of the brain.

Within the eye's coats, there are the **aqueous humour**, the **vitreous body**. The **aqueous humour** is a watery fluid that fills the chamber called the "anterior chamber of the eye" which is located immediately behind the **cornea** and in front of the **lens**, and also the "posterior chamber of the eye" which is a very narrow compartment located between the peripheral part of the **iris**, the suspensory ligament of the lens, and the **ciliary processes**. The **vitreous humour** (which is also known as the **vitreous body**) is located in the large area that occupies approx. 80% of each eye in the human body. The **vitreous humour** is a transparent thin-jelly-like substance that fills the chamber behind ('posterior to') the **lens** of the eye. It is an albuminous fluid enclosed in a delicate transparent membrane called the **hyaloid membrane**.

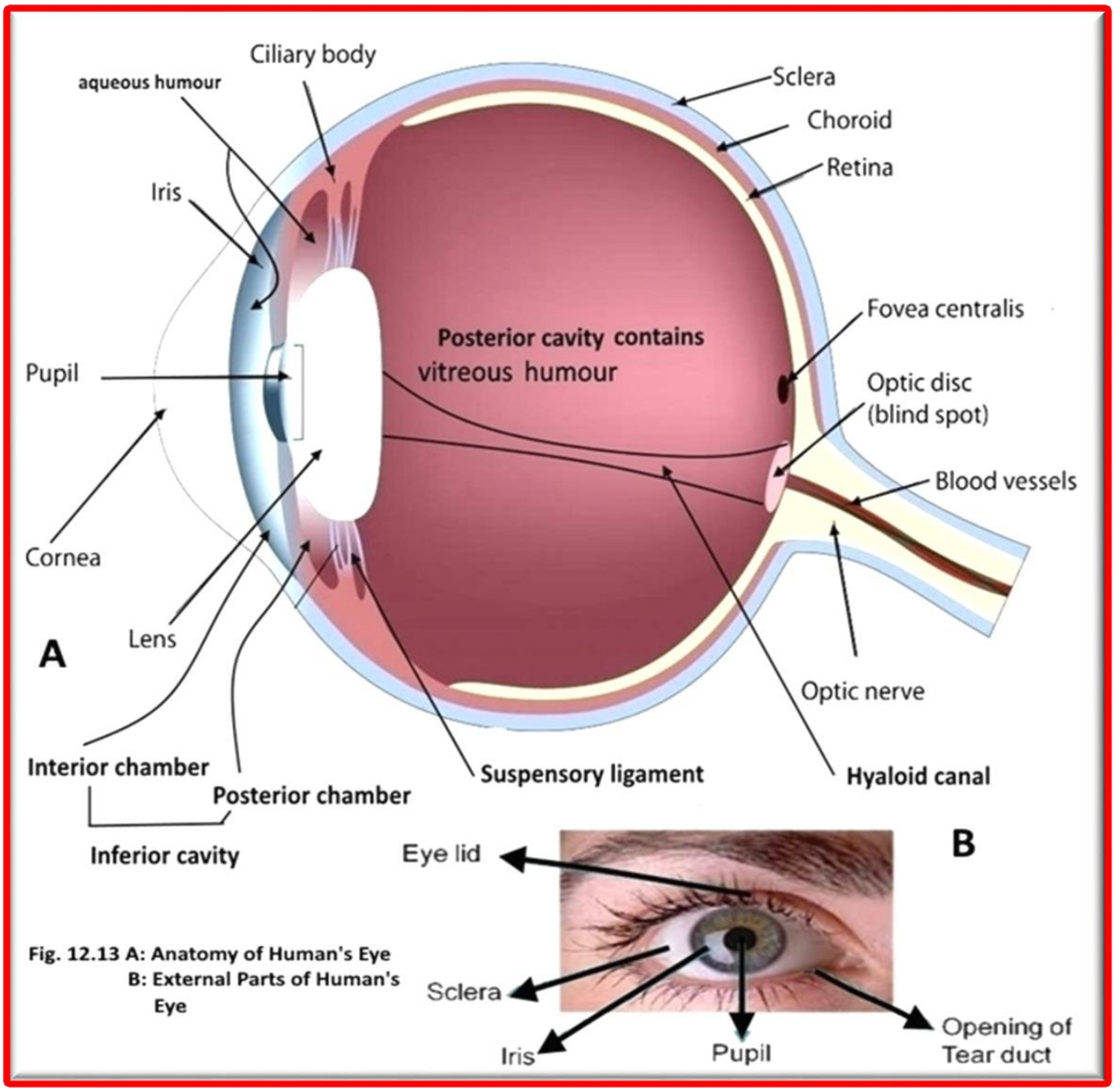


Fig. 12.13 A: Anatomy of Human's Eye
B: External Parts of Human's Eye

The Ear:

The ear is part of the **auditory system**. It is the **organ** that detects **sound**. It not only receives sound, but also aids in **balance** and body position. The ear has external, middle, and inner portions.

Outer ear

The outer ear (fig.12.14A) is the most external portion of the ear. The outer ear includes the fleshy visible outer ear, called the **pinna** or auricle, it shaped like a cup to direct sounds toward the tympanic membrane through the ear canal into the tympanic membrane, also known as the ear drum. The skin surrounding the auditory canal contains glands that produce **ear wax (cerumen)** and has hair to filter the air.

Middle ear

The middle ear (fig.12.14A & fig.12.15) is an air-filled cavity behind the tympanic membrane, includes three bones (**ossicles**): the **malleus (or hammer), incus (or anvil), and stapes (or stirrup)**. The middle ear also connects to the upper throat via the **Eustachian tube.** The three ossicles transmit sound from the tympanic membrane to the **secondary tympanic membrane** which is situated within the **oval window** and is one of the two windows between the middle ear and the inner ear. When the stapes footplate pushes on the oval window, it causes movement of fluid within the cochlea (a portion of the inner ear). The ossicles help in amplification of sound waves by nearly thirty times. The **round window** is the second of the two windows between the middle ear and the inner ear. As the stapes pushes the tympanic membrane in the inner ear fluid moves and pushes the **round window membrane** which allows for the fluid within the inner ear to move.

Inner ear

The inner ear (fig.12.14B) is split anatomically into bony and membranous labyrinths. This contains the sensory organs for balance and motion, namely the vestibules of the ear (**utricle** and **saccule**), and the **semicircular canals**. The cochlea consists of three fluid-filled spaces: the **scala tympani,** the **scala vestibuli** and the **scala media**, which together are responsible for hearing.

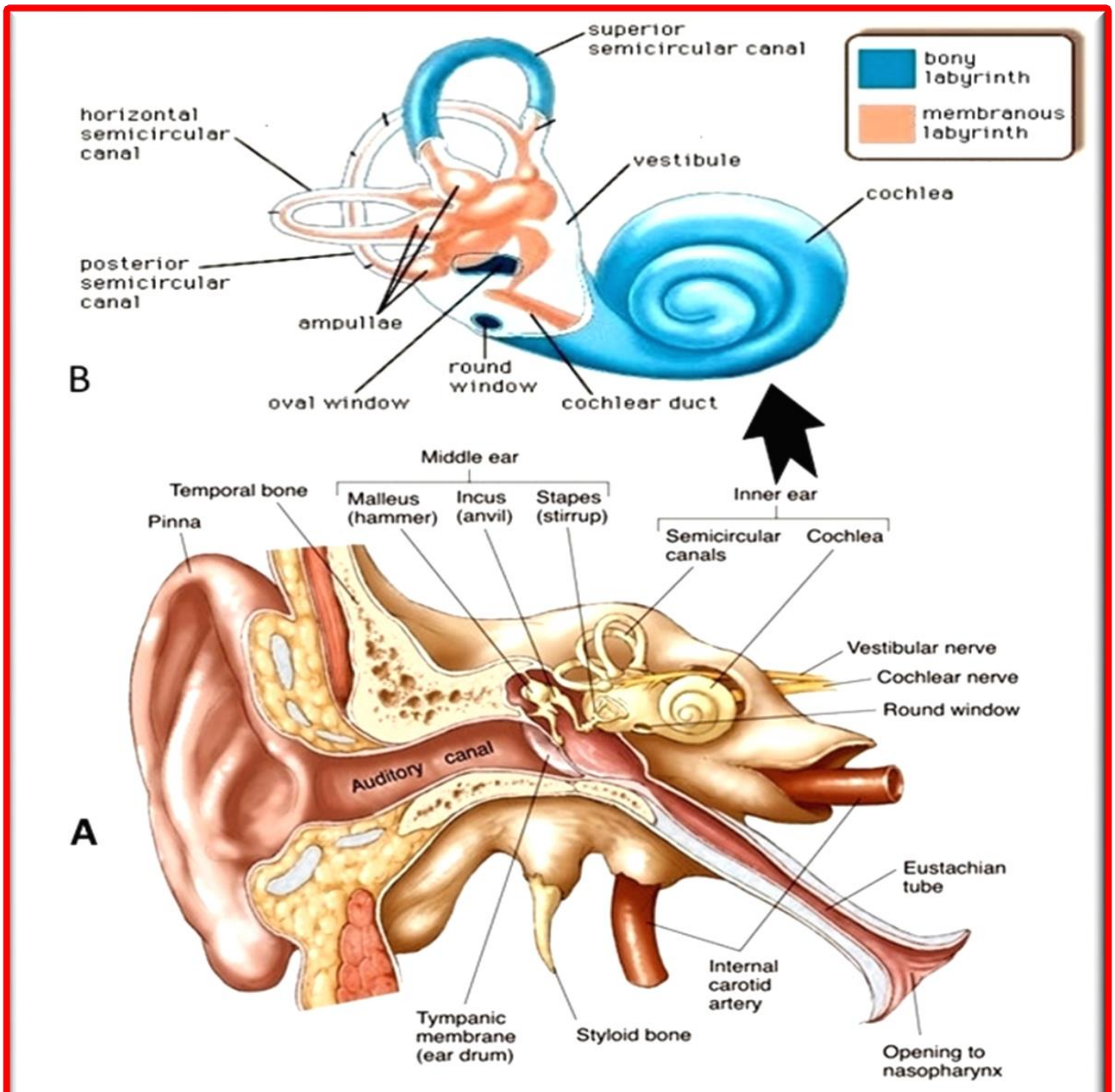


Fig. 12.14: A) Anatomy of Human's Ear; B) Annatomy of Human's inner Ear

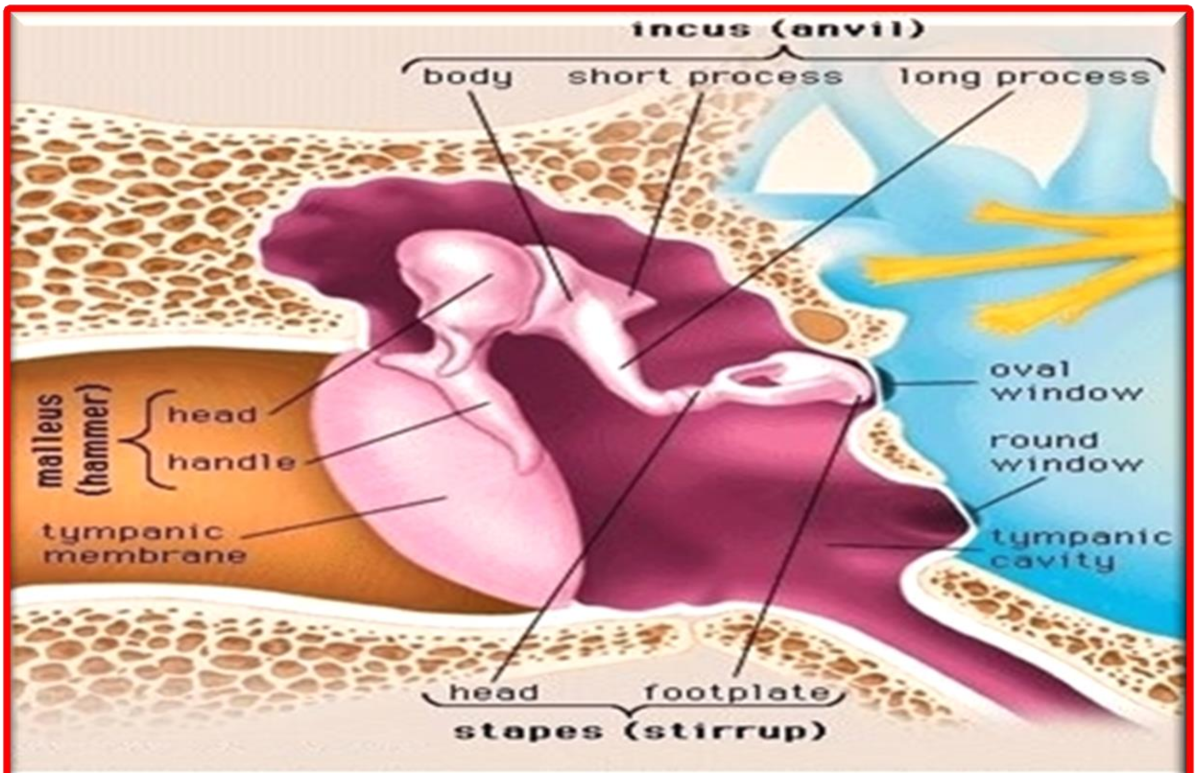


Fig. 12.15: Anatomy of Human's Middle Ear

Practical: Study available model of the following and identify the main parts of each:

- Human Heart
- Human digestive system
- Human reproductive system (male and female)
- Human muscular system
- Human respiratory system
- Human brain
- Human skeletal system with the skull
- Human Urinary system
- Human sense organs (eye and ear)

LABORATORY REPORT

Lab Safety, Scientific Writing, and Microscopy

NOTE: Draw as you see under the microscope

Name:________________________________ **Student No.:**______________________ **Section No.:** __________

****

Part D: Viewing a prepared slide (Amoeba):

4 X **10 X** **40 X** **100 X**

Calculate how much Amoeba magnify under the following objective lenses?

4X	10X	40X	100X	Objective lenses
				Total magnification

Part E: Preparing of a Wet Mount (Letter e):

4X **10X** **40 X**

** Conclusion about the following in compound microscope:	
Resolution	______________________
Microscopic field	______________________
Characteristic of the image	______________________
Working distance	______________________
Resolution	______________________

******* When the slide Move slowly to the right as you view the image in the field of view, the letters appear to move____, and the slide Move slowly away from you as you view the image in the field of view, the letters appear to move _______________.**

******* The image formed by Dissecting microscope is:**

1- ____

2- ____

******* The function of contractile vacuole in *Amoeba* is ____.**

******** Experiment: A researcher wants to find out if spraying fruits and vegetables with pesticides affects the vitamin levels in those foods.*** **For the experiment listed above, identify:**

a) the independent variable: ____

b) the dependent variable: ____

c) the control treatment: ____

e) the experimental treatment: ____

GOOD LUCK

LABORATORY REPORT
Cell Structure and Function

NOTE: Draw as you see under the Microscope

Name:________________________________ **Student No.:**______________________ **Section No.:** ___________

****

Bacteria slide	**Onion Epidermis**	**Geranium leaf**

Cheek Epithelium	**Paramecium**

1- Why was the cell stained?

____

____

2- List five differences between animal and plant cells:

Animal cells	Plant cells

2- List four differences between Prokaryotic and Eukaryotic cells:

Prokaryotic cells	Eukaryotic cells

GOOD LUCK

LABORATORY REPORT

Macromolecules and Living Things

Name:_________________________________ Student No.:_______________________ Section No.: ___________

****

Carbohydrates

Table 1: Benedict's Test for Reducing Sugars

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	5% Glucose			
2	5% Sucrose			
3	5% Starch			
4	Onion juice			
5	Potato juice			
6	Distilled water			

Table 2: Lugol's Test for Starch

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	5% Glucose			
2	5% Sucrose			
3	5% Starch			
4	Onion juice			
5	Potato juice			
6	Distilled water			

Lipids and Fats

Table 3: Solubility Test for fat

Tube	Solution	Solubility of oil	Remarks
1	Ether		
2	Acetone		
3	Ethanol		
4	Chloroform		
5	Distilled water		

Proteins

Table 4: Biuret Test for Peptide bond

Tube	Solution	Initial Color	Color of the mixture	Remarks
1	**Egg albumin solution**			
2	**2.5% Glucose**			
3	**1% Starch**			
4	**Distilled water**			

Vitamin C

Table 5: qualitative test of vitamin C in different juices:

Juices	No. of juice need to make the indophenol colorless	Rank of Juice according to [vitamin C]	
1- ____________________		The highest	
2- ____________________		The medium	
3- ____________________		The lowest	

GOOD LUCK

LABORATORY REPORT
Enzymes

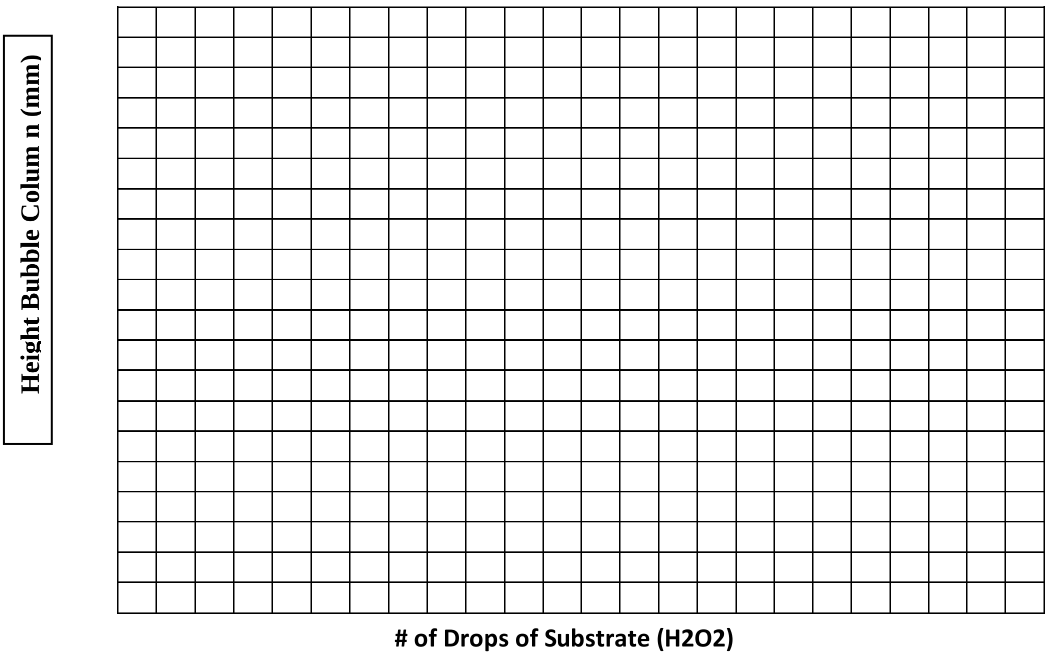
Name:______________________________________ **Student No.:**__________________________ **Section No.:** ____________

****

Table 4.1 Reaction Rates for Catalase at Various Substrate Concentrations

Test Tube No.	Potato extract (# drops)	H_2O_2 (# drops)	Height of the foam Column (mm)
1	15	0.0	
2	15	2	
3	15	4	
4	15	6	
5	15	8	
6	15	10	
7	15	12	
8	15	14	

Fig. 4.1 Reaction Rate of Catalase as a Function of Substrate Concentration

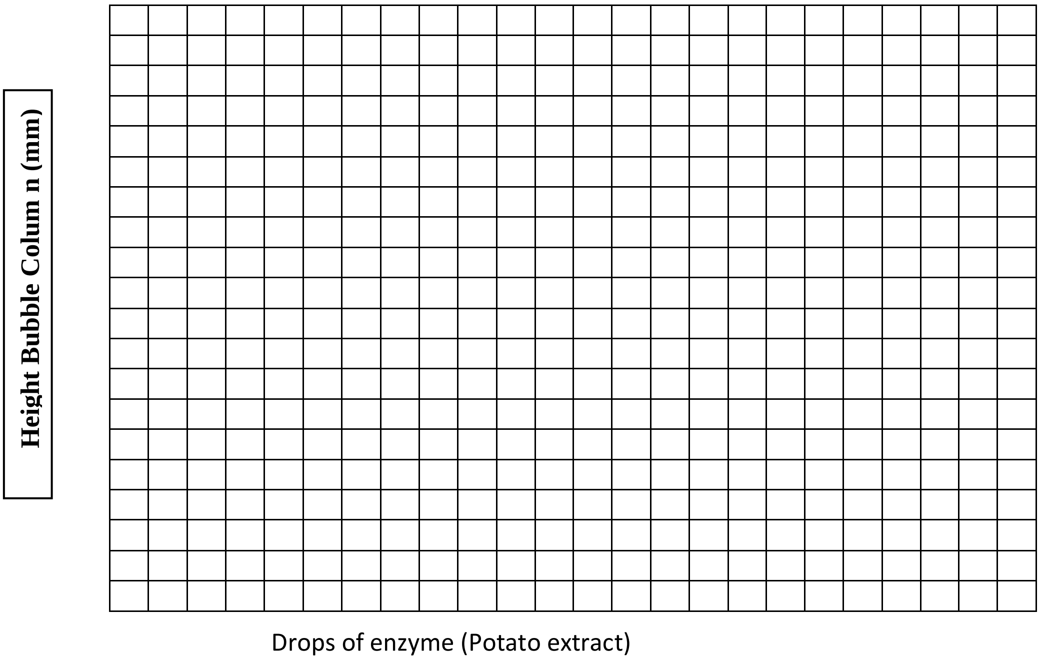


Make a general statement regarding the effect of substrate concentration on enzymatic reaction rate:

Table 4.2 Reaction Rate for Catalase at Various Enzyme Concentrations

Test Tube	H_2O_2 (# drops)	Potato extract (#drops)	Height of the foam Column (mm)
1	**20**	**0.0**	
2	**20**	**3**	
3	**20**	**6**	
4	**20**	**9**	
5	**20**	**12**	
6	**20**	**15**	

Fig. 4.2 Reaction Rate of Catalase as a Function of Enzyme Concentration

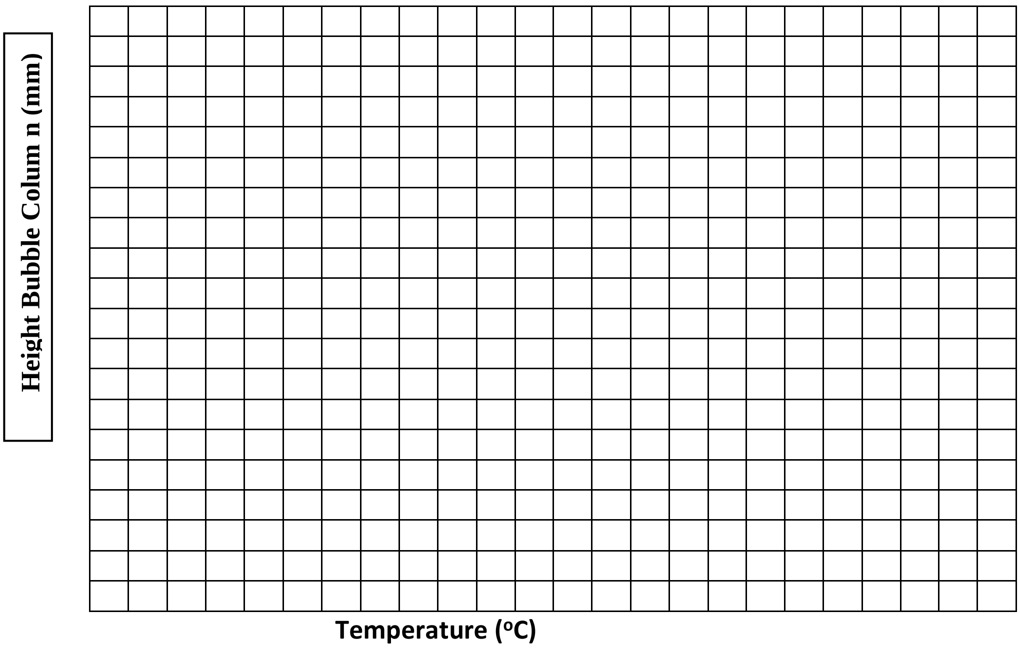


Make a general statement regarding the effect of enzyme concentration on enzymatic reaction rate:

Table 4.3 Reaction Rates for Catalase at Various Temperatures

Condition	Test Tube	H_2O_2 (# drops)	Test Tube	Potato extract (# drops)	Height of the foam after pouring the test tubes together
Ice pocket	1	**20**	6	5	
Refrigerator	2	**20**	7	5	
Room Temperature	3	**20**	8	5	
Water bath (50 ^{o}C)	4	**20**	**9**	5	
Boiling bath	5	**20**	**10**	5	

Fig. 4.3 Reaction Rate of Catalase as a Function of Temperature



Make a general statement regarding the effect of temperature on enzymatic reaction rate:

Fig. 4.4 Reaction Rate of Catalase as a Function of pH V

Test Tube	10 drops of	Potato extract (# drops)	After 10 minutes	H_2O_2 (#drops)	Height of the foam Column (mm)
1	Lemon juice	15		5	
2	H_2O	15		5	
3	NaOH	15		5	

Make a general statement regarding the effect of pH on enzymatic reaction rate:

GOOD LUCK

LABORATORY REPORT

Cellular Respiration and Fermentation Laboratory

Name:__________________________________ **Student No.:**________________________ **Section No.:** ___________

****

A. Carbon Dioxide (CO_2) Liberation exercise:

1- Record the color and the changes in the exhale and inhale bottles.

____

2- Explain your observation and Write the chemical equation of the reaction.

____

____

B. Water liberation

1- Record what happened on the surface of the glass test tube after you exhale in it several times, explain your observation.

____

C. Heat of Respiration

1- Record the temperature in each chamber at the end of your lab and explain the differences in the temperature reading.

____

D. Dehydrogenase activity

1- Record the results of your exercise by filling the table below:

Tube #	Contents	Time for methylene blue to go colourless
1	Glucose + dye + yeast	
2	Glucose + dye + boiled yeast	
3	Glucose + dye + buffer	

1- Record the results of your exercise by filling the table below:

Tube #	Contents	Final Air space length (mm) in the vial	Color change
1	Water + yeast		
2	Glucose + yeast		
3	Sucrose + yeast		
4	Starch+ yeast		

2- Explain what gas was accumulated in the vials? Why the color of the dye changed?.

____

____

GOOD LUCK

LABORATORY REPORT
Diffusion and Osmosis

Name:________________________________ **Student No.:**______________________ **Section No.:** ___________

****

Part A: Gas- Gas Diffusion

$$HCl\,(g) + NH_3(g) \rightarrow \_\_\_\_\_\_\_\_\_\_$$

a. The faster diffusing gas is:____

b. The precipitate was formed closest to ________________Explain your reasoning ______________

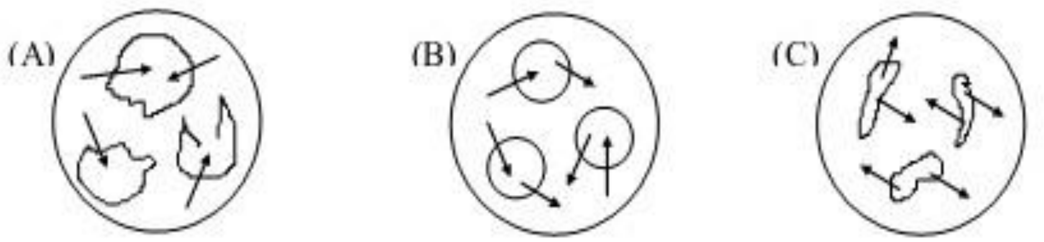
____

PART B: Dialysis tubing bag

Substance	Color		Direction of Net Movement	Reason for the net movement of substance
	Bag	Beaker		
Chloride ion				
Egg albumin				
Sulfate ion				
Starch				

PART C: Osmotic Behavior of animal Cells.

Given the following three microscopic fields containing animal cells; identify which field contains each solution indicated below. Use the letters of the fields of view as your choices and place them in the blanks.



a. Which field contains a hypertonic extracellular solution? __________

b. Which field contains an isotonic extracellular solution? ___________

c. Which field contains a hypotonic extracellular solution? __________

PART D: Osmotic Behavior of plant Cells (Onion Cells) under oil immersion lens { 100 X)

Onion cells appearance		
0.85 % NaCl	**Distilled water**	**10% NaCl**

GOOD LUCK

LABORATORY REPORT

Mitosis and Meiosis

Name:___________________________________ **Student No.:**_______________________ **Section No.:** ___________

****

*** Study the Onion Root tip then draw the following and label each part:** Draw as you see under the Microscope only

Interphase	Prophase	Metaphase

Anaphase	Telophase	Cytokinesis

- **What are the function of Mitosis and Meiosis?**

Mitosis	Meiosis
____	____
____	____
____	____

- **List six differences between Mitosis and Meiosis:**

Mitosis	Meiosis
1. ______________________	1. ______________________
2. ______________________	2. ______________________
3. ______________________	3. ______________________
4. ______________________	4. ______________________
5. ______________________	5. ______________________
6. ______________________	6. ______________________

- **Compare between Spermatogenesis and Oogenesis:**

According to	Spermatogenesis	Oogenesis
Cell division		
Human sex		
Primary cell		
No. of chromosomes in Primary cells		
End products		
No. of chromosomes in each cell produced		
Genetic Variation		

GOOD LUCK

LABORATORY REPORT
Human genetics

Name:__________________________________ **Student No.:**________________________ **Section No.:** ___________

****

A- Single Gene Traits

Single Gene traits.			
Dominant	Symbol of Recessive	Phenotype	Genotype (there may be several)
(T) **Tongue roller**	T		
(W) Widow's Peak	W		
(A) **Attached earlobes**	A		
(F) **Freckles**	F		
(H) **Has dimples**	H		
(C) **Color blind**	C		
(L) **Cleft chin**	L		
(I) **Hitchhiker's thumb**	I		
(B) **baldness (Hair loss)**	b		
(E) **Bent Pinky**	E		
(M) **Mid-digital hair**	M		
(X) Index finger shorter than ring finger	x		

B- Blood Type	
Your blood type	
Possible genotypes of your Blood	
Type(s) of blood could you safely receive in a transfusion	
Type(s) of blood could you safely donate in a transfusion	
Antigen in your Blood	

- What is the cross match of the blood?

____

____

GOOD LUCK

LABORATORY REPORT

Plant Tissues and Organ

Draw only as you see under the Microscope

Name:________________________________ Student No.:______________________ Section No.: ___________

****

Dicot ROOT

Monocot ROOT

ROOT HAIR

Stele of Dicot ROOt

Stele of Monocot ROOt

Dicot LEAF

FLOWER

LABORATORY REPORT

Plant Tissues and Organ

Draw only as you see under the Microscope

Name:________________________________ **Student No.:**______________________ **Section No.:** ___________

****

Dicot STEM	Monocot STEM	Old Woody STEM

Vascular bundle of dicot STEM	Vascular bundle of Monocot STEM

GOOD LUCK

LABORATORY REPORT

Animal Tissue

Draw only as you see under the Microscope

Name:________________________________ Student No.:______________________ Section No.: ___________

****

Epithelium Tissue

Simple Columnar (Villus of Illeum)	Simple Cuboidal (Kidney Tubule)	Pseudostratified (Trachea)

Epithelium / Connective tissue

Stratified Squamous	Reticular C. T.	Adipose C.T.

Connective Tissue

Bone C.T	Blood C.T.	Hyaline Cartilage

LABORATORY REPORT
Animal Tissue

Draw only as you see under the Microscope

Name:______________________________ Student No.:____________________ Section No.: __________

****

Human Muscle

Skeletal Muscle	Cardiac M	Smooth M

Nervous Tissue

Spinal Cord

GOOD LUCK

LABORATORY REPORT
Histology of Human organs

Draw only as you see under the Microscope

Name:________________________________ Student No.:______________________ Section No.: ___________

Human Skin	Human Intestine	Human Kidney
Human Liver	Human Trachea	Human Spinal Cord

White Matter	Gray Matter

GOOD LUCK